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Andrabi et al.

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(54) **SIV ENVELOPE TRIMER**

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(51) **Int. Cl.**

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A61K 9/127 (2006.01)
A61K 9/51 (2006.01)
A61K 39/00 (2006.01)
A61K 39/21 (2006.01)
A61K 39/39 (2006.01)
A61K 47/08 (2006.01)
A61K 47/10 (2017.01)
A61P 31/18 (2006.01)
C12N 7/00 (2006.01)
C12N 15/86 (2006.01)
G01N 33/569 (2006.01)

(52) **U.S. Cl.**

CPC **C07K 5/08** (2013.01); **A61K 9/127** (2013.01); **A61K 9/51** (2013.01); **A61K 39/21** (2013.01); **A61K 39/39** (2013.01); **A61K 47/08** (2013.01); **A61K 47/10** (2013.01); **A61P 31/18** (2018.01); **C12N 7/00** (2013.01); **C12N 15/86** (2013.01); **G01N 33/56988** (2013.01); **A61K 2039/545** (2013.01); **A61K 2039/55505** (2013.01); **A61K 2039/55555** (2013.01); **A61K 2039/55566** (2013.01); **C12N 2740/15022** (2013.01); **C12N 2740/15034** (2013.01); **C12N 2750/14143** (2013.01)

(58) **Field of Classification Search**

CPC **C07K 5/08**; **A61P 31/18**; **A61K 9/127**;
A61K 9/51; **A61K 39/21**; **A61K 39/39**;
A61K 47/08; **A61K 47/10**; **A61K**

2039/545; **A61K 2039/55505**; **A61K 2039/55555**; **A61K 2039/55566**; **C12N 7/00**; **C12N 15/86**; **C12N 2740/15022**; **C12N 2740/15034**; **C12N 2750/14143**; **G01N 33/56988**

See application file for complete search history.

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(57) **ABSTRACT**

The present application relates to epitope-targeted SIV and HIV vaccines. The invention provides novel envelope glycoproteins which may be utilized as HIV-1 vaccine immunogens, antigens for crystallization, and for identification of broadly neutralizing antibodies. The invention encompasses preparation and purification of immunogenic compositions which are formulated into vaccines of the present invention.

18 Claims, 62 Drawing Sheets
(57 of 62 Drawing Sheet(s) Filed in Color)
Specification includes a Sequence Listing.

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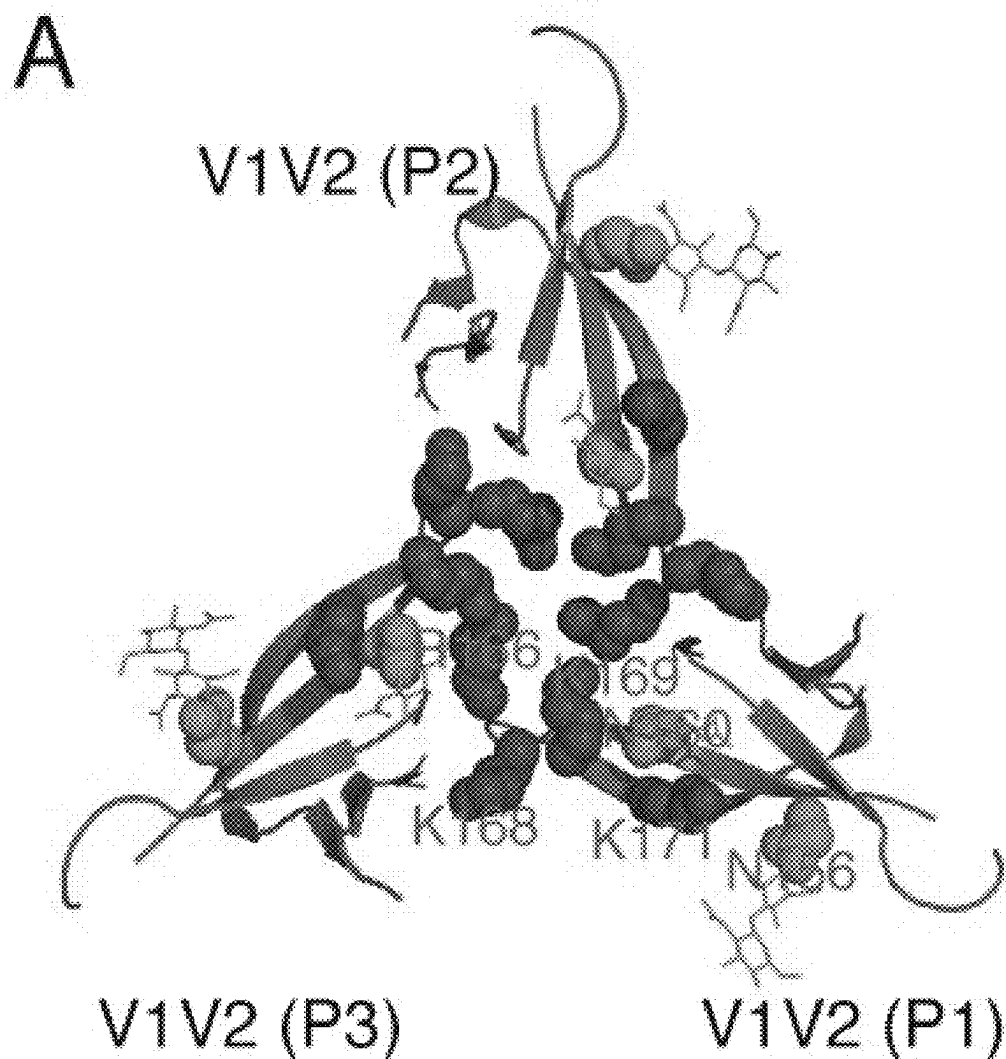
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B

	156	160	166	171
HXB2	NCSF	ISTSI	RGKVQ	KEYAFFY
CRF250	...	VT.EL.	D.KK.	...S...
CAP256.SU	...	AT.EL.	D.KK.	...L..
MT145-WT	...	VT.EL.	D.KRQV.	SL..
MT145K	...	VT.EL.	D.KR.V.	SL..

Fig. 1A-1B

C

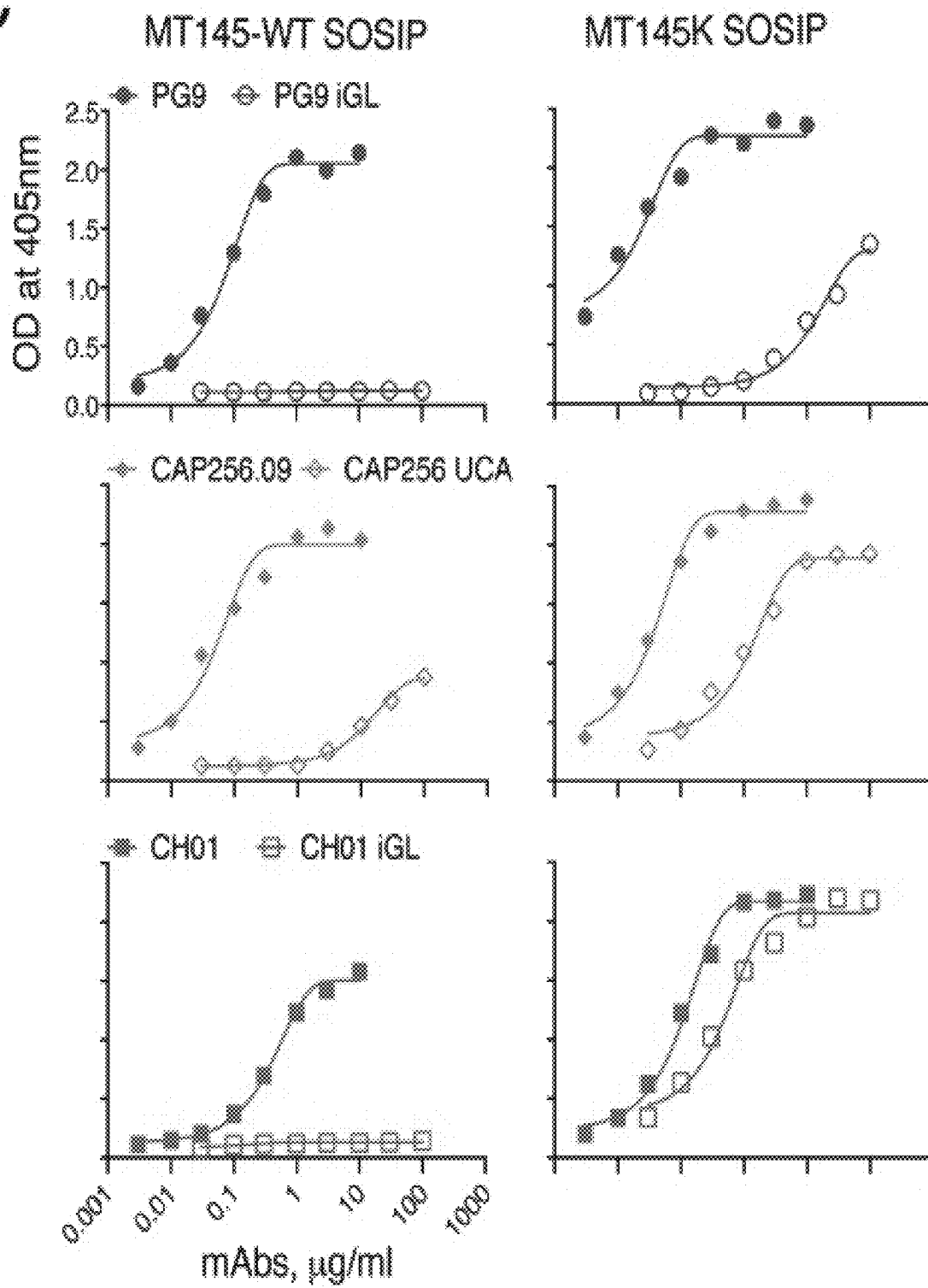


Fig. 1C

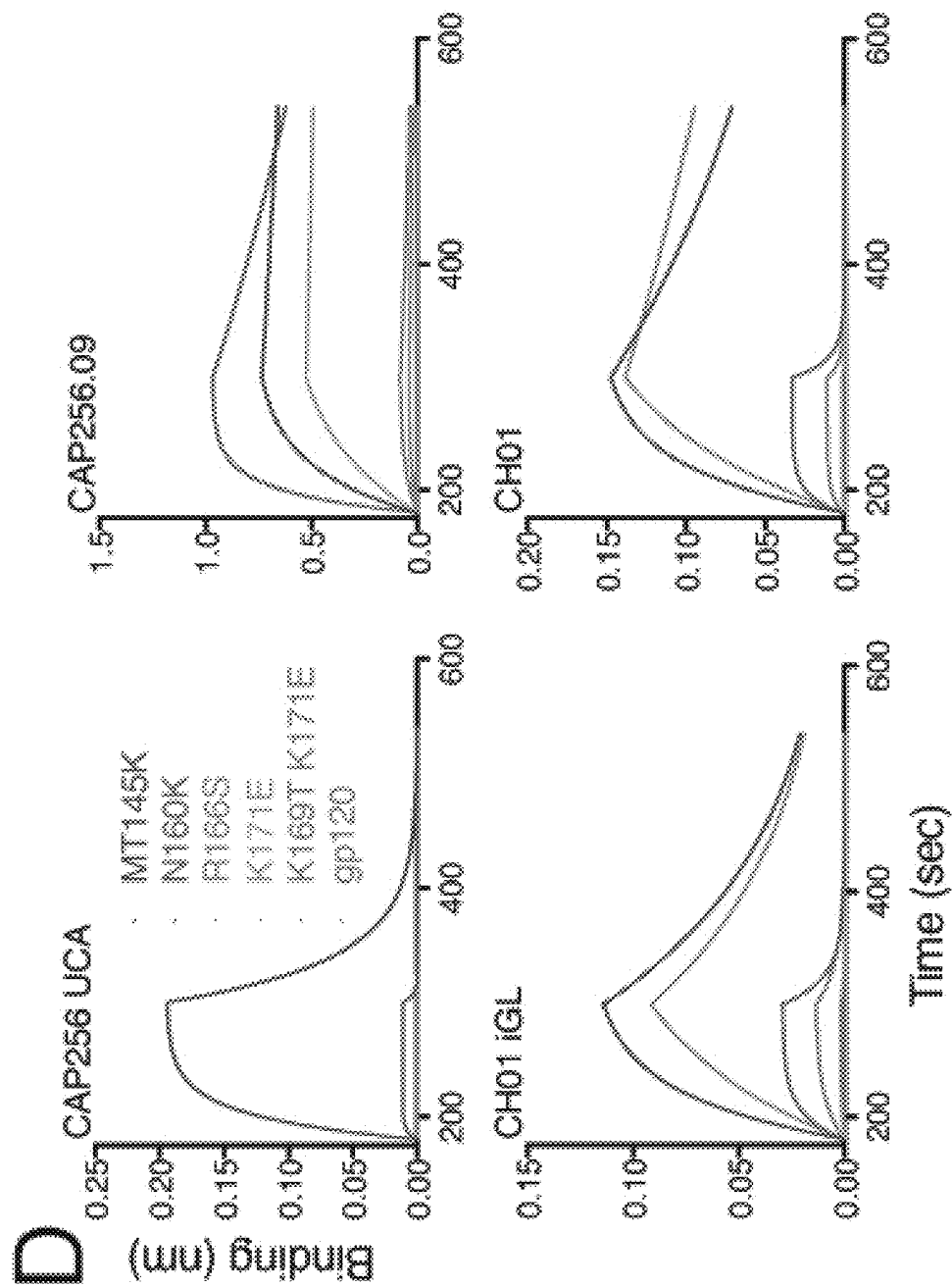


Fig. 1D

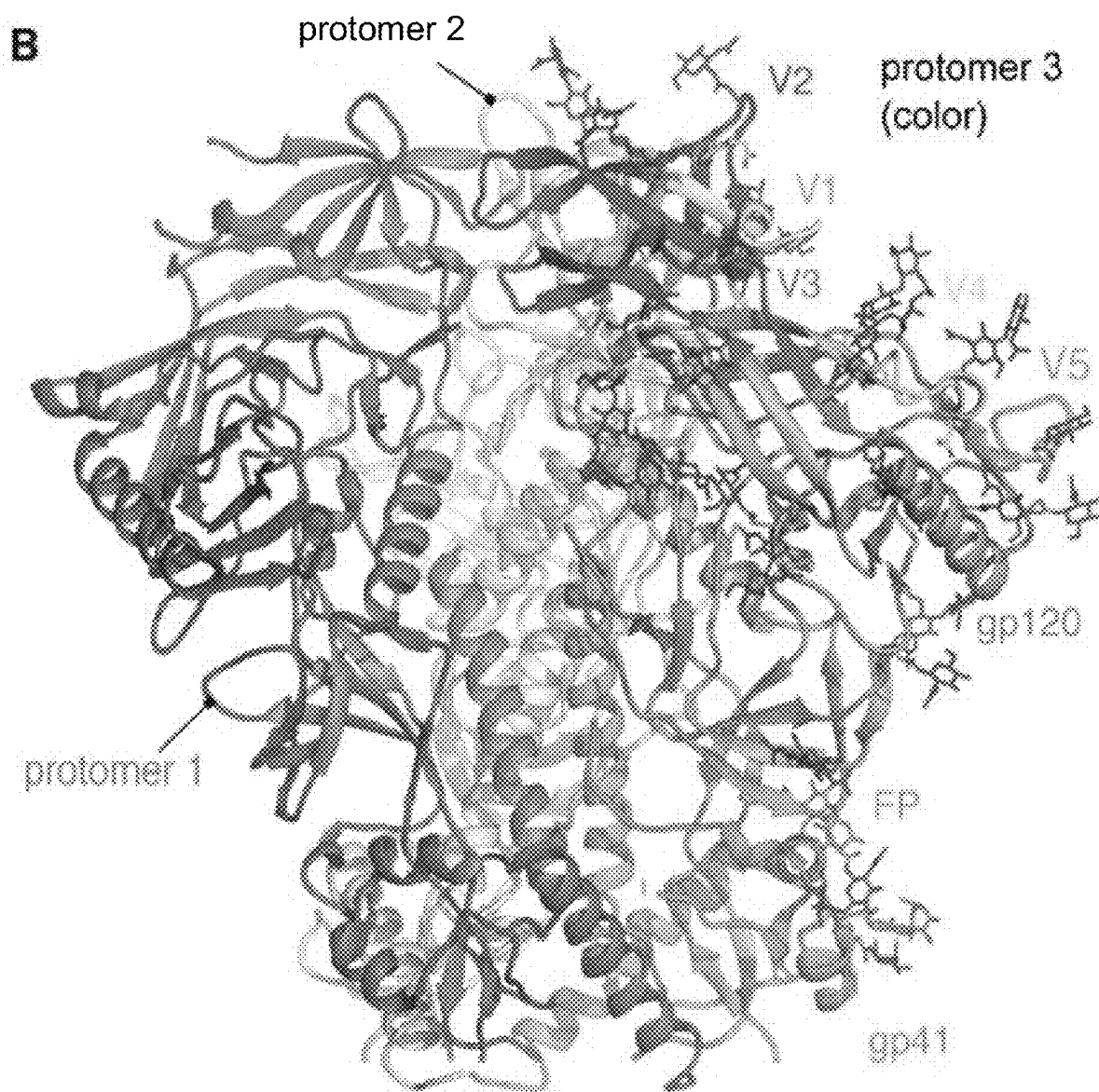


Fig. 2B

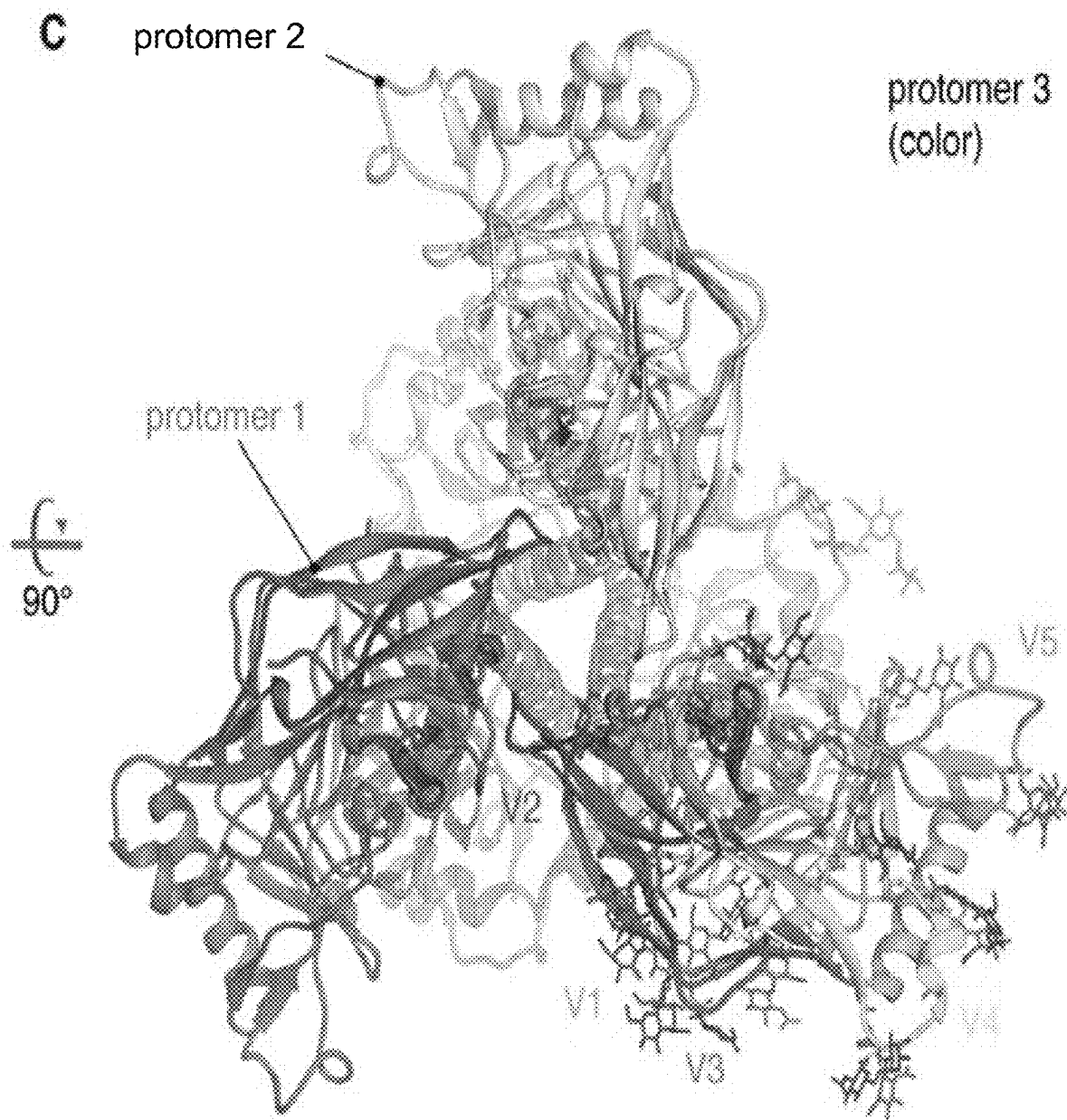


Fig. 2C

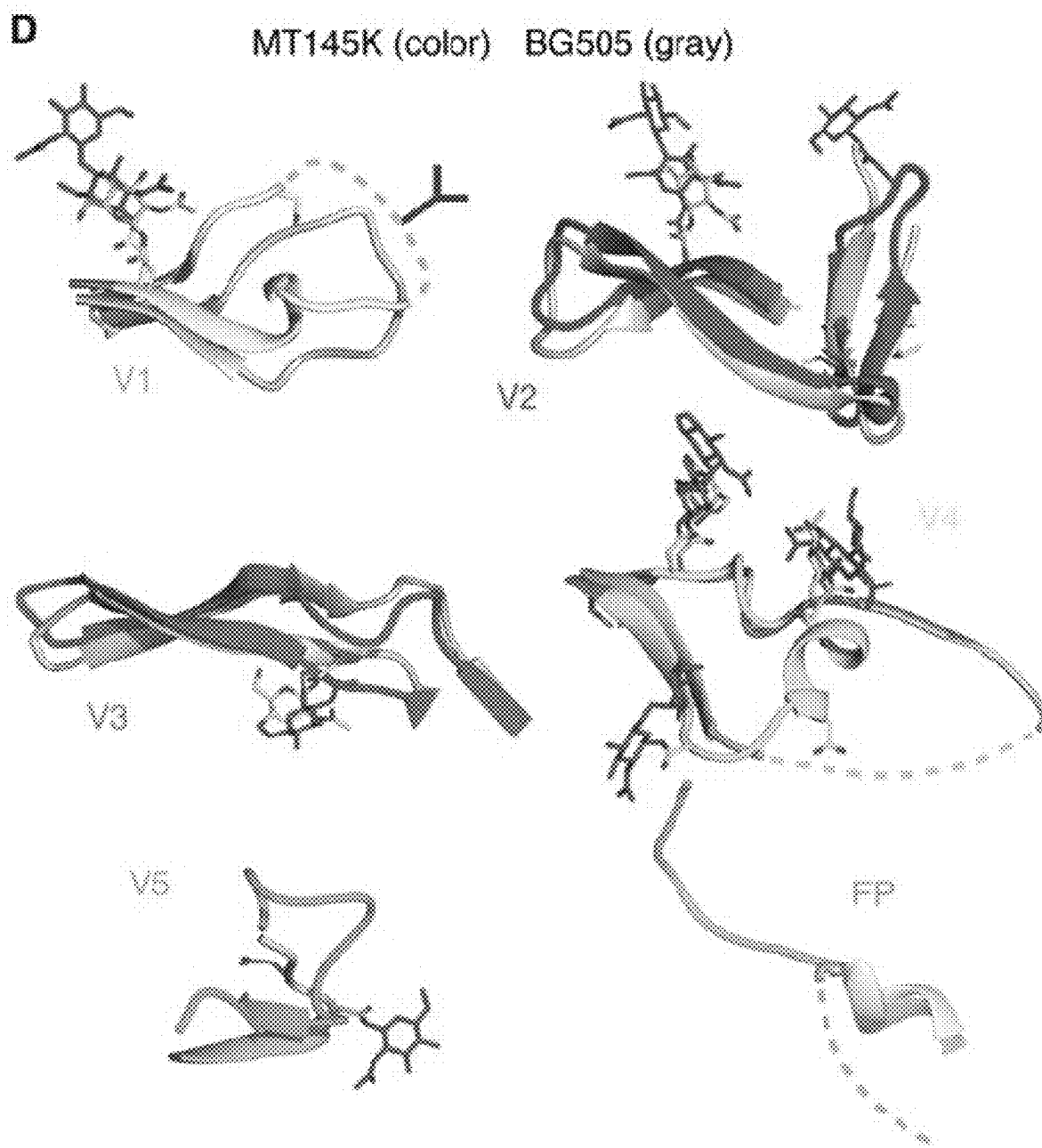


Fig. 2D

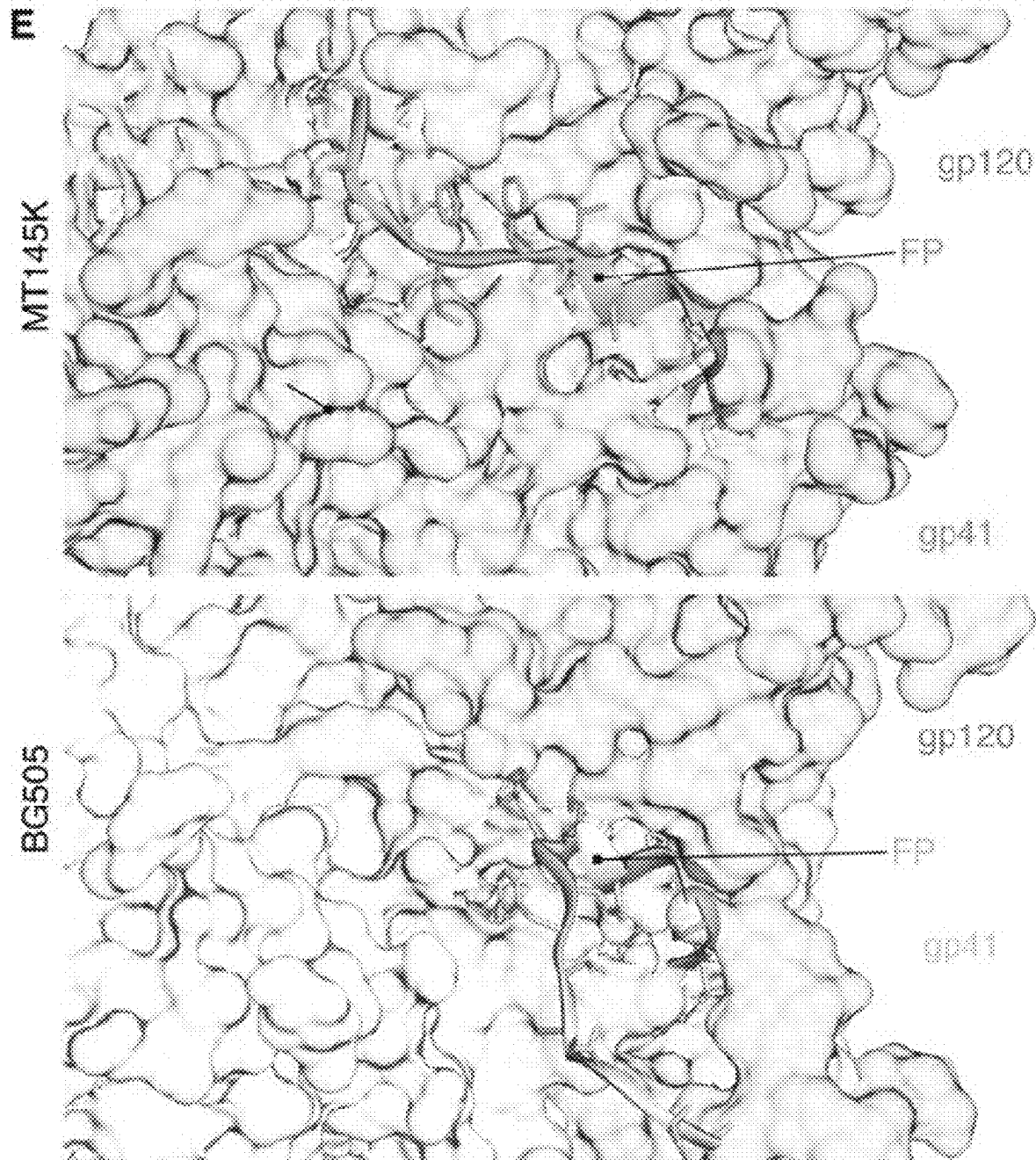


Fig. 2E

A

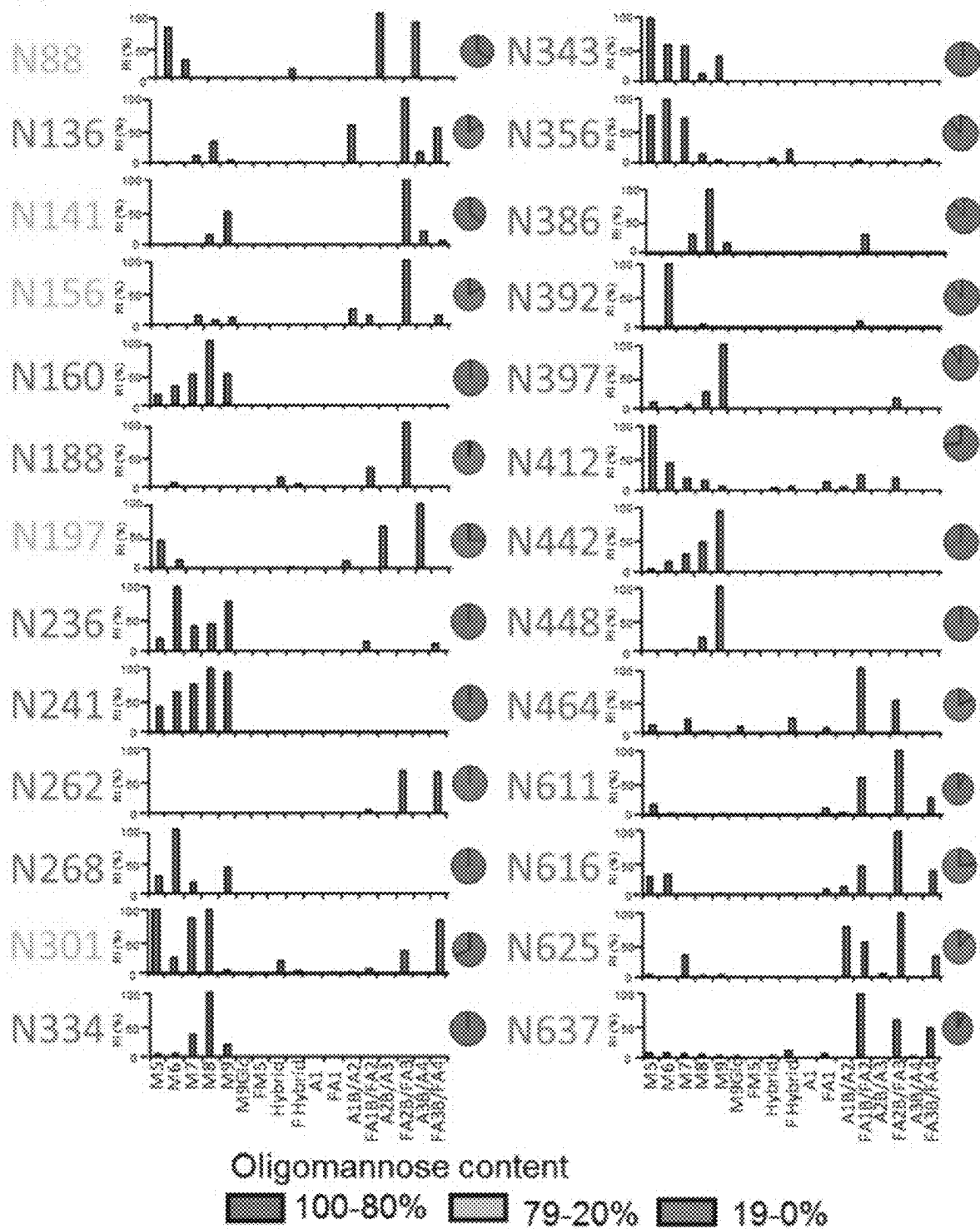


Fig. 3A

B

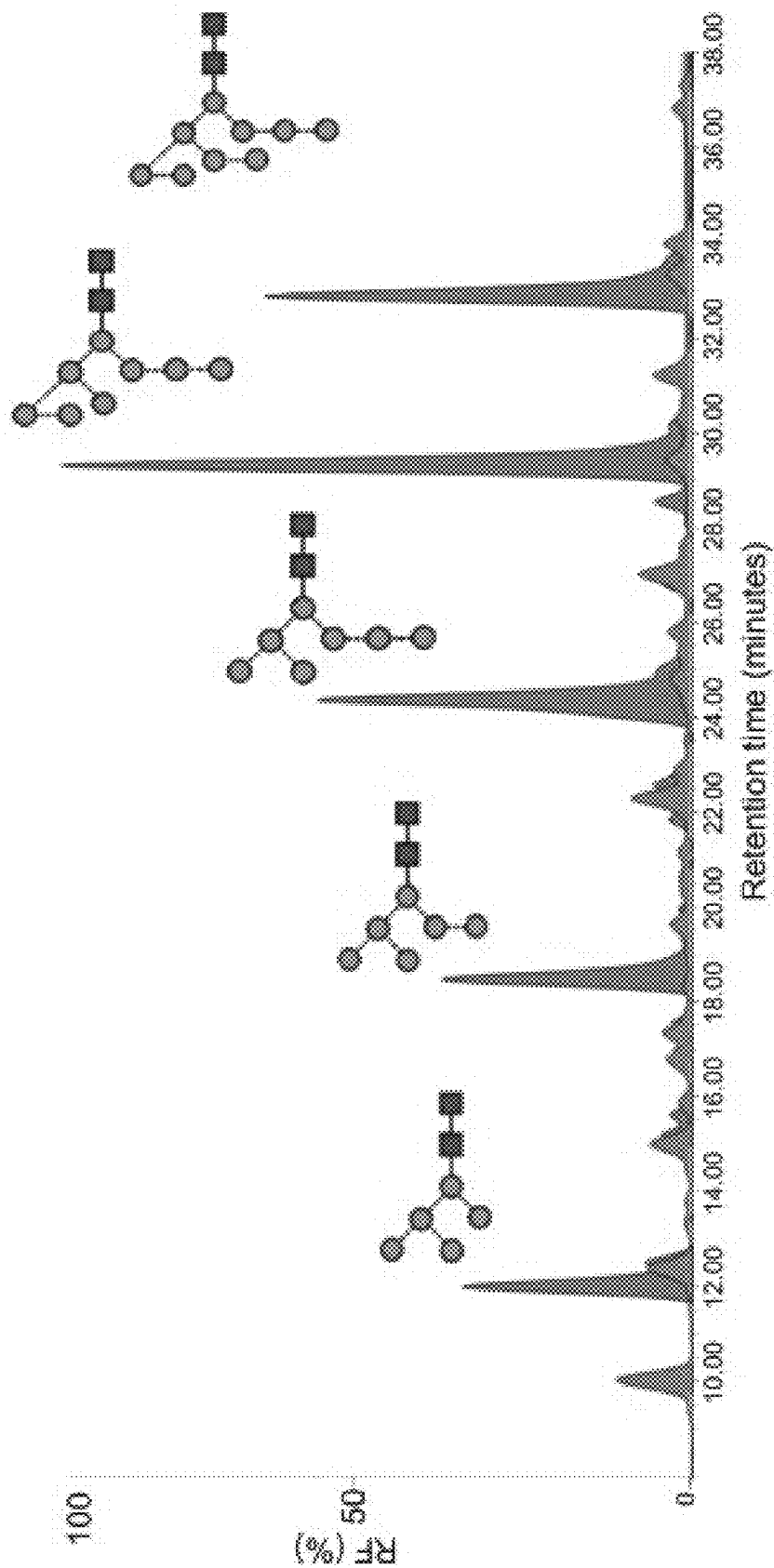
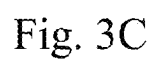


Fig. 3B



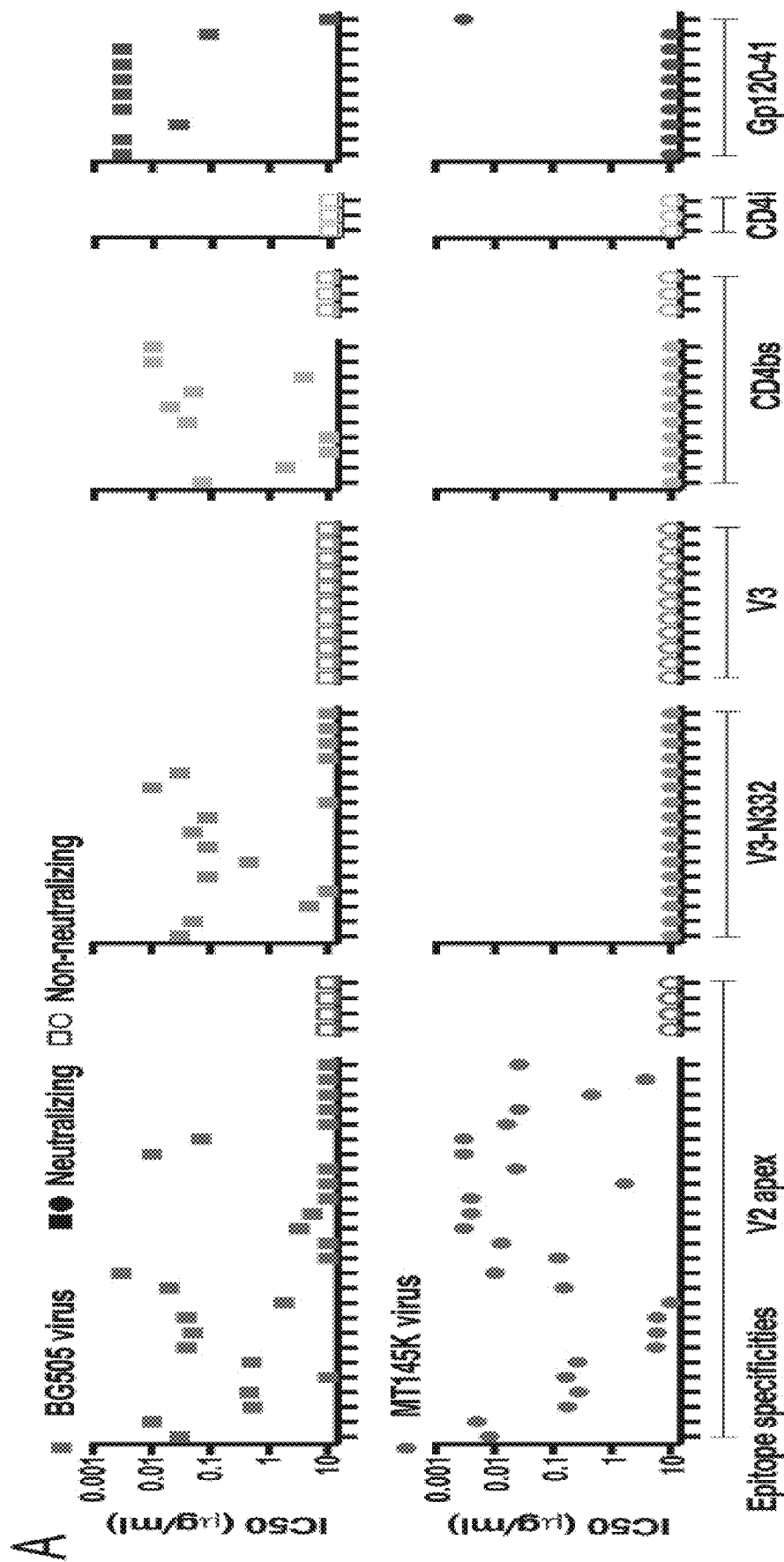


Fig. 4A

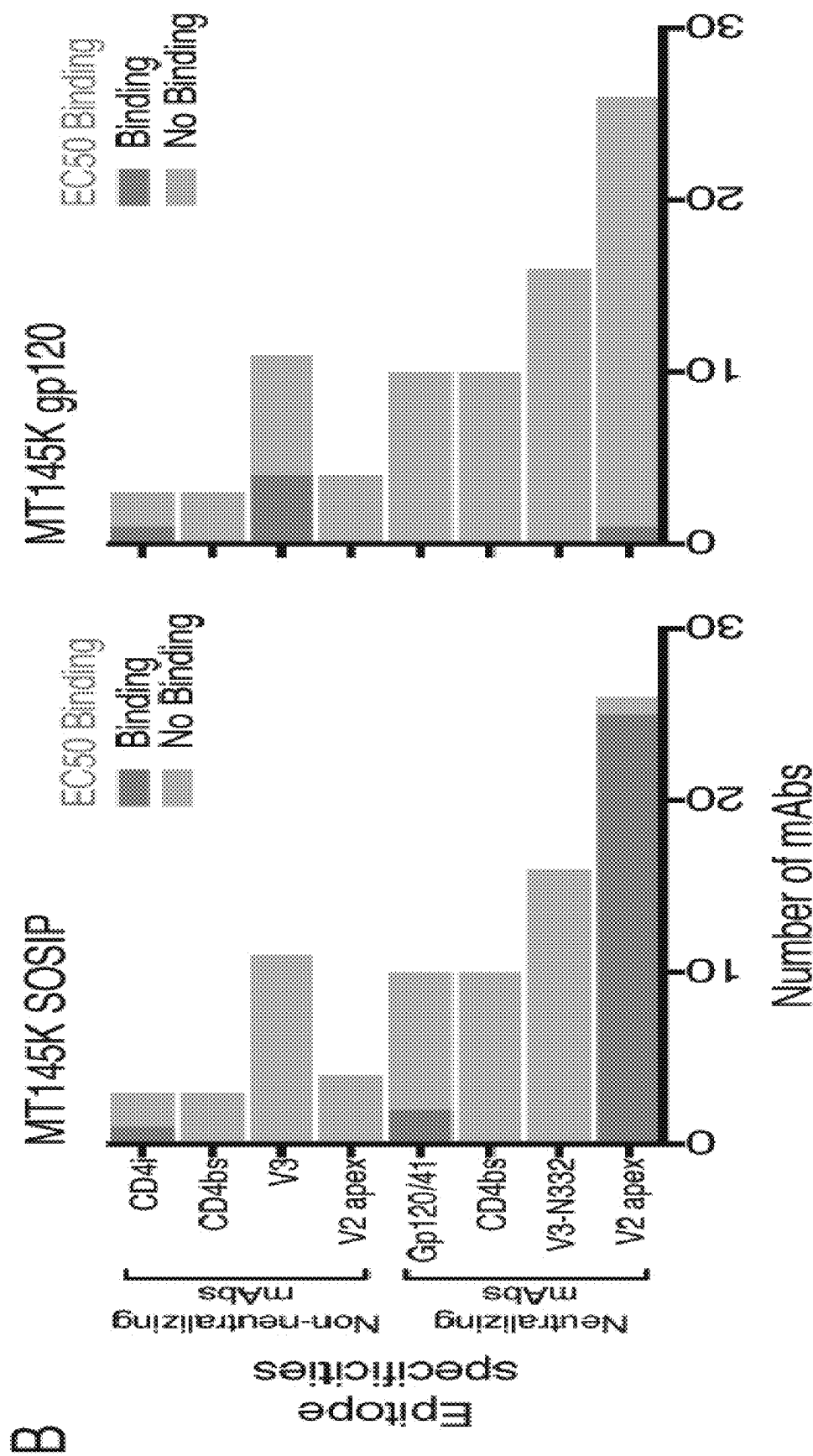


Fig. 4B

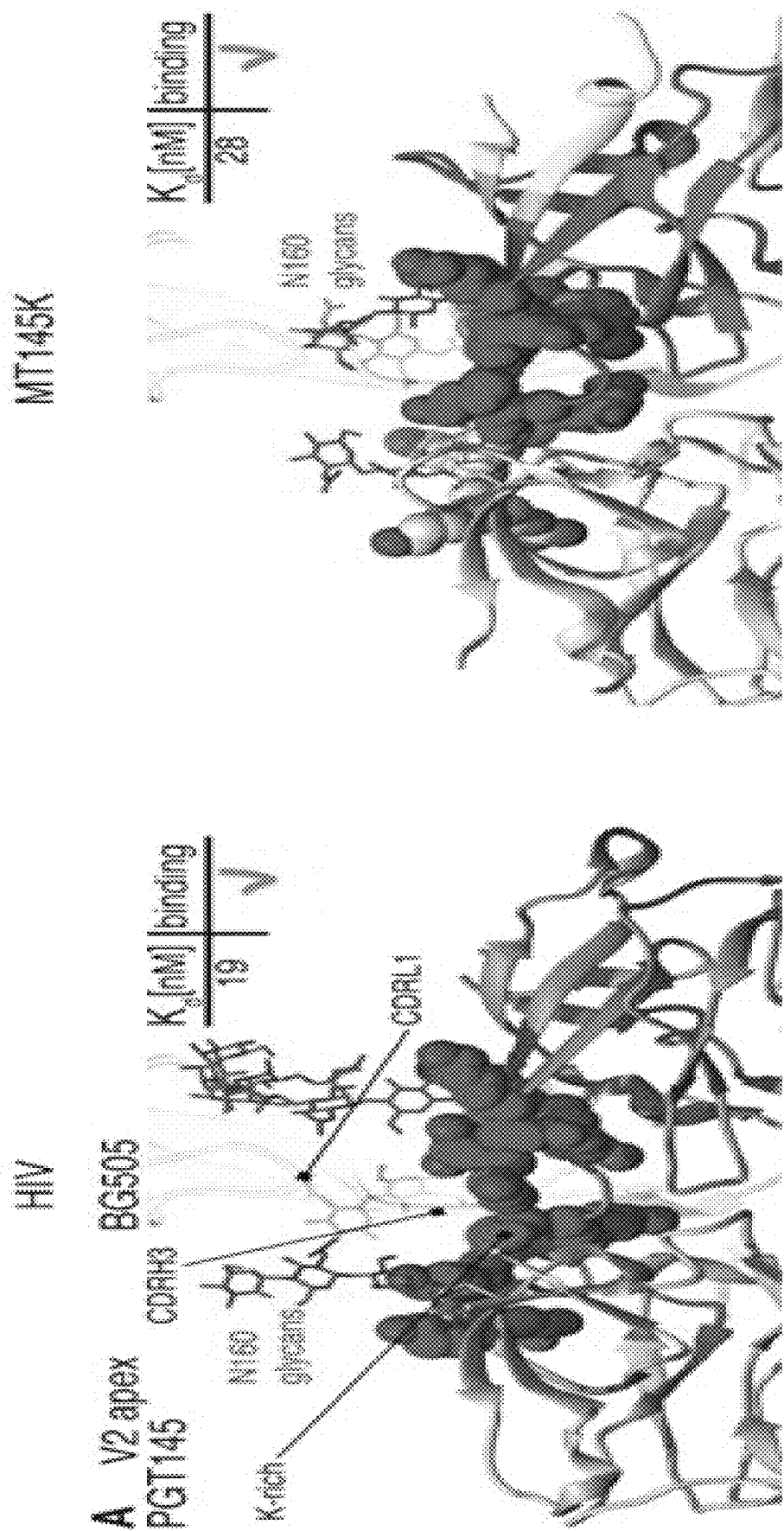


Fig.5A

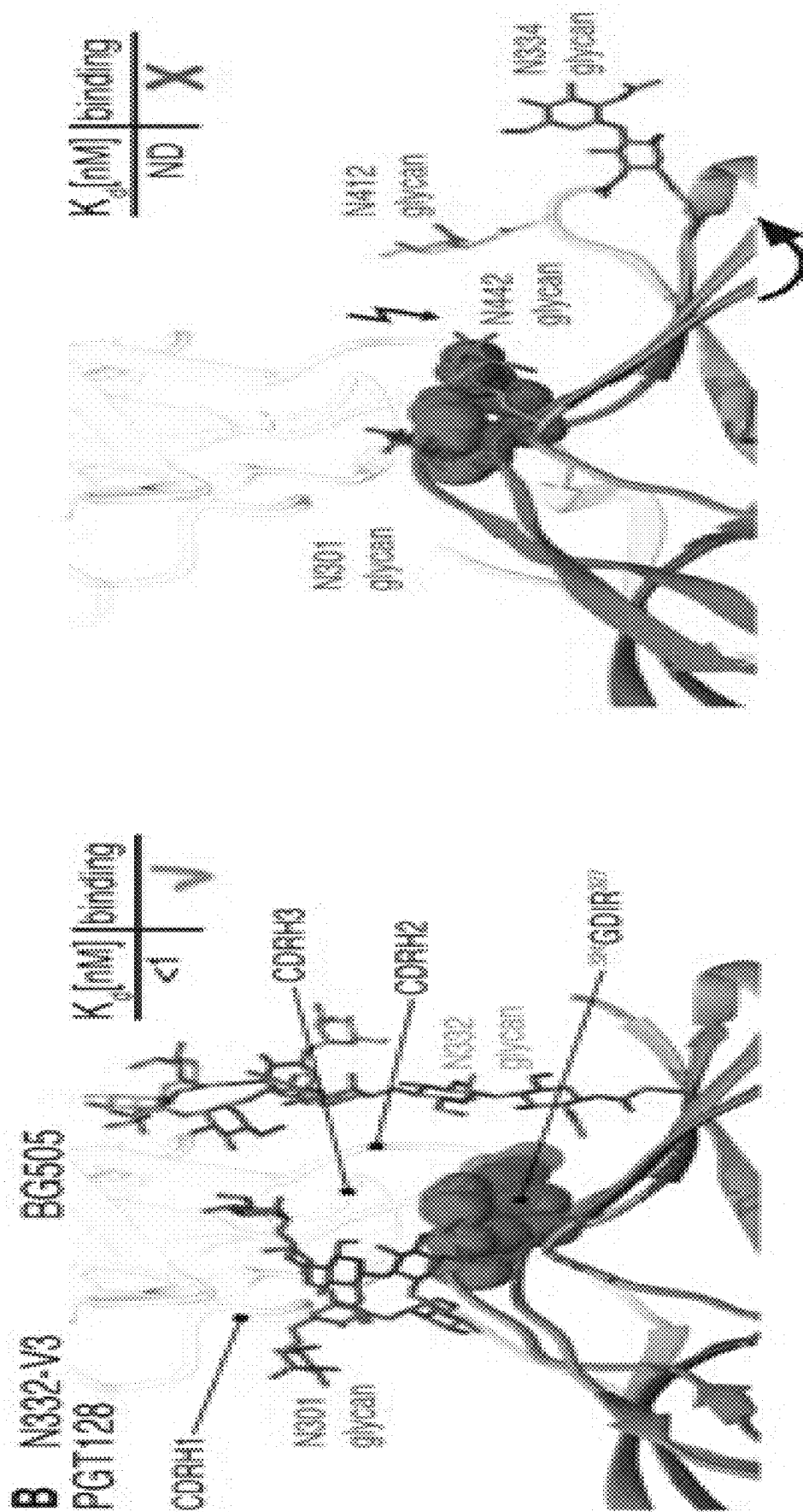


Fig. 5B

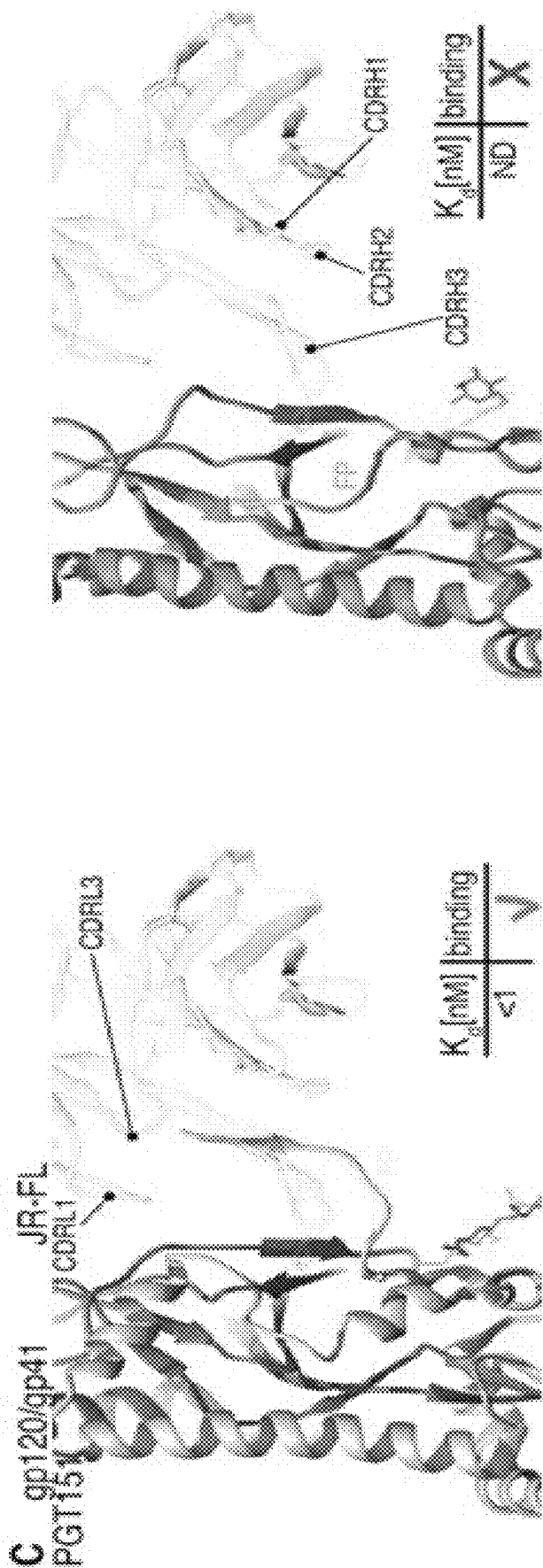


Fig. 5C

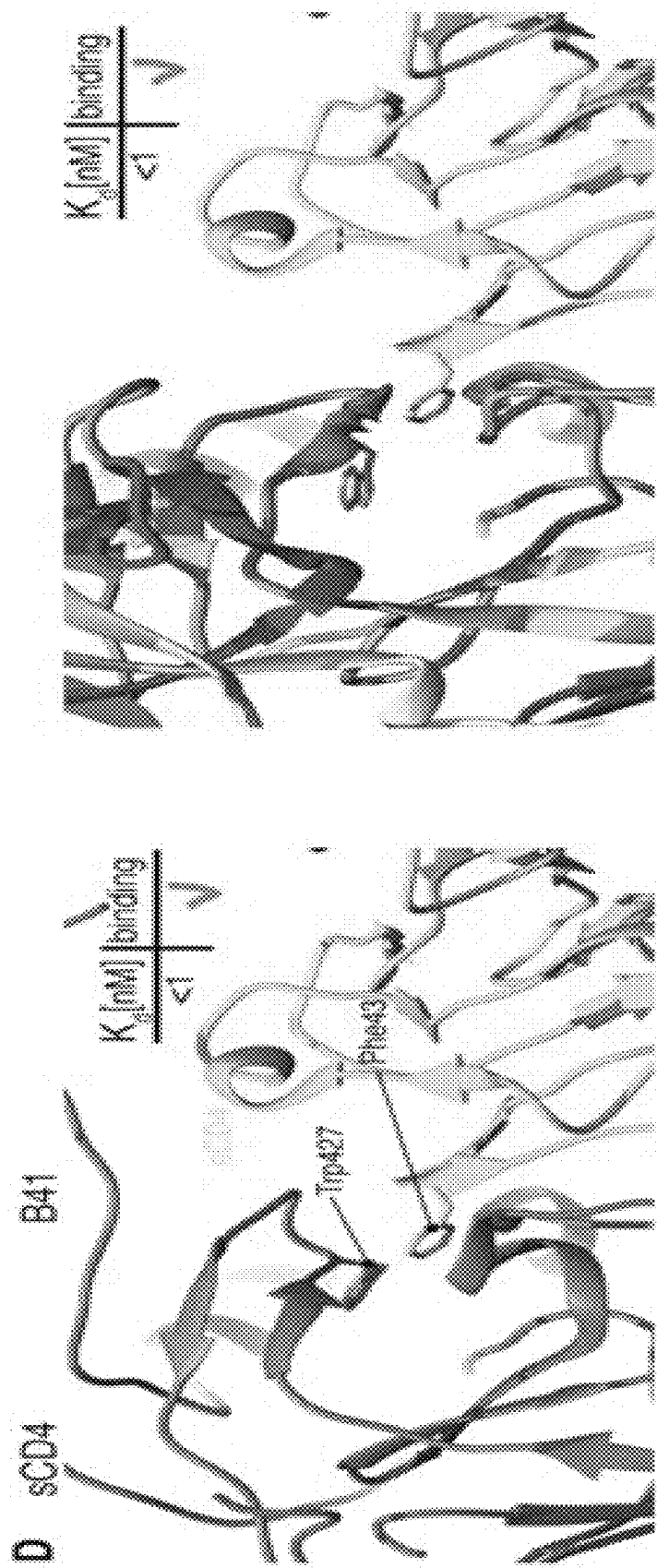


Fig. 5D

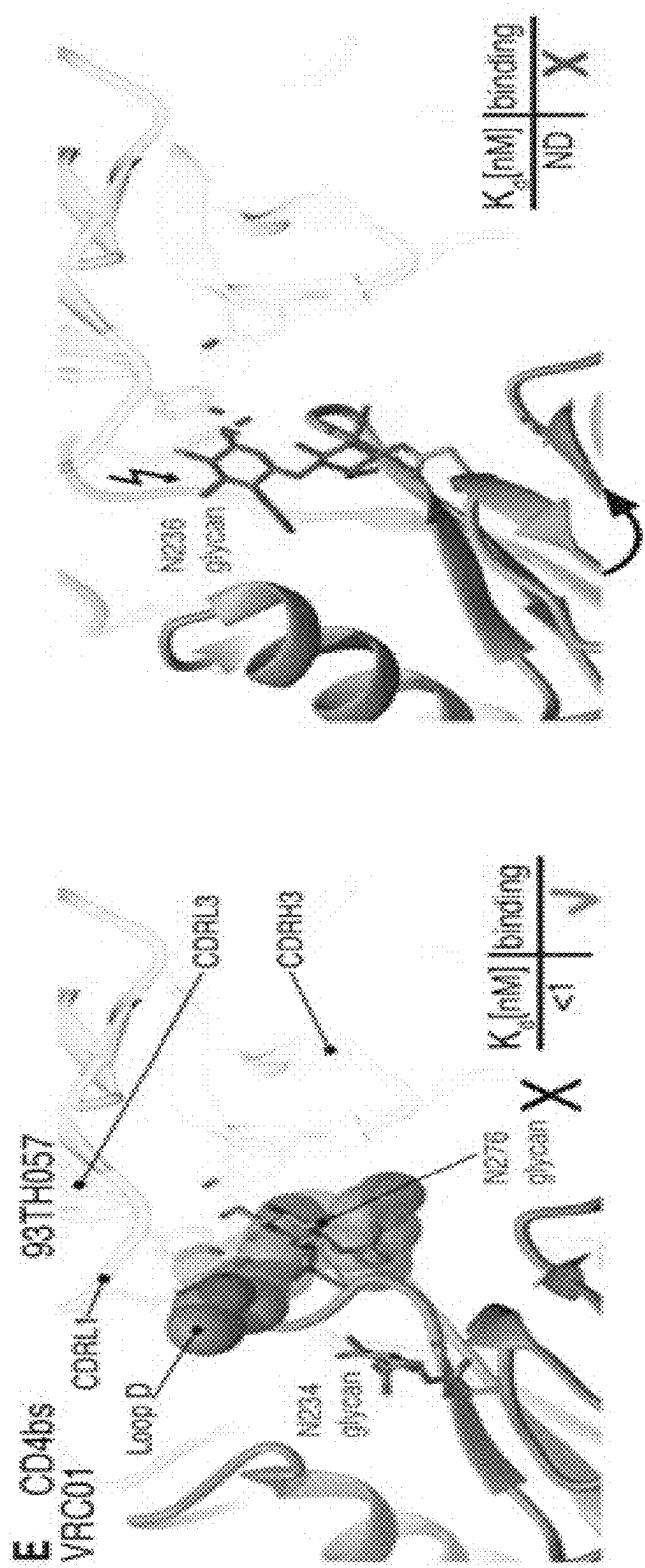


Fig. 5E

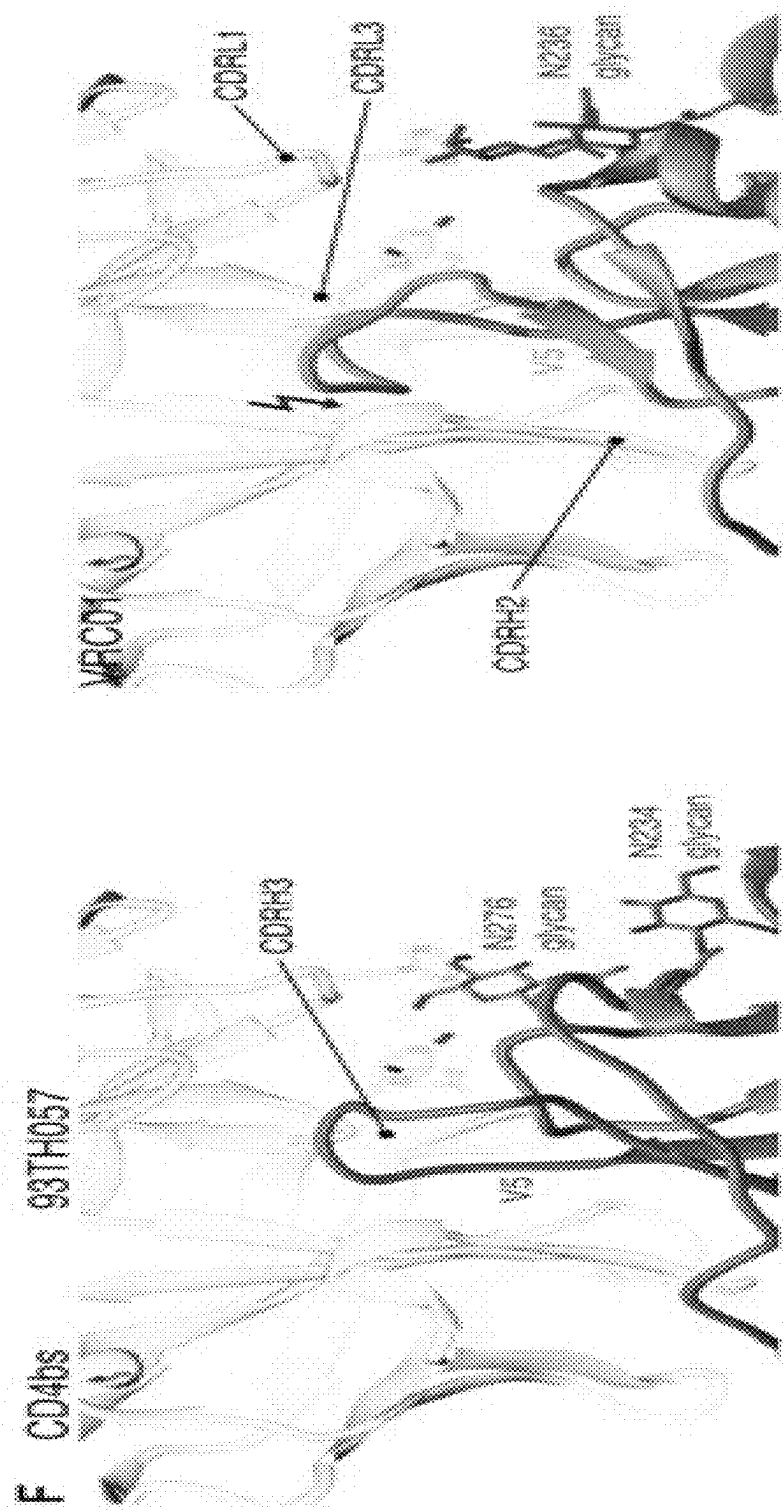


Fig. 5F

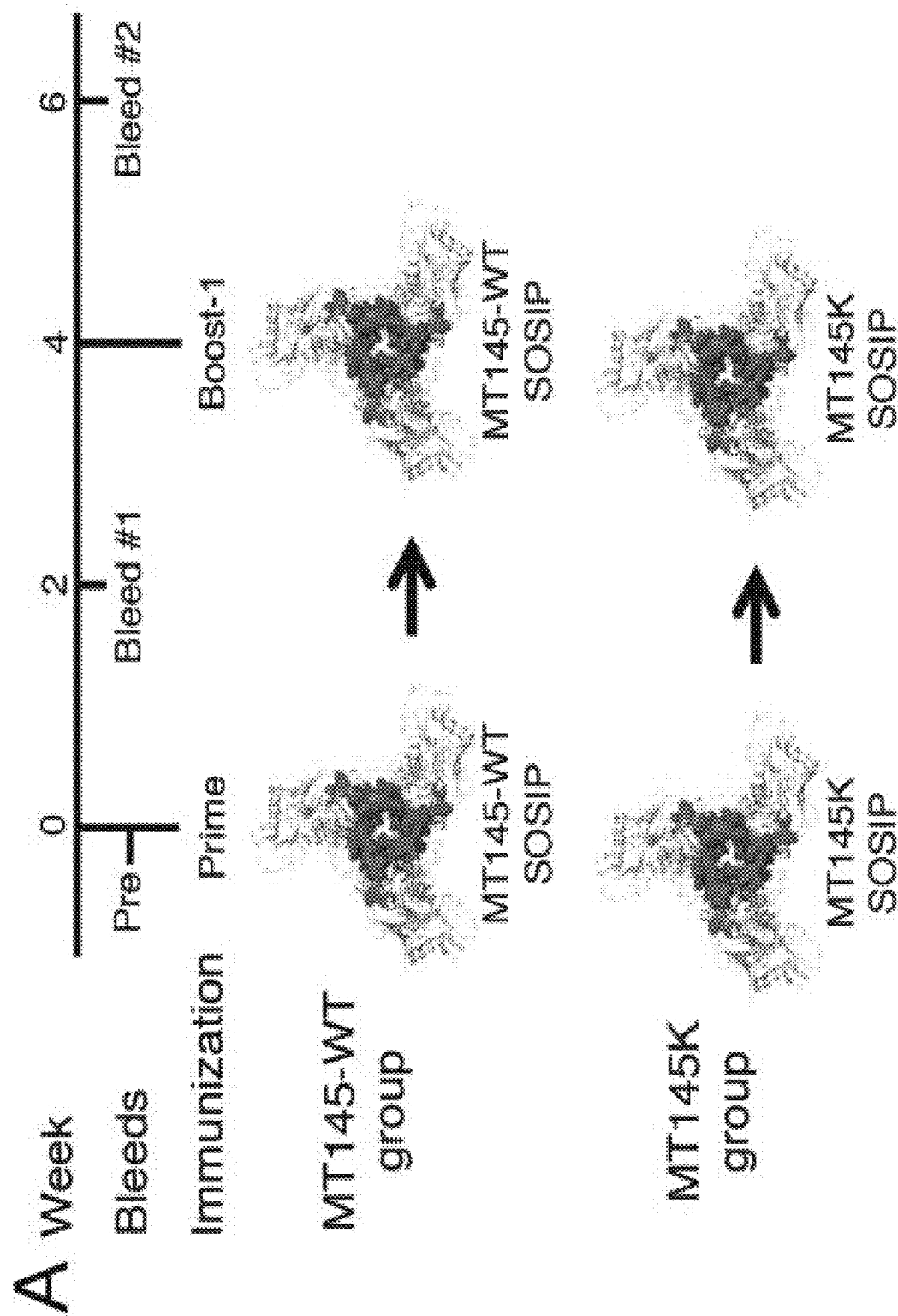


Fig. 6A

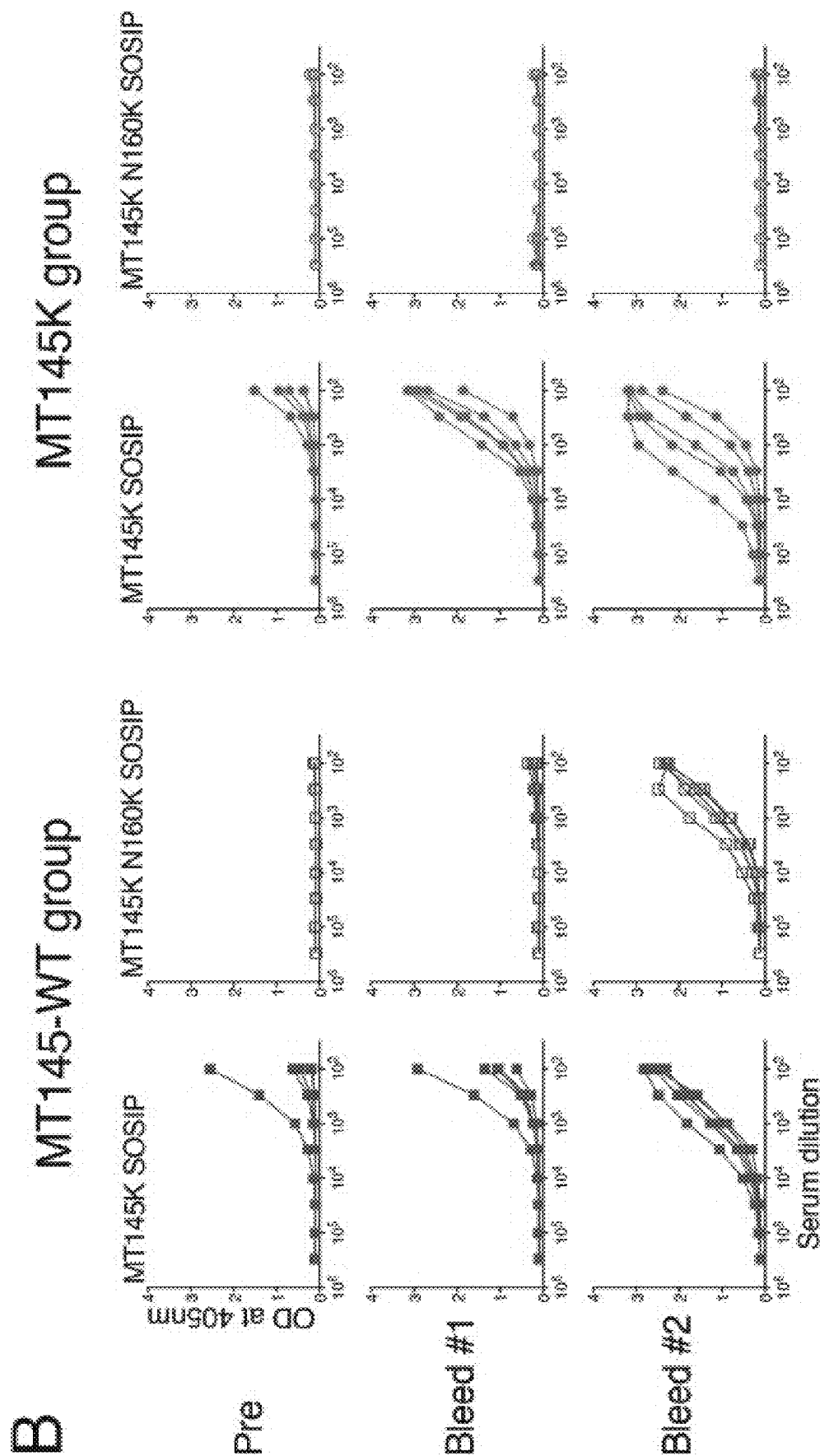


Fig. 6B

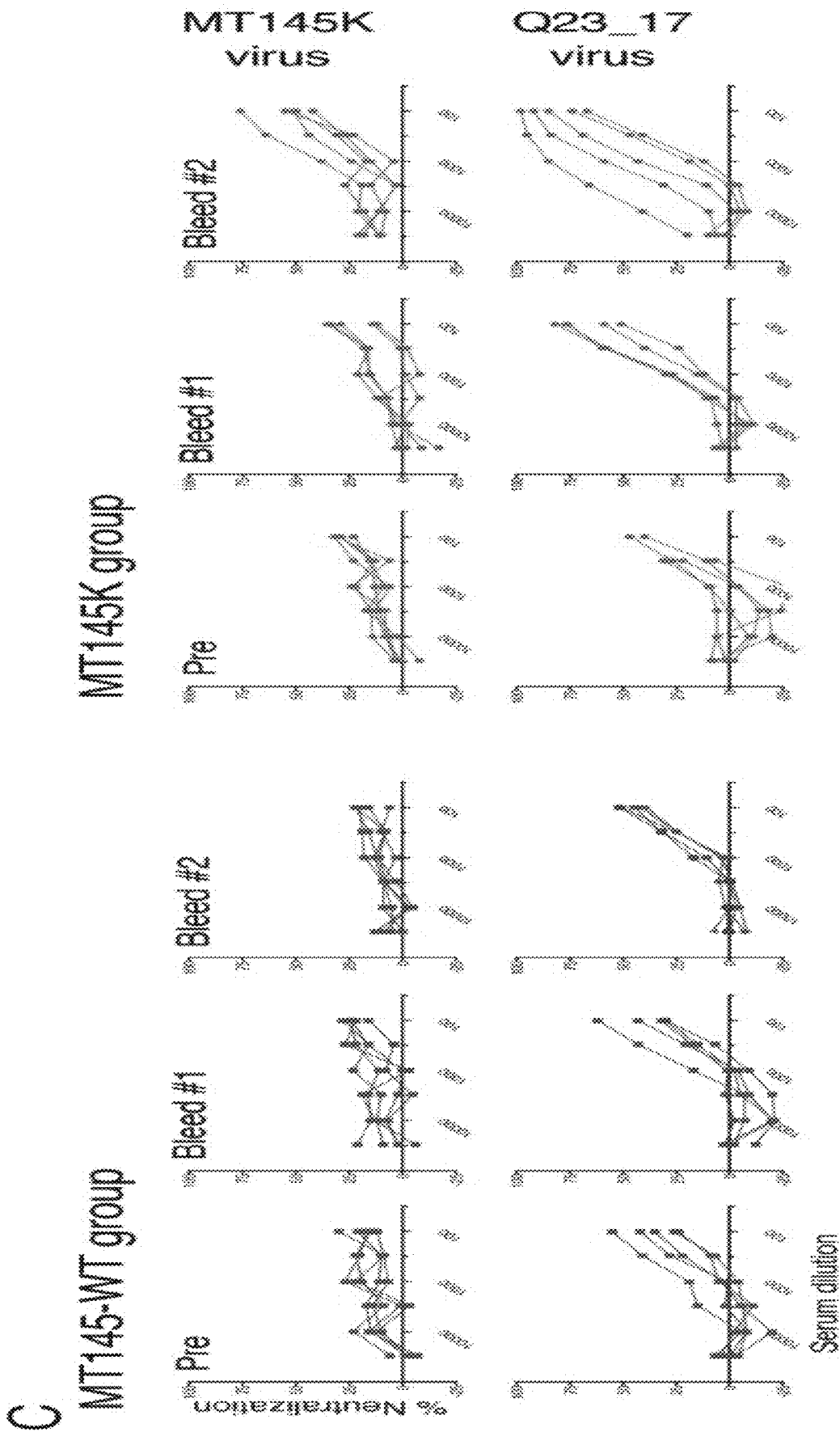


Fig. 6C

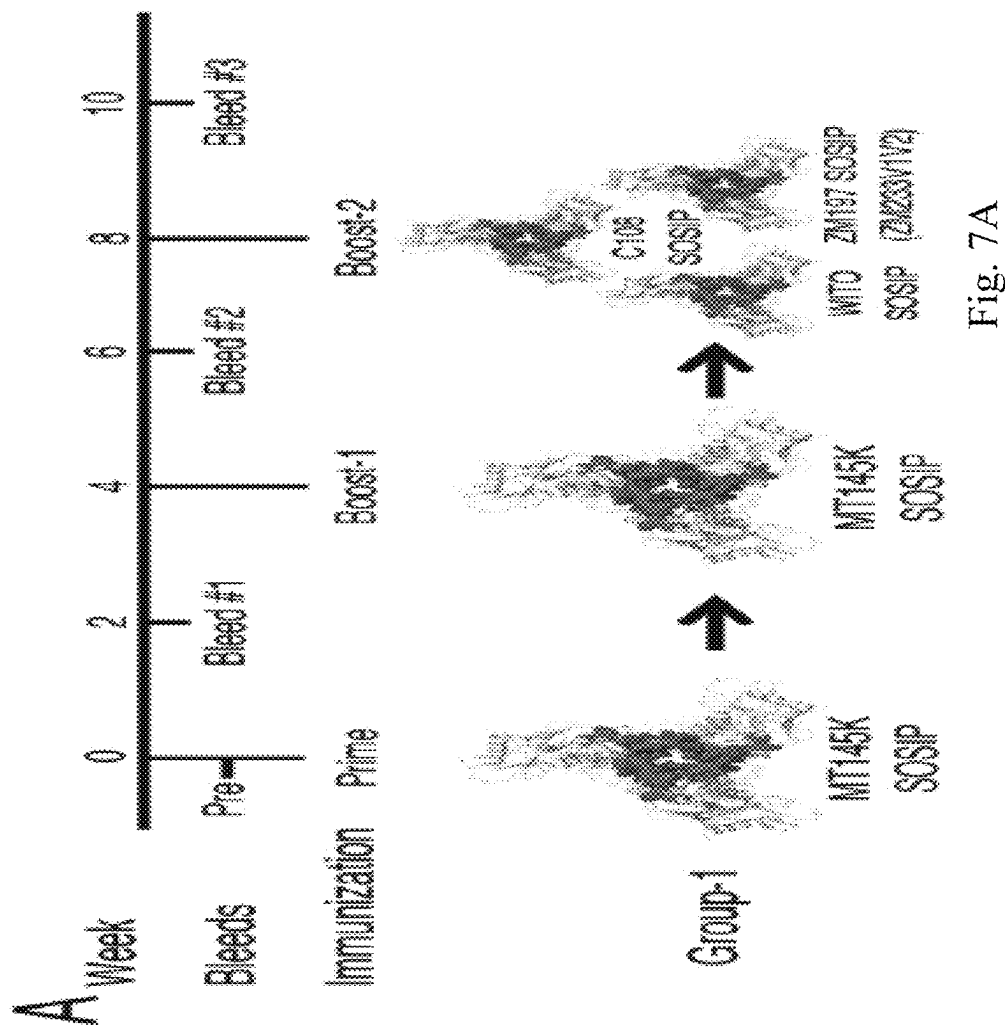
D

D

MT145-WT group															
	Pre-bleed					Bleed #1					Bleed #2				
	13315	13316	13317	13318	13319	13315	13316	13317	13318	13319	13315	13316	13317	13318	13319
Virus/mutant	13315	13316	13317	13318	13319	13315	13316	13317	13318	13319	13315	13316	13317	13318	13319
MT145K	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100
MT145K N160A	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100
MT145-WT	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100
Q23_17	<100	150	<100	<100	<100	<100	<100	<100	<100	160	<100	<100	<100	<100	<100

MT145K group															
	Pre-bleed					Bleed #1					Bleed #2				
	13200	13201	13202	13300	13301	13200	13201	13202	13300	13301	13200	13201	13202	13300	13301
Virus/mutant	13200 <td>13201<td>13202<td>13300<td>13301<td>13200<td>13201<td>13202<td>13300<td>13301<td>13200<td>13201<td>13202<td>13300<td>13301</td></td></td></td></td></td></td></td></td></td></td></td></td></td>	13201 <td>13202<td>13300<td>13301<td>13200<td>13201<td>13202<td>13300<td>13301<td>13200<td>13201<td>13202<td>13300<td>13301</td></td></td></td></td></td></td></td></td></td></td></td></td>	13202 <td>13300<td>13301<td>13200<td>13201<td>13202<td>13300<td>13301<td>13200<td>13201<td>13202<td>13300<td>13301</td></td></td></td></td></td></td></td></td></td></td></td>	13300 <td>13301<td>13200<td>13201<td>13202<td>13300<td>13301<td>13200<td>13201<td>13202<td>13300<td>13301</td></td></td></td></td></td></td></td></td></td></td>	13301 <td>13200<td>13201<td>13202<td>13300<td>13301<td>13200<td>13201<td>13202<td>13300<td>13301</td></td></td></td></td></td></td></td></td></td>	13200 <td>13201<td>13202<td>13300<td>13301<td>13200<td>13201<td>13202<td>13300<td>13301</td></td></td></td></td></td></td></td></td>	13201 <td>13202<td>13300<td>13301<td>13200<td>13201<td>13202<td>13300<td>13301</td></td></td></td></td></td></td></td>	13202 <td>13300<td>13301<td>13200<td>13201<td>13202<td>13300<td>13301</td></td></td></td></td></td></td>	13300 <td>13301<td>13200<td>13201<td>13202<td>13300<td>13301</td></td></td></td></td></td>	13301 <td>13200<td>13201<td>13202<td>13300<td>13301</td></td></td></td></td>	13200 <td>13201<td>13202<td>13300<td>13301</td></td></td></td>	13201 <td>13202<td>13300<td>13301</td></td></td>	13202 <td>13300<td>13301</td></td>	13300 <td>13301</td>	13301
MT145K	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	109	121	<100	135	576
MT145K N160A	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100
MT145-WT	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100
Q23_17	<100	<100	<100	<100	<100	116	383	165	369	427	197	635	260	1397	6506

Fig. 6D



B

Type	Virus/infectant	Subtype	Tier	Ab	Pre-bleed				Bleed #1				Bleed #2				Bleed #3																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
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Autologous	MT145K	SIV	2	0.173	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100	<100</

Fig. 7B

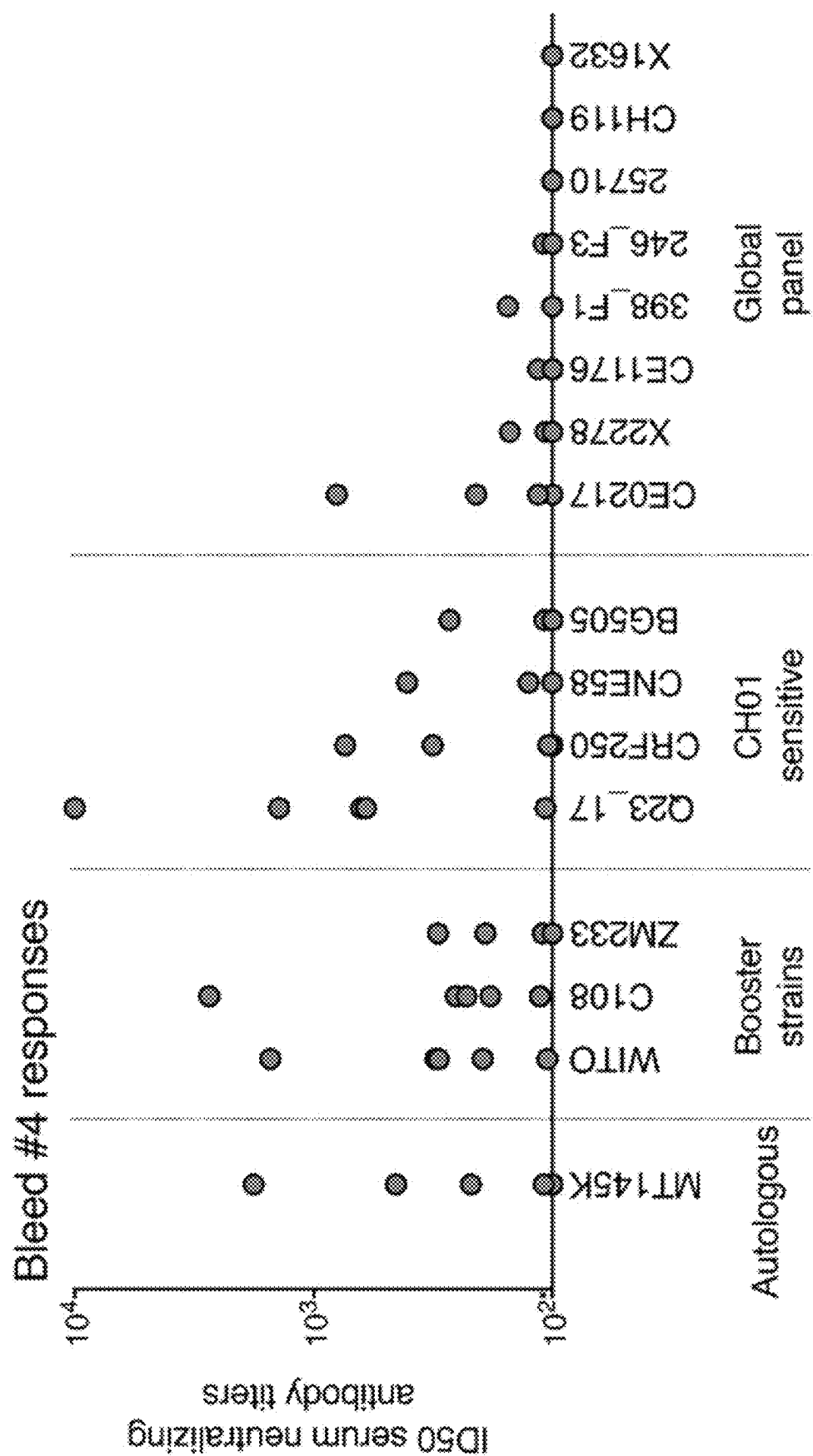


Fig. 7C

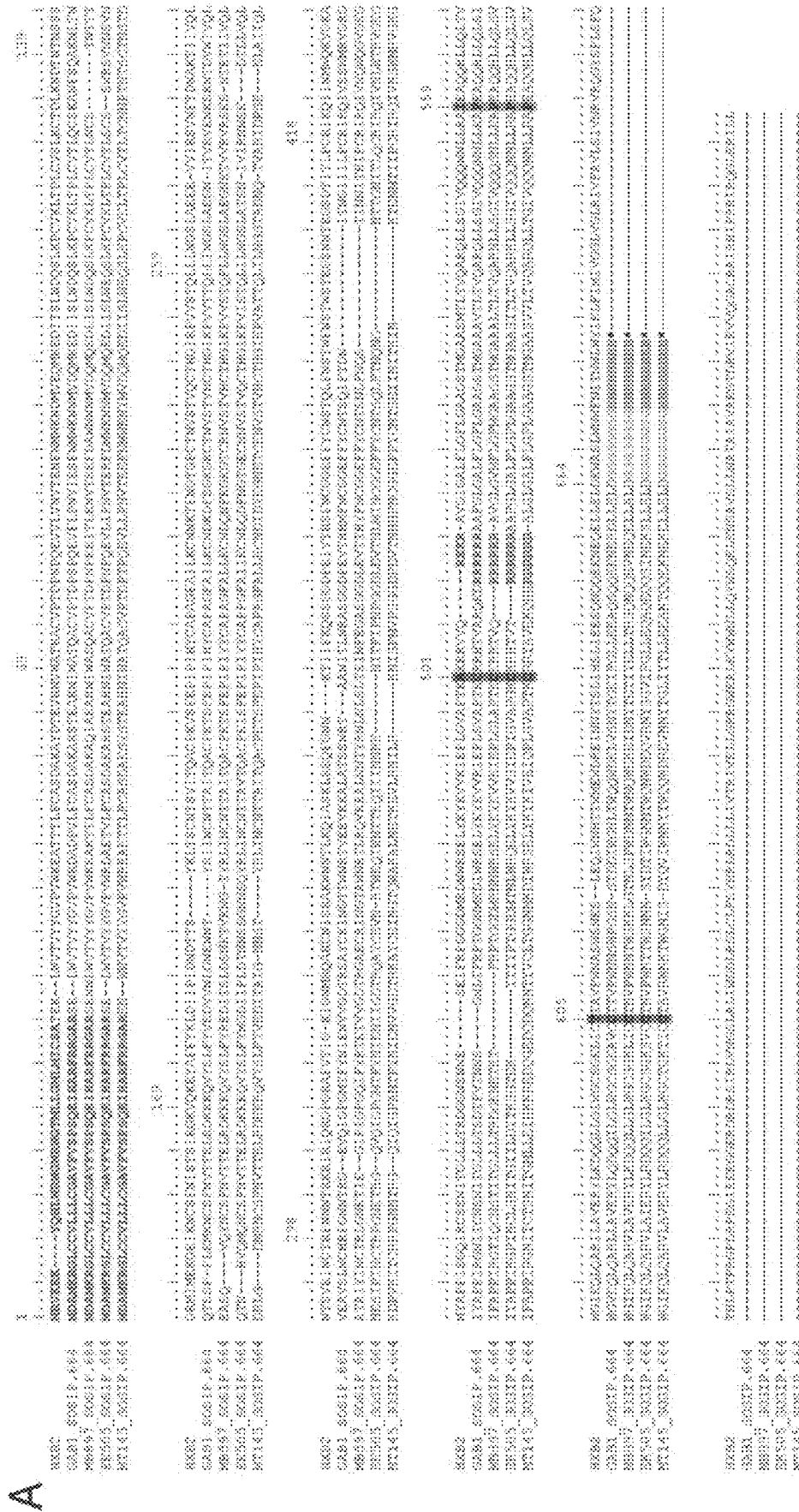


Fig. 8A

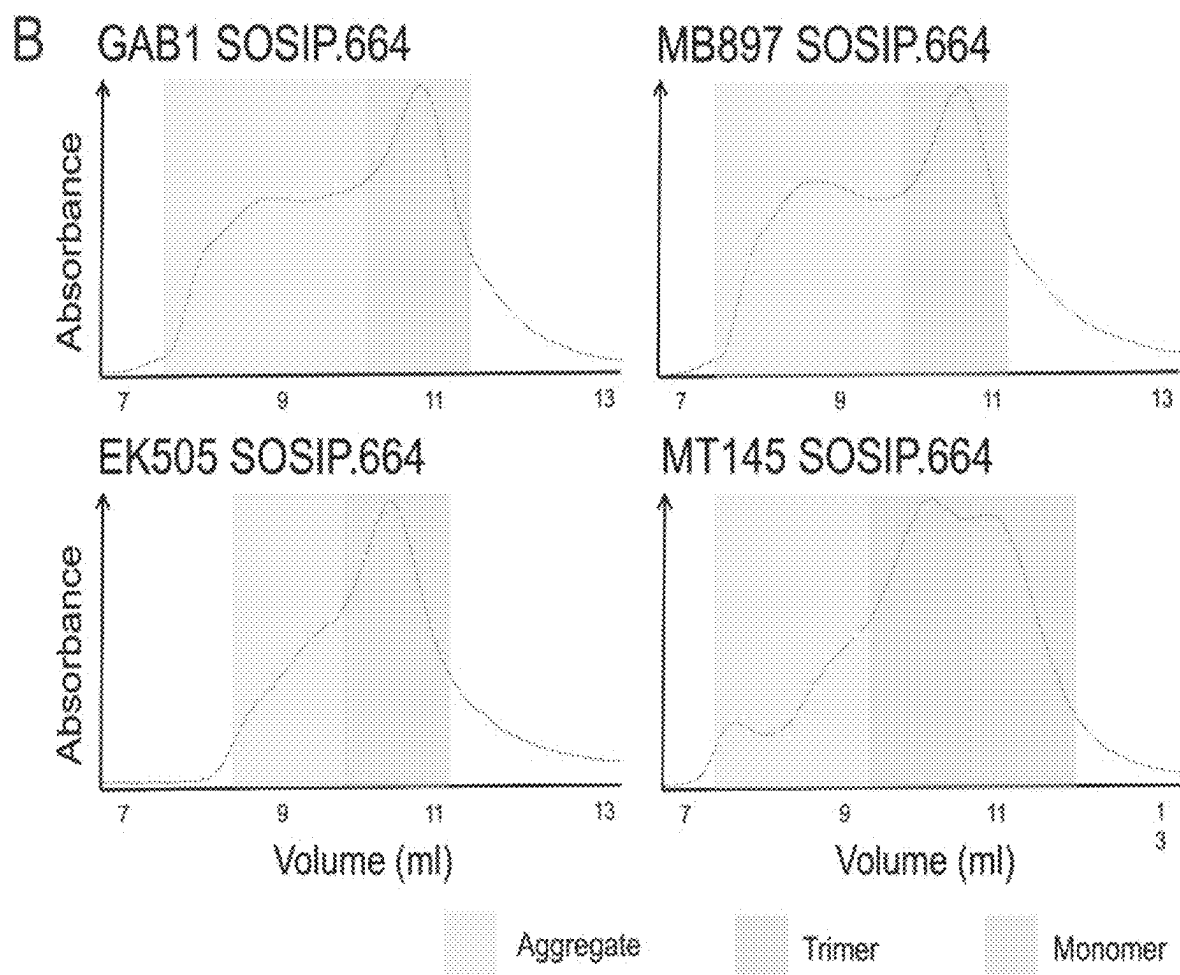


Fig. 8B

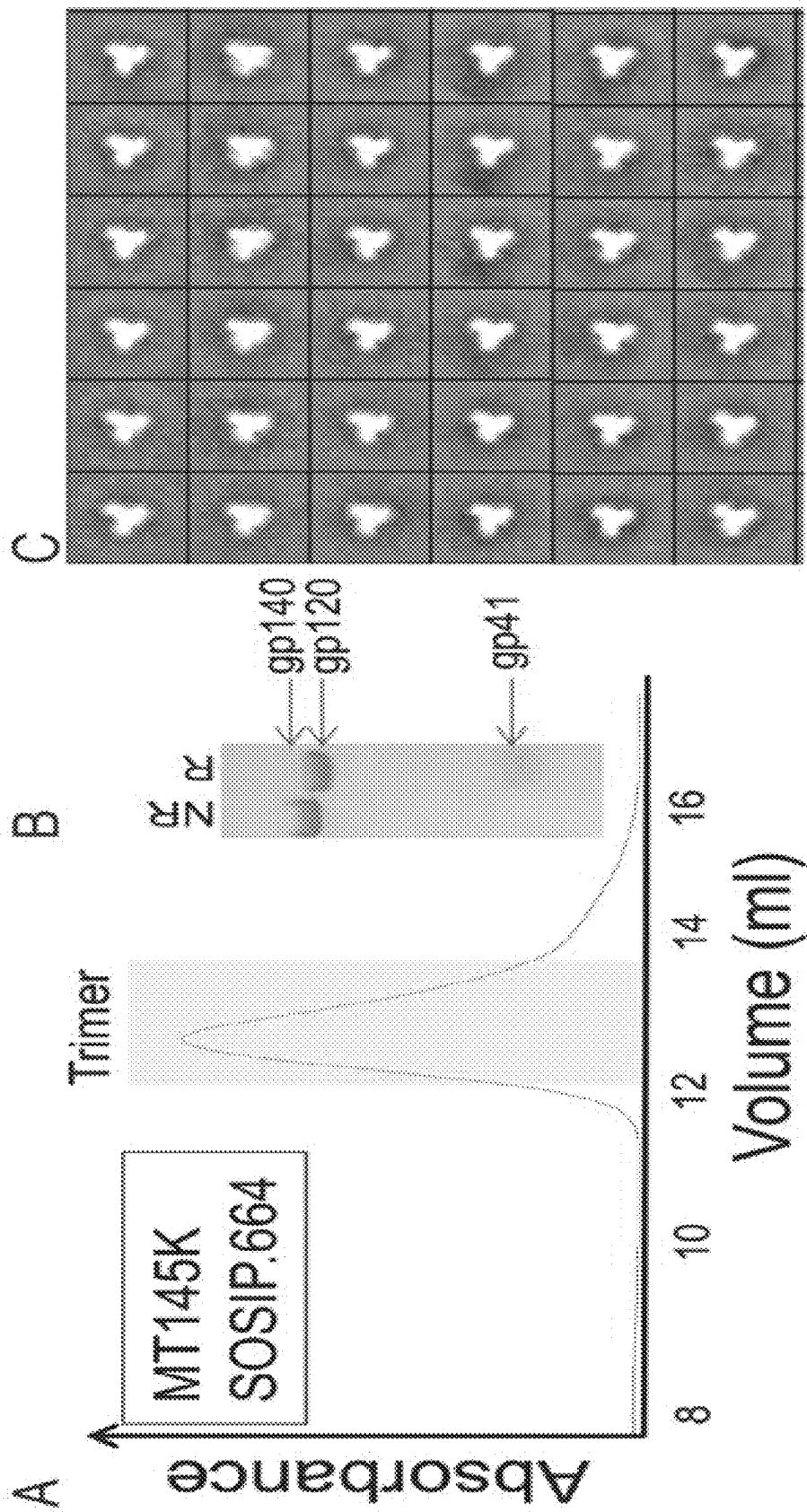


Fig. 9A-9C

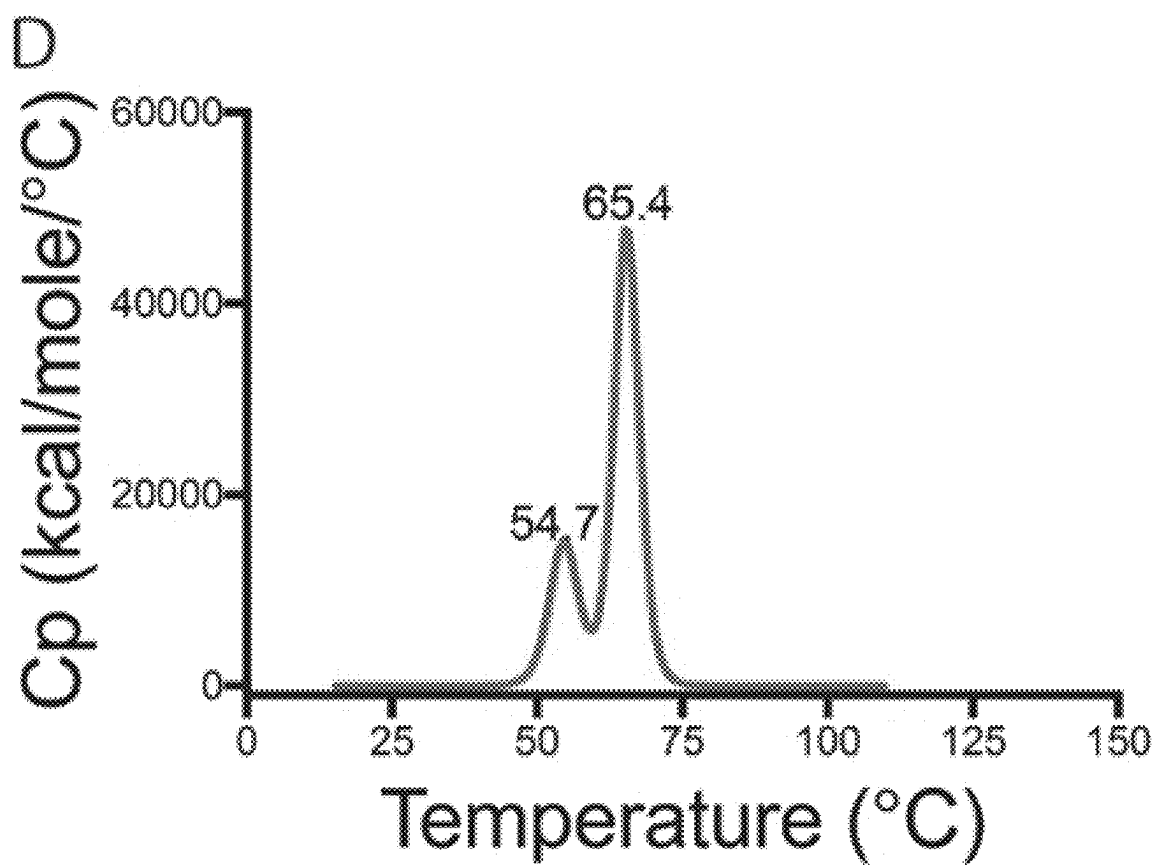


Fig. 9D

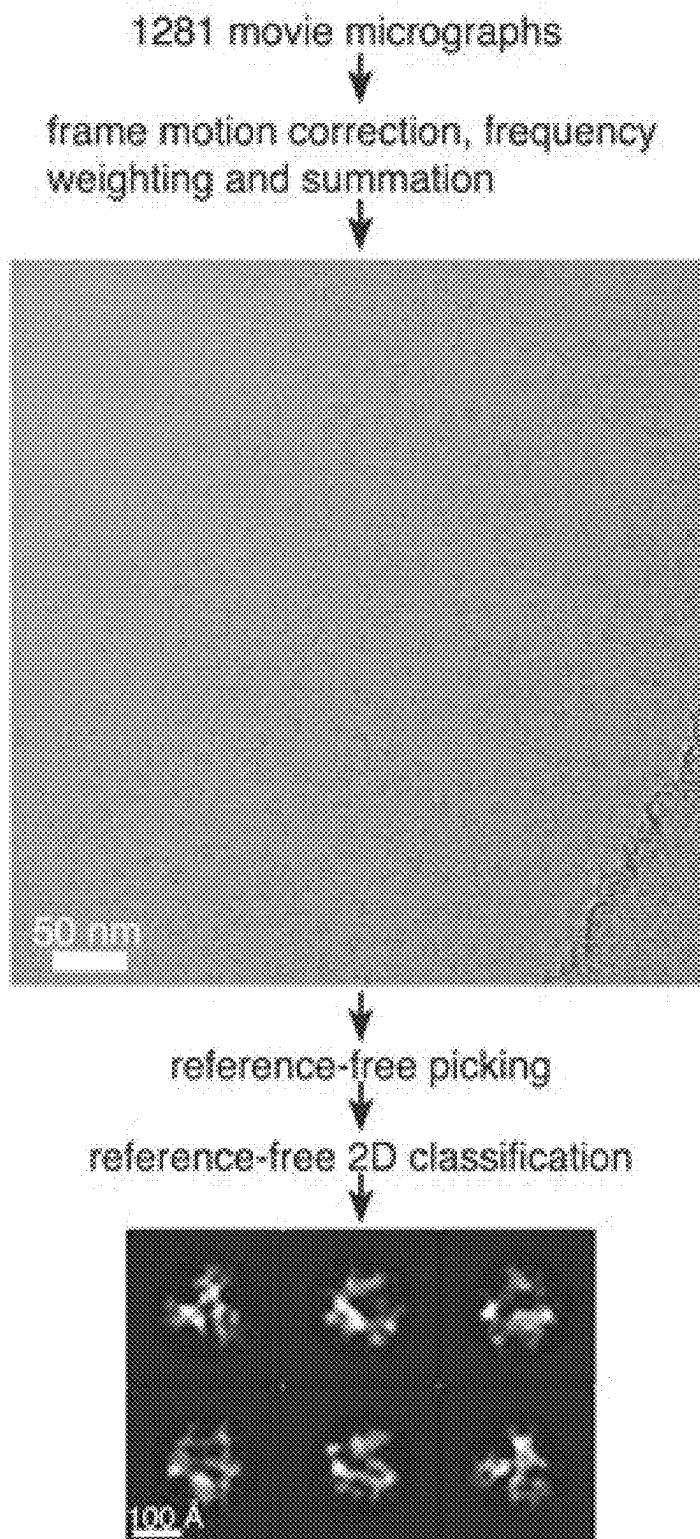
A

Fig. 11A

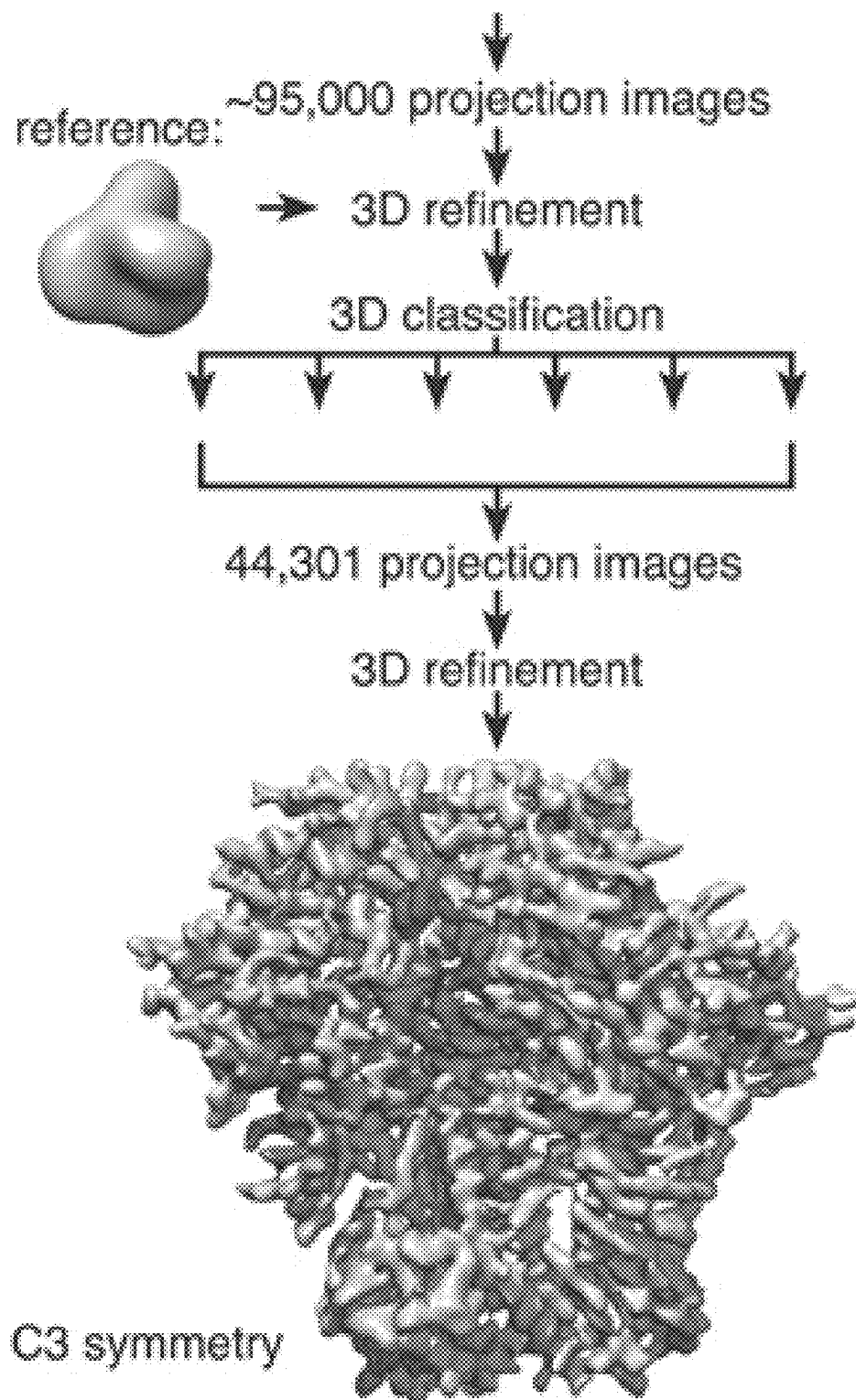


Fig. 11A (Cont'd)

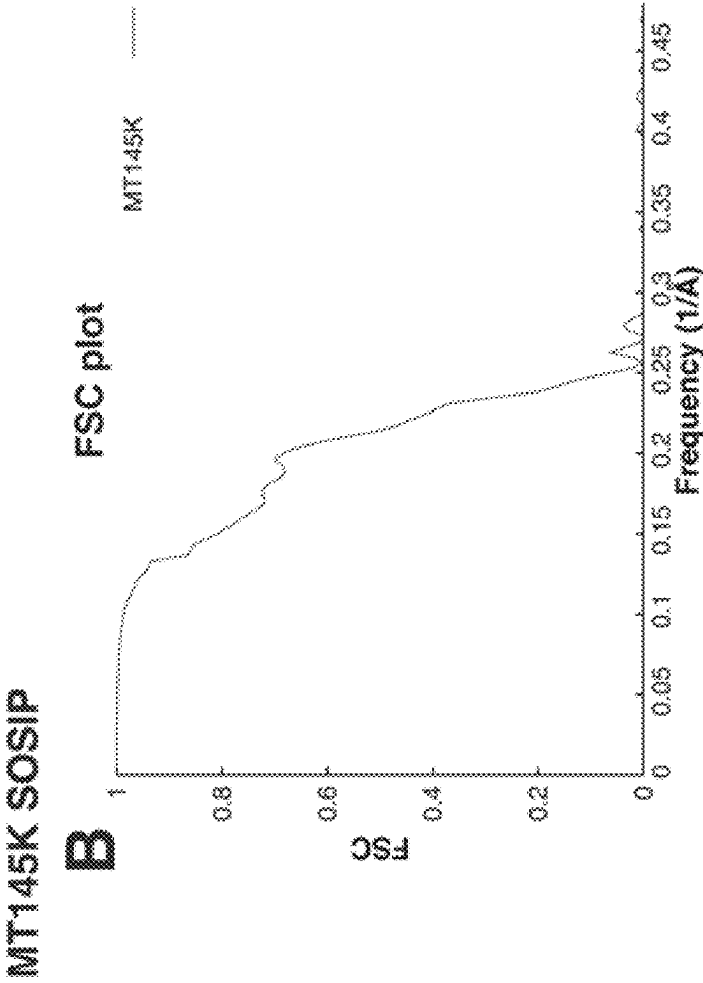


Fig. 11B

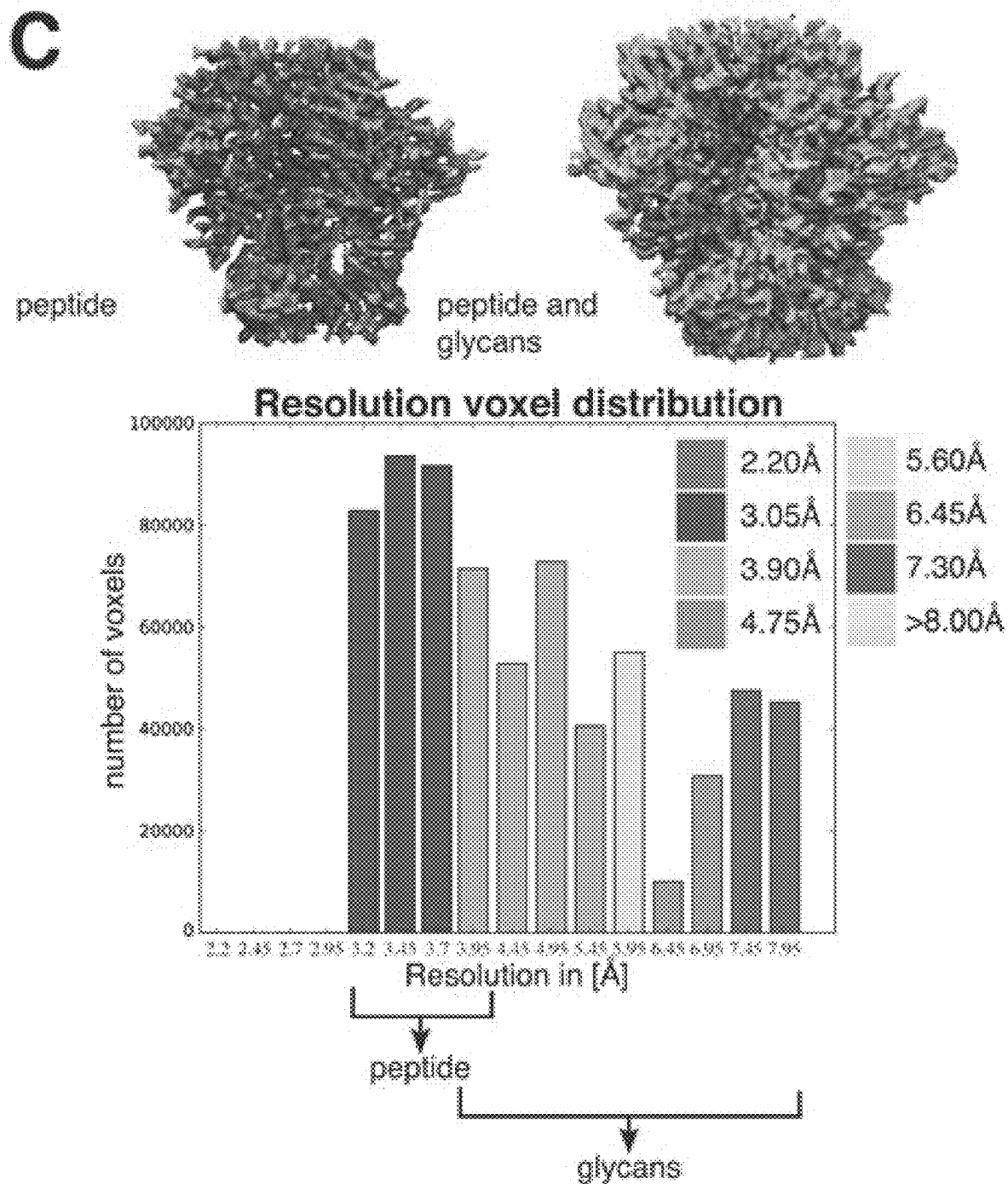


Fig. 11C

D Relative angular distribution of projection images

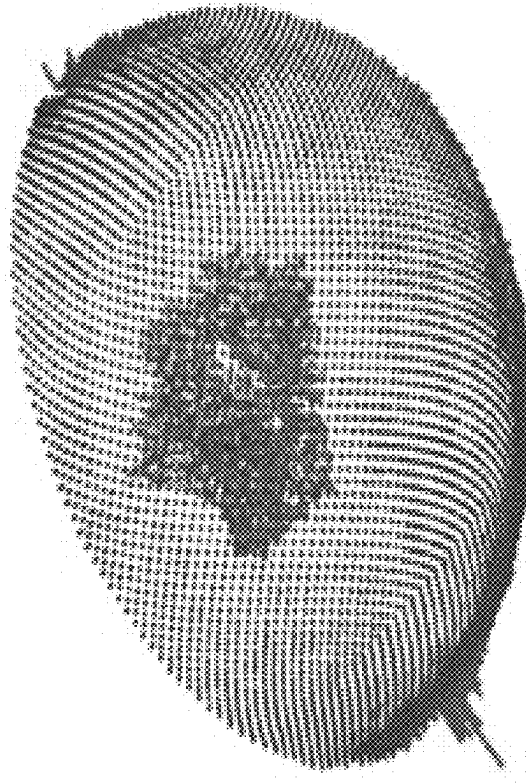


Fig. 11D

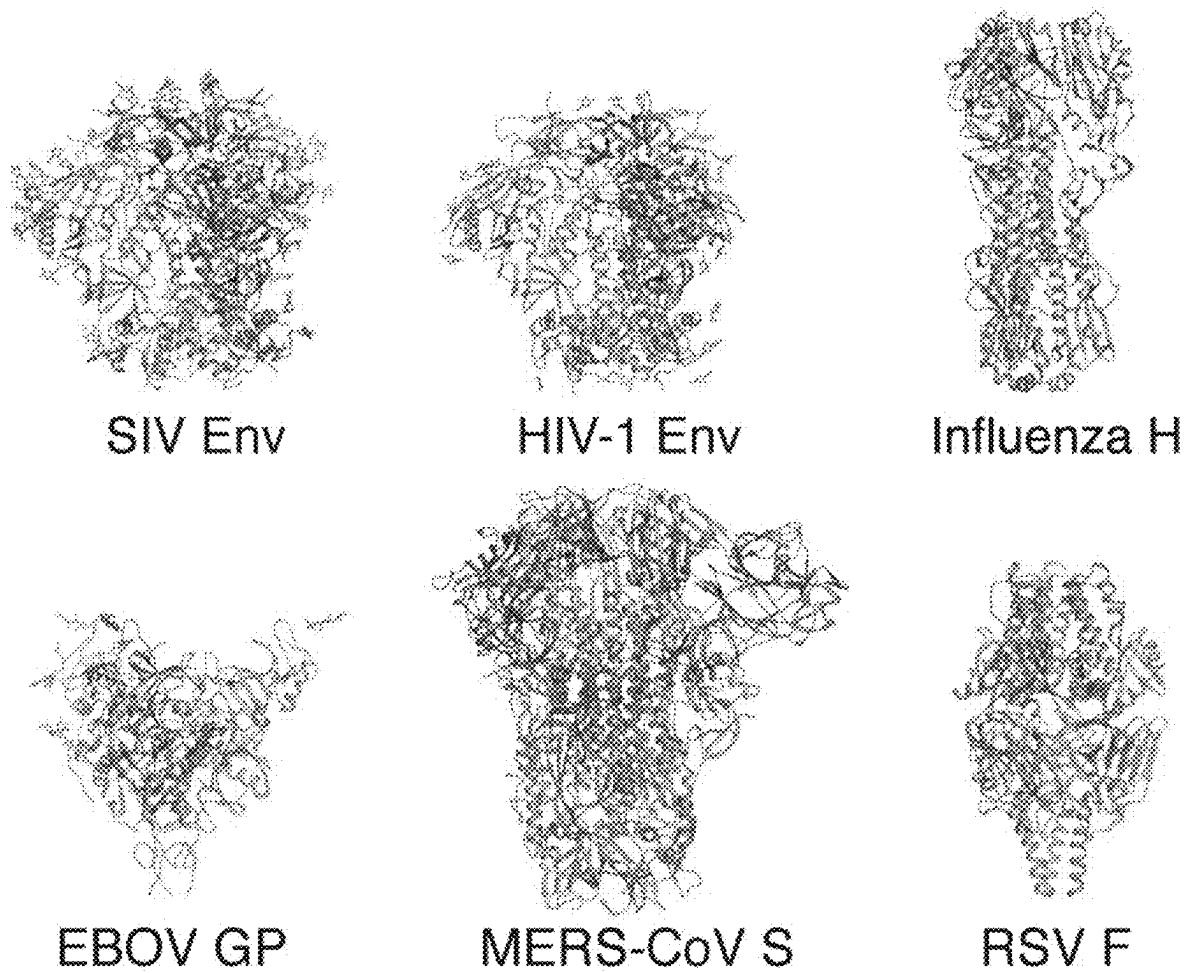


Fig. 12

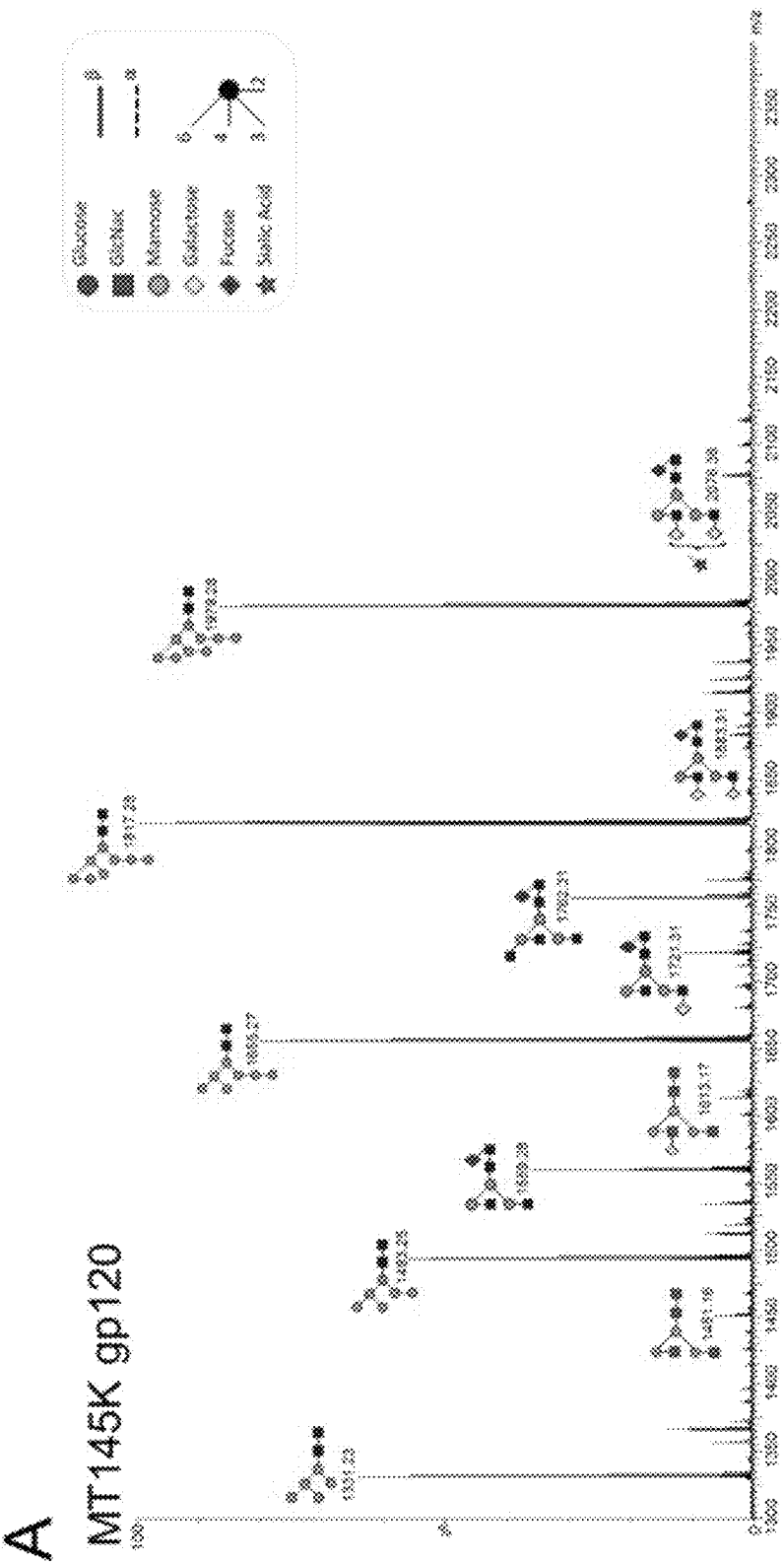


Fig. 13A

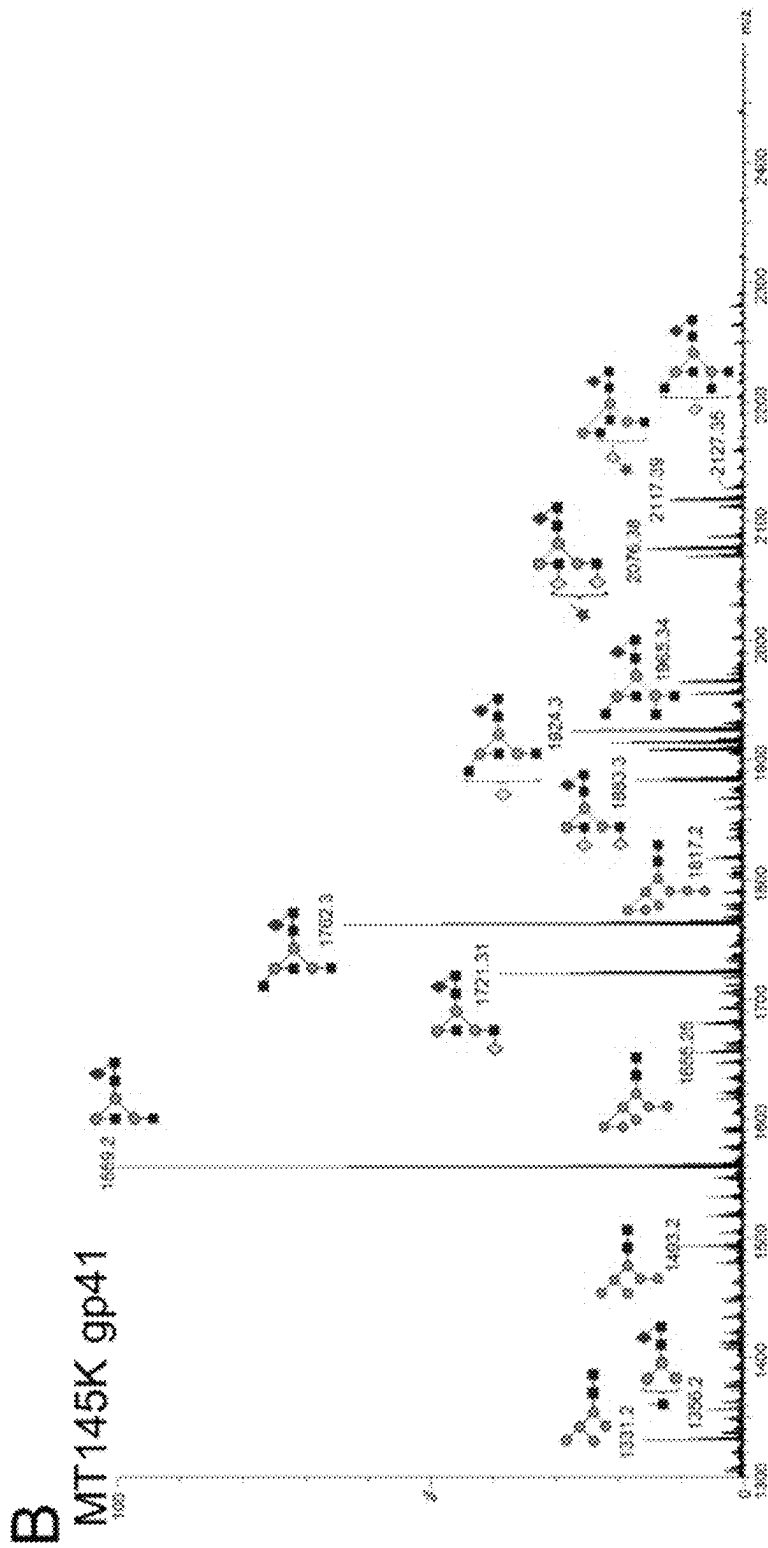


Fig. 13B

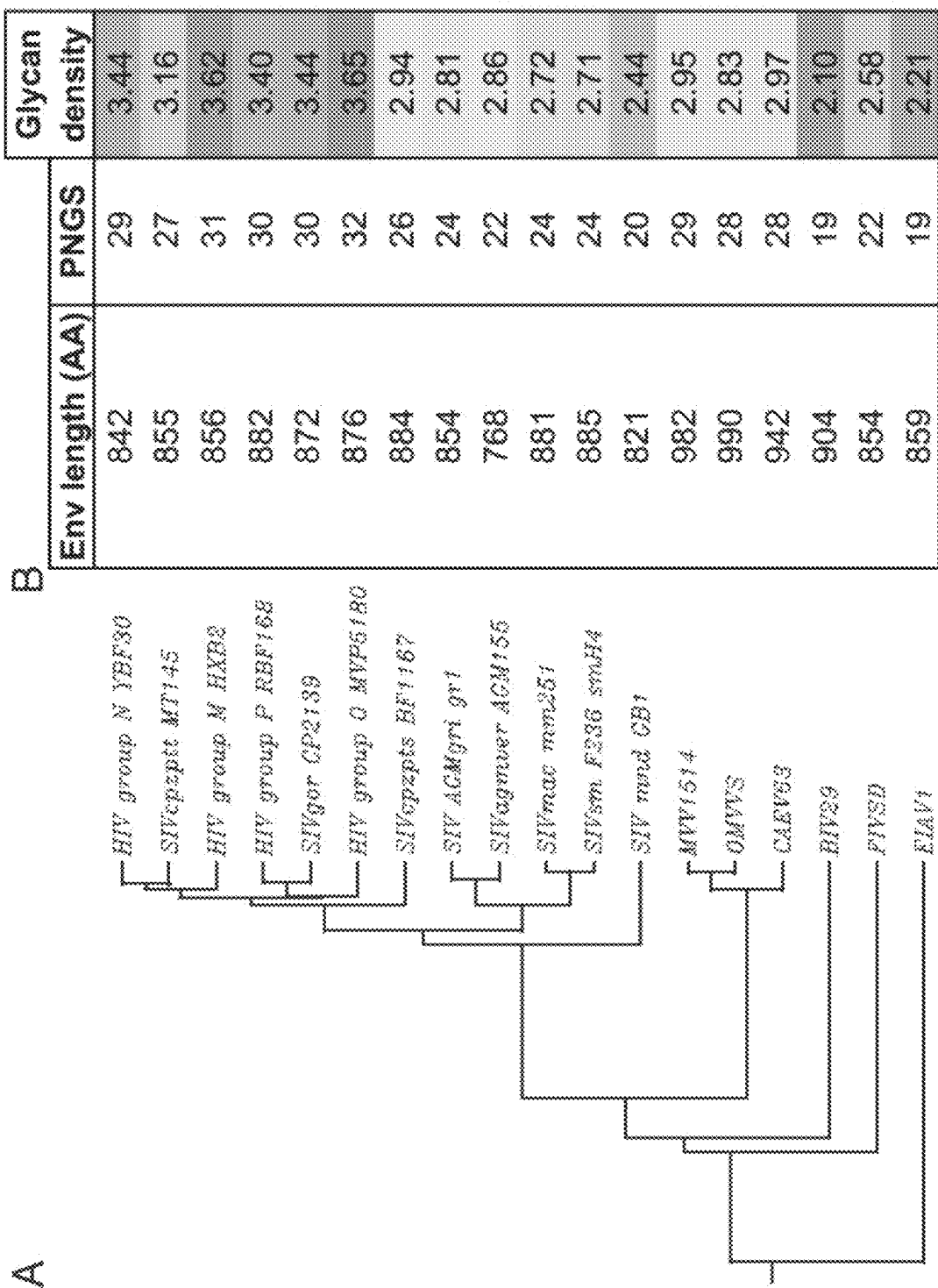


Fig. 14A-14B

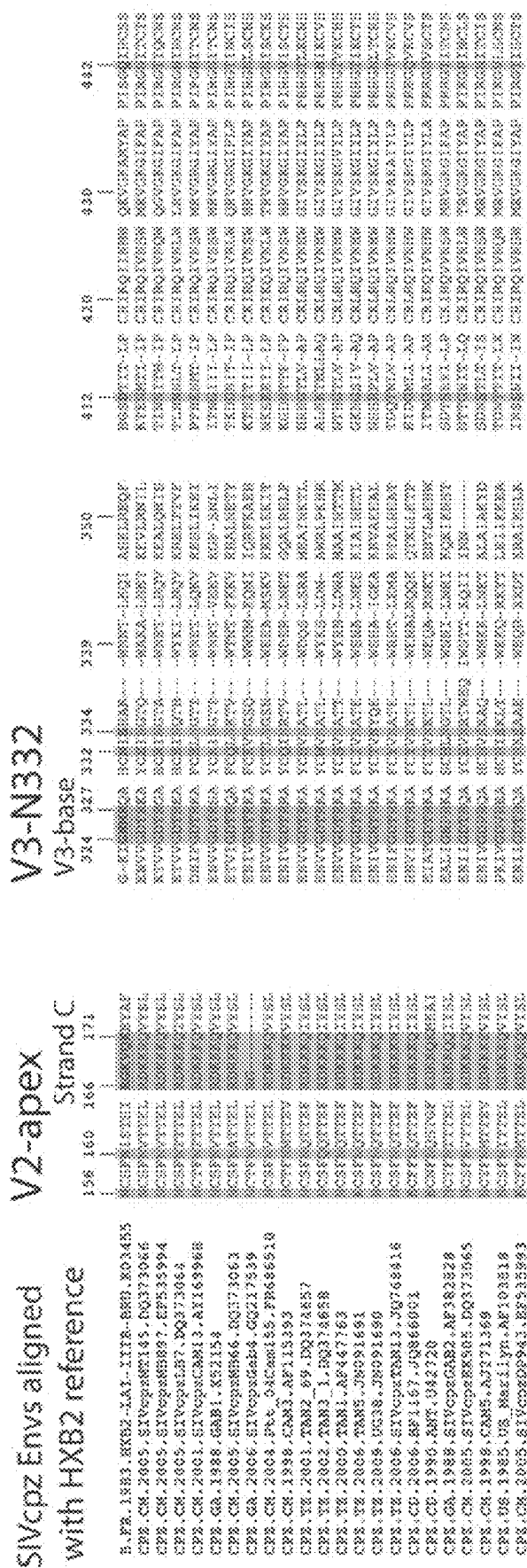


Fig. 15

	CD4bs	Loop D	V5 loop
	262 268 276 280	289	448 456
B.PB.1983.HR82-LAT-TTB-N80.K03455	STQILLNGSL AEEK-VYIR	---EETPTN---A-K	---EETPTN---A-K
CP2.CM.2005.SIVcpMT245.DQ373066	STQILLNGST ADEQ-TVARI	---DPFERRIA	---TIIVQLATK
CP2.CM.2005.SIVcpMB937.RF535994	STQILLNGSL AEEKTVAR	---VSEES-K-T-E	---TIIVQLATK
CP2.CM.2005.SIVcpZLB7.DQ373084	STQILLNGSL AEEK-IRIR	---YKA-D	---EVIQLSTK
CP2.CM.2001.SIVcpCAB13.AF169968	STQILLNGSL ATEK-ITVR	---EKAQSD--T-E	---EVIQLSTK
CP2.CA.1989.CAB1.K52154	STQILLNGSL AEEK-ITVR	---EKAQSD--T-E	---EVIQLSTK
CP2.CM.2005.SIVcpMB86.DQ373063	STQILLNGSL EE-K-VTVL	---EETPTN---A-K	---EETPTN---A-K
CP2.CA.2006.SIVcpGub4.GQ217539	STQILLNGSL TQ-Q-IMIR	---EETPTN---A-K	---EETPTN---A-K
CP2.CM.2004.Ptc.94Cm155.RF646510	STQILLNGSL AEEK-VTIR	---EETPTN---A-K	---EETPTN---A-K
CP2.CM.1998.CAM3.KY115393	STQILLNGSL AEEK-TIAR	---EETPTN---A-K	---EETPTN---A-K
CP2.TE.2001.TAN2.69.KQ374657	ATALLNGSL EEEK-TVAY	---EETPTN---A-K	---EETPTN---A-K
CP2.TE.2002.TAN3.1.DQ374658	ATALLNGSL EEEK-TVAY	---EETPTN---A-K	---EETPTN---A-K
CP2.TE.2000.TAN1.AF467763	ATALLNGSL EEEK-TVAY	---EETPTN---A-K	---EETPTN---A-K
CP2.TE.2006.TAN5.JH031691	ATALLNGSL EEEK-TVAY	---EETPTN---A-K	---EETPTN---A-K
CP2.TE.2009.QC38.JH031690	ATALLNGSL EEEK-TVAY	---EETPTN---A-K	---EETPTN---A-K
CP2.TE.2006.SIVcpTAN13.SQ768416	ATALLNGSL EEEK-TVAY	---EETPTN---A-K	---EETPTN---A-K
CP2.CD.2006.RP1167.JQ966001	ATALLNGSL EEEK-TVAY	---EETPTN---A-K	---EETPTN---A-K
CP2.CO.1990.KMT.042720	ATALLNGSL EEEK-TVAY	---EETPTN---A-K	---EETPTN---A-K
CP2.CA.1988.SIVcpGAR2.AF382828	ATALLNGSL EEEK-TVAY	---EETPTN---A-K	---EETPTN---A-K
CP2.CM.2005.SIVcpEEK95.DQ373065	STQILLNGSL ATEK-ITVR	---EETPTN---A-K	---EETPTN---A-K
CP2.CM.1998.CAM5.AJ271369	STQILLNGSL AEEKTVAR	---EETPTN---A-K	---EETPTN---A-K
CP2.OS.1985.OS.Mar119.AF103818	STQILLNGSL ATEK-VTVR	---EETPTN---A-K	---EETPTN---A-K
CP2.CM.2005.SIVcpDP943.RF535993	STQILLNGSL AEEK-TAKR	---EETPTN---A-K	---EETPTN---A-K

Fig. 15 (Cont'd)

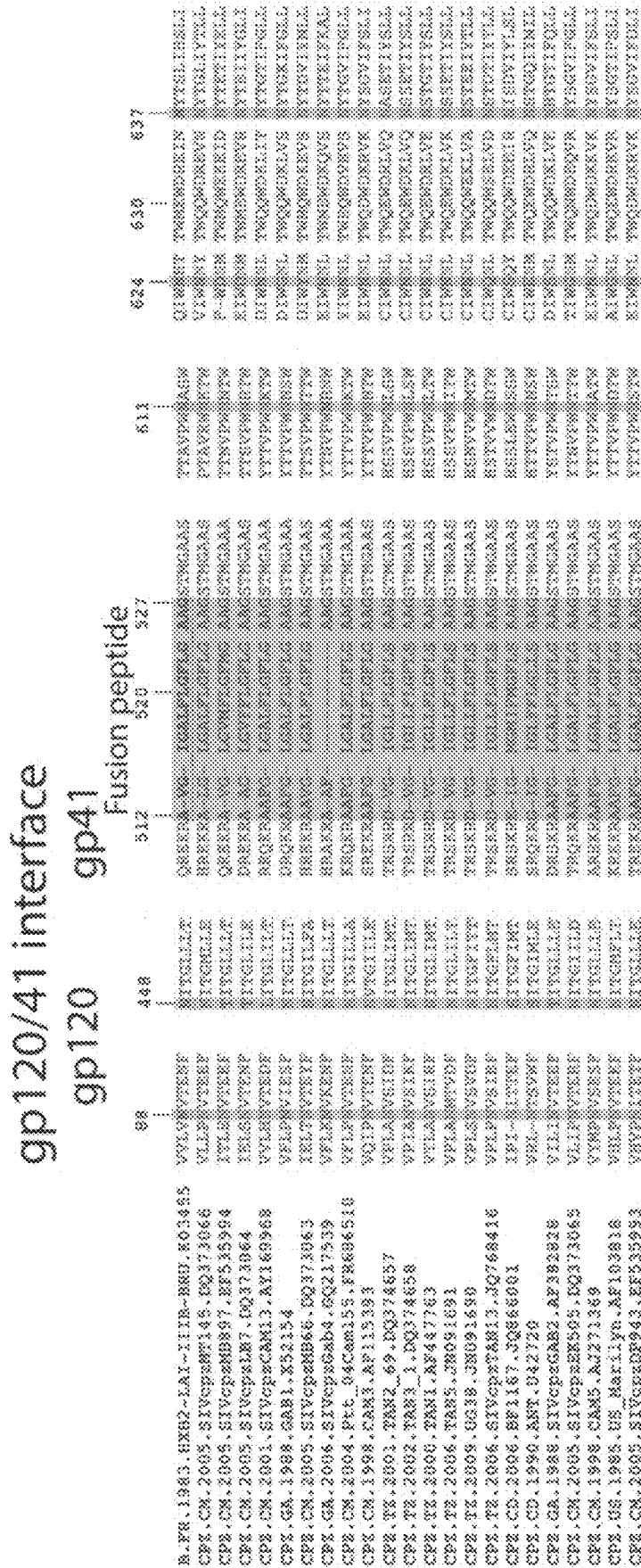


Fig. 15 (Cont'd)

[illegible]

Fig. 16

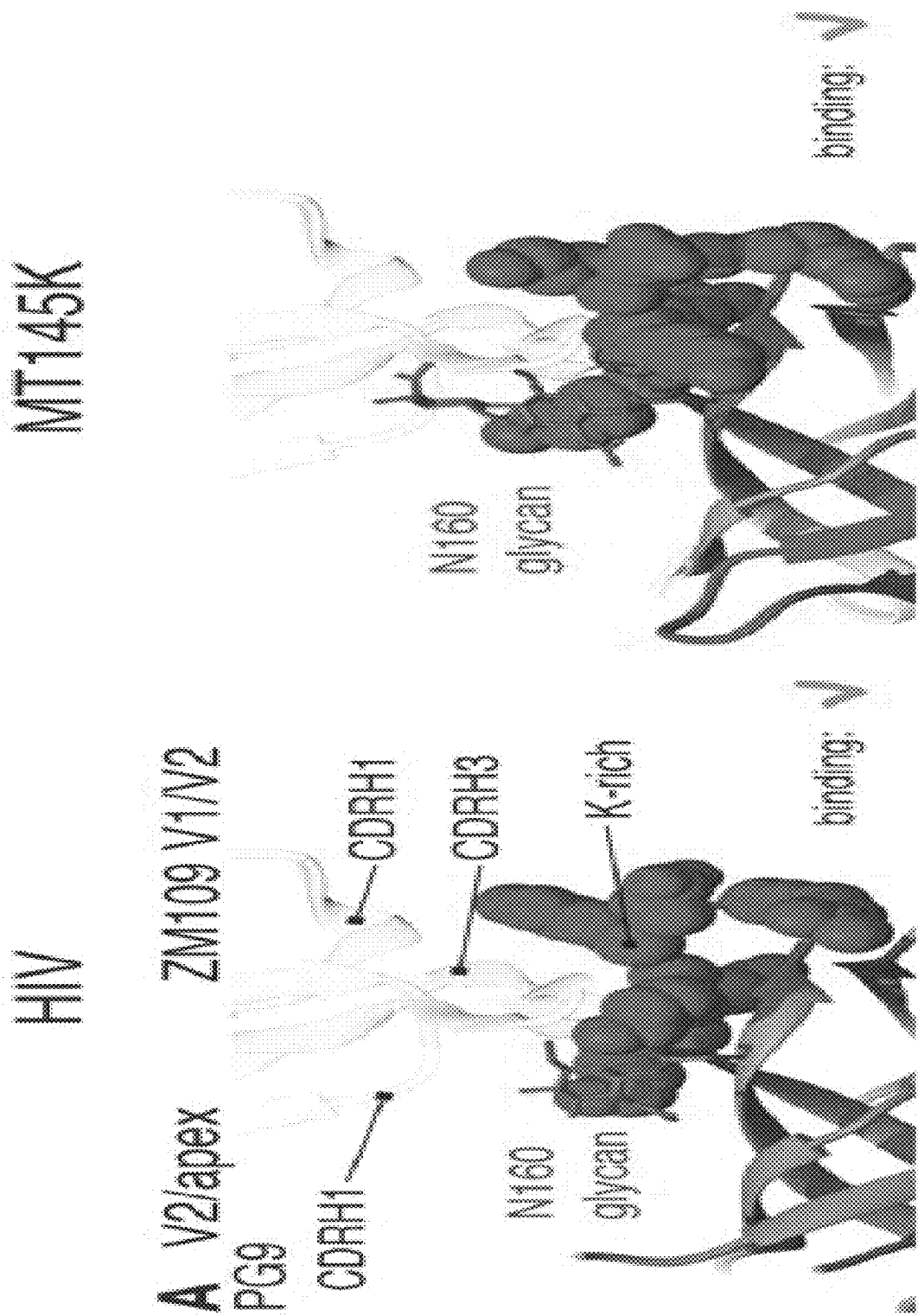


Fig. 17A

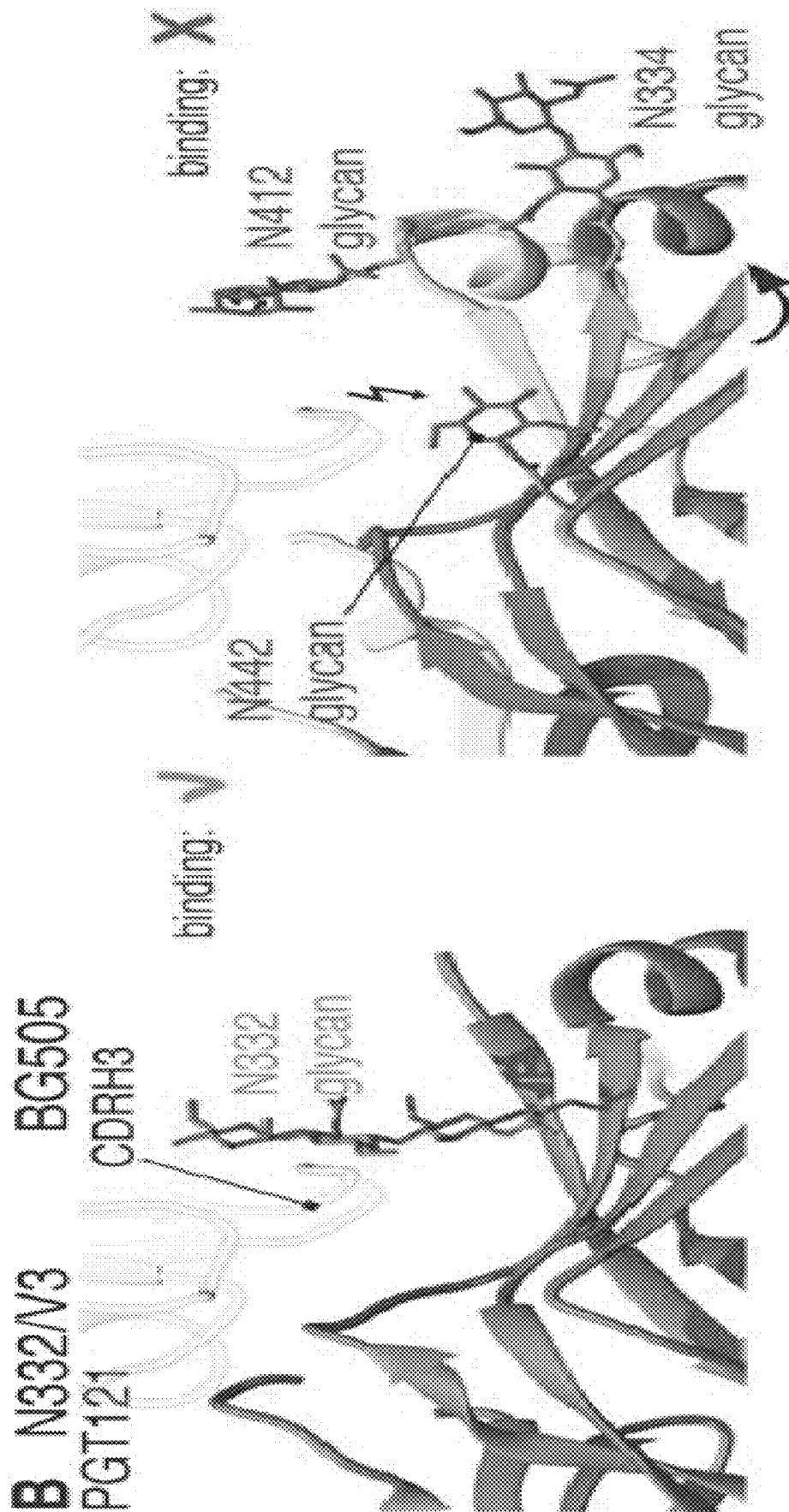


Fig. 17B

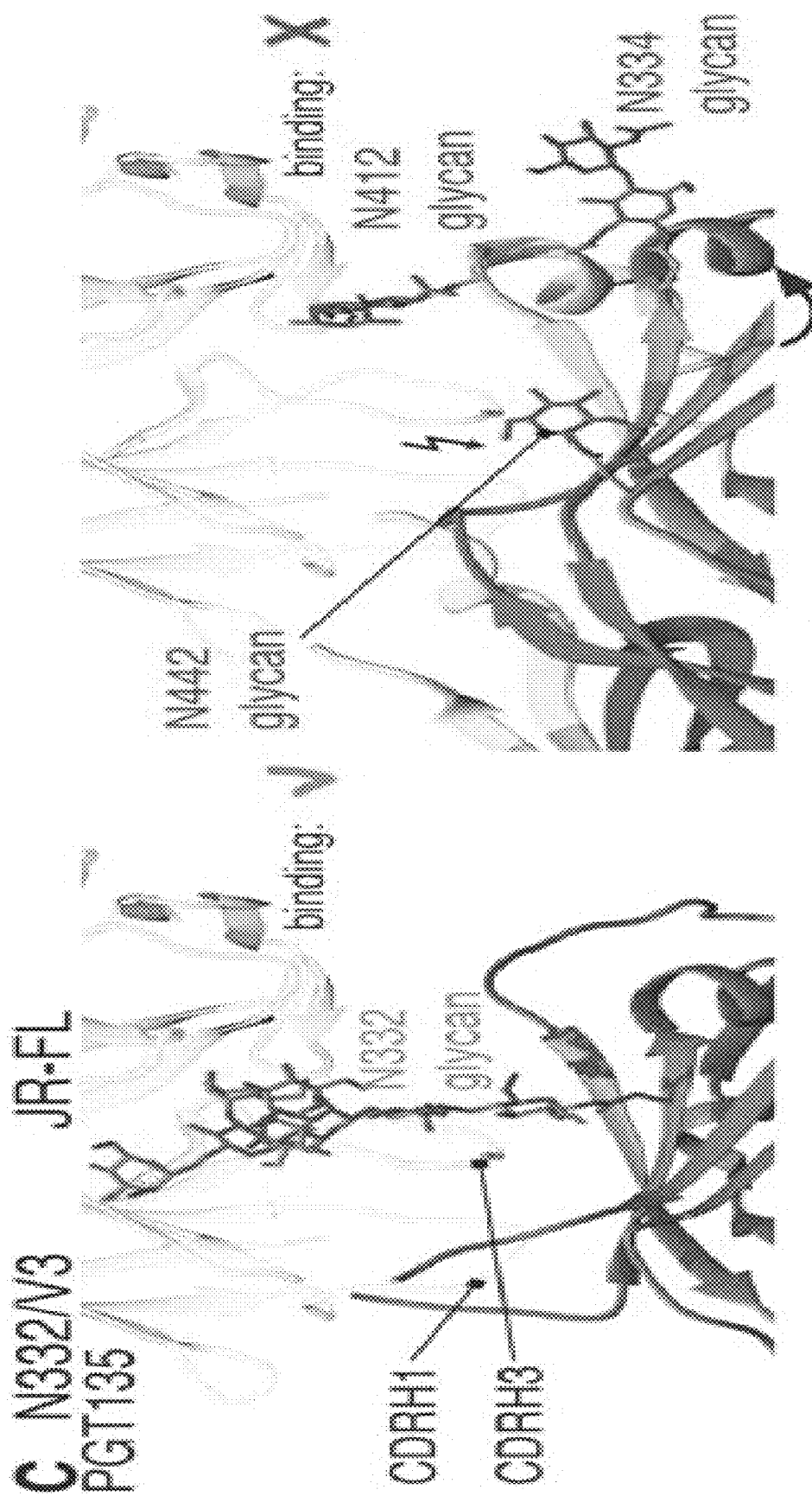


Fig. 17C

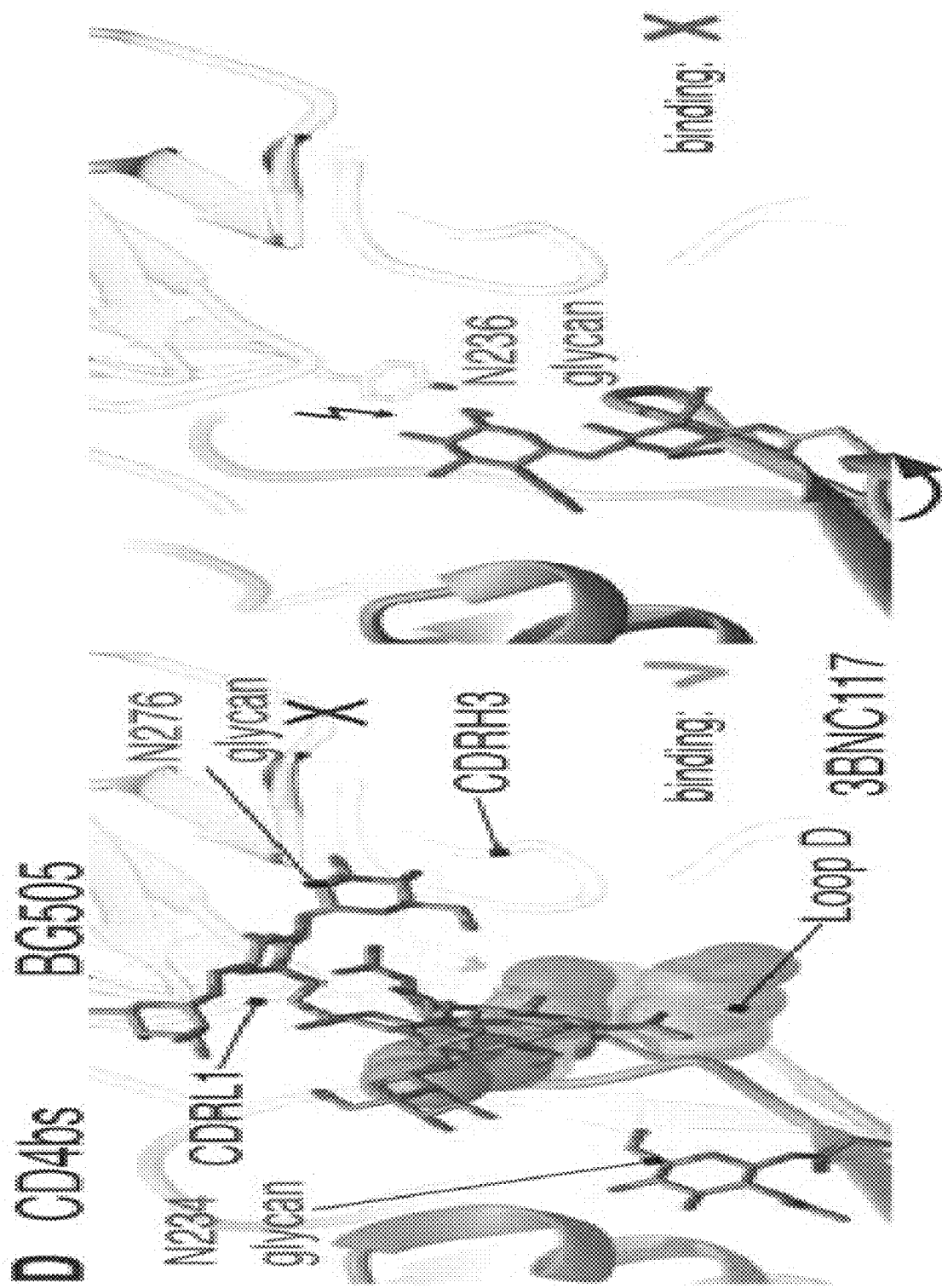


Fig. 17D

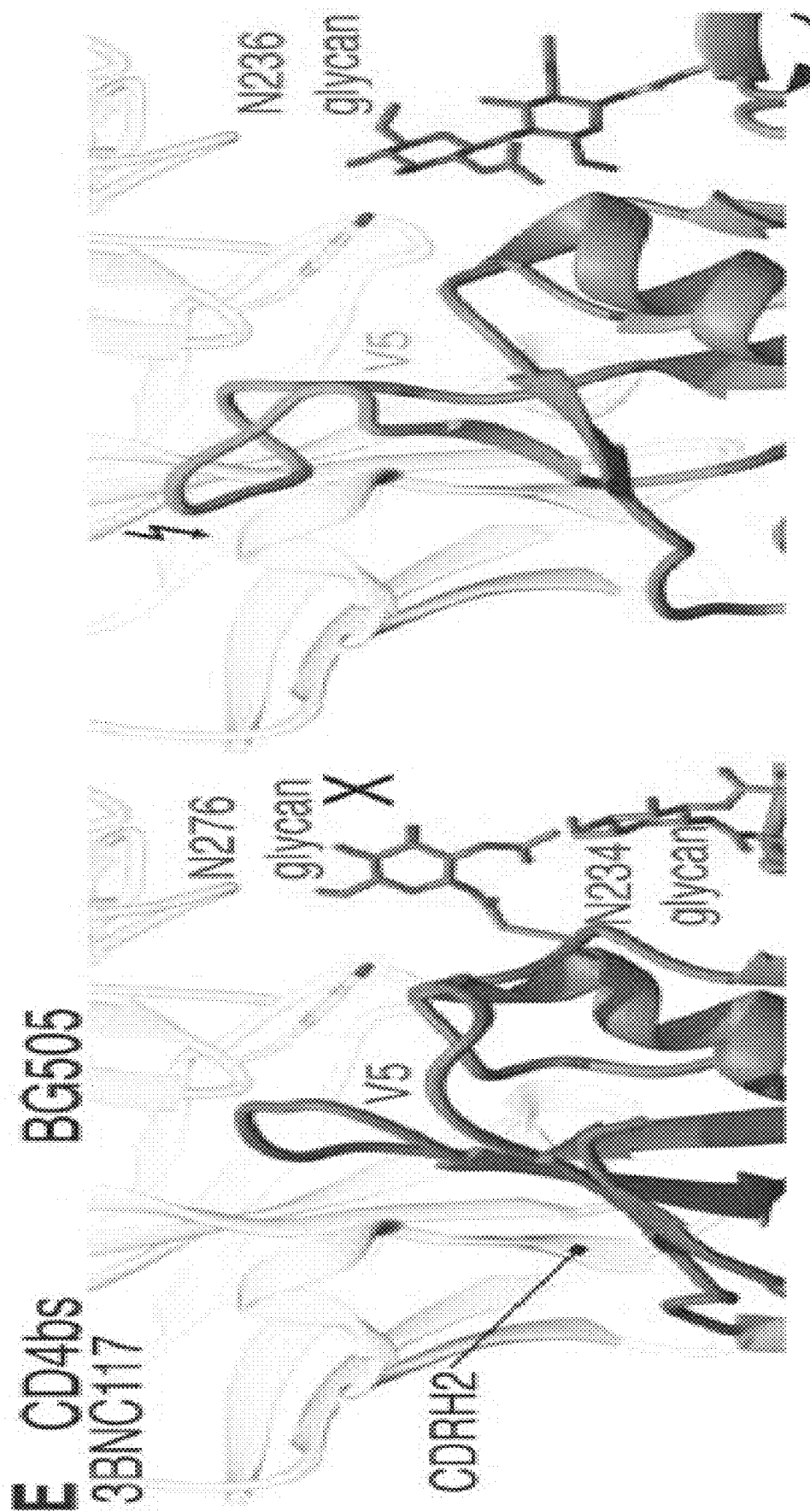


Fig. 17E

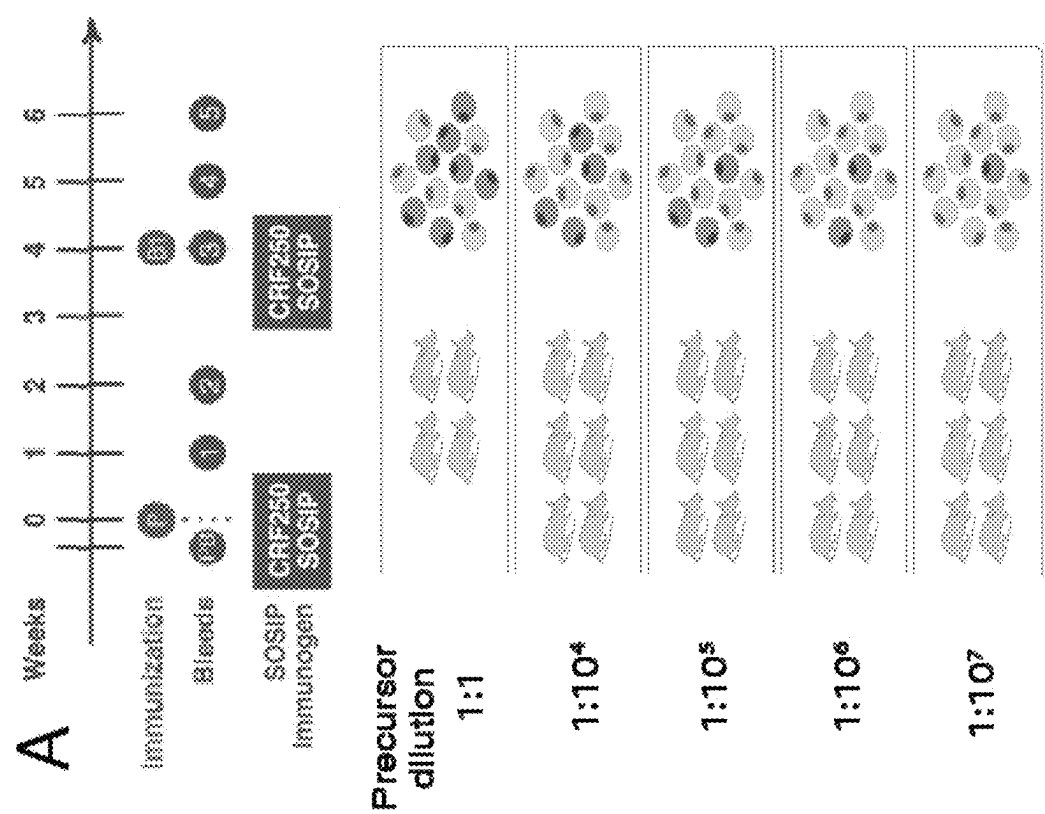


Fig. 18A

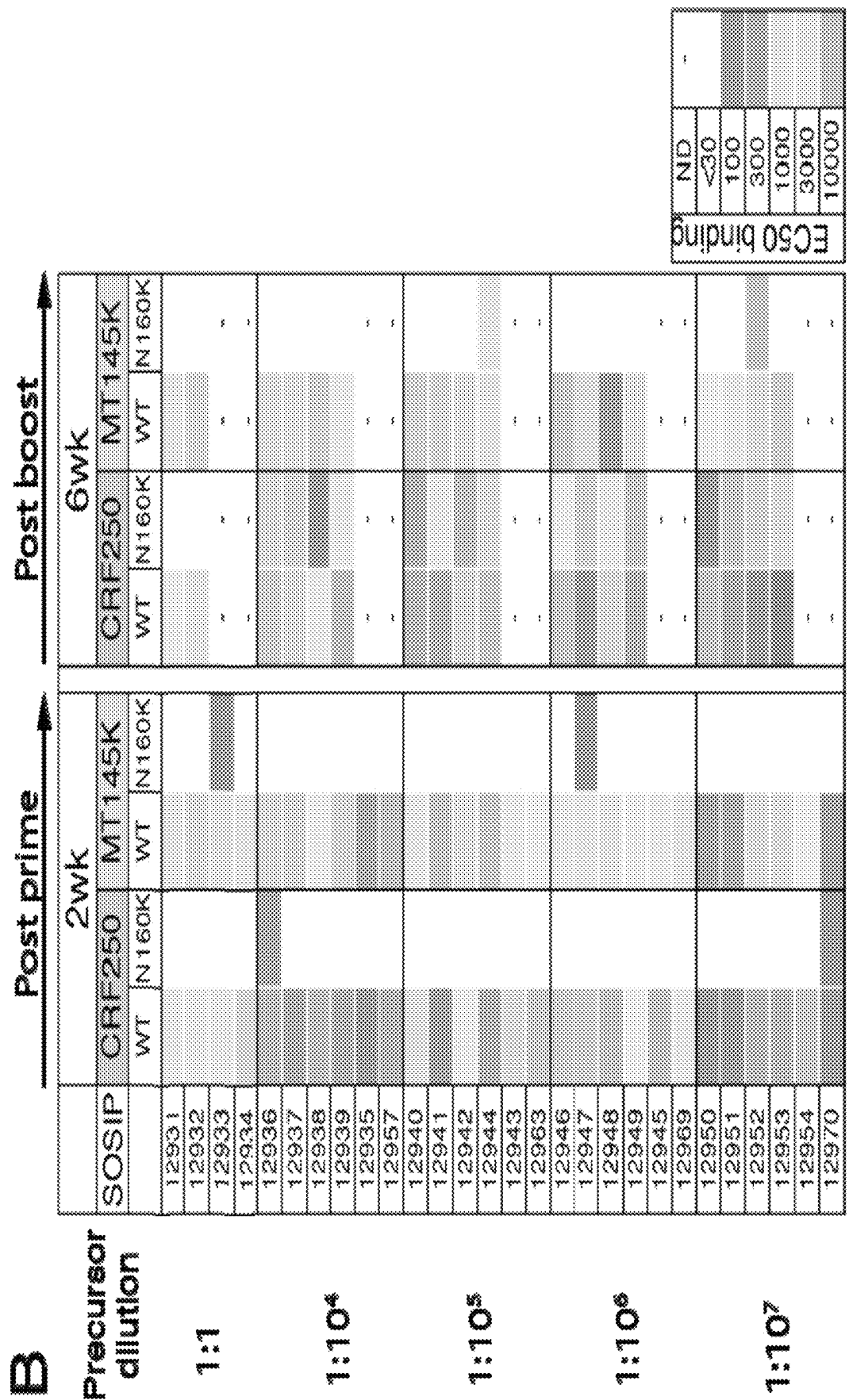


Fig. 18B

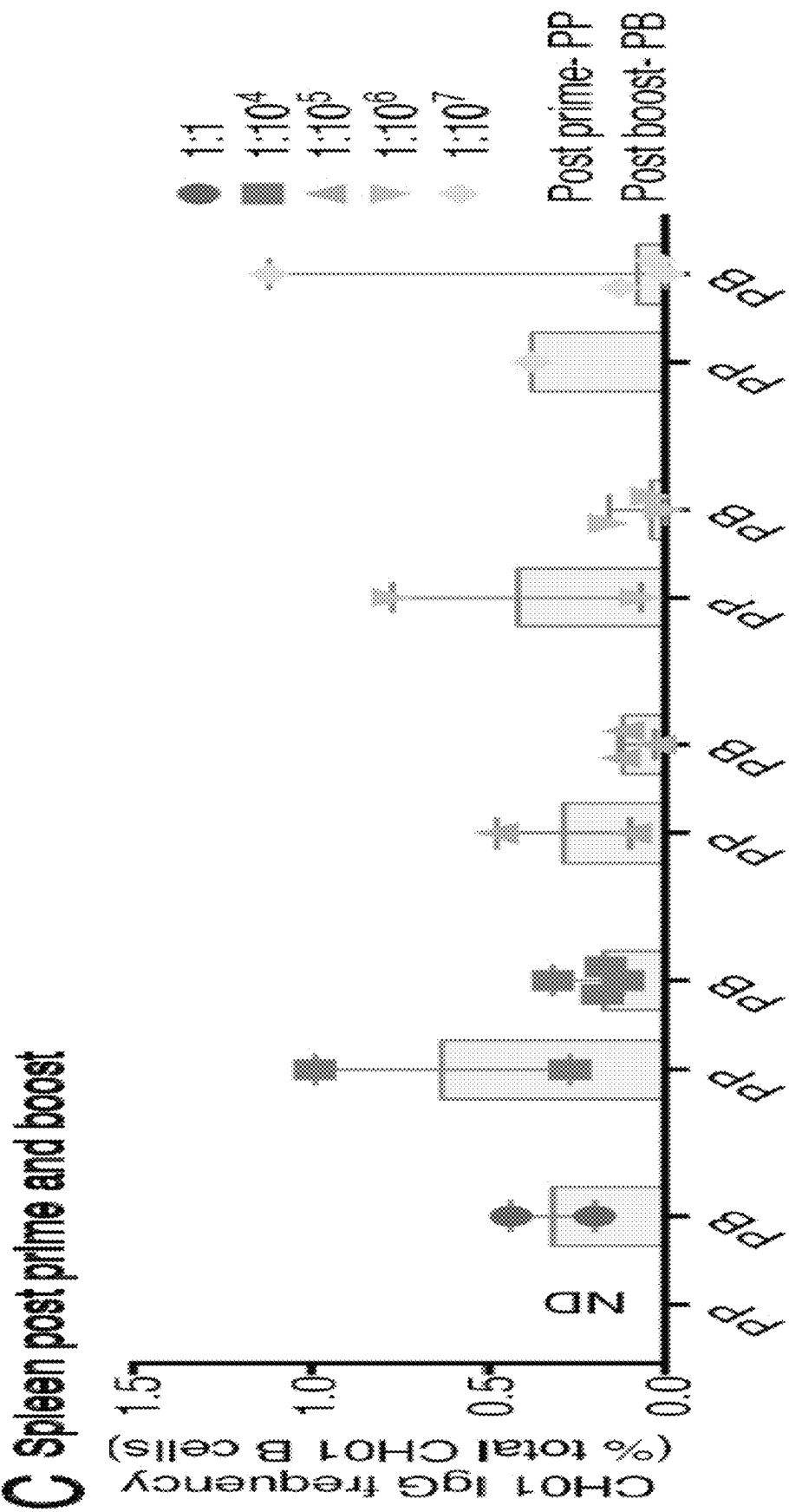


Fig. 18C

>MT145K_dv5_SOSIP.664
MDAMKRGGLCCVLLLCGAVFVSPSQEIHARFRRGAR
ENWWVTVYVGVPVWREAKTTLFCASDAKSYSTEAHNIWATQACVPTDPTPQEVLLPNVTEEFNM
WENYMVDQMEDIISLWEQSLKPCVKLTPLCVTLTCNNPTNTSCTNSTDDRLGDMRNCSENVTT
ELRDKKRKVYSLFYVEDIT AIGNNSTYRLINCNTT AITQACPKTSFEPIPIHYCAPAGFALLKC
NDIDYKGNETCKNVSTVHCTHGIKPVATTQLILNGSTADNQTVARIDPSENLAIIQLKDPVKIT
CRRPGNNTRGQIQIPAMTFYNIENVVGDTRKAYCEINGTQWAKALNETKEVLRNLRKNISFM
VPSGGDPEVTNHHFENCGGEFFYCNTSEIINITKINKTENMTIIPCRIRQIVNSWMRVGKGIFAP
PIRGNITCTSNITGMILLEIHKDQNNNTYVCLTGGNMKDIWRSELYKYKIVEIQPLGVAPTCCRYY
AVEKQHRRRRRRALGLGALFLGAAGSTMGAASVVLTVQARQLLTGIVQQQNNLLRAPEAQ
QHLLQLSVWGKQLQARVLAVERYLRDQQLLGLWGCTGKTICCTAVRWNKTWGNISDYQVIWNN
YTWQQWDREVNNTGLIYTLLEEANTQQEKEKELELD*

Fig. 19

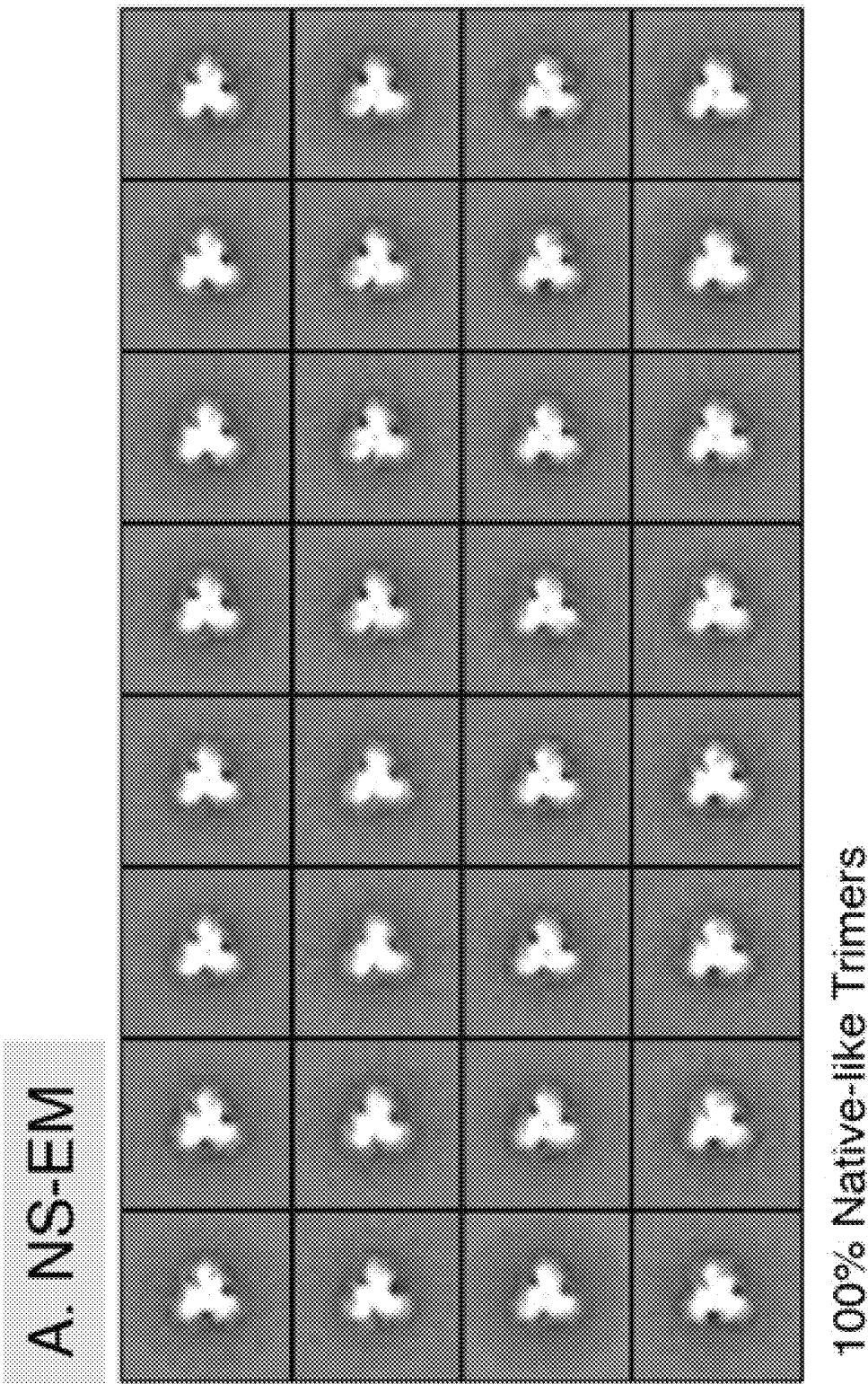


Fig. 20A

B. BLI binding

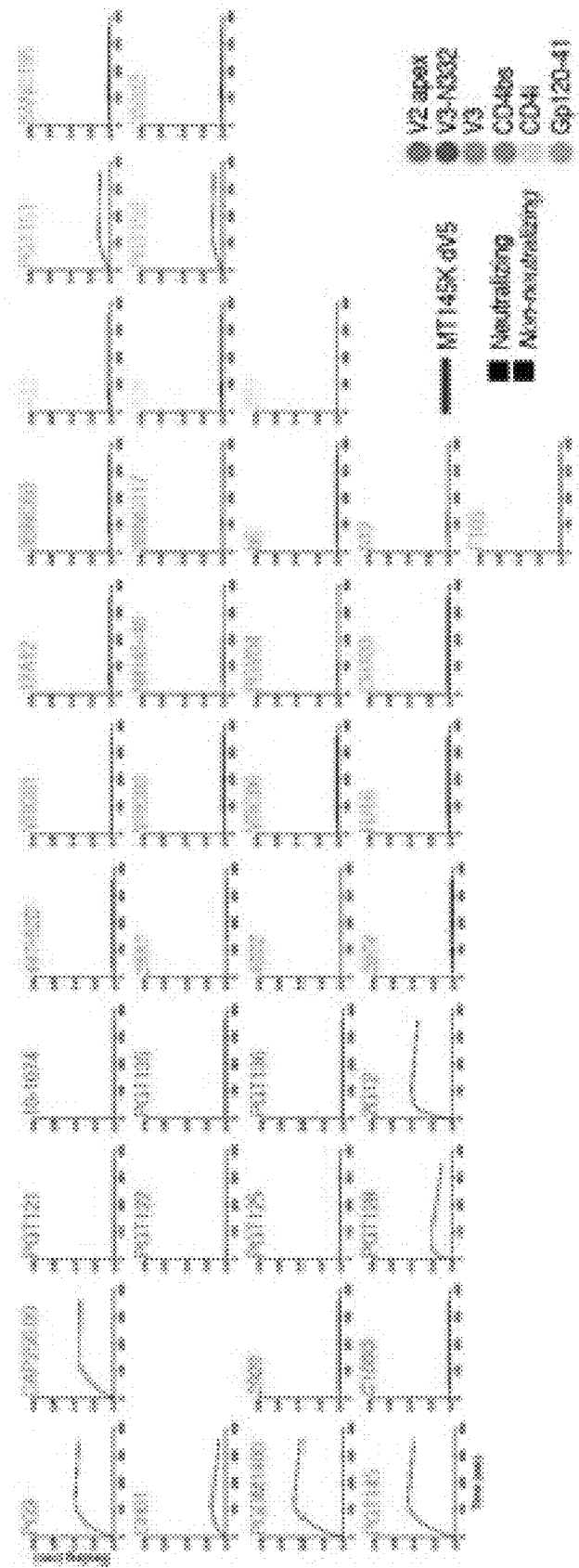


Fig. 20B

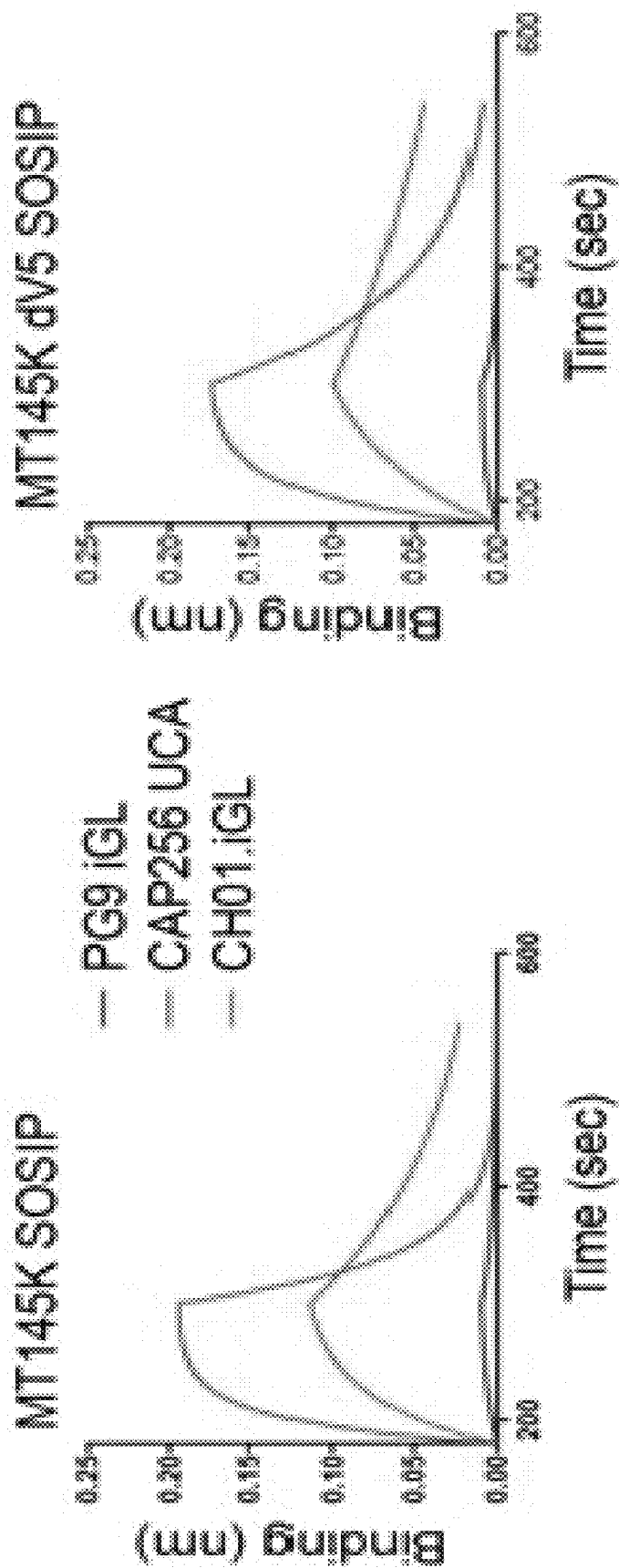


Fig. 20B Cont'd

C. Immunization schedule

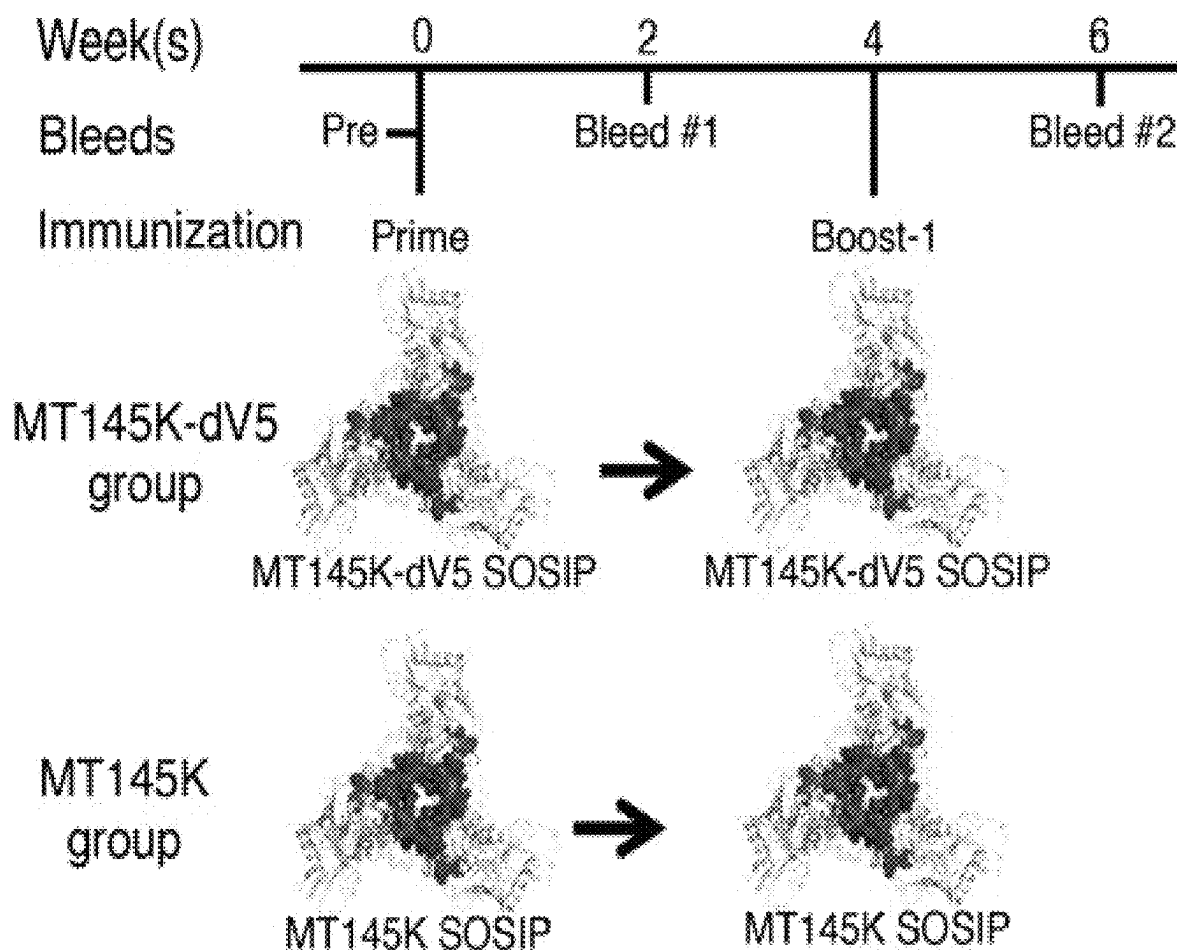


Fig. 20C

D. Neutralizing Ab responses

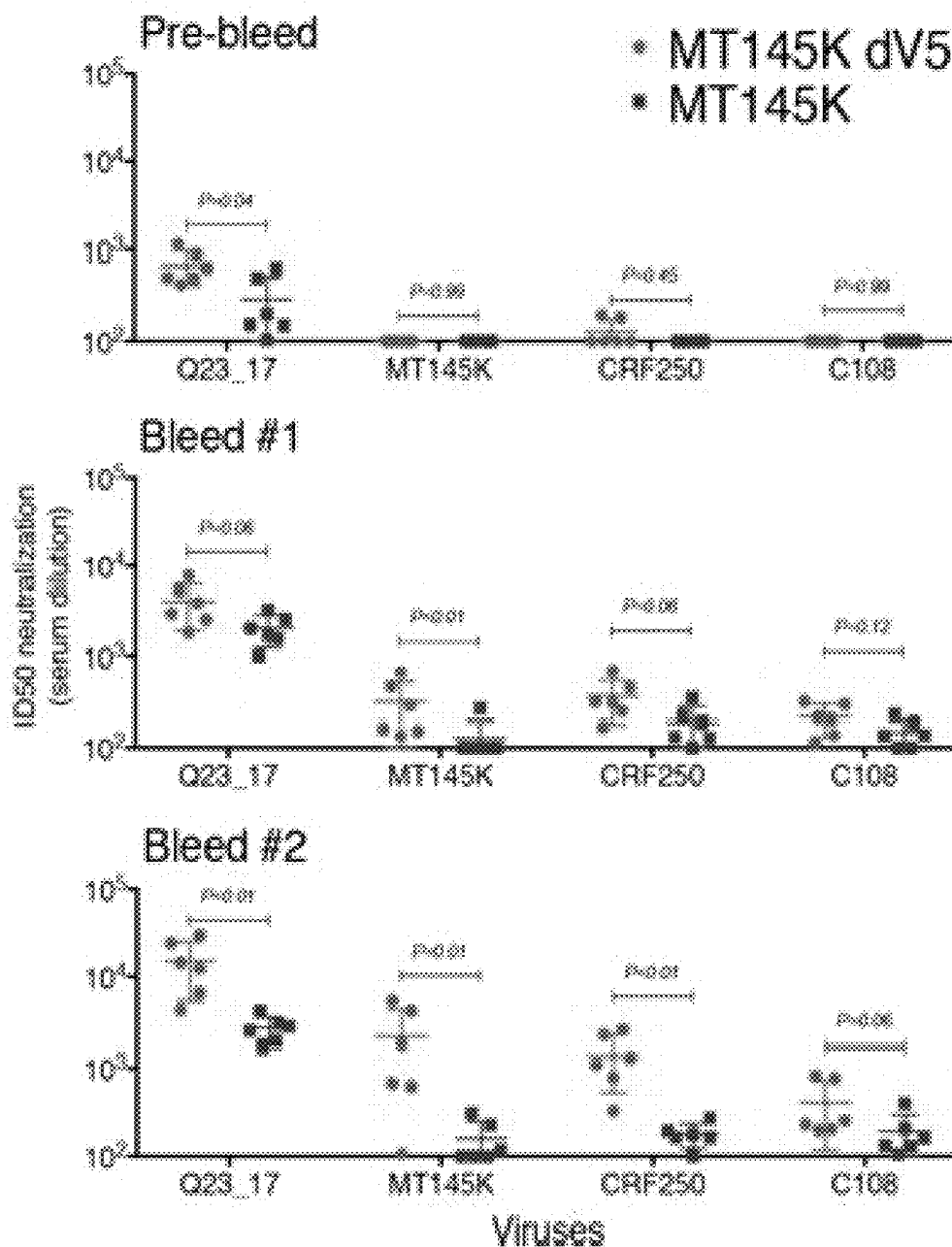


Fig. 20D

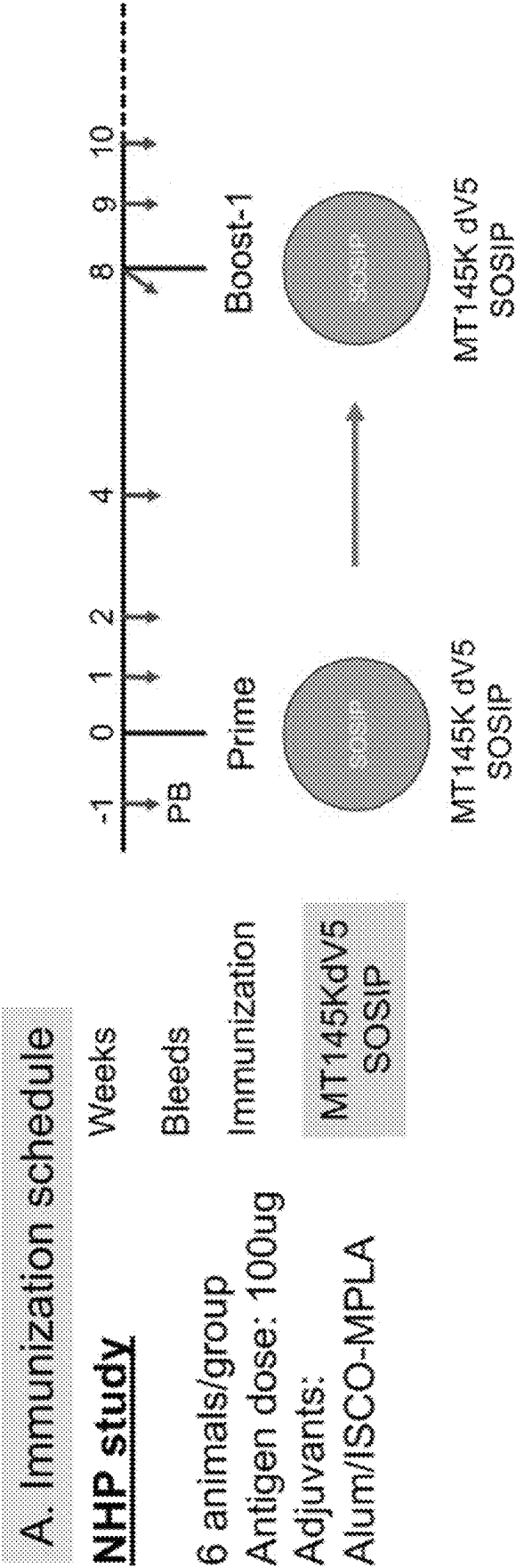


Fig. 21A

B. ELISA binding

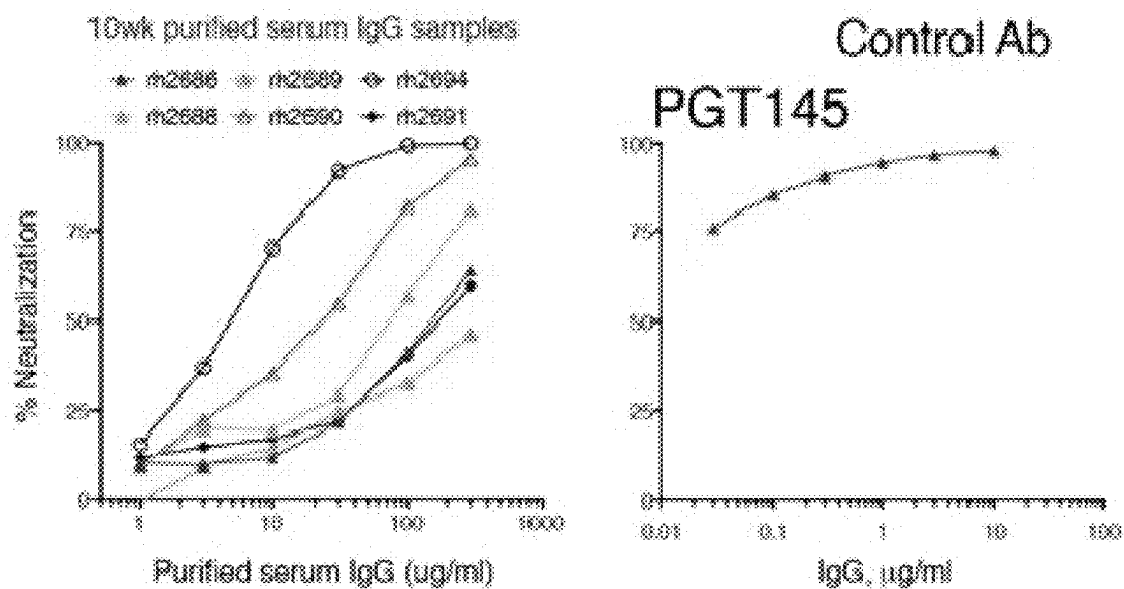
MT145KdV5 SOSIP						
	rh2686	rh2688	rh2689	rh2690	rh2691	rh2694
PB	<30	<30	<30	<30	<30	<30
1wk	<30	<30	<30	<30	<30	<30
2wk	267	832	217	749	344	658
4wk	722	2262	1650	2897	2167	962
8wk	240	1793	858	2743	1356	733
9wk	33926	41971	58730	43796	65915	49644
10wk	76452	83972	102384	65255	110291	69398

EC50 serum IgG ELISA binding titers with
MT145KdV5 SOSIP

Fig. 21B

C. Virus neutralization

MT145KdV5 virus



	MT145KdV5 SOSIP					
Virus	rh2686	rh2688	rh2689	rh2690	rh2691	rh2694
MT145KdV5	152	64	>300	19	189	5

IC50 virus neutralizing Ab titers
(IC50: purified serum IgG titer (ug/ml))

Fig. 21C

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SIV ENVELOPE TRIMER**RELATED APPLICATIONS AND
INCORPORATION BY REFERENCE**

This application is a continuation-in-part of international patent application Serial No. PCT/US2019/056867 filed Oct. 18, 2019, which published as PCT Publication No. WO 2020/081895 on Apr. 23, 2020, and which claims priority to U.S. provisional application Ser. No. 62/747,650, filed on Oct. 18, 2018.

The foregoing applications, and all documents cited therein or during their prosecution (“appln cited documents”) and all documents cited or referenced in the appln cited documents, and all documents cited or referenced herein (“herein cited documents”), and all documents cited or referenced in herein cited documents, together with any manufacturer’s instructions, descriptions, product specifications, and product sheets for any products mentioned herein or in any document incorporated by reference herein, are hereby incorporated herein by reference, and may be employed in the practice of the invention. More specifically, all referenced documents are incorporated by reference to the same extent as if each individual document was specifically and individually indicated to be incorporated by reference.

FEDERAL FUNDING LEGEND

This invention was made with government support under grant number AI100663 awarded by the National Institutes of Health. The government has certain rights in the invention.

SEQUENCE STATEMENT

The instant application contains a Sequence Listing which has been submitted electronically and is hereby incorporated by reference in its entirety. Said ASCII copy, was created Nov. 18, 2024, and amended on Nov. 20, 2024, is named Y7969_03051SL.txt and is 432,909 bytes in size.

FIELD OF THE INVENTION

The present invention relates to epitope-targeted SIV and HIV vaccines. The invention relates to epitopes that elicit antibodies that are broadly neutralizing against HIV. The invention provides epitopes that efficiently stimulate germ-line antibody production. The invention provides novel envelope glycoproteins which may be utilized as HIV-1 vaccine immunogens, antigens for crystallization, and for identification of broadly neutralizing antibodies. The invention encompasses preparation and purification of immunogenic compositions which are formulated into vaccines of the present invention.

BACKGROUND OF THE INVENTION

AIDS, or Acquired Immunodeficiency Syndrome, is caused by human immunodeficiency virus (HIV) and is characterized by several clinical features including wasting syndromes, central nervous system degeneration and profound immunosuppression that results in opportunistic infections and malignancies. HIV is a member of the lentivirus family of animal retroviruses, which include the visna virus of sheep and the bovine, feline, and simian immunodeficiency viruses (SIV). Two closely related types of HIV,

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designated HIV-1 and HIV-2, have been identified thus far, of which HIV-1 is by far the most common cause of AIDS. However, HIV-2, which differs in genomic structure and antigenicity, causes a similar clinical syndrome.

5 An infectious HIV particle consists of two identical strands of RNA, each approximately 9.2 kb long, packaged within a core of viral proteins. This core structure is surrounded by a phospholipid bilayer envelope derived from the host cell membrane that also includes virally-encoded membrane proteins (Abbas et al., Cellular and Molecular Immunology, 4th edition, W.B. Saunders Company, 2000, p. 454). The HIV genome has the characteristic 5'-LTR-Gag-Pol-Env-LTR-3' organization of the retrovirus family. Long terminal repeats (LTRs) at each end of the viral genome 10 serve as binding sites for transcriptional regulatory proteins from the host and regulate viral integration into the host genome, viral gene expression, and viral replication.

The HIV genome encodes several structural proteins. The gag gene encodes structural proteins of the nucleocapsid core and matrix. The pol gene encodes reverse transcriptase (RT), integrase (IN), and viral protease (PR) enzymes required for viral replication. The tat gene encodes a protein that is required for elongation of viral transcripts. The rev gene encodes a protein that promotes the nuclear export of incompletely spliced or unspliced viral RNAs. The vif gene product enhances the infectivity of viral particles. The vpr gene product promotes the nuclear import of viral DNA and regulates G2 cell cycle arrest. The vpu and nef genes encode proteins that down regulate host cell CD4 expression and enhance release of virus from infected cells. The env gene encodes the viral envelope glycoprotein that is translated as a 160-kilodalton (kDa) precursor (gp160) and cleaved by a cellular protease to yield the external 120-kDa envelope glycoprotein (gp120) and the transmembrane 41-kDa envelope glycoprotein (gp41), which are required for the infection of cells (Abbas et al., Cellular and Molecular Immunology, 4th edition, W.B. Saunders Company, 2000, pp. 454-456). gp140 is a modified form of the Env glycoprotein, which contains the external 120-kDa envelope glycoprotein portion and the extracellular part of the gp41 portion of Env and has characteristics of both gp120 and gp41. The nef gene is conserved among primate lentiviruses and is one of the first viral genes that is transcribed following infection. In vitro, several functions have been described, including down-regulation of CD4 and MHC class I surface expression, altered T-cell signaling and activation, and enhanced viral infectivity.

HIV infection initiates with gp120 on the viral particle binding to the CD4 and chemokine receptor molecules (e.g., CXCR4, CCR5) on the cell membrane of target cells such as CD4+ T-cells, macrophages and dendritic cells. The bound virus fuses with the target cell and reverse transcribes the RNA genome. The resulting viral DNA integrates into the cellular genome, where it directs the production of new viral RNA, and thereby viral proteins and new virions. These virions bud from the infected cell membrane and establish productive infections in other cells. This process also kills the originally infected cell. HIV can also kill cells indirectly because the CD4 receptor on uninfected T-cells has a strong affinity for gp120 expressed on the surface of infected cells. In this case, the uninfected cells bind, via the CD4 receptor-gp120 interaction, to infected cells and fuse to form a syncytium, which cannot survive. Destruction of CD4+T-lymphocytes, which are critical to immune defense, is a major cause of the progressive immune dysfunction that is the hallmark of AIDS disease progression. The loss of CD4+ T cells seriously impairs the body’s ability to fight most

invaders, but it has a particularly severe impact on the defenses against viruses, fungi, parasites and certain bacteria, including mycobacteria.

Research on the Env glycoprotein has shown that the virus has many effective protective mechanisms with few vulnerabilities (Wyatt & Sodroski, *Science*. 1998 Jun. 1 9; 280(5371):1884-8). For fusion with its target cells, HIV-I uses a trimeric Env complex containing gp120 and gp41 subunits (Burton et al., *Nat Immunol*. 2004 March; 5(3): 233-6). The fusion potential of the Env complex is triggered by engagement of the CD4 receptor and a coreceptor, usually CCR5 or CXCR4. Neutralizing antibodies seem to work either by binding to the mature trimer on the virion surface and preventing initial receptor engagement events, or by binding after virion attachment and inhibiting the fusion process (Parren & Burton, *Adv Immunol*. 200 1; 77:195-262). In the latter case, neutralizing antibodies may bind to epitopes whose exposure is enhanced or triggered by receptor binding. However, given the potential antiviral effects of neutralizing antibodies, it is not unexpected that HIV-I has evolved multiple mechanisms to protect it from antibody binding (Johnson & Desrosiers, *Annu Rev Med*. 2002; 53:499-518).

Most experimental HIV-I vaccines tested in human and/or non-human primate suggests that a successful vaccine incorporate immunogens that elicit broad neutralizing antibodies (bnAbs) and robust cell-mediated immunity. HIV-I envelope glycoprotein (Env) is the main viral protein involved in the entry of the virus and is also the primary target for neutralizing antibodies, but due to immune evasion strategies and extreme sequence variability of Envs, generation of bnAbs has been a daunting task (Phogat S, Wyatt R. *Curr Pharm Des*. 2007; 13:213-27, Phogat S, et al. *J Intern Med*. 2007 262:26-43, Karlsson Hedestam G B, et al *Nat Rev Microbial*. 2008 6:143-55).

SUMMARY OF THE INVENTION

HIV and chimpanzee SIV Envs, the target of potentially protective neutralizing antibodies, display about 60% sequence conservation at the amino acid level, and HIV V2-apex bnAbs have been shown to neutralize certain chimpanzee SIV isolates, including the SIVcpzPtt isolate MT145, suggesting cross-species conservation of this epitope. The invention provides engineered molecules derived from SIV isolates that bind to broadly neutralizing HIV antibodies. In certain embodiments, the engineered molecules mimic structures and epitopes such as but not limited to V2 or V3.

In certain examples herein, the engineered molecules include SIVcpzPtt-based trimers including from the MT145 isolate and mutants of the MT145 isolate comprising amino acid substitutions, such as Asn at position 171 by Lys, which display native trimer-like properties, and selectively binds V2 apex bnAbs. In certain embodiments, the engineered molecules are designed to bind to germline or germline-reverted antibodies. For example, molecules comprising the V2 apex epitope of MT145K are effective in activating V2-apex-specific B cell precursors.

The Applicants determined the structure of the MT145K trimer by cryo-EM at a global resolution of 4.1 Å. The overall architecture displays similarity to HIV Env trimers (Julien et al., 2013a; Lyumkis et al., 2013; Ozorowski et al., 2017; Pancera et al., 2014). In addition, the glycan shield composition of MT145K closely resembled that of HIV Env glycans but is sufficiently different in detailed positioning of the glycans to exclude binding of all HIV bnAbs except for

those directed to the V2 apex. MT145K trimer immunization in a V2 apex unmutated common ancestor (UCA)-expressing knock-in mouse model reveals induction of a predominantly V2-apex-site neutralizing Ab response that is reproducible and cross-neutralizes a related set of HIV isolates.

The engineered molecules are useful as immunogens in HIV vaccination strategies. In V2-apex bnAb unmutated common ancestor (UCA) H chain only knock-in mice, MT145K trimer-derived molecules reproducibly elicited Abs able to neutralize the autologous virus and several V2-apex Ab sensitive viruses. The specificities of the nAbs are dependent on the glycan at N160 and a lysine on strand C of the V2. Boosting with a cocktail of HIV Env trimers successfully recalled the V2 apex specific nAb responses and generated some enhanced heterologous neutralization.

In an aspect, the invention provides an engineered or non-naturally occurring molecule from Simian Immunodeficiency Virus (SIV) that binds to a broadly neutralizing antibody (bnAb) that binds to HIV. In an embodiment of the invention, the bnAb binds to the V2 apex region of HIV. In an embodiment, the bnAb is a germline or germline reverted HIV bnAb. In an embodiment, the bnAb comprises one, two, three, four, five, or six CDRs of a bnAB described herein, including without limitation sequences in Table 2, or comprises one or more complementarity determining regions (CDRs) of a heavy chain variable domain, and/or one or more CDRs of a light chain variable domain.

In certain embodiments, the engineered or non-naturally occurring molecule binds to a HIV bnAB which comprises an amino acid sequence at least 50% identical, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identical to PG9 or to CH01 antibody.

In an embodiment of the invention, the molecule comprises a complex of gp120 and gp41. In certain embodiments, the molecules comprise all or part of a subunit, such as the Env protein of SIV. Env subunits and epitopes therefrom, particularly immunogenic epitopes are useful in the invention. Thus, in certain embodiments, the molecules comprise amino acids from SIV grafted into another protein or peptide, for example but without limitation HIV, HIV env, or other molecule capable of presenting an SIV V2 apex region and/or V2 apex epitope of the molecules herein. In an embodiment, the molecule comprises a gp140. In an embodiment, the molecule comprises a complex of gp120 and gp41. In an embodiment, the molecule comprises gp120 and gp140 that are covalently linked. In an embodiment, the molecule comprises gp120 and gp140 fusion. In an embodiment, the molecule comprises an SIV trimer. In an embodiment, the molecule comprises a chimeric trimer, such as but not limited to a chimeric trimer combining an SIV sequence within the V2 apex region that binds to a bnAb and HIV sequences elsewhere. In an embodiment, the molecule is a stabilized trimer, such as without limitation, a SOSIP, NFL, or UFO trimer.

In an embodiment, the SIV, V2 apex region, or V2 apex epitope comprises or is from SIVcpzPtt, SIVcpzPts, or SIVgor. In an embodiment, the V2 apex region, or V2 apex epitope comprises or is from the SIVcpzPtt isolate MT145 or other V2-apex sequence set forth herein. In an embodiment, the V2 apex region, or V2 apex epitope comprises one or more amino acid substitutions that improve binding to a HIV broadly neutralizing antibody (bnAb) or to a germline precursor of the HIV bnAb. In an embodiment, the V2 apex

epitope is carried on an env protein or fragments altered by deletion, insertion, or substitution of amino acids to disrupt an immunogenic or immunodominant epitope other than the epitope that binds to a bnAb. In an embodiment, the V5 loop is altered. In an embodiment, one or more amino acids is substituted with a basic amino acid, e.g., arginine or lysine. In an embodiment, the amino acid corresponding to position 171 of the MT145 isolate is or is substituted with lysine.

As described herein, in certain embodiments of the invention, useful V2 apex sequences comprise two glycans, thus two glycosylated asparagine residues, and basic amino acid residues, preferably three or four basic amino acid residues, in a four-amino acid sequence, for example, without limitation, Xxx-Lys-Lys-Lys (SEQ ID NO:1), Lys-Xxx-Lys-Lys (SEQ ID NO:2), Lys-Lys-Xxx-Lys (SEQ ID NO:3), Lys-Lys-Lys-Xxx (SEQ ID NO:4), or Lys-Lys-Lys-Lys (SEQ ID NO:5). In certain embodiments, the molecule comprises Asn-Xxx_a-Asn-Xxx_b-Xxx_c-Xxx_d, wherein Xxx_a comprises two or three amino acids, Xxx_b comprises five to seven amino acids, and Xxx_c comprises four amino acids, each of the four amino acid selected from Lys or Arg (SEQ ID NO:6).

In certain embodiments, molecules of the invention comprise the amino acid sequence N-C-X₁-F-N-X₂-X₃-X₄-X₅-X₆-X₇-X₈-X₉-X₁₀-X₁₁-X₁₂-X₁₃-X₁₄-X₁₅-X₁₆, wherein X₁ comprises S, T, N, or F; X₂ comprises I, V, T, Q, or M; X₃ comprises S or T; X₄ comprises S or T; X₅ comprises S, E, or G; X₆ comprises I, L, V or F; X₇ comprises K or R; X₈ comprises G or D; X₉ comprises K, R, E, or Q; X₁₀ comprises K, R, E, or Q; X₁₁ comprises K, R, E, or Q; X₁₂ comprises K, E, or Q; X₁₃ comprises E, T, V, I, or M; X₁₄ comprises Y or K; X₁₅ comprises A or S; and X₁₆ comprises F, I, or L (SEQ ID NO:7). In certain embodiments, at least three of X₉, X₁₀, X₁₁, and X₁₂ comprise basic amino acid residues (SEQ ID NO:8). In certain embodiments, all of X₉, X₁₀, X₁₁, and X₁₂ comprise basic amino acid residues (SEQ ID NO:9). In certain embodiments, X₉, X₁₀, X₁₁, and X₁₂ each comprise K (SEQ ID NO:10). In one such embodiment, the amino acid sequence comprises NCSFNVTTELDRDKKRQVYSL (SEQ ID NO:11). In another embodiment, the amino acid sequence comprises NCSFNVTTELDRDKKRKVYSL (SEQ ID NO:12). In another embodiment, the amino acid sequence comprises NCSFNVTTELDRDKKKKEYSF (SEQ ID NO:13). In another embodiment, the amino acid sequence comprises NCSFNATTELDRDKKKKEYAL (SEQ ID NO:14).

In certain embodiments, molecules of the invention comprise Asn-Cys-Ser-Phe-Asn-Val-Thr-Thr-Glu-Leu-Arg-Asp-Lys-Lys-Arg-Gln-Val-Tyr-Ser-Leu-Phe-Phe-Tyr (SEQ ID NO:15), or Asn-Cys-Ser-Phe-Asn-Val-Thr-Thr-Glu-Leu-Arg-Asp-Lys-Lys-Arg-Gln-Val-Tyr-Ser-Leu-Phe-Tyr (SEQ ID NO:16), or Asn-Cys-Ser-Phe-Asn-Val-Thr-Thr-Glu-Leu-Arg-Asp-Lys-Lys-Arg-Lys-Val-Tyr-Ser-Leu-Phe-Phe-Tyr (SEQ ID NO:17), or Asn-Cys-Ser-Phe-Asn-Val-Thr-Thr-Glu-Leu-Arg-Asp-Lys-Lys-Arg-Lys-Val-Tyr-Ser-Leu-Phe-Tyr (SEQ ID NO:18), or Asn-Cys-Ser-Phe-Asn-Val-Thr-Thr-Glu-Leu-Arg-Asp-Lys-Lys-Lys-Glu-Tyr-Ser-Phe-Phe-Tyr (SEQ ID NO:19); or Asn-Cys-Ser-Phe-Asn-Val-Thr-Thr-Glu-Leu-Arg-Asp-Lys-Lys-Lys-Lys-Glu-Tyr-Ser-Phe-Phe-Tyr (SEQ ID NO:20); or Asn-Cys-Ser-Phe-Asn-Ala-Thr-Thr-Glu-Leu-Arg-Asp-Lys-Lys-Lys-Lys-Glu-Tyr-Ala-Leu-Phe-Phe-Tyr (SEQ ID NO:21), or Asn-Cys-Ser-Phe-Asn-Ala-Thr-Thr-Glu-Leu-Arg-Asp-Lys-Lys-Lys-Lys-Glu-Tyr-Ala-Leu-Phe-Tyr (SEQ ID NO:22).

In an aspect, the invention provides a nucleic acid encoding the molecule from Simian Immunodeficiency Virus (SIV) that binds to a broadly neutralizing antibody (bnAb)

that binds to HIV. In an embodiment, the nucleic acid comprises RNA. In an embodiment, the nucleic acid comprises DNA. In an embodiment, the nucleic acid is operably linked to a regulatory element operable in a cell. In certain such embodiments, the regulatory element is operable in a eukaryotic cell. In certain such embodiments, the regulatory element is operable in a prokaryotic cell. The invention further provides vectors, such as but not limited to viral vectors, comprising the nucleic acids and regulatory elements. In an embodiment of the invention, the viral vector is AAV. The invention further provides cell which comprise the nucleic acids, including cells that express the molecules of the invention.

In an aspect, the invention provides a method of eliciting an immune response in a mammal comprising administering an immunogenic composition comprising a molecule of the invention. In an embodiment, a method of stimulating an HIV bnAb is provided. In certain embodiments, the bnAb is a germline antibody or a germline reverted antibody. In an embodiment of the invention, the mammal is a human. In an embodiment, the mammal is a non-human primate, such as but not limited to a chimpanzee or gorilla. In an embodiment, the mammal is one capable of expressing one or more elements of a human immune system, for example a transgenic animal such as but not limited to a mouse. In one such embodiment, the mammal comprises and is capable of expressing a human germline or germline reverted antibody or germline or germline reverted heavy chain variable domain or light chain variable domain. In another embodiment, the mammal comprises and is capable of expressing a repertoire of human germline genes.

In certain embodiments, a molecule of the invention is administered one time. In other embodiments, a molecule of the invention is administered multiple times, such as twice, three times, or more. In certain embodiments, more than one molecule is administered, such as two, three, four, or more molecules. When immunogenic compositions are administered at more than one time point, the molecules administered at each time can be the same or different. For example, two or more that are different molecules can be administered concurrently or sequentially. Alternatively, the same molecule can be administered on two or more occasions. Without being bound by theory, the molecules are administered in a course of immunization designed to favor production of bnAbs. bnAbs can be favored when a selected epitope that elicits bnAbs is present in multiple immunizations while other epitopes of the immunogen differ from one immunization to another.

In certain embodiments, one molecule of the invention is administered at a first time point to initiate an immune response, such as, for example, to stimulate germline antibody-expressing B cells, and again at a later time to boost an immune response. In certain embodiments, one molecule of the invention is administered at a first time point and another different molecule of the invention is administered at a later time to selectively boost the antibody response against an epitope common to both molecules that binds to a HIV bnAb.

In certain embodiments, a molecule of the invention is administered at a first time point to initiate an immune response, such as, for example, to stimulate germline antibody-expressing B cells, and an HIV molecule, antigen, or vaccine is administered at a later time point to boost the response against an HIV epitope that binds to a HIV bnAb.

In certain embodiments, the germline antibodies that are stimulated are bnAbs. In certain embodiments, the germline antibodies are precursors of bnAbs. In certain such embodi-

ments, the germline antibodies undergo somatic hypermutation and selection during the course of an immune response to higher affinity and/or specificity bnAbs.

In certain embodiments, a molecule of the invention is administered to boost or reactivate memory B cells.

The invention provides immunogenic compositions comprising the molecules of the invention. In certain embodiments, a molecule of the invention is comprised in a composition or administered with an adjuvant. In certain embodiments, the adjuvant comprises a lecithin. In certain embodiments, the lecithin is (a) combined with an acrylic polymer, (b) in a coated oil droplet in an oil-in-water emulsion, or (c) in an acrylic polymer in an oil-in-water emulsion. In certain embodiments the adjuvant is ISCOMATRIX or Adjuvax. In certain embodiments the adjuvant comprises alum.

In certain embodiments, the immunogenic composition comprises a liposome or a nanoparticle. In certain embodiments, the molecule of the invention is fixed. In an embodiment, the fixing agent can be glutaraldehyde. In an embodiment, the trimer or immunogen comprising the trimer is quenched with glycine.

In an aspect, the invention provides a method of engineering a molecule or immunogen capable of eliciting a broadly neutralizing antibody (bnAb) directed to HIV. The method comprises identifying a viral structure such as a viral epitope of SIV or other immunodeficiency virus that is conserved in HIV, selecting or designing bnAb that binds to the epitope, and designing a molecule or immunogen that comprises the conserved epitope, optionally modified to bind to a germline or germline derived antibody precursor of the bnAb. In certain embodiments, the structure, region or epitope is that of V2 of SIV.

In an aspect of the invention, there is provided a method of identifying or selecting a broadly neutralizing component of binding pair wherein such component binds to HIV and is broadly neutralizing. The method comprises contacting a molecule of the invention with a candidate binding pair component and selecting or identifying the component as broadly neutralizing if it binds to a molecule of the invention. In certain embodiments, the broadly neutralizing component comprises a bnAb. In certain embodiments, the broadly neutralizing component comprises a germline or germline reverted bnAb. In certain embodiments, the broadly neutralizing component is a germline or germline reverted antibody that is a precursor of a bnAb.

Accordingly, it is an object of the invention not to encompass within the invention any previously known product, process of making the product, or method of using the product such that Applicants reserve the right and hereby disclose a disclaimer of any previously known product, process, or method. It is further noted that the invention does not intend to encompass within the scope of the invention any product, process, or making of the product or method of using the product, which does not meet the written description and enablement requirements of the USPTO (35 U.S.C. § 112, first paragraph) or the EPO (Article 83 of the EPC), such that Applicants reserve the right and hereby disclose a disclaimer of any previously described product, process of making the product, or method of using the product. It may be advantageous in the practice of the invention to be in compliance with Art. 53(c) EPC and Rule 28(b) and (c) EPC. All rights to explicitly disclaim any embodiments that are the subject of any granted patent(s) of applicant in the lineage of this application or in any other lineage or in any prior filed application of any third party is explicitly reserved. Nothing herein is to be construed as a promise.

It is noted that in this disclosure and particularly in the claims and/or paragraphs, terms such as “comprises”, “comprising”, “includes”, “included”, “including”, and the like; and that terms such as “consisting essentially of” and “consists essentially of” have the meaning ascribed to them in U.S. Patent law, e.g., they allow for elements not explicitly recited, but exclude elements that are found in the prior art or that affect a basic or novel characteristic of the invention.

These and other embodiments are disclosed or are obvious from and encompassed by, the following Detailed Description.

BRIEF DESCRIPTION OF THE DRAWINGS

The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

The following detailed description, given by way of example, but not intended to limit the invention solely to the specific embodiments described, may best be understood in conjunction with the accompanying drawings.

FIG. 1A-1D. Design of a chimpanzee Env-stabilized trimer and binding to V2 apex bnAb iGL Abs. A. Structural arrangement of the V2 apex bnAb core epitope region on BG505.664 soluble Env trimer (modified from (Garces et al., 2015) (PDB: 5CEZ)). The ribbon representation of V1V2 loop strands that form the trimer apex show a cluster of positively charged lysine-rich peptide regions (HXB2-R166-K171: R or K residues shown as blue spheres) and the two glycans N156 and N160 (depicted in green spheres with lines). The side chains of the positively charged residues intersperse with the side chains of residues from adjacent protomers to form a continuous positively charged surface at the tip of the trimer to provide a minimal V2 apex bnAb epitope. B. Amino-acid sequence alignment of strand B and C V2 of HXB2 (SEQ ID NO:23), HIV CRF250 (SEQ ID NO:20), CAP256. SU (SEQ ID NO:22), chimpanzee SIV MT145 WT (SEQ ID NO:16) and its V2-modified variant (Q171K), MT145K (SEQ ID NO:28). Glutamine (Q) at position 171 (shown in red) was substituted with lysine (K) in MT145 Env to gain binding to V2 apex bnAb inferred germline (iGL) Abs. C. ELISA binding of mature V2 apex bnAbs, PG9, CAP256.09 and CH01 and their iGL versions to WT MT145 (red) and MT145K SOSIP trimers. D. Octet binding curves (association: 120s (180-300) and dissociation: 240s (300-540)) of CAP256 UCA and CH01 iGL Abs and their respective mature Ab versions (CAP256.09 and CH01) to MT145K trimer, its glycan knock-out (N160K) variant, K-rich core epitope substituted variants and the corresponding monomeric gp120. The Abs were immobilized on human IgG Fc capture biosensors and 1 μ M trimer or gp120 proteins used as analytes. The binding response is shown as nanometer (nm).

FIG. 2A-2E. Cryo-EM structure of the MT145K trimer. A. Schematic showing MT145K SOSIP soluble trimer design from its full-length gp160 Env sequence. The gp120 constant (C1-C5) and variable (V1-V5) regions and the gp41 regions (fusion peptide (FP), heptad repeat (HR1 and HR2), membrane proximal external region (MPER), transmembrane (TM) and cytoplasmic tail (CT)) are indicated. The N-linked glycan positions for each NXT/S residue are labeled according to the HIV HXB2 numbering scheme. The SOSIP trimer stabilizing modifications include: (i) disulfide bond: A501C-T605C, (ii) R6 cleavage site, (iii) I559P, and

(iv) 664-residue truncation in gp41 MPER. The substitution to incorporate a K-residue at position 171 (Q171K) to gain binding for V2 apex iGL Abs is indicated in blue. B-C. Side and top views of the unliganded MT145K trimer model based on the cryo-EM density map at ~4.1 Å resolution. Ribbon representations of the MT145K trimer spike, in which the subunits gp120 (cornflower blue) and gp41 (orange) are depicted on one protomer. The gp120 variable loops (V1-V5) positioned to the trimer periphery are depicted in different colors (V1: khaki, V2: red, V3: magenta, V4: yellow and V5: chartreuse). The fusion peptide region of gp41 is shown in cyan. Glycan sugar residues modeled based on density are represented in forest green stick form. D. Superimposition of variable loops (V1-V5) and fusion peptide region for MT145K and unliganded HIV clade A BG505 (PDB: 4ZMJ) SOSIP trimers. The dotted lines indicate regions in the V-loops or FP for which the observed electron density was absent or unclear. E. Structural comparison of gp41 regions of MT145K (orange) and BG505 (grey) trimers. The gp41 structural elements overall show a similar arrangement except for the fusion peptide region (colored cyan on MT145K and pink on BG505), that is exposed on the BG505 trimer but remains hidden in a pocket inside the MT145K trimer.

FIG. 3A-3C. Site-specific glycoform composition of MT145K trimer. A. Site-specific glycoform quantification of the MT145K SOSIP soluble trimer. MT145K trimers from transiently transfected HEK293T cell expressed supernatants were affinity purified by the quaternary trimer-specific antibody, PGT145. The purified MT145K trimers were treated separately with three proteases: trypsin, chymotrypsin and elastase and the digests were enriched for glycopeptides and analysed by LC-ESI MS. The individual glycan composition of the N-linked glycan sites (n=26) is represented by bar graphs that indicate the relative abundance of each glycoform species and are derived from the mean of two analytical replicates. The pie charts summarize the proportion of glycoforms for each site and this information is color coded; oligomannose-type (green) and complex/hybrid glycans (pink). B. HILIC-UPLC profiles of the total N-linked glycans released from MT145K trimers. The proportions of oligomannose plus hybrid glycan contents and complex-type glycans are represented in green and pink colors, respectively. C. Modeled glycan shields for MT145K and BG505 SOSIP trimers. $\text{Man}_9\text{GlcNAc}_2$ oligomannose-type glycans were docked and rigid-body-fitted at each of the corresponding Env glycan positions using the MT145K structure (determined in this study (PDB: submit)) and the unliganded BG505 SOSIP.664 trimer structure ((Kwon et al., 2015) PDB: 4ZMJ). Top and side views of the trimers are shown and the individual glycan sites are labelled and color-coded based on the content of oligomannose; green (100-80%), orange (79-20%) and pink (19-0%).

FIG. 4A-4B. Antigenic profile of the MT145K trimer. A. HIV-1 Env-specific mAbs were used to characterize the antigenicity of the MT 145K Env trimer. MAb targeting neutralizing and non-neutralizing epitope specificities, including V2-apex, N332-V3, linear V3, CD4bs, CD4i and gp120-41 interface were tested with MT145K and BG505 Env-encoding pseudoviruses in a TZM-bl cell-based reporter assay. The reciprocal IC_{50} neutralization titers for each virus are indicated as dot plots; plots for individual epitope specificities are depicted separately. The neutralization sensitivity comparison of BG505 and MT145K viruses against the mAb panel shows a selectively potent neutralization of MT145K by V2 apex bnAbs but no other bnAbs, except a single gp120-gp41 interface bnAb, 35022. BG505

virus was neutralized by bnAbs targeting diverse Env sites. B. The above mAb panel was further tested with PGT145 Ab-purified MT145K trimer and GNL-purified MT145K gp120 monomer by ELISA. The binding, represented as EC_{50} binding titers, shows selective binding of MT145K by V2 apex bnAbs. Two of the gp120-gp41 interface bnAbs and a CD4i mAb also showed significant binding to MT145K trimer. Four of the non-neutralizing mAbs specific to a linear V3 epitope exhibited binding to MT145K gp120, but not to the trimer.

FIG. 5A-5F. A close-up view of regions on the MT145K trimer that correspond to those recognized by HIV bnAbs on HIV trimers. A. V2 apex bnAb binding region: cryo-EM model of PGT145 bnAb (HC: transparent sandy brown; LC: transparent orchid) in complex with BG505 SOSIP trimer depicting V1V2 loops in ribbon representation ((Lee et al., 2017) PDB: 5V8L). The strand C K-rich region ($^{166}\text{RDKKQK}^{171}$ (SEQ ID NO:355); red spheres) and the glycan N160 (forest green sticks) that form the epitope for PGT145 bnAb are indicated. The elements in the core epitope interact with the CDRL1 loop and the long CDRH3 loop that penetrates through glycans to reach the positively charged surface underneath. Both glycan N160 and the positively charged protein residues are conserved between BG505 HIV-1 and MT145K SIV Env trimers. B. V3-glycan bnAb binding region: cryo-EM model of PGT128 bnAb (HC: transparent sandy brown; LC: transparent orchid) in complex with the BG505 SOSIP trimer ((Lee et al., 2015) PDB: 5ACO). The V3 loop protein backbone residues ($^{324}\text{GDIR}^{327}$ (SEQ ID NO:356); depicted in purple spheres) and the glycans N301 and N332 form the bnAb epitope and are shown to interact with the antibody CDR loops. The MT145K trimer has a glycan at N334 rather than N332 and the glycan points away from the expected location of the PGT128 Ab paratope. In addition, MT145K Env has glycans at two positions N412, (positioned differently on HIV Env) and N442 (absent on HIV Envs) and particularly the latter will clash with PGT128 CDRH2 and prevent it from interacting with the protein part of the epitope. C. The gp120-gp41 interface bnAb binding region: cryo-EM model of PGT151 bnAb bound to a membrane-extracted clade B JRFL Env trimer. The structure depicts PGT151 bnAb CDRs interacting with gp120 and the gp41 interface regions ((Lee et al., 2016); PDB: 5FUU). PGT151 CDRH3 interacts with the epitope formed by the protein backbone (in both gp120 and gp41) including the fusion peptide (depicted in pink) and the gp120 (N88, N448) and gp41 (N611 and N637) glycans (not shown). PGT151 Ab CDR loops interact with the FP region on the BG505 trimer. The MT145K trimer FP region (cyan) remains hidden inside the trimer. D. Cryo-EM model of 2-domain human sCD4 with B41 SOSIP trimer ((Ozoroski et al., 2017); PDB: 5VN3). The structure shows how the Phe43 residue on sCD4 stacks into the Env cavity lining Trp427. This Trp427 cavity is conserved between HIV-1 and MT145K Envs to accommodate CD4 binding. E-F. CD4bs bnAb binding region: crystal structure of VRC01 bnAb in complex with 93TH057 gp120 ((Zhou et al., 2010) PDB: 3NGB). The structure depicts VRC01 CDRH3, CDRL3 and CDRL1 loops interacting with the protein residues in loop D (HXB2: 278-282) and the glycan at N276. The MT145K trimer lacks the N276 glycan and bears glycan N236 (unique to SIV Env) in place of N234 that would clash with the VRC01 CDRL1 loop. Additionally, the MT145K Env trimer has a longer gp120-V5 loop due to a 6-amino acid insertion at HIV HXB2-456 residue that would shift the loop such that it clashes with the CDRH2 the VRC01 Ab.

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FIG. 6A-6D. Immunogenicity of MT145-WT compared to MT145K trimers in CH01 UCA HC-only knock-in mice. A. Schematic showing immunization schedule of CH01 UCA "HC-only" KI mice with MT145-WT and engineered MT145K trimers. The CH01 UCA "HC-only" KI mice were immunized twice with 25 μ g of the soluble trimer with GLA-SA as adjuvant. Time points for immunization and bleeds are indicated. B. ELISA binding of the MT145-WT and MT145K group trimer-immunized CH01 UCA "HC-only" KI mice serum samples (pre-bleed (Pre), two-weeks post prime (Bleed #1) and two-weeks post boost-1 (Bleed #2)) with soluble MT145K SOSIP and its glycan knock-out variant (MT145K N160K) trimers. C. Neutralization titrations of the MT145-WT and MT145K group trimer immunized CH01 UCA "HC-only" KI mice sera (pre-bleed (Pre), post prime (Bleed #1) and post boost-1 (Bleed #2)) with MT145K virus and a CH01-sensitive virus (Q23_17). 3-fold diluted sera were tested against the viruses in a TZM-bl reporter cell assay. D. ID₅₀ neutralization titers of the MT145-WT and MT145K group trimer-immunized CH01 UCA "HC-only" KI mice sera (pre- and post-immunization bleed time points). Neutralization was assessed against the priming immunogen-matched autologous viruses in each group (MT145-WT group: MT145-WT virus, and MT145K group: MT145K virus), the N160 glycan knock-out variant of MT145K virus (MT145K N160A) and a CH01-sensitive virus, Q23_17. The numerical values shown in the table represent the ID₅₀ neutralization titers of the immune serum samples and were calculated by non-linear regression method from the percent neutralizations of serum titrations with virus.

FIG. 7A-7C. Immunizations combining chimpanzee SIV MT145K trimer with HIV trimers in CH01 UCA HC-only knock-in mice. A. Schematic showing immunization schedule of CH01 UCA "HC-only" KI mice with a combination of chimpanzee SIV MT145K trimer and HIV Env-derived trimers. A group of 5 animals was immunized with two doses of MT145K trimer (prime: week-0 and boost-1: week-4) followed by further boosting (boost-2 at week-8) with an HIV Env derived 3-trimer cocktail (C108, WITO and ZM197-ZM233V1V2). The V1V2 loops on trimer cartoons are depicted in red to highlight the region is shared across HIV and SIV Env trimers. The CH01 UCA "HC-only" KI mice were immunized with 25 μ g of the soluble MT145K trimer or HIV trimer cocktail (25 μ g total protein) with GLA-SA as adjuvant. Time points for the immunization and the bleeds are indicated. B. Neutralization of the pre- and post-immune sera were carried out against a panel of viruses in the TZM-bl reporter cell assay. The numerical values in the table represent ID₅₀ neutralization titers of the pre-bleed (Pre) and post-prime (Bleed #1), post boost-1 (Bleed #2) and post boost-2 (Bleed #3) serum samples. Neutralization was assessed against the priming immunogen-matched autologous virus, MT145K, its N160 glycan knock-out variant, boosting immunogen-matched, CH01 sensitive, and global panel HIV Env-encoding viruses. The numerical values shown in the table represent the ID₅₀ serum neutralization titers and were calculated by non-linear regression method from the percent neutralizations of serum titrations with virus. The IC₅₀ neutralization titers for the CH01 mature Ab against the virus panel are shown. C. The ID₅₀ virus neutralization titers of the Bleed #4 serum samples collected post boost-2 immunization with an HIV three-trimer cocktail. The neutralization of the post-immune sera was assessed against the priming immunogen-matched autologous virus, MT145K, the boosting immunogen-matched, CH01 sensitive viruses, and global panel HIV

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Env-encoding viruses. Each filled circle in the plot represents virus ID₅₀ neutralization values for the individual animals; an asterisk (*) indicates that 50% neutralization was not reached at a 1:100 serum dilution.

FIG. 8A-8B. Design and purification of SIVcpzPtt envelope derived SOSIP.664 trimers. A. Amino-acid sequence alignment of 4 SIVcpzPtt envelope sequences (GAB1 (SEQ ID NO:25), MB897 (SEQ ID NO:26), EK505 (SEQ ID NO:27) and MT145 (SEQ ID NO:28)) with reference HIV-1 Env sequence, HXB2 (SEQ ID NO:24), showing SOSIP.664 trimer stabilizing modifications. The soluble SOSIP.664 trimer modifications include: (i) incorporation of a disulfide bond between residue 501 of gp120 (A501C) and residue 605 of gp41 (T605C), (ii) replacing naturally occurring gp120 and gp41 cleavage site with R6-cleavage site (shown in blue), (iii) I559P substitution in the gp41, (iv) and a truncation in the gp41 subunit at residue 664. B. Size Exclusion Chromatography (SEC) profiles of *Gallanthus nivalis* lectin (GNL) purified trimers on Superdex 200 Increase 10/300 GL column. The SEC profiles show the aggregate/trimer-dimer, trimer and monomer peaks of the GNL-purified proteins. The MT145 SOSIP.664 trimer showed substantial protein fractions that assemble as a trimer.

FIG. 9A-9D. MT145K SOSIP.664 forms native-like trimers that are thermostable. Size exclusion chromatography (SEC) of PGT145 antibody purified MT145K SOSIP.664 trimer reveals trimer eluting as a single peak with no aggregation. B. SDS-PAGE of MT145K trimer protein under non-reducing (NR) and reducing (R) conditions (reducing agent DTT was added). The MT145K trimer is efficiently cleaved into gp120 and gp41 subunits. C. Negative Stain Electron-Microscopy (NS-EM) of MT145K trimer: 2D class averages show that trimers adopt well-ordered, native-like conformations. D. Thermostability of the MT145K trimer. Thermal denaturation of the MT145K trimer by Differential Scanning Calorimetry (DSC) reveals a major trimer population melting at a T_m of 65.4° C.

FIG. 10A-10B. Design of inferred germline (iGL) versions of V2 apex bnAbs, PG9 and CH01. The heavy chain (VH), light chain (VL) and the diversity (D) and the joining (J) regions in the mature antibodies were reverted to their corresponding inferred germline genes. The amino-acid reversions from mature VH (panel A) and VL (panel B) chains to germline residues are indicated by asterisks. The VH_{iGL} and the VL_{iGL} chains were paired and expressed as soluble antibodies. The GenBank accession numbers and SEQ ID NOS for the heavy and light chain inferred germline versions of PG9 and CH01 broadly neutralizing antibodies are PG9_VDJH_{iGL}: Accession MK825341 (SEQ ID NO:32); PG9_VJL_{iGL}: Accession MK825342 (SEQ ID NO:41); CH01VDJH_{iGL}: Accession MK825343 (SEQ ID NO:49); and CH01VJL_{iGL}: Accession MK825344 (SEQ ID NO:58). Depicted PG9-related sequences are identified in the Sequence Listing as follows: IGHV3-33-05: SEQ ID NO:29; PG9_HC: SEQ ID NO:30; PG9VH_{iGL}: SEQ ID NO:31; IGHJ6-03: SEQ ID NO:36; IGHD3-3-01: SEQ ID NO:37; IGLV2-14-01: SEQ ID NO:38; PG9_LC: SEQ ID NO:39; PG9VL_{iGL}: SEQ ID NO:40; IGLJ3-02: SEQ ID NO:45. Depicted CH01-related sequences are identified in the Sequence Listing as follows: IGHV3-20-01: SEQ ID NO:46; CH01_HC: SEQ ID NO:47; CH01VH_{iGL}: SEQ ID NO:48; IGHJ2-01: SEQ ID NO:53; IGHD3-10-01: SEQ ID NO:54; IGKV3-20-01: SEQ ID NO:55; CH01_LC: SEQ ID NO:56; CH01VL_{iGL}: SEQ ID NO:57; IGKJ1-01: SEQ ID NO:62.

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FIG. 11A-11D. Structural analysis of MT145K by cryo-electron microscopy. A. Cryo-EM data processing flow diagram resulting in a density map at ~ 4.1 Å global resolution. B. FSC between two independently refined data half sets. C. Local resolution in the MT145K density map. The peptide part of MT145K is resolved primarily in the 3.0-4.0 Å range, whereas the glycan shield overall is resolved to significantly lower resolutions. D. Angular distribution of the angularly refined data class giving rise to the final reconstruction.

FIG. 12. Comparison of Env trimer architectures of class I fusion proteins. Prefusion SIV Env (MT145K; current study) follows an overall organization similar to other known prefusion class I fusion proteins. Fusion subunits (shown in orange) are most commonly organized membrane-proximally (RSV F deviating somewhat from this trend). Fusion subunits are largely shielded and capped by receptor-recognizing subunits (shown in cornflower blue) in the meta-stable prefusion state. SIV Env is most similar to HIV-1 Env; however, MT145K is less compact than the displayed HIV-1 Env trimer (BG505 isolate). The more open organization of MT145K may be the result of its fusion peptide being inserted into a pocket at the gp120/gp41 interface as opposed to the fusion peptide found on the outside of the HIV-1 Env trimer (MT145K: current study, HIV-1: PDB 4ZMJ (Kwon et al., 2015); Influenza A H1N1: PDB 1RD8 (Stevens et al., 2004); Ebola virus GP: PDB 5KEL (Pallesen et al., 2016); MERS-CoV S: PDB 5W9J (Pallesen et al., 2017); RSV F: PDB 4JHW (McLellan et al., 2013)).

FIG. 13A-13B. Negative ion TOF MS of released glycans from Chimpanzee SIV, MT145K gp120 and gp41. A. Driftscope extracted singly charged ions of released glycans from MT145K-gp120. B. Driftscope extracted singly charged ions of released glycans from MT145K-gp41.

FIG. 14A-14B. Phylogeny of lentiviruses and the glycan shield density. A. Phylogenetic relationships among the envelope sequences derived from various lentiviruses; tree constructed by maximum likelihood method. These lentiviruses infect host species that include equine (EIAV), feline (FIV), bovine (BIV), caprine/ovine (CAEV, OMVVS and MVV), simian (SIV) and humans (HIV). B. The number of amino acids and the number of Potential N-linked Glycan Sites (PNGS: predicted by N-GlycoSite tool at Los Alamos HIV database) in the full-length envelopes are listed for each lentivirus. The glycan shield density for each virus is represented as % Env residues that encode PNGS motifs and was calculated by (number of PNGS/Env length $\times$ 100). The glycan shield density shows a gradual increase from EIAV through CAEV, and plateaus in some of the SIV species including SIVcpzPtt, the one that has been shown to have crossed into humans.

FIG. 15. Amino-acid sequence alignment of SIV envelopes with HXB2 HIV-1 Env as a reference depicting SIV Env residues equivalent to HIV bnAb epitopes. The amino-acid alignment of SIVcpzPtt envelope sequences compared to the HXB2 reference sequence. The protein residues (highlighted in grey) and the N-glycosylation sites (depicted in red) for each bnAb epitope region (V2-apex, V3-N332, CD4bs and gp120-gp41 interface) are shown. Eleven regions of 23 aligned Env isolates are depicted, indicated by amino acid numbers: i) 156-175, ii) 320-353, iii) 409-447, iv) 256-291, v) 228-245, vi) 448-467, vii) 84-93, viii) 448-455, ix) 507-534, x) 605-614, and xi) 621-646. The eleven regions of each of the 23 isolates are represented in the Sequence Listing as follows: 1) B. FR.1983. HXB2-LAI-III-BRU.K03455: SEQ ID NOS: 63-73; 2)

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CPZ.CM.2005. SIVcpzMT145. DQ373066: SEQ ID NOS: 74-84; 3) CPZ.CM.2005. SIVcpzMB897. EF535994: SEQ ID NOS:85-95; 4) CPZ.CM.2005. SIVcpzLB7. DQ373064: SEQ ID NOS:96-106; 5) CPZ.CM.2001. SIVcpzCAM13. AY169968: SEQ ID NOS: 107-117; 6) CPZ.GA.1988. GAB1. X52154: SEQ ID NOS:118-128; 7) CPZ.CM.2005. SIVcpzMB66. DQ373063: SEQ ID NOS: 129-139; 8) CPZ.GA.2006. SIVcpzGab4. GQ217539: SEQ ID NOS: 140-150; 9) CPZ.CM.2004. Ptt_04Cam155. FR686510: SEQ ID NOS:151-161; 10) CPZ.CM.1998. CAM3. AF115393: SEQ ID NOS:162-172; 11) CPZ.TZ.2001. TAN2_69. DQ374657: SEQ ID NOS: 173-183; 12) CPZ.TZ.2002. TAN3_1. DQ374658: SEQ ID NOS:184-194; 13) CPZ.TZ.2000. TAN1. AF447763: SEQ ID NOS: 195-205; 14) CPZ.TZ.2006. TAN5. JN091691: SEQ ID NOS:206-216; 15) CPZ.TZ.2009.UG38. JN091690: SEQ ID NOS:217-227; 16) CPZ.TZ.2006. SIVcpzTAN13. JQ768416: SEQ ID NOS:228-238; 17) CPZ. CD.2006. BF1167. JQ866001: SEQ ID NOS:239-249; 18) CPZ. CD.1990. ANT.U42720: SEQ ID NOS:250-260; 19) CPZ.GA.1988. SIVcpzGAB2. AF382828: SEQ ID NOS: 261-271; 20) CPZ.CM.2005. SIVcpzEK505. DQ373065: SEQ ID NOS:272-282; 21) CPZ.CM.1998. CAM5. AJ271369: SEQ ID NOS:283-293; 22) CPZ.US.1985.US_Marilyn.AF103818: SEQ ID NOS:294-304; 23) CPZ.CM.2005. SIVcpzDP943. EF535993: SEQ ID NOS:305-315.

FIG. 16. Antigenic profile of various MT145K Env forms with HIV Env-specific mAbs. Neutralizing and non-neutralizing HIV Env-specific mAbs targeting various epitope specificities, including V2-apex, V3-N332, linear V3, CD4bs, CD4i and gp120-gp41 interface were tested with MT145K and BG505 Env-encoding pseudoviruses in a neutralization assay and against the MT145K SOSIP trimer and MT145K gp120 monomer by ELISA. The reciprocal IC₅₀ neutralization titers against the MT145K and BG505 viruses and the 50% ELISA binding titers (EC₅₀ binding with trimer and gp120 proteins) for each mAb are shown.

FIG. 17A-17E. Molecular details of the regions on the MT145K trimer that correspond to bnAb epitopes on HIV trimers. A. Crystal structure of PG9 (PDB: 3U2S (McLellan et al., 2011)) bound to ZM109 V1V2 apex and docked onto BG505. The long CDRH3 of PG9 interacts with the K-rich region of the V2 apex and the glycans at N160. Both glycan/peptide epitope elements are conserved between MT145K and HIV-1 BG505 Env and, hence, PG9 binds strongly to both. B-C. Crystal structures of V3-N332 bnAbs PGT121 (5T3Z (Gristick et al., 2016)) and PGT135 (PDB: 4JM2 (Kong et al., 2013)) interacting with their epitopes at the base of the V3 loop. The N332 glycan required by these bnAbs is absent on MT145K Env and the N334 glycan is projecting away from the epitopes and is not a functional substitute. In addition, the glycans N412 and N442, the latter being unique to the SIV Env, would clash with the CDRs of these bnAbs, thus preventing interaction. D-E. Cryo-EM reconstruction of CD4bs bnAb 3BNC117, bound to BG505 trimer (PDB: 5V8M (Lee et al., 2017)). The lack of 3BNC117 binding with MT145K SIV Env trimer correlates with significant sequence variation in loop D and a clash of bnAb CDR loops with the N236 glycan that is unique to SIV. Moreover, a longer MT145 V5 loop would introduce a clash with 3BNC117 CDRH2.

FIG. 18A-18C. Cross-reactivity of CRF250 trimer-induced "on" and "off" target Ab responses with chimpanzee SIV MT145K trimer. A. CD45.2-encoded CH01 UCA B cells were transferred into CD45.1 wild-type (WT) mice at 5 dilutions (1:1, 1:10⁴, 1:10⁵, 1:10⁶ and 1:10⁷). Mice were

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immunized with CRF250 trimer at week-0 (prime) and boosted at week-4 (boost). A dose of 25 ug of the trimer protein/mouse/immunization was administered with adjuvant GLA-SE. B. Heatmap showing ELISA binding of the CRF250 trimer prime and boosted serum Ab responses of CH01 UCA B cell diluted mice with CRF250 (HIV subtype AG) and MT145K (chimpanzee SIV) Env trimers and their N160K variants. Chimpanzee SIV MT145K trimer shares cross-reactivity only at the V2-apex bnAb epitope with HIV trimers. EC50 binding titers for each serum/protein combination are shown; red color indicates stronger binding; green indicates weaker binding. CRF250 trimer-induced post prime (2 wk) Ab responses exhibit cross-reactive binding with MT 145K trimer almost exclusively through the V2-apex bnAb site; no binding with MT 145K N160K trimer variant. CRF250 trimer-induced post boost (6 wk) Ab responses, in addition cross-reacting at the V2-apex bnAb site, show minimal cross-reactive binding of the CRF250 trimer Env backbone off-target (N160K-dependent) Ab responses with MT145K trimer backbone (2/16 animals). C. Scatter dot plots with bars showing the frequencies of CH01 specific class-switched IgG B cells from CRF250 trimer immunization post-prime and post-boost splenic samples as determined by next generation sequencing (NGS). The CH01 IgG frequencies are derived from Ig sequencing data of the total splenic CH01 specific B cells (IgM+IgG). Each dot represents an individual mouse and the bar height in each dilution groups indicates median CH01 IgG frequencies.

FIG. 19. Amino-acid sequence of the Chimpanzee Simian Immunodeficiency virus (SIVcpz) envelope trimer construct, MT145K dV5 SOSIP.664 (SEQ ID NO:316), showing SOSIP.664 trimer stabilizing modifications. The soluble SOSIP.664 trimer modifications include: (i) replacement of N-terminus envelope gp120 signal peptide sequence with tissue plasminogen activator (TPA) signal sequence for higher protein expression (shown in yellow) (ii) incorporation of a disulfide bond between residue 501 of gp120 (A501C) and residue 605 of gp41 (T605C) (shown in turquoise), (iii) replacing naturally occurring gp120 and gp41 cleavage site with R6-cleavage site (shown in red), (iv) I559P substitution in the gp41 (shown in grey), (v) and a truncation in the gp41 subunit at residue 664 by inserting a stop codon (indicated by an asterisk (*)). Underlined residues (KD) indicate V5 deletion position.

FIG. 20A-20D. Immunogenicity of the MT145KdV5 SOSIP trimer version versus MT145K SOSIP trimer in CH01 HC-only KI mouse model. A. Negative Stain Electron-Microscopy (NS-EM) of MT145KdV5 trimer: 2D class averages show trimers adopting 100% native-like conformations. B. BioLayer Interferometry (BLI) binding responses of the HIV Env specific mAbs and V2-apex bnAb iGLs PG9, CH01, CAP256 UCA to the soluble MT145KdV5 SOSIP trimer. C. Schematic showing immunization schedule of CH01 UCA HC-only KI mice with MT 145KdV5 and MT 145K trimers. The mice were immunized twice with 25 ug of the soluble trimer with glucopyranosyl lipid adjuvant stable emulsion (GLA-SE) as adjuvant. Time points for immunization and bleeds are indicated. D. Neutralizing antibody responses induced by MT145KdV5 and MT145K SOSIP trimer immunogens in CH01 UCA HC-only KI mice. Neutralizing activities of serum samples (pre-bleed; 2 weeks post prime, Bleed #1; and 2 weeks post boost-1, Bleed #2) were tested with MT145K and CH01 Ab sensitive viruses (Q23_17, CRF250 and C108) in TZM-bl reporter cell assay. Each dot in the plots represents virus ID50 neutralization titers that were calculated by a non-linear regression method from the percent neutralizations of

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serum titrations with the corresponding viruses. The ID50 neutralization titers between MT145KdV5 (red) and MT145K (blue) trimer immunized animal groups were compared by two-tailed unpaired Mann-Whitney test and the P-value for each comparison is indicated. P-values of less than 0.05 are treated as significant. Overall MT145KdV5 SOSIP trimer induced superior nAb responses as compared to the parent MT145K SOSIP trimer.

FIG. 21A-21C: Immunogenicity of MT145KdV5 SOSIP trimer immunogen in non-human primate model. A. Schematic showing immunization schedule of non-human primates (NHPs) vaccinated with MT145KdV5 SOSIP trimer. A group of 6 animals was immunized twice with MT145KdV5 trimer (prime, week 0; boost-1, week 8). The NHPs were immunized with 100 ug of the soluble MT145KdV5 trimer with Alum/ISCO-MPLA adjuvants. Time points for the immunizations and the bleeds are indicated. B. ELISA binding of polyclonal serum antibodies to MT145KdV5 SOSIP trimer. EC50 binding titers of pre-bleed (PB), post prime (1 wk, 2 wk, 4 wk and 8 wk) and post boost (9 wk and 10 wk) serum samples with MT145KdV5 SOSIP trimer. C. Neutralization curves of 10 wk immune serum IgG samples against the autologous MT 145KdV5 virus. Purified IgGs from 10 wk immune serum samples were tested against the MT145KdV5 virus in TZM-bl reporter cell-based assay. IC50 virus neutralization titers for each serum IgG sample are shown. V2-apex bnAb, PGT145 was control for the neutralization assay.

DETAILED DESCRIPTION OF THE INVENTION

The invention provides molecules which comprise regions and epitopes based on Env of various SIV isolates and advantageously bind to and are capable of stimulating production of bnAbs of HIV and display V2 apex epitopes associated with that. Virus isolates may display epitopes that are conserved across species boundaries. For example, SIV displays epitopes observed to bind to SIV and HIV antibodies, and it is theorized that such epitopes are both important for virus function and provide a means for crossing of species boundaries. Nevertheless, such epitopes may not be highly immunogenic, thus immune responses may be dominated by antibodies that bind to other more variable viral epitopes. Such variable epitopes may not be present on a broad range of viral isolates such that vaccines presenting those epitopes will not be protective against a broad range of viral isolates. Moreover, epitopes that might stimulate bnAbs may be inaccessible to antibody binding, one consequence being limited stimulation of germline B cells making antibodies capable of recognizing such epitopes. Accordingly, the invention provides engineered virus components and vaccines that display epitopes that bind to and are capable of efficiently stimulating antibodies that broadly neutralize HIV. Moreover, the invention compositions and methods of making and using immunogenic molecules that efficiently stimulate germline B cells and antibody production by those cells.

One of the Env sites that has shown great promise for vaccine targeting is the V2 apex bnAb epitope (Andrabi et al., 2015; Gorman et al., 2016; Voss et al., 2017). This bnAb epitope sits at the 3-fold axis of the trimer and is primarily formed by a patch rich in positively charged lysine residues and protected by two glycans at HXB2 HIV-1 reference positions N160 and N156/N173 that are part of the Env glycan shield. The bnAb precursors targeting this site possess a long anionic heavy-chain complementarity-determin-

ing region 3 (CDRH3) that penetrates the glycan shield to reach the protein epitope surface underneath. BnAb prototypes within this class interact with the V2 apex bnAb protein-glycan core epitope through common germline-encoded motifs and are, thus, targetable by unique trimers that bind with their germline Ab versions. Hence, the germline-priming immunogens to this site could be based directly on native-like trimer configurations. Other features of this site include: a) V2 apex bnAbs are elicited frequently in humans that make bnAbs, b) they emerge early in infection, and c) they possess relatively low levels of somatic mutation compared to most other HIV Env bnAbs.

Among the major HIV Env bnAb specificities the V2 apex site is a bnAb site that consistently exhibits cross-group neutralizing activity with virus Envs derived from HIV-1 group M, N, O and P, and in addition, V2-apex bnAbs display cross-neutralizing activity with the Simian Immunodeficiency Virus (SIV) isolates that infect chimpanzees (SIVcpzPtt [Pan troglodytes troglodytes], SIVcpzPts [Pan troglodytes *schweinfurthii*] and gorillas (SIVgor). The sought to design molecules useful as immunogens, based on features of SIVcpzPtt Env, that could guide an immunofocused response to the HIV V2 apex bnAb epitope. The applicants elected a stable trimer as a useful format for the design of SIVcpzPtt-based molecules and immunogens. The SIV based immunogens of the invention are capable of enriching V2 apex-specific B cells, and owing to V2 apex species cross-conservation, help guide a V2-focused neutralizing Ab response when coupled with HIV trimers in a sequential prime/boost immunization strategy. In such an immunization scheme, overall Env backbone sequence diversity in combination with conservation of the V2 apex bnAb epitope in sequentially administered immunogens also reduces germinal center competition for V2 apex-specific B cells, favors a B cell recall response to the V2 apex bnAb epitope, and reduces off-target Env-specific responses.

The invention pertains to the identification, design, synthesis and isolation of molecules from SIV proteins, peptides, fragments, and mutants thereof that bind to HIV directed antibodies. The invention also pertains to molecules from SIV proteins, peptides, fragments, and mutants thereof used as immunogens to stimulate HIV directed antibodies, especially bnAbs that bind to HIV as well as nucleic acids encoding the same. The immunogens may comprise proteins, peptides, fragments, such as the various trimers disclosed herein. The present invention also relates to homologues, derivatives and variants of the sequences of the mutant trimers and nucleic acids encoding the same, wherein it is preferred that the homologue, derivative or variant have at least 50%, at least 60%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 93%, at least 95%, at least 97%, at least 98% or at least 99% homology or identity with the sequence of the mutant trimers and nucleic acids encoding the same. It is noted that within this specification, homology to sequences of the mutant proteins and nucleic acids encoding the same refers to the homology of the homologue, derivative or variant to the binding site of the mutant proteins and nucleic acids encoding the same.

The invention still further relates to nucleic acid sequences expressing the proteins, peptides, fragments, trimers, and mutants disclosed herein, or homologues, variants or derivatives thereof and the epitopes presented thereon. One of skill in the art will know, recognize and understand techniques used to create such. Additionally, one of skill in the art will be able to incorporate such a nucleic acid sequence into an appropriate vector, allowing for pro-

duction of the amino acid sequence of mutant proteins and nucleic acids encoding the same or a homologue, variant or derivative thereof.

The proteins, peptides, fragments, trimers, and mutants disclosed herein, or homologues, variants or derivatives thereof and the epitopes presented thereon may also be useful in screening for ligands that bind to said epitopes. Such ligands preferably block the epitopes and thus prevent infection.

Where used herein and unless specifically indicated otherwise, the following terms are intended to have the following meanings in addition to any broader (or narrower) meanings the terms might enjoy in the art:

The term "isolated" or "non-naturally occurring" is used herein to indicate that the isolated moiety (e.g. peptide or compound) exists in a physical milieu distinct from that in which it occurs in nature. For example, the isolated peptide may be substantially isolated with respect to the complex cellular milieu in which it naturally occurs. The absolute level of purity is not critical, and those skilled in the art may readily determine appropriate levels of purity according to the use to which the peptide is to be put. The term "isolating" when used a step in a process is to be interpreted accordingly.

In many circumstances, the isolated moiety will form part of a composition (for example a more or less crude extract containing many other molecules and substances), buffer system, matrix or excipient, which may for example contain other components (including proteins, such as albumin).

In other circumstances, the isolated moiety may be purified to essential homogeneity, for example as determined by PAGE or column chromatography (for example HPLC or mass spectrometry). In preferred embodiments, the isolated peptide or nucleic acid of the invention is essentially the sole peptide or nucleic acid in a given composition.

In an advantageous embodiment, a tag may be utilized for purification or biotinylation. The tag for purification may be a his tag. In another embodiment, the tag for biotinylation may be an avi-tag. Other tags are contemplated for purification, however, purification may be accomplished without a tag. In another embodiment, antibody (such as, not limited to, a broadly neutralizing antibody) affinity columns are contemplated. In another embodiment, lectin columns are contemplated.

Native-like soluble trimers can be made by several methods that all involve stabilizing associations between envelope protein subunits. See, e.g., P. Dosenovic et al, "Immunization for HIV-1 broadly neutralizing antibodies in human Ig knockin mice," *Cell*, 161:1-11, 2015; J. G. Jardine et al, "Priming a broadly neutralizing antibody response to HIV-1 using a germline targeting immunogen," *Science*, doi: 10.1126/science.aac5894, 2015 and R. W. Sanders et al, "HIV-1 neutralizing antibodies induced by native-like envelope trimers," *Science*, doi: 10.1126/science.aac4223, 2015.

The proteins and compounds of the invention need not be isolated in the sense defined above, however.

Compounds which have a chemical structure selected using the invention, wherein said compounds are neutralizing antibody binders, form a further aspect of the invention; and, such compounds may be used in methods of medical treatments, such as for diagnosis, preventing or treating HIV or for eliciting antibodies for diagnosis of HIV, including use in vaccines. Further, such compounds may be used in the preparation of medicaments for such treatments or prevention, or compositions for diagnostic purposes. The compounds may be employed alone or in combination with other

treatments, vaccines or preventatives; and, the compounds may be used in the preparation of combination medicaments for such treatments or prevention, or in kits containing the compound and the other treatment or preventative.

The invention relates to immunogenic molecules that bind to and stimulate antibody production, particularly molecules that efficiently stimulate germline antibodies that are broadly neutralizing and/or precursors to antibodies that are broadly neutralizing against epitopes common to a variety of HIV isolates. The common epitopes may be evolutionarily selected and not especially immunogenic. Thus it can be beneficial to present antigenic molecules to the immune system that efficiently stimulate antibodies that bind to the common epitope while avoiding antibodies directed at epitopes that are not broadly neutralizing. Accordingly, the invention provides molecules engineered from SIV env regions designed to bind to and stimulate germline antibodies that are broadly neutralizing HIV antibodies or are precursors of broadly neutralizing HIV antibodies.

As described herein, in certain embodiments of the invention, useful V2 apex sequences comprise two glycans, thus two glycosylated asparagine residues, and basic amino acid residues, preferably three or four basic amino acid residues, in a four-amino acid sequence, for example, without limitation, Xxx-Lys-Lys-Lys (SEQ ID NO:1), Lys-Xxx-Lys-Lys (SEQ ID NO:2), Lys-Lys-Xxx-Lys (SEQ ID NO:3), Lys-Lys-Lys-XXX (SEQ ID NO:4), or Lys-Lys-Lys-Lys (SEQ ID NO:5). In certain embodiments, the molecule comprises Asn-Xxx_a-Asn-Xxx_b-Xxx_c-Xxx_d, wherein Xxx_a comprises two or three amino acids, Xxx_b comprises five to seven amino acids, and Xxx_c comprises four amino acids, each of the four amino acid selected from Lys or Arg.

In certain embodiments, molecules of the invention comprise the amino acid sequence N-C-X₁-F-N-X₂-X₃-X₄-X₅-X₆-X₇-X₈-X₉-X₁₀-X₁₁-X₁₂-X₁₃-X₁₄-X₁₅-X₁₆ (SEQ ID NO:7); wherein X₁ comprises S, T, N, or F; X₂ comprises I, V, T, Q, or M; X₃ comprises S or T; X₄ comprises S or T; X₅ comprises S, E, or G; X₆ comprises I, L, V or F; X₇ comprises K or R; X₈ comprises G or D; X₉ comprises K, R, E, or Q; X₁₀ comprises K, R, E, or Q; X₁₁ comprises K, R, E, or Q; X₁₂ comprises K, E, or Q; X₁₃ comprises E, T, V, I, or M; X₁₄ comprises Y or K; X₁₅ comprises A or S; and X₁₆ comprises F, I, or L. In certain embodiments, at least three of X₉, X₁₀, X₁₁, and X₁₂ comprise basic amino acid residues. (SEQ ID NO: 8) In certain embodiments, all of X₉, X₁₀, X₁₁, and X₁₂ comprise basic amino acid residues. (SEQ

ID NO: 9) In certain embodiments, X₉, X₁₀, X₁₁, and X₁₂ each comprise K. (SEQ ID NO: 10) In one such embodiment, the amino acid sequence comprises NCSFNVTTELRLDKKRQVYSL (SEQ ID NO:11). In another embodiment, the amino acid sequence comprises NCSFNVTTELRLDKKRKVYSL (SEQ ID NO:12). In another embodiment, the amino acid sequence comprises NCSFNVTTELRLDKKKKEYSF (SEQ ID NO:13). In another embodiment, the amino acid sequence comprises NCSFNATTELRLDKKKKEYAL (SEQ ID NO:14).

In certain embodiments, molecules of the invention comprise Asn-Cys-Ser-Phe-Asn-Val-Thr-Thr-Glu-Leu-Arg-Asp-Lys-Lys-Arg-Gln-Val-Tyr-Ser-Leu-Phe-Tyr (SEQ ID NO:15), or Asn-Cys-Ser-Phe-Asn-Val-Thr-Thr-Glu-Leu-Arg-Asp-Lys-Lys-Arg-Gln-Val-Tyr-Ser-Leu-Phe-Tyr (SEQ ID NO:16), or Asn-Cys-Ser-Phe-Asn-Val-Thr-Thr-Glu-Leu-Arg-Asp-Lys-Lys-Arg-Lys-Val-Tyr-Ser-Leu-Phe-Phe-Tyr (SEQ ID NO:17), or Asn-Cys-Ser-Phe-Asn-Val-Thr-Thr-Glu-Leu-Arg-Asp-Lys-Lys-Arg-Lys-Val-Tyr-Ser-Leu-Phe-Tyr (SEQ ID NO:18), or Asn-Cys-Ser-Phe-Asn-Val-Thr-Thr-Glu-Leu-Arg-Asp-Lys-Lys-Lys-Lys-Glu-Tyr-Ser-Phe-Phe-Phe-Tyr (SEQ ID NO:19); or Asn-Cys-Ser-Phe-Asn-Val-Thr-Thr-Glu-Leu-Arg-Asp-Lys-Lys-Lys-Lys-Glu-Tyr-Ser-Phe-Phe-Tyr (SEQ ID NO:20); or Asn-Cys-Ser-Phe-Asn-Ala-Thr-Thr-Glu-Leu-Arg-Asp-Lys-Lys-Lys-Lys-Glu-Tyr-Ala-Leu-Phe-Phe-Tyr (SEQ ID NO:21), or Asn-Cys-Ser-Phe-Asn-Ala-Thr-Thr-Glu-Leu-Arg-Asp-Lys-Lys-Lys-Lys-Glu-Tyr-Ala-Leu-Phe-Tyr (SEQ ID NO:22).

The bnAb stimulating engineered V2 apex region sequences of the invention, such as but not limited to MT145K, may be incorporated in engineered env proteins and protein fragments. Exemplary, non-limiting examples of env proteins with GenBank sequence accession numbers are set forth in the table below. See also public databases such as hiv.lanl.gov/components/sequence/HIV/search/search.html. Along with the V2 apex region sequences, the env proteins and fragments can comprise sequences set forth in the table below (e.g., env sequences of naturally occurring SIV variants and escape mutants isolated from infected individuals) or be similar in sequence to such isolated variants. Thus, alternative embodiments of env proteins and fragments have at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identity to the SIV env proteins in Table 1. (See hiv.lanl.gov/components/sequence/HIV/search/search.html)

TABLE 1

Exemplary SIV ENV proteins					
Patient ID	Accession	SEQ ID NO:	Name	Subtype	Country
Marylin(4288)	AF103818	317	US_Marilyn	CPZ	United States
CAM3(4290)	AF115393	318	CAM3	CPZ	Cameroon
GAB2(3694)	AF382828	319	SIVcpzGAB2	CPZ	Gabon
Ch-06(27172)	AF447763	320	TAN1	CPZ	Tanzania
CAM5(4289)	AJ271369	321	CAM5	CPZ	Cameroon
CAM13(6922)	AY169968	322	SIVcpzCAM13	CPZ	Cameroon
MB66(23595)	DQ373063	323	SIVcpzMB66	CPZ	Cameroon
LB7(41638)	DQ373064	324	SIVcpzLB7	CPZ	Cameroon
EK505(50579)	DQ373065	325	SIVcpzEK505	CPZ	Cameroon
MT145(23809)	DQ373066	326	SIVcpzMT145	CPZ	Cameroon
Ch-64(3056)	DQ374657	327	TAN2_69	CPZ	Tanzania
CH-45(3137)	DQ374658	328	TAN3_1	CPZ	Tanzania
Ch-06(27172)	EF394356	329	TAN1	CPZ	Tanzania
Ch-64(3056)	EF394357	330	TAN2	CPZ	Tanzania
CH-45(3137)	EF394358	331	TAN3	CPZ	Tanzania
DP943(23593)	EF535993	332	SIVcpzDP943	CPZ	Cameroon
MB897(23594)	EF535994	333	SIVcpzMB897	CPZ	Cameroon

TABLE 1-continued

Exemplary SIV ENV proteins					
Patient ID	Accession	SEQ ID NO:	Name	Subtype	Country
ID10(30898)	FJ424863	334	SIVgorCP2135con	GOR	Cameroon
ID9(30899)	FJ424864	335	SIVgorCP2139_1con	GOR	Cameroon
ID9(30899)	FJ424865	336	SIVgorCP2139_2con	GOR	Cameroon
ID9(30899)	FJ424866	337	SIVgor2139_287	GOR	Cameroon
ID1(13725)	FJ424871	338	SIVgorCP684con	GOR	Cameroon
Cam155(38651)	FR686510	339	Ptt_04Cam155	CPZ	Cameroon
Cam155(38651)	FR686511	340	Ptt_09Cam155	CPZ	Cameroon
Gab4(47562)	GQ217539	341	SIVcpzGab4	CPZ	Gabon
Ch-58(44459)	JN091690	342	UG38	CPZ	Tanzania
Ch-071(32642)	JN091691	343	TAN5	CPZ	Tanzania
EK505(50579)	JN835460	344	SIVcpzEK505.c2	CPZ	Cameroon
MB897(23594)	JN835461	345	SIVcpzMB897.c2	CPZ	Cameroon
MT145(23809)	JN835462	346	SIVcpzMT145.c2	CPZ	Cameroon
Ch-080(32641)	JQ768416	347	SIVcpzTAN13	CPZ	Tanzania
BF1167(32614)	JQ866001	348	BF1167	CPZ	Congo
LB715(41737)	JX178450	349	LB715	CPZ	Cameroon
BPID1(78681)	KP004989	350	SIVgor-BPID1	GOR	Cameroon
BPID15(78682)	KP004990	351	SIVgor-BPID15	GOR	Cameroon
BQID2(78683)	KP004991	352	SIVgor-BQID2	GOR	Cameroon
ANT(65)	U42720	353	ANT	CPZ	Congo
GAB1(66)	X52154	354	GAB1	CPZ	Gabon

Molecules of the invention may be further engineered to remove or reduce the immunogenicity of dominant epitopes that might otherwise interfere with a desired immune response. Such dominant env regions have been characterized and include, without limitation, epitopes located in the Vi (a.a. 111-130) and V3 (a.a. 311-330) domains and near the amino terminal part (a.a. 601-619) of the transmembrane gp36 of SIVmac251. Many immunodominant regions are similar among SIVs and HIVs (see e.g., Benichou, S. et al., 1993, Virology 194:870). As demonstrated herein, deletion of a portion of the immunodominant V5-loop from an engineered MT145K molecule results in improved induction of cross-neutralizing antibody responses compared to the parent SIV molecule.

As disclosed and exemplified herein, germline reverted antibodies can be designed by comparing the heavy and light chains of an isolated bnAb that binds to an epitope of interest to nucleotide sequences and/or the amino acid sequences potentially encoded by germline V gene segments. For

example, one can obtain a germline reverted antibody starting from the amino acid sequence of an antibody directed to an antigen or epitope of choice and identifying germline encoded antibody segments (i.e., heavy chain V, D, and J gene segments and light chain V and J gene segments) close in sequence to the starting antibody. Germline variable genes are cataloged, for example at VBASE2 (Retter I, Althaus H H, Münch R, Müller W: VBASE2, an integrative V gene database. Nucleic Acids Res. 2005 Jan. 1; 33(Database issue):D671-4.). Starting from a known or preselected antibody and moving to such a database can be advantageous as the paired heavy and light chains will be compatible and likely to be well represented in the repertoire of expressed germline genes.

The invention provides germline reverted heavy and light chain variable domains, including heavy and light chain CDR regions that bind to molecules of the invention. The following are exemplary and bind to certain engineered V2 regions described herein.

TABLE 2

Germline reverted heavy and light chains and Kabat CDRs			
	SEQ ID NO:		
PG9_VDJH_iGL GenBank Accession: MK825341	32		QVQLVESGGGVVQPGRSLRLSCAASGFTFSSYGMHWVRQAPGKGLEWVAV ISYDGSNKYYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCAREAF GGPDYRNGYNYYDFWSGYYYYYMDVWGKGTITVTVSS
CDR-H1	33	SYGMH	
CDR-H2	34	VISYDGSNKYYADSVKG	
CDR-H3	35	EAGGPDYRNGYNYYDFWSGYYYYYMDV	
PG9VJL_iGL GenBank Accession: MK825342	41		QSALTQPASVSGSPGQSITISCTGTSSDVGGYNYVSWYQQHPGKAPKLMI YEVSNRPSGVSNRFSGSKSGNTASLTISGLQAEDEADYYCSSYTSSTWV FGGGTKLTVL
CDR-L1	42	TGTSSDVGGYNYVS	
CDR-L2	43	EVSNRPS	

TABLE 2-continued

Germline reverted heavy and light chains and Kabat CDRs		
	SEQ ID NO:	
CDR-L3	44	SSYTSSTWV
CH01VDJH_iGL GenBank Accession: MK825343	49	EVQLVESGGGVVPRPGGSLRLSCAASGFTFDDYGMWVRQAPGKGLEWVSG INWNGGSTGYADSVKGRFTISRDAKNSLYLQMNSLRAEDTALYHCARGT DYTIDDAIGIHYYGSGSYWYFDLWGRGTLVTVSS
CDR-H1	50	DYGMS
CDR-H2	51	GINWNGGSTGYADSVKG
CDR-H3	52	GTDTIDDAIGIHYYGSGSYWYFDL
CH01VJL_iGL GenBank Accession: MK825344	58	EIVLTQSPGTLISLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIY GASSRATGIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCQQYGSSSPWTFG QGTKVEIK
CDR-L1	59	RASQSVSSSYLA
CDR-L2	60	GASSRAT
CDR-L3	61	QQYGSSSPWT

Germline reverted antibodies such as the above provide for antibody design or selection. Using the CDRs of the germline reverted antibodies, bnAbs can be designed, or evolved by mutation and selection, that bind to the V2 apex region of HIV and SIV.

Broadly neutralizing antibodies of the invention include those set forth above and antibodies which comprise one, two, three, four, five, or all six of the CDRs set forth in Table 2, or comprise a heavy chain comprising one, two or three V_H CDRs, or comprise a light chain comprising one, two or three V_L CDRs. In certain embodiments, in bnAbs that bind to the V2 apex of HIV and/or SIV the CDRs, there can be conservative substitutions, as set forth elsewhere herein. The invention further includes bnAbs in which the heavy chains and/or light chains have at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to germline reverted antibodies set forth. In certain such embodiments, variation in the CDRs is conservative.

Germline genes that bind to an epitope of interest can also be identified by screening libraries of human heavy and light chain genes, for example by phage display. Another method is by immunization and preparation of monoclonal antibodies from a mouse that expresses human antibody genes. Preferably hybridomas are prepared before there is substantial somatic mutation and selection. As mentioned, depending on the method employed, it is advantageous to select germline antibody gene segments that are well represented in the repertoire of expressed human antibodies, also taking into account compatibility of various heavy chain—light chain pairings. In certain embodiments, it is advantageous to select V-D, D-J, V-D-J (heavy chain) or V-J (light chain) combinations that are well represented.

Amino acid sequences of the V2 apex regions of many SIV isolates have been determined and can be used or

compared with HIV isolates to identify useful conserved epitopes that would be bound by broadly neutralizing antibodies. FIG. 15 herein, for example provides a comparison of SIVcpz Envs with an HBX2 HIV reference. An analysis may also include modeling or determination of 3D structure to identify interactions with bnAbs. Similar analysis can be done for other HIV reference isolates and SIV counterparts based on conserved structures, regions, and epitopes of the V2 apex, as well other conserved structures, regions, and epitopes that comprise bnAb binding sites to find suitable bases for development of SIV-based molecules and immunogens that bind to and stimulate HIV antibodies.

The HIV envelope protein (Env) is the target of broadly neutralizing antibodies (bnAbs) in natural infection. Env is a membrane protein composed of a trimer of gp120 and gp41 subunits that contains a high degree of sequence diversity and a surface that is shielded by N-linked glycans. The bnAbs that target Env often have unusual features such as a long complementarity-determining region (CDR) H3, high levels of somatic hypermutation (SHM), and insertions and deletions (INDELS). Furthermore, most of the bnAbs recognize complex epitopes that are typically non-linear and have both protein and glycan components.

HIV type I (HIV-I) envelope is a noncovalent trimer of gp120-gp41 heterodimers, and its lability has hindered structural studies. SOSIP gp140 is a soluble, proteolytically mature form of the HIV-I envelope wherein gp120-gp41 interactions are stabilized via a disulfide bond and gp41 contains an additional trimer-stabilizing point mutation. The isolation of a substantially pure preparation of SOSIP gp140 trimers derived from KHN1144, a subtype A isolate was described in Iyer S P et al., AIDS Res Hum Retroviruses. 2007 June; 23(6):817-28. Following initial purification, the only significant contaminant was higher-order gp140 aggregates; however, 0.05% Tween 20 quantitatively converted these aggregates into trimers. The surfactant effect was rapid, dose dependent, and similarly effective for a subtype B SOSIP gp140.

Surfactant-treated SOSIP gp140 retained favorable antigenicity and formed compact trimers 12-13 nm in size as determined by electron microscopy. Iyer S P et al., *AIDS Res Hum Retroviruses*. 2007 June; 23(6):817-28 provides a description of homogeneous, cleaved HIV-I envelope trimers. These proteins may be useful as vaccine immunogens and for studying structure-function relationships within the HIV-I envelope glycoproteins.

Soluble, stabilized, proteolytically cleaved, trimeric proteins may be generated by engineering an intermolecular disulphide bond between gp120 and gp41 (SOS), combined with a single residue change, I559P, within gp41 (SOSIP). SOSIP gp140 proteins based on the subtype A HIV-I strain KNIH 144 form particularly homogenous trimers compared to a prototypic strain (JR-FL, subtype B). Described in U.S. Pat. No. 7,939,083 are the determinants of this enhanced stability which are located in the N-terminal region of KNIH 1144 gp41 and that, when substituted into heterologous Env sequences (e.g., JR-FL and Ba-L) they have a similarly beneficial effect on trimer stability. These stabilized trimers retain the epitopes for several neutralizing antibodies and related agents (CD4-IgG2, b12, 2G12, 2F5 and 4E10) and the CD4-IgG2 molecule, so that the overall antigenic structure of the gp140 protein has not been adversely impaired by the trimer-stabilizing substitutions.

The HIV-I envelope glycoprotein (Env) is a trimer of heterodimers composed of two non-covalently associated subunits; the receptor-binding gp120, and the fusion machinery-containing gp41. Each subunit is derived from a gp160 precursor glycoprotein following cleavage by cellular furins (Wyatt R & Sodroski J (1998) *Science* 280(5371): 1884-1888). HIV-I gp120 binds the CD4 molecule on the surface of human target T cells to initiate the viral entry process, and following co-receptor engagement, fusion is mediated by gp41 (Dalglish A G, et al. (1984) *Nature* 312(5996):763-767; McDougal J S, et al. (1986) *J Immunol* 137(9):2937-2944; mKarlsson Hedestam G B, et al. (2008) *Nat Rev Microbial* 6(2):143-155). The surface-exposed HIV-I Env trimer is the sole target for antibodies capable of neutralizing the virus (Burton D R, et al. (2004) *Nat Immunol* 5(3):233-236). Recently, a myriad of Env-directed broadly neutralizing antibodies (bnAbs) were isolated from numerous HIV-I-infected individuals, demonstrating that the human B cell response can effectively inhibit this variable pathogen (Wu X, et al. (2010) *Science* 329(5993):856-861; Walker L M, et al. (2009) *Science* 326(5950):285-289; Walker L M, et al. (2011) *Nature* 477(7365):466-470; Huang J, et al. (2012) *Nature* 491(7424):406-412; Scharf L, et al. (2014) Antibody 8ANC195 reveals a site of broad vulnerability on the HIV-I envelope spike. *Cell reports* 7(3):785-795; Klein F, et al. (2012) *J Exp Med* 209(8):1469-1479). Infection of macaques by a chimeric model virus, SHIV, can be prevented by prior passive immunization of all bnAbs so far tested, confirming the capacity of neutralizing antibodies to prevent HIV infection (Mascola J R, et al. (1999) *J Virol* 73(5):4009-4018; Hessel A J, et al. (2009) *PLoS Pathog* 5(5):e1000433; Moldt B, et al. (2012) *Proc Natl Acad Sci US A* 109(46):18921-18925; Barouch D H, et al. (2013) Therapeutic efficacy of potent neutralizing HIV-I-specific monoclonal antibodies in SHIV-infected rhesus monkeys. *Nature* 503(7475):224-228).

Along with virus-specific T cells, an efficacious HIV-I vaccine therefore would likely need to generate bnAbs targeting Env. Although the premise is simple, in actuality, it is a tremendous challenge without precedent in the history of vaccinology. The difficulty to vaccinate against HIV arises from the extensive variability of Env present on the

large number of HIV-I isolates simultaneously circulating in the human population as well as other mechanisms of immune evasion selected for by strong pressure from the human immune system.

Generally, vaccine-generated antibodies using either or both gp120 or gp41 sequences do not recognize native Env on the surface of cells or virus, do not neutralize primary isolates in vitro, and do not prevent infection in laboratory animals (Burton D R, et al. (2011) *Proc Natl Acad Sci US A* 108(27):11181-11186; Sundling C, et al. (2012) *Science translational medicine* 4(142):142ra196; Tran K, et al. (2014) Vaccine-elicited primate antibodies use a distinct approach to the HIV-I primary receptor binding site informing vaccine redesign. *Proc Natl Acad Sci USA* 111(7):E738-747). Non-neutralizing antibodies directed to the major variable region two (V2) of gp120 are associated with modest efficacy in a single human clinical trial (Haynes B F, et al. (2012) *N Engl J Med* 366(14):1275-1286; Zolla-Pazner S, et al. (2014) Vaccine-induced IgG antibodies to VIV2 regions of multiple HIV-I subtypes correlate with decreased risk of HIV-I infection. *PLoS One* 9(2):e87572), while, in general, Env-elicited antibodies fail to demonstrate protection in previous human clinical trials (Jones N G, et al. (2009) *Vaccine* 27(7):1136-1140; Rekrs-Ngarm S, et al. (2009) *N Engl J Med* 361(23):2209-2220; Yates N L, et al. (2014) Vaccine-induced Env V1-V2 IgG3 correlates with lower HIV-I infection risk and declines soon after vaccination. *Science translational medicine* 6(228):228ra239).

Many Env-based trimeric candidate immunogens are engineered to eliminate cleavage between gp120 and gp41 (so called uncleaved gp140 trimers), usually generating imperfect mimetics of the functional spike based on antigenic profiling or EM analysis (Tran K, et al. (2014) *Proc Natl Acad Sci US A* 111(7):E738-747; Ringe R P, et al. (2013) *Proc Natl Acad Sci USA* 110(45):18256-18261). As a group, the defined, or presumed to be, disordered trimers (in adjuvant) generate high self-binding antibody titers. However, these vaccine-elicited antibodies do not efficiently neutralize most HIV-I primary isolates, that is, strains representative of those circulating in the human population (Sundling C, et al. (2012) *Science translational medicine* 4(142):142ra196; Chakrabarti B K, et al. (2013) *J Virol* 87(24):13239-13251; Kovacs J M, et al. (2012) *Proc Natl Acad Sci US A* 109(30):12111-12116; Nkolola J P, et al. (2014) Comparison of multiple adjuvants on the stability and immunogenicity of a clade C HIV-I gp140 trimer. *Vaccine* 32(18):2109-2116). Antibodies elicited by these immunogens target epitopes exposed only on the free gp120 and trimeric post-fusion forms of gp41 or disordered gp140s and thus are ineffective at accessing their epitopes buried within the ordered, quaternary structure achieved in the native Env spike. Applicants recently described the limitations of two CD4bs-directed non-bnAbs, (GE148 and GE136) generated following immunization of uncleaved gp140 trimers (YU2 gp140-foldon) in non-human primates (NHP). Non-bnAbs, represented by GE136 and 148, can only neutralize the sensitive so-called "tier I viruses" that are not representative of the more neutralization resistant tier 2-like primary isolates circulating in the human population. Using crystallography, EM reconstructions, paratope scanning and molecular modeling, Applicants determined that these vaccine-elicited antibodies fail to reach the CD4bs due to steric barriers imposed by quaternary packing of the native Env on neutralization resistant primary isolates, a property that Applicants use to Applicants' advantage in the negative-selection strategy presented here (Tran K, et al. (2014) *Proc Natl Acad Sci US A* 111(7):E738-747).

The cumulative historical data have led to the hypothesis that a more faithful mimic of the HIV-I spike that better recapitulates the native, pre-fusion form of Env, selectively displaying neutralizing determinants while occluding non-neutralizing determinants, may better elicit antibodies capable of accessing the native spike. A soluble Env mimetic, containing a disulfide linkage between gp120 and gp41 (SOS), first described in the 2000s, and further developed over the next decade, displays many of these properties, culminating in the determination of the high resolution structures of the well-ordered BG505 SOSIP trimers by crystallography and EM (Lyumkis D, et al. (2013) *Science* 342(6165):1484-1490; Julien J P, et al. (2013) *Science* 342(6165):1477-1483; Sanders R W, et al. (2013) *PLoS Pathog* 9(9):e1003618; Depetris R S, et al. (2012) *J Biol Chem* 287(29):24239-24254). A sub-nanometer EM reconstruction of KNHI 144 SOSIP is also available but does not provide atomic level details (Bartesaghi A, Merk A, Borgnia M J, Milne J L, & Subramaniam S (2013) *Nat Struct Mol Biol* 20(12):1352-1357). The BG505 SOSIP and KNH144 SOSIP trimers are derived from the Env sequences of the subtype A BG505 and KNHI 144 strains. These soluble trimers possess an engineered disulfide linkage between the gp120 and gp41 (at residues 501C and 605C, respectively) and an additional mutation in the heptad repeat 1 (HRI) of gp41 (I559P) that facilitates trimerization (Binley J M, et al. (2000) *J Virol* 74(2):627-643; Sanders R W, et al. (2002) *J Virol* 76(17):8875-8889). A truncation of the membrane proximal external region (MPER) at residue 664 enhances expression while decreasing aggregation is incorporated into the so-called BG505 SOSIP.664 trimers (Sanders R W, et al. (2013) *PLoS Pathog* 9(9):e1003618; Sanders R W, et al. (2002) *J Virol* 76(17):8875-8889). Although SOSIP molecules based on other HIV-I primary strains were attempted over the past decade, the BG505- and KNHI 144-derived SOSIP trimers are the two limited examples of SOSIPs that yield homogeneous trimers suitable for high resolution biophysical and structural analysis. The structural explanation for the difficulty to readily transfer the SOSIP design to other HIV-I strain-derived sequences is not yet fully understood and would be valuable information to broaden the trimer design horizon.

Since the initial soluble native-like BG505 SOSIP.664 Env trimer was confirmed to adopt a near-native conformation by high-resolution structural analysis, multiple efforts to produce stable, soluble Env mimetics derived from multiple HIV-1 strains were pursued [Javiers, sanders]. Multiple solutions to this objective include the improved cleavage-independent NFL trimers, UFOs and modified SOSIPs. Both the SOSIP and NFL well-ordered trimers are efficiently recognized by broadly neutralizing antibodies (bnAbs) which arise sporadically during the course of natural infection. In some cases, including the important advances described here, have been used to isolate such bnAbs. One approach to elicit tier 2 neutralizing Abs has been to immunize the existing well-ordered trimers using prime:boosting in selected animal models. For BG505 and 16055 native-like trimers this approach does elicit tier 2 neutralizing antibodies, but of limited cross-reactive breadth (REFS).

Most cross-conserved sites on the HIV Env spike are occluded by evolved, incorporated self-N-glycans, limiting naïve B cell recognition of the underlying polypeptide surface. The exceptions are the protein surfaces of the primary receptor CD4 binding site (CD4bs) and the furin cleavage site (proximal to the gp120:41 interface). Infrequently, during the course of the natural HIV infection process, bnAbs are elicited to these aforementioned sites of

vulnerability. In addition, other bnAbs directed to the V2 apex, the 332N-glycan supersite and to the fusion peptide or the high-mannose patch are elicited during the course of chronic HIV infection (REFS). However, prior to the present invention, rarely, if ever, have such bnAbs been elicited by vaccination of Env formulated with adjuvant.

After decades of development, advances in soluble HIV-1 Env mimics design permits the generation of a diverse array of native-like trimers (Ward and Wilson, 2017. The HIV-1 envelope glycoprotein structure: nailing down a moving target. *Immunol Rev* 275:21-32; Karlsson et al., 2017. Evolution of B cell analysis and Env trimer redesign. *Immunol Rev* 275:183-202). The successful development of the soluble SOSIP trimers provided proof-of-principle (Sanders et al, 2013. A next-generation cleaved, soluble HIV-1 Env trimer, BG505 SOSIP.664 gp140, expresses multiple epitopes for broadly neutralizing but not non-neutralizing antibodies. *PLoS Pathog* 9:e1003618) forming a prefusion native-like conformation (Lyumkis et al., 2013. Cryo-EM structure of a fully glycosylated soluble cleaved HIV-1 envelope trimer. *Science* 342:1484-1490; Julien et al, 2013. Crystal structure of a soluble cleaved HIV-1 envelope trimer. *Science* 342:1477-1483; Garces et al., 2015. Affinity Maturation of a Potent Family of HIV Antibodies Is Primarily Focused on Accommodating or Avoiding Glycans. *Immunity* 43:1053-1063; Pancera et al., 2014. Structure and immune recognition of trimeric pre-fusion HIV-1 Env. *Nature* 514:455-461). The SOSIP gp140 trimer is proteolytically cleaved by cellular furins to gp120 and gp41 subunits and covalently linked by an engineered intra-protomer disulfide bond A501C-T605C (SOS). These trimers also require mutation (I559P) in the gp41 heptad repeat 1 (HRI) to maintain well-ordered oligomers, as well as expression of exogenous furin for full conformational integrity (Sanders et al., 2013. A next-generation cleaved, soluble HIV-1 Env trimer, BG505 SOSIP.664 gp140, expresses multiple epitopes for broadly neutralizing but not non-neutralizing antibodies. *PLoS Pathog* 9:e1003618; Guenaga et al., 2015. Well-Ordered Trimeric HIV-1 Subtype B and C Soluble Spike Mimetics Generated by Negative Selection Display Native-like Properties. *PLoS Pathog* 11:e1004570; Julien et al., 2015. Design and structure of two HIV-1 clade C SOSIP.664 trimers that increase the arsenal of native-like Env immunogens. *Proc Natl Acad Sci USA* 112:11947-11952; de Taeye et al. 2015. Immunogenicity of Stabilized HIV-1 Envelope Trimers with Reduced Exposure of Non-neutralizing Epitopes. *Cell* 163:1702-1715; Pugach et al. 2015. A native-like SOSIP.664 trimer based on an HIV-1 subtype B env gene. *J Virol* 89:3380-3395; Ringe et al. 2013. Cleavage strongly influences whether soluble HIV-1 envelope glycoprotein trimers adopt a native-like conformation. *Proc Natl Acad Sci USA* 110:18256-18261; Ringe et al. 2015. Influences on the Design and Purification of Soluble, Recombinant Native-Like HIV-1 Envelope Glycoprotein Trimers. *J Virol* 89:12189-12210; Ringe et al. 2017. Reducing V3 Antigenicity and Immunogenicity on Soluble, Native-Like HIV-1 Env SOSIP Trimers. *J Virol* 91; Ahmed et al. 2017. Stabilization of a soluble, native-like trimeric form of an efficiently cleaved Indian HIV-1 clade C envelope glycoprotein. *J Biol Chem* 292:8236-8243; Sanders et al. 2002. Stabilization of the soluble, cleaved, trimeric form of the envelope glycoprotein complex of human immunodeficiency virus type 1. *J Virol* 76:8875-8889; Binley et al. 2000. A recombinant human immunodeficiency virus type 1 envelope glycoprotein complex stabilized by an intermolecular disulfide bond between the gp120 and gp41 subunits is an antigenic mimic of the trimeric virion-associated

structure. *J Virol* 74:627-643). In the past years, Applicants developed an improved native-like trimer design, generating well-ordered soluble Env mimics that are fully cleavage-independent, termed native flexibly linked (NFL) trimers. This design uses a flexible linker (two copies of Gly4-Ser, “G4S” (SEQ ID NO:357)) to replace the natural cleavage site and sequence (Sharma et al. 2015. Cleavage-independent HIV-1 Env trimers engineered as soluble native spike mimetics for vaccine design. *Cell Rep* 11:539-550). The flexible linker between the natural C-terminus of gp120 and N-terminus of gp41, allows the un-cleaved trimers to achieve a native-like conformation without the need of furin for precursor processing. However, the original NFL trimer design contains the I559P mutation (Sharma et al. 2015. Cleavage-independent HIV-1 Env trimers engineered as soluble native spike mimetics for vaccine design. *Cell Rep* 11:539-550) that was initially identified in the SOSIP context to disfavor the post fusion state (Sanders et al. 2013. A next-generation cleaved, soluble HIV-1 Env trimer, BG505 SOSIP.664 gp140, expresses multiple epitopes for broadly neutralizing but not non-neutralizing antibodies. *PLoS Pathog* 9:e1003618). Both the original SOSIP and NFL designs do not form a high percentage of well-ordered trimers in all Env contexts. In the original NFL design, it is relatively inefficient in generating high yields of trimers derived from clade C strains, such as 16055 (Guenaga et al. 2015. Structure-Guided Redesign Increases the Propensity of HIV Env To Generate Highly Stable Soluble Trimers. *J Virol* 90:2806-2817). To improve trimer design, Applicants incorporated residues from BG505 (called trimer-derived (TD) residues) into 16055 NFLs, substantially improving the propensity to form native-like trimers (Guenaga et al. 2015. Structure-Guided Redesign Increases the Propensity of HIV Env To Generate Highly Stable Soluble Trimers. *J Virol* 90:2806-2817) and the elicitation of tier 2 clade C neutralizing antibodies (Martinez-Murillo et al., GB. 2017. Particulate Array of Well-Ordered HIV Clade C Env Trimers Elicits Neutralizing Antibodies that Display a Unique V2 Cap Approach. *Immunity* 46:804-817 e807; Dubrovskaya et al. 2017. Targeted N-glycan deletion at the receptor-binding site retains HIV Env NFL trimer integrity and accelerates the elicited antibody response. *PLoS Pathog* 13:e1006614). Further improvements on the TD design by targeted glycine substitutions at helix-to-coil transitions that disfavor the post-fusion state of Env (TD CC+, namely “TD+”), significantly improve trimer homogeneity, yield, stability and antigenicity, resulting in the first high-resolution clade C Env structure (Guenaga et al. 2017. Glycine Substitution at Helix-to-Coil Transitions Facilitates the Structural Determination of a Stabilized Subtype C HIV Envelope Glycoprotein. *Immunity* 46:792-803 e793).

Applicants believe that the glycine changes may lower the activation potential of the gp41 (and Env) to change conformation, and therefore results in better behaved trimers in a lower energy well from the “activation state” to spring to the next conformation. In a simple model, gp41 is essentially spring-loaded and constrained by gp120 until receptor binding. These mutations may contribute to reducing the springiness.

Assays for screening for neutralizing antibodies are known in the art. A neutralization assay approach has been described previously (Binley J M, et al., (2004). Comprehensive Cross-Clade Neutralization Analysis of a Panel of Anti-Human Immunodeficiency Virus Type 1 Monoclonal Antibodies. *J Viral*. 78: 13232-13252). Pseudotyped viruses may be generated by co-transfecting cells with at least two plasmids encoding the soluble Env cDNA of the present

invention and the rest of the HIV genome separately. In the HIV genome encoding vector, the Env gene may be replaced by the firefly luciferase gene. Transfectant supernatants containing pseudotyped virus may be co-incubated overnight with B cell supernatants derived from activation of an infected donor’s primary peripheral blood mononuclear cells (PBMCs). Cells stably transfected with and expressing CD4 plus the CCR5 and CXCR4 coreceptors may be added to the mixture and incubated for 3 days at 37° C. Infected cells may be quantified by luminometry.

In another embodiment of the present invention, the soluble envelope glycoproteins of the present invention may be crystallized in combination with any neutralizing antibodies, including those identified by the above methods, to determine the exact molecular surface where the soluble envelope glycoprotein binds with the neutralizing antibody to design HIV-I immunogens.

It is to be understood that this invention is not limited to particular methods, components, products or combinations described, as such methods, components, products and combinations may, of course, vary. It is also to be understood that the terminology used herein is not intended to be limiting, since the scope of the present invention will be limited only by the appended claims.

As used herein, the singular forms “a”, “an”, and “the” include both singular and plural referents unless the context clearly dictates otherwise.

The terms “comprising”, “comprises” and “comprised of” as used herein are synonymous with “including”, “includes” or “containing”, “contains”, and are inclusive or open-ended and do not exclude additional, non-recited members, elements or method steps. It will be appreciated that the terms “comprising”, “comprises” and “comprised of” as used herein comprise the terms “consisting of”, “consists” and “consists of, as well as the terms “consisting essentially of”, “consists essentially” and “consists essentially of. It is noted that in this disclosure and particularly in the claims and/or paragraphs, terms such as “comprises”, “comprised”, “comprising” and the like can have the meaning attributed to it in U. S. Patent law; e.g., they can mean “includes”, “included”, “including”, and the like; and that terms such as “consisting essentially of” and “consists essentially of have the meaning ascribed to them in U. S. Patent law, e.g., they allow for elements not explicitly recited, but exclude elements that are found in the prior art or that affect a basic or novel characteristic of the invention. It may be advantageous in the practice of the invention to be in compliance with Art. 53(c) EPC and Rule 28(b) and (c) EPC. Nothing herein is intended as a promise.

The recitation of numerical ranges by endpoints includes all numbers and fractions subsumed within the respective ranges, as well as the recited endpoints.

The term “about” or “approximately” as used herein when referring to a measurable value such as a parameter, an amount, a temporal duration, and the like, is meant to encompass variations of +/-20% or less, preferably +/-10% or less, more preferably +/-5% or less, and still more preferably +/-1% or less of and from the specified value, insofar such variations are appropriate to perform in the disclosed invention. It is to be understood that the value to which the modifier “about” or “approximately” refers is itself also specifically, and preferably, disclosed.

Whereas the terms “one or more” or “at least one” or “X or more”, where X is a number and understand to mean X or increases one by one of X, such as one or more or at least one member(s) or “X or more” of a group of members, is clear per se, by means of further exemplification, the term

encompasses inter alia a reference to any one of said members, or to any two or more of said members, such as, e.g., any >3, >4, >5, >6 or >7 etc. of said members, and up to all said members.

All references cited in the present specification are hereby incorporated by reference in their entirety. In particular, the teachings of all references herein specifically referred to are incorporated by reference. Andrabi et al., 2019 is incorporated by reference.

Unless otherwise defined, all terms used in disclosing the invention, including technical and scientific terms, have the meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. By means of further guidance, term definitions are included to better appreciate the teaching of the present invention. In the following passages, different aspects of the invention are defined in more detail. Each aspect so defined may be combined with any other aspect or aspects unless clearly indicated to the contrary. In particular, any feature indicated as being preferred or advantageous may be combined with any other feature or features indicated as being preferred or advantageous.

Standard reference works setting forth the general principles of recombinant DNA technology include Molecular Cloning: A Laboratory Manual, 2nd ed., vol. 1-3, ed. Sambrook et al., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y., 1989; Current Protocols in Molecular Biology, ed. Ausubel et al., Greene Publishing and Wiley-Interscience, New York, 1992 (with periodic updates) ("Ausubel et al. 1992"); the series Methods in Enzymology (Academic Press, Inc.); Innis et al., PCR Protocols: A Guide to Methods and Applications, Academic Press: San Diego, 1990; PCR 2: A Practical Approach (M. J. MacPherson, B. D. Hames and G. R. Taylor eds. (1995); Harlow and Lane, eds. (1988) Antibodies, a Laboratory Manual; and Animal Cell Culture (R. I. Freshney, ed. (1987). General principles of microbiology are set forth, for example, in Davis, B. D. et al., Microbiology, 3rd edition, Harper & Row, publishers, Philadelphia, Pa. (1980).

Reference throughout this specification to "one embodiment" or "an embodiment" means that a particular feature, structure or characteristic described in connection with the embodiment is included in at least one embodiment of the present invention. Thus, appearances of the phrases "in one embodiment" or "in an embodiment" in various places throughout this specification are not necessarily all referring to the same embodiment, but may. Furthermore, the particular features, structures or characteristics may be combined in any suitable manner, as would be apparent to a person skilled in the art from this disclosure, in one or more embodiments. Furthermore, while some embodiments described herein include some but not other features included in other embodiments, combinations of features of different embodiments are meant to be within the scope of the invention, and form different embodiments, as would be understood by those in the art. For example, in the appended claims, any of the claimed embodiments can be used in any combination.

In this description of the invention, reference is made to the accompanying drawings that form a part hereof, and in which are shown by way of illustration only of specific embodiments in which the invention may be practiced. It is to be understood that other embodiments may be utilized and structural or logical changes may be made without departing from the scope of the present invention. The

description, therefore, is not to be taken in a limiting sense, and the scope of the present invention is defined by the appended claims.

Preferred statements (features) and embodiments of this invention are set herein below. Each statements and embodiments of the invention so defined may be combined with any other statement and/or embodiments unless clearly indicated to the contrary. In particular, any feature indicated as being preferred or advantageous may be combined with any other feature or features or statements indicated as being preferred or advantageous.

Unless specifically stated or obvious from context, as used herein, the term "about" is understood as within a range of normal tolerance in the art, for example within 2 standard deviations of the mean. About can be understood as within 50%, 45%, 40%, 35%, 30%, 25%, 20%, 15%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, 0.5%, 0.1%, 0.05%, or 0.01% of the stated value. Unless otherwise clear from context, all numerical values provided herein are modified by the term about.

Unless specifically stated or obvious from context, as used herein, the term "or" is understood to be inclusive. Unless specifically stated or obvious from context, as used herein, the terms "a," "an," and "the" are understood to be singular or plural.

The terms "protein", "peptide", "polypeptide", and "amino acid sequence" are used interchangeably herein to refer to polymers of amino acid residues of any length. The polymer may be linear or branched, it may comprise modified amino acids or amino acid analogs, and it may be interrupted by chemical moieties other than amino acids. The terms also encompass an amino acid polymer that has been modified naturally or by intervention; for example disulfide bond formation, glycosylation, lipidation, acetylation, phosphorylation, or any other manipulation or modification, such as conjugation with a labeling or bioactive component.

As used herein, the terms "antigen" or "immunogen" are used interchangeably to refer to a substance, typically a protein, which is capable of inducing an immune response in a subject. The term also refers to proteins that are immunologically active in the sense that once administered to a subject (either directly or by administering to the subject a nucleotide sequence or vector that encodes the protein) is able to evoke an immune response of the humoral and/or cellular type directed against that protein.

The term "antibody" includes intact molecules as well as fragments thereof, such as Fab, F(ab')₂, Fv and scFv which are capable of binding the epitope determinant. These antibody fragments retain some ability to selectively bind with its antigen or receptor and include, for example:

Fab, the fragment which contains a monovalent antigen-binding fragment of an antibody molecule can be produced by digestion of whole antibody with the enzyme papain to yield an intact light chain and a portion of one heavy chain;

Fab', the fragment of an antibody molecule can be obtained by treating whole antibody with pepsin, followed by reduction, to yield an intact light chain and a portion of the heavy chain; two Fab' fragments are obtained per antibody molecule;

F(ab')₂, the fragment of the antibody that can be obtained by treating whole antibody with the enzyme pepsin without subsequent reduction; F(ab')₂ is a dimer of two Fab' fragments held together by two disulfide bonds;

scFv, including a genetically engineered fragment containing the variable region of a heavy and a light chain as a fused single chain molecule.

General methods of making these fragments are known in the art. (See for example, Harlow and Lane, *Antibodies: A Laboratory Manual*, Cold Spring Harbor Laboratory, New York (1988), which is incorporated herein by reference).

A "neutralizing antibody" may inhibit the entry of HIV-I virus F with a neutralization index >1.5 or >2.0. Broad and potent neutralizing antibodies may neutralize greater than about 50% of HIV-I viruses (from diverse clades and different strains within a clade) in a neutralization assay. The inhibitory concentration of the monoclonal antibody may be less than about 25 mg/ml to neutralize about 50% of the input virus in the neutralization assay.

It should be understood that the proteins, including the antibodies and/or antigens of the invention may differ from the exact sequences illustrated and described herein. Thus, the invention contemplates deletions, additions and substitutions to the sequences shown, so long as the sequences function in accordance with the methods of the invention. In this regard, particularly preferred substitutions may generally be conservative in nature, i.e., those substitutions that take place within a family of amino acids. For example, amino acids are generally divided into four families: (1) acidic—aspartate and glutamate; (2) basic—lysine, arginine, histidine; (3) non-polar—alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, tryptophan; and (4) uncharged polar—glycine, asparagine, glutamine, cysteine, serine, threonine, tyrosine. Phenylalanine, tryptophan, and tyrosine are sometimes classified as aromatic amino acids. It is reasonably predictable that an isolated replacement of leucine with isoleucine or valine, or vice versa; an aspartate with a glutamate or vice versa; a threonine with a serine or vice versa; or a similar conservative replacement of an amino acid with a structurally related amino acid, will not have a major effect on the biological activity. Proteins having substantially the same amino acid sequence as the sequences illustrated and described but possessing minor amino acid substitutions that do not substantially affect the immunogenicity of the protein are, therefore, within the scope of the invention.

As used herein the terms "nucleotide sequences" and "nucleic acid sequences" refer to deoxyribonucleic acid (DNA) or ribonucleic acid (RNA) sequences, including, without limitation, messenger RNA (mRNA), DNA/RNA hybrids, or synthetic nucleic acids. The nucleic acid can be single-stranded, or partially or completely double-stranded (duplex). Duplex nucleic acids can be homoduplex or heteroduplex.

As used herein the term "transgene" may be used to refer to "recombinant" nucleotide sequences that may be derived from any of the nucleotide sequences encoding the proteins of the present invention. The term "recombinant" means a nucleotide sequence that has been manipulated "by man" and which does not occur in nature, or is linked to another nucleotide sequence or found in a different arrangement in nature. It is understood that manipulated "by man" means manipulated by some artificial means, including by use of machines, codon optimization, restriction enzymes, etc.

For example, in one embodiment the nucleotide sequences may be mutated such that the activity of the encoded proteins in vivo is abrogated. In another embodiment the nucleotide sequences may be codon optimized, for example the codons may be optimized for human use. In preferred embodiments the nucleotide sequences of the invention are both mutated to abrogate the normal in vivo

function of the encoded proteins, and codon optimized for human use. For example, each of the Gag, Pol, Env, Nef, RT, and Int sequences of the invention may be altered in these ways.

As regards codon optimization, the nucleic acid molecules of the invention have a nucleotide sequence that encodes the antigens of the invention and can be designed to employ codons that are used in the genes of the subject in which the antigen is to be produced. Many viruses, including HIV and other lentiviruses, use a large number of rare codons and, by altering these codons to correspond to codons commonly used in the desired subject, enhanced expression of the antigens can be achieved. In a preferred embodiment, the codons used are "humanized" codons, i.e., the codons are those that appear frequently in highly expressed human genes (Andre et al., *J. Virol.* 72:1497-1503, 1998) instead of those codons that are frequently used by HIV. Such codon usage provides for efficient expression of the transgenic HIV proteins in human cells. Any suitable method of codon optimization may be used. Such methods, and the selection of such methods, are well known to those of skill in the art. In addition, there are several companies that will optimize codons of sequences, such as Genart. Thus, the nucleotide sequences of the invention can readily be codon optimized.

The invention further encompasses nucleotide sequences encoding functionally and/or antigenically equivalent variants and derivatives of the antigens of the invention and functionally equivalent fragments thereof. These functionally equivalent variants, derivatives, and fragments display the ability to retain antigenic activity. For instance, changes in a DNA sequence that do not change the encoded amino acid sequence, as well as those that result in conservative substitutions of amino acid residues, one or a few amino acid deletions or additions, and substitution of amino acid residues by amino acid analogs are those which will not significantly affect properties of the encoded polypeptide. Conservative amino acid substitutions are glycine/alanine; valine/isoleucine/leucine; asparagine/glutamine; aspartic acid/glutamic acid; serine/threonine/methionine; lysine/arginine; and phenylalanine/tyrosine/tryptophan. In one embodiment, the variants have at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98% or at least 99% homology or identity to the antigen, epitope, immunogen, peptide or polypeptide of interest.

For the purposes of the present invention, sequence identity or homology is determined by comparing the sequences when aligned so as to maximize overlap and identity while minimizing sequence gaps. In particular, sequence identity may be determined using any of a number of mathematical algorithms. A nonlimiting example of a mathematical algorithm used for comparison of two sequences is the algorithm of Karlin & Altschul, *Proc. Natl. Acad. Sci. USA* 1990; 87: 2264-2268, modified as in Karlin & Altschul, *Proc. Natl. Acad. Sci. USA* 1993; 90: 5873-5877.

Another example of a mathematical algorithm used for comparison of sequences is the algorithm of Myers & Miller, *CABIOS* 1988; 4: 11-17. Such an algorithm is incorporated into the ALIGN program (version 2.0) which is part of the GCG sequence alignment software package. When utilizing the ALIGN program for comparing amino acid sequences, a PAM120 weight residue table, a gap length penalty of 12, and a gap penalty of 4 can be used. Yet another useful

algorithm for identifying regions of local sequence similarity and alignment is the FASTA algorithm as described in Pearson & Lipman, Proc. Natl. Acad. Sci. USA 1988; 85: 2444-2448. Advantageous for use according to the present invention is the WU-BLAST (Washington University BLAST) version 2.0 software. WU-BLAST version 2.0 executable programs for several UNIX platforms can be downloaded from the Washington University BLAST web-site. This program is based on WU-BLAST version 1.4, which in turn is based on the public domain NCBI-BLAST version 1.4 (Altschul & Gish, 1996, Local alignment statistics, Doolittle ed., Methods in Enzymology 266: 460-480; Altschul et al., Journal of Molecular Biology 1990; 215: 403-410; Gish & States, 1993; Nature Genetics 3: 266-272; Karlin & Altschul, 1993; Proc. Natl. Acad. Sci. USA 90: 5873-5877; all of which are incorporated by reference herein).

The various recombinant nucleotide sequences and antibodies and/or antigens of the invention are made using standard recombinant DNA and cloning techniques. Such techniques are well known to those of skill in the art. See for example, "Molecular Cloning: A Laboratory Manual", second edition (Sambrook et al. 1989).

The nucleotide sequences of the present invention may be inserted into "vectors." The term "vector" is widely used and understood by those of skill in the art, and as used herein the term "vector" is used consistent with its meaning to those of skill in the art. For example, the term "vector" is commonly used by those skilled in the art to refer to a vehicle that allows or facilitates the transfer of nucleic acid molecules from one environment to another or that allows or facilitates the manipulation of a nucleic acid molecule.

Any vector that allows expression of the antibodies and/or antigens of the present invention may be used in accordance with the present invention. In certain embodiments, the antigens and/or antibodies of the present invention may be used in vitro (such as using cell-free expression systems) and/or in cultured cells grown in vitro in order to produce the encoded HIV-antigens and/or antibodies which may then be used for various applications such as in the production of proteinaceous vaccines. For such applications, any vector that allows expression of the antigens and/or antibodies in vitro and/or in cultured cells may be used.

For applications where it is desired that the antibodies and/or antigens be expressed in vivo, for example when the transgenes of the invention are used in DNA or DNA-containing vaccines, any vector that allows for the expression of the antibodies and/or antigens of the present invention and is safe for use in vivo may be used. In preferred embodiments the vectors used are safe for use in humans, mammals and/or laboratory animals.

For the antibodies and/or antigens of the present invention to be expressed, the protein coding sequence should be "operably linked" to regulatory or nucleic acid control sequences that direct transcription and translation of the protein. As used herein, a coding sequence and a nucleic acid control sequence or promoter are said to be "operably linked" when they are covalently linked in such a way as to place the expression or transcription and/or translation of the coding sequence under the influence or control of the nucleic acid control sequence. The "nucleic acid control sequence" can be any nucleic acid element, such as, but not limited to promoters, enhancers, IRES, introns, and other elements described herein that direct the expression of a nucleic acid sequence or coding sequence that is operably linked thereto. The term "promoter" will be used herein to refer to a group of transcriptional control modules that are clustered around

the initiation site for RNA polymerase II and that when operationally linked to the protein coding sequences of the invention lead to the expression of the encoded protein. The expression of the transgenes of the present invention can be under the control of a constitutive promoter or of an inducible promoter, which initiates transcription only when exposed to some particular external stimulus, such as, without limitation, antibiotics such as tetracycline, hormones such as ecdysone, or heavy metals. The promoter can also be specific to a particular cell-type, tissue or organ. Many suitable promoters and enhancers are known in the art, and any such suitable promoter or enhancer may be used for expression of the transgenes of the invention. For example, suitable promoters and/or enhancers can be selected from the Eukaryotic Promoter Database (EPDB).

The present invention relates to a recombinant vector expressing a foreign epitope. Advantageously, the epitope is an SIV or HIV epitope or combination thereof. In an advantageous embodiment, the SIV or HIV epitope is a soluble envelope glycoprotein, however, the present invention may encompass additional HIV antigens, epitopes or immunogens. Advantageously, the additional epitope is an HIV antigen, HIV epitope or an HIV immunogen, such as, but not limited to, the HIV antigens, HIV epitopes or HIV immunogens of U.S. Pat. Nos. 7,341,731; 7,335,364; 7,329,807; 7,323,553; 7,320,859; 7,311,920; 7,306,798; 7,285,646; 7,285,289; 7,285,271; 7,282,364; 7,273,695; 7,270,997; 7,262,270; 7,244,819; 7,244,575; 7,232,567; 7,232,566; 7,223,844; 7,223,739; 7,223,534; 7,223,368; 7,220,554; 7,214,530; 7,211,659; 7,211,432; 7,205,159; 7,198,934; 7,195,768; 7,192,555; 7,189,826; 7,189,522; 7,186,507; 7,179,645; 7,175,843; 7,172,761; 7,169,550; 7,157,083; 7,153,509; 7,147,862; 7,141,550; 7,129,219; 7,122,188; 7,118,859; 7,118,855; 7,118,751; 7,118,742; 7,105,655; 7,101,552; 7,097,971; 7,097,842; 7,094,405; 7,091,049; 7,090,648; 7,087,377; 7,083,787; 7,070,787; 7,070,781; 7,060,273; 7,056,521; 7,056,519; 7,049,136; 7,048,929; 7,033,593; 7,030,094; 7,022,326; 7,009,037; 7,008,622; 7,001,759; 6,997,863; 6,995,008; 6,979,535; 6,974,574; 6,972,126; 6,969,609; 6,964,769; 6,964,762; 6,958,158; 6,956,059; 6,953,689; 6,951,648; 6,946,075; 6,927,031; 6,919,319; 6,919,318; 6,919,077; 6,913,752; 6,911,315; 6,908,617; 6,908,612; 6,902,743; 6,900,010; 6,893,869; 6,884,785; 6,884,435; 6,875,435; 6,867,005; 6,861,234; 6,855,539; 6,841,381; 6,841,345; 6,838,477; 6,821,955; 6,818,392; 6,818,222; 6,815,217; 6,815,201; 6,812,026; 6,812,025; 6,812,024; 6,808,923; 6,806,055; 6,803,231; 6,800,613; 6,800,288; 6,797,811; 6,780,967; 6,780,598; 6,773,920; 6,764,682; 6,761,893; 6,753,015; 6,750,005; 6,737,239; 6,737,067; 6,730,304; 6,720,310; 6,716,823; 6,713,301; 6,713,070; 6,706,859; 6,699,722; 6,699,656; 6,696,291; 6,692,745; 6,670,181; 6,670,115; 6,664,406; 6,657,055; 6,657,050; 6,656,471; 6,653,066; 6,649,409; 6,649,372; 6,645,732; 6,641,816; 6,635,469; 6,613,530; 6,605,427; 6,602,709; 6,602,705; 6,600,023; 6,596,477; 6,596,172; 6,593,103; 6,593,079; 6,579,673; 6,576,758; 6,573,245; 6,573,040; 6,569,418; 6,569,340; 6,562,800; 6,558,961; 6,551,828; 6,551,824; 6,548,275; 6,544,780; 6,544,752; 6,544,728; 6,534,482; 6,534,312; 6,534,064; 6,531,572; 6,531,313; 6,525,179; 6,525,028; 6,524,582; 6,521,449; 6,518,030; 6,518,015; 6,514,691; 6,514,503; 6,511,845; 6,511,812; 6,511,801; 6,509,313; 6,506,384; 6,503,882; 6,495,676; 6,495,526; 6,495,347; 6,492,123; 6,489,131; 6,489,129; 6,482,614; 6,479,286; 6,479,284; 6,465,634; 6,461,615; 6,458,560; 6,458,527; 6,458,370; 6,451,601; 6,451,592; 6,451,323; 6,436,407; 6,432,633; 6,428,970; 6,428,952; 6,428,790; 6,420,139; 6,416,997;

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6,410,318; 6,410,028; 6,410,014; 6,407,221; 6,406,710;
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5,541,057; 5,534,406; 5,529,765; 5,523,232; 5,516,895;
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 5,317,009; 5,312,902; 5,304,466; 5,296,347; 5,286,852;
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 5,256,561; 5,252,556; 5,230,998; 5,230,887; 5,227,159;
 5,225,347; 5,221,610; 5,217,861; 5,208,321; 5,206,136;
 5,198,346; 5,185,147; 5,178,865; 5,173,400; 5,173,399;
 5,166,050; 5,156,951; 5,135,864; 5,122,446; 5,120,662;
 5,103,836; 5,100,777; 5,100,662; 5,093,230; 5,077,284;
 5,070,010; 5,068,174; 5,066,782; 5,055,391; 5,043,262;
 5,039,604; 5,039,522; 5,030,718; 5,030,555; 5,030,449;
 5,019,387; 5,013,556; 5,008,183; 5,004,697; 4,997,772;
 4,983,529; 4,983,387; 4,965,069; 4,945,082; 4,921,787;
 4,918,166; 4,900,548; 4,888,290; 4,886,742; 4,885,235;
 4,870,003; 4,869,903; 4,861,707; 4,853,326; 4,839,288;
 4,833,072 and 4,795,739.

In another embodiment, HIV, or immunogenic fragments thereof, may be utilized as the HIV epitope. For example, the HIV nucleotides of U.S. Pat. Nos. 7,393,949, 7,374,877, 7,306,901, 7,303,754, 7,173,014, 7,122,180, 7,078,516, 7,022,814, 6,974,866, 6,958,211, 6,949,337, 6,946,254, 6,896,900, 6,887,977, 6,870,045, 6,803,187, 6,794,129, 6,773,915, 6,768,004, 6,706,268, 6,696,291, 6,692,955, 6,656,706, 6,649,409, 6,627,442, 6,610,476, 6,602,705, 6,582,920, 6,557,296, 6,531,587, 6,531,137, 6,500,623, 6,448,078, 6,429,306, 6,420,545, 6,410,013, 6,407,077, 6,395,891, 6,355,789, 6,335,158, 6,323,185, 6,316,183, 6,303,293, 6,300,056, 6,277,561, 6,270,975, 6,261,564, 6,225,045, 6,222,024, 6,194,391, 6,194,142, 6,162,631, 6,114,167, 6,114,109, 6,090,392, 6,060,587, 6,057,102, 6,054,565, 6,043,081, 6,037,165, 6,034,233, 6,033,902, 6,030,769, 6,020,123, 6,015,661, 6,010,895, 6,001,555, 5,985,661, 5,980,900, 5,972,596, 5,939,538, 5,912,338, 5,869,339, 5,866,701, 5,866,694, 5,866,320, 5,866,137, 5,864,027, 5,861,242, 5,858,785, 5,858,651, 5,849,475, 5,843,638, 5,840,480, 5,821,046, 5,801,056, 5,786,177, 5,786,145, 5,773,247, 5,770,703, 5,756,674, 5,741,706, 5,705,612, 5,693,752, 5,688,637, 5,688,511, 5,684,147, 5,665,577, 5,585,263, 5,578,715, 5,571,712, 5,567,603, 5,554,528, 5,545,726, 5,527,895, 5,527,894, 5,223,423, 5,204,259, 5,144,019, 5,051,496 and 4,942,122 are useful for the present invention.

Any epitope recognized by an HIV antibody may be used in the present invention. For example, the anti-HIV antibodies of U.S. Pat. Nos. 6,984,721, 6,972,126, 6,949,337, 6,946,465, 6,919,077, 6,916,475, 6,911,315, 6,905,680, 6,900,010, 6,825,217, 6,824,975, 6,818,392, 6,815,201, 6,812,026, 6,812,024, 6,797,811, 6,768,004, 6,703,019, 6,689,118, 6,657,050, 6,608,179, 6,600,023, 6,596,497, 6,589,748, 6,569,143, 6,548,275, 6,525,179, 6,524,582, 6,506,384, 6,498,006, 6,489,131, 6,465,173, 6,461,612, 6,458,933, 6,432,633, 6,410,318, 6,406,701, 6,395,275, 6,391,657, 6,391,635, 6,384,198, 6,376,170, 6,372,217, 6,344,545, 6,337,181, 6,329,202, 6,319,665, 6,319,500, 6,316,003, 6,312,931, 6,309,880, 6,296,807, 6,291,239, 6,261,558, 6,248,514, 6,245,331, 6,242,197, 6,241,986, 6,228,361, 6,221,580, 6,190,871, 6,177,253, 6,146,635, 6,146,627, 6,146,614, 6,143,876, 6,132,992, 6,124,132, RE36,866, 6,114,143, 6,103,238, 6,060,254, 6,039,684, 6,030,772, 6,020,468, 6,013,484, 6,008,044, 5,998,132, 5,994,515, 5,993,812, 5,985,545, 5,981,278, 5,958,765, 5,939,277, 5,928,930, 5,922,325, 5,919,457, 5,916,806,

5,914,109, 5,911,989, 5,906,936, 5,889,158, 5,876,716, 5,874,226, 5,872,012, 5,871,732, 5,866,694, 5,854,400, 5,849,583, 5,849,288, 5,840,480, 5,840,305, 5,834,599, 5,831,034, 5,827,723, 5,821,047, 5,817,767, 5,817,458, 5,804,440, 5,795,572, 5,783,670, 5,776,703, 5,773,225, 5,766,944, 5,753,503, 5,750,373, 5,747,641, 5,736,341, 5,731,189, 5,707,814, 5,702,707, 5,698,178, 5,695,927, 5,665,536, 5,658,745, 5,652,138, 5,645,836, 5,635,345, 5,618,922, 5,610,035, 5,607,847, 5,604,092, 5,601,819, 5,597,896, 5,597,688, 5,591,829, 5,558,865, 5,514,541, 5,510,264, 5,478,753, 5,374,518, 5,374,516, 5,344,755, 5,332,567, 5,300,433, 5,296,347, 5,286,852, 5,264,221, 5,260,308, 5,256,561, 5,254,457, 5,230,998, 5,227,159, 5,223,408, 5,217,895, 5,180,660, 5,173,399, 5,169,752, 5,166,050, 5,156,951, 5,140,105, 5,135,864, 5,120,640, 5,108,904, 5,104,790, 5,049,389, 5,030,718, 5,030,555, 5,004,697, 4,983,529, 4,888,290, 4,886,742 and 4,853,326, are also useful for the present invention.

The vectors used in accordance with the present invention should typically be chosen such that they contain a suitable gene regulatory region, such as a promoter or enhancer, such that the antigens and/or antibodies of the invention can be expressed.

For example, when the aim is to express the antibodies and/or antigens of the invention in vitro, or in cultured cells, or in any prokaryotic or eukaryotic system for the purpose of producing the protein(s) encoded by that antibody and/or antigen, then any suitable vector can be used depending on the application. For example, plasmids, viral vectors, bacterial vectors, protozoal vectors, insect vectors, baculovirus expression vectors, yeast vectors, mammalian cell vectors, and the like, can be used. Suitable vectors can be selected by the skilled artisan taking into consideration the characteristics of the vector and the requirements for expressing the antibodies and/or antigens under the identified circumstances.

When the aim is to express the antibodies and/or antigens of the invention in vivo in a subject, for example in order to generate an immune response against an HIV-I antigen and/or protective immunity against HIV-I, expression vectors that are suitable for expression on that subject, and that are safe for use in vivo, should be chosen. For example, in some embodiments it may be desired to express the antibodies and/or antigens of the invention in a laboratory animal, such as for pre-clinical testing of the HIV-I immunogenic compositions and vaccines of the invention. In other embodiments, it will be desirable to express the antibodies and/or antigens of the invention in human subjects, such as in clinical trials and for actual clinical use of the immunogenic compositions and vaccine of the invention. Any vectors that are suitable for such uses can be employed, and it is well within the capabilities of the skilled artisan to select a suitable vector. In some embodiments it may be preferred that the vectors used for these in vivo applications are attenuated to vector from amplifying in the subject. For example, if plasmid vectors are used, preferably they will lack an origin of replication that functions in the subject so as to enhance safety for in vivo use in the subject. If viral vectors are used, preferably they are attenuated or replication-defective in the subject, again, so as to enhance safety for in vivo use in the subject.

In preferred embodiments of the present invention viral vectors are used. Viral expression vectors are well known to those skilled in the art and include, for example, viruses such as adenoviruses, adeno-associated viruses (AAV), alphaviruses, herpesviruses, retroviruses and poxviruses, including avipox viruses, attenuated poxviruses, vaccinia viruses, and

particularly, the modified vaccinia Ankara virus (MVA; ATCC Accession No. VR-I566). Such viruses, when used as expression vectors are innately non-pathogenic in the selected subjects such as humans or have been modified to render them non-pathogenic in the selected subjects. For example, replication-defective adenoviruses and alphaviruses are well known and can be used as gene delivery vectors.

The nucleotide sequences and vectors of the invention can be delivered to cells, for example if aim is to express and the HIV-I antigens in cells in order to produce and isolate the expressed proteins, such as from cells grown in culture. For expressing the antibodies and/or antigens in cells any suitable transfection, transformation, or gene delivery methods can be used. Such methods are well known by those skilled in the art, and one of skill in the art would readily be able to select a suitable method depending on the nature of the nucleotide sequences, vectors, and cell types used. For example, transfection, transformation, microinjection, infection, electroporation, lipofection, or liposome-mediated delivery could be used. Expression of the antibodies and/or antigens can be carried out in any suitable type of host cells, such as bacterial cells, yeast, insect cells, and mammalian cells. The antibodies and/or antigens of the invention can also be expressed using including in vitro transcription/translation systems. All of such methods are well known by those skilled in the art, and one of skill in the art would readily be able to select a suitable method depending on the nature of the nucleotide sequences, vectors, and cell types used.

In preferred embodiments, the nucleotide sequences, antibodies and/or antigens of the invention are administered in vivo, for example where the aim is to produce an immunogenic response in a subject. A "subject" in the context of the present invention may be any animal. For example, in some embodiments it may be desired to express the transgenes of the invention in a laboratory animal, such as for pre-clinical testing of the HIV-I immunogenic compositions and vaccines of the invention. In other embodiments, it will be desirable to express the antibodies and/or antigens of the invention in human subjects, such as in clinical trials and for actual clinical use of the immunogenic compositions and vaccine of the invention. In preferred embodiments the subject is a human, for example a human that is infected with, or is at risk of infection with, HIV-1.

For such in vivo applications the nucleotide sequences, antibodies and/or antigens of the invention are preferably administered as a component of an immunogenic composition comprising the nucleotide sequences and/or antigens of the invention in admixture with a pharmaceutically acceptable carrier. The immunogenic compositions of the invention are useful to stimulate an immune response against HIV-1 and may be used as one or more components of a prophylactic or therapeutic vaccine against HIV-1 for the prevention, amelioration or treatment of AIDS. The nucleic acids and vectors of the invention are particularly useful for providing genetic vaccines, i.e. vaccines for delivering the nucleic acids encoding the antibodies and/or antigens of the invention to a subject, such as a human, such that the antibodies and/or antigens are then expressed in the subject to elicit an immune response.

The compositions of the invention may be injectable suspensions, solutions, sprays, lyophilized powders, syrups, elixirs and the like. Any suitable form of composition may be used. To prepare such a composition, a nucleic acid or vector of the invention, having the desired degree of purity, is mixed with one or more pharmaceutically acceptable

carriers and/or excipients. The carriers and excipients must be "acceptable" in the sense of being compatible with the other ingredients of the composition. Acceptable carriers, excipients, or stabilizers are nontoxic to recipients at the dosages and concentrations employed, and include, but are not limited to, water, saline, phosphate buffered saline, dextrose, glycerol, ethanol, or combinations thereof; buffers such as phosphate, citrate, and other organic acids; antioxidants including ascorbic acid and methionine; preservatives (such as octadecyldimethylbenzyl ammonium chloride; hexamethonium chloride; benzalkonium chloride, benzethonium chloride; phenol, butyl or benzyl alcohol; alkyl parabens such as methyl or propyl paraben; catechol; resorcinol; cyclohexanol; 3-pentanol; and m-cresol); low molecular weight (less than about 10 residues) polypeptide; proteins, such as serum albumin, gelatin, or immunoglobulins; hydrophilic polymers such as polyvinylpyrrolidone; amino acids such as glycine, glutamine, asparagine, histidine, arginine, or lysine; monosaccharides, disaccharides, and other carbohydrates including glucose, mannose, or dextrins; chelating agents such as EDTA; sugars such as sucrose, mannitol, trehalose or sorbitol; salt-forming counter-ions such as sodium; metal complexes (e.g., Zn-protein complexes); and/or non-ionic surfactants such as TWEENTM, PLURON-ICSM or polyethylene glycol. An immunogenic or immunological composition can also be formulated in the form of an oil-in-water emulsion. The oil-in-water emulsion can be based, for example, on light liquid paraffin oil (European Pharmacopeia type); isoprenoid oil such as squalane, squalene, EICOSANETM or tetratetracontane; oil resulting from the oligomerization of alkene(s), e.g., isobutene or decene; esters of acids or of alcohols containing a linear alkyl group, such as plant oils, ethyl oleate, propylene glycol di(caprylate/caprate), glyceryl tri(caprylate/caprate) or propylene glycol dioleate; esters of branched fatty acids or alcohols, e.g., isostearic acid esters. The oil advantageously is used in combination with emulsifiers to form the emulsion. The emulsifiers can be nonionic surfactants, such as esters of sorbitan, mannide (e.g., anhydromannitol oleate), glycerol, polyglycerol, propylene glycol, and oleic, isostearic, ricinoleic, or hydroxystearic acid, which are optionally ethoxylated, and polyoxypropylene-polyoxyethylene copolymer blocks, such as the Pluronic[®] products, e.g., L121. The adjuvant can be a mixture of emulsifier(s), micelle-forming agent, and oil such as that which is commercially available under the name Provax[®] (IDEC Pharmaceuticals, San Diego, CA). (PEG).

The immunogenic compositions of the invention can contain additional substances, such as wetting or emulsifying agents, buffering agents, or adjuvants to enhance the effectiveness of the vaccines (Remington's Pharmaceutical Sciences, 18th edition, Mack Publishing Company, (ed.) 1980).

Adjuvants may also be included. Adjuvants include, but are not limited to, mineral salts (e.g., $\text{AlK}(\text{SO}_4)_2$, $\text{AlNa}(\text{SO}_4)_2$, $\text{AlNH}(\text{SO}_4)_2$, silica, alum, $\text{Al}(\text{OH})_3$, $\text{Ca}_3(\text{PO}_4)_2$, kaolin, or carbon), polynucleotides with or without immune stimulating complexes (ISCOMs) (e.g., CpG oligonucleotides, such as those described in Chuang, T. H. et al (2002) J. Leuk. Biol. 71(3): 538-44; Ahmad-Nejad, P. et al (2002) Eur. J. Immunol. 32(7): 1958-68; poly IC or poly AU acids, polyarginine with or without CpG (also known in the art as IC31; see Schellack, C. et al (2003) Proceedings of the 34th Annual Meeting of the German Society of Immunology; Lingnau, K. et al (2002) Vaccine 20(29-30): 3498-508), JuvaVaxTM (U.S. Pat. No. 6,693,086), certain natural substances (e.g., wax D from *Mycobacterium tuberculosis*, sub-

stances found in *Corynebacterium parvum*, *Bordetella pertussis*, or members of the genus *Brucella*), flagellin (Toll-like receptor 5 ligand; see McSorley, S. J. et al (2002) J. Immunol. 169(7): 3914-9), saponins such as QS21, QS17, and QS7 (U.S. Pat. Nos. 5,057,540; 5,650,398; 6,524,584; 6,645,495), monophosphoryl lipid A, in particular, 3-de-O-acetylated monophosphoryl lipid A (3D-MPL), imiquimod (also known in the art as IQM and commercially available as Aldara[®]; U.S. Pat. Nos. 4,689,338; 5,238,944; Zuber, AK. et al (2004) 22(13-14): 1791-8), and the CCR5 inhibitor CMPD167 (see Veazey, R. S. et al (2003) J. Exp. Med. 198: 1551-1562).

Aluminum hydroxide or phosphate (alum) are commonly used at 0.05 to 0.1% solution in phosphate buffered saline. Other adjuvants that can be used, especially with DNA vaccines, are cholera toxin, especially CTA1-DD/ISCOMs (see Mowat, A. M. et al (2001) J. Immunol. 167(6): 3398-405), polyphosphazenes (Allcock, H. R. (1998) App. Organometallic Chem. 12(10-11): 659-666; Payne, L. G. et al (1995) Pharm. Biotechnol. 6: 473-93), cytokines such as, but not limited to, IL-2, IL-4, GM-CSF, IL-12, IL-15 IGF-1, IFN- α , IFN- γ , and IFN- γ (Boyer et al., (2002) J. Liposome Res. 121:137-142; WO01/095919), immunoregulatory proteins such as CD40L (ADX40; see, for example, WO03/063899), and the CD1a ligand of natural killer cells (also known as CRONY or α -galactosyl ceramide; see Green, T. D. et al, (2003) J. Virol. 77(3): 2046-2055), immunostimulatory fusion proteins such as IL-2 fused to the Fe fragment of immunoglobulins (Barouch et al., Science 290:486-492, 2000) and co-stimulatory molecules B7.1 and B7.2 (Boyer), all of which can be administered either as proteins or in the form of DNA, on the same expression vectors as those encoding the antigens of the invention or on separate ex In an advantageous embodiment, the adjuvants may be lecithin combined with an acrylic polymer (Adjuplex-LAP), lecithin coated oil droplets in an oil-in-water emulsion (Adjuplex-LE) or lecithin and acrylic polymer in an oil-in-water emulsion (Adjuplex-LAO) (Advanced BioAdjuvants (ABA)).pression vectors.

The immunogenic compositions can be designed to introduce the nucleic acids or expression vectors to a desired site of action and release it at an appropriate and controllable rate. Methods of preparing controlled-release formulations are known in the art. For example, controlled release preparations can be produced by the use of polymers to complex or absorb the immunogen and/or immunogenic composition. A controlled-release formulation can be prepared using appropriate macromolecules (for example, polyesters, polyamino acids, polyvinyl, pyrrolidone, ethylenevinylacetate, methylcellulose, carboxymethylcellulose, or protamine sulfate) known to provide the desired controlled release characteristics or release profile. Another possible method to control the duration of action by a controlled-release preparation is to incorporate the active ingredients into particles of a polymeric material such as, for example, polyesters, polyamino acids, hydrogels, polylactic acid, polyglycolic acid, copolymers of these acids, or ethylene vinylacetate copolymers. Alternatively, instead of incorporating these active ingredients into polymeric particles, it is possible to entrap these materials into microcapsules prepared, for example, by coacervation techniques or by interfacial polymerization, for example, hydroxymethylcellulose or gelatin-microcapsule and poly-(methylmethacrylate) microcapsule, respectively, in colloidal drug delivery systems (for example, liposomes, albumin microspheres, microemulsions, nano-particles and nanocapsules) or in macroemulsions. Such techniques are disclosed in New

Trends and Developments in Vaccines, Voller et al. (eds.), University Park Press, Baltimore, Md., 1978 and Remington's Pharmaceutical Sciences, 16th edition.

Suitable dosages of the nucleic acids and expression vectors of the invention (collectively, the immunogens) in the immunogenic composition of the invention can be readily determined by those of skill in the art. For example, the dosage of the immunogens can vary depending on the route of administration and the size of the subject. Suitable doses can be determined by those of skill in the art, for example by measuring the immune response of a subject, such as a laboratory animal, using conventional immunological techniques, and adjusting the dosages as appropriate. Such techniques for measuring the immune response of the subject include but are not limited to, chromium release assays, tetramer binding assays, IFN- γ ELISPOT assays, IL-2 ELISPOT assays, intracellular cytokine assays, and other immunological detection assays, e.g., as detailed in the text "Antibodies: A Laboratory Manual" by Ed Harlow and David Lane.

When provided prophylactically, the immunogenic compositions of the invention are ideally administered to a subject in advance of HIV infection, or evidence of HIV infection, or in advance of any symptom due to AIDS, especially in high-risk subjects. The prophylactic administration of the immunogenic compositions can serve to provide protective immunity of a subject against HIV-1 infection or to prevent or attenuate the progression of AIDS in a subject already infected with HIV-1. When provided therapeutically, the immunogenic compositions can serve to ameliorate and treat AIDS symptoms and are advantageously used as soon after infection as possible, preferably before appearance of any symptoms of AIDS but may also be used at (or after) the onset of the disease symptoms.

The immunogenic compositions can be administered using any suitable delivery method including, but not limited to, intramuscular, intravenous, intradermal, mucosal, and topical delivery. Such techniques are well known to those of skill in the art. More specific examples of delivery methods are intramuscular injection, intradermal injection, and subcutaneous injection. However, delivery need not be limited to injection methods. Further, delivery of DNA to animal tissue has been achieved by cationic liposomes (Watanabe et al., (1994) Mol. Reprod. Dev. 38:268-274; and WO 96/20013), direct injection of naked DNA into animal muscle tissue (Robinson et al., (1993) Vaccine 11:957-960; Hoffman et al., (1994) Vaccine 12: 1529-1533; Xiang et al., (1994) Virology 199: 132-140; Webster et al., (1994) Vaccine 12: 1495-1498; Davis et al., (1994) Vaccine 12: 1503-1509; and Davis et al., (1993) Hum. Mol. Gen. 2: 1847-1851), or intradermal injection of DNA using "gene gun" technology (Johnston et al., (1994) Meth. Cell Biol. 43:353-365). Alternatively, delivery routes can be oral, intranasal or by any other suitable route. Delivery also be accomplished via a mucosal surface such as the anal, vaginal or oral mucosa.

Immunization schedules (or regimens) are well known for animals (including humans) and can be readily determined for the particular subject and immunogenic composition. Hence, the immunogens can be administered one or more times to the subject. Preferably, there is a set time interval between separate administrations of the immunogenic composition. While this interval varies for every subject, typically it ranges from 10 days to several weeks, and is often 2, 4, 6 or 8 weeks. For humans, the interval is typically from 2 to 6 weeks. The immunization regimes typically have from 1 to 6 administrations of the immunogenic composition, but

may have as few as one or two or four. The methods of inducing an immune response can also include administration of an adjuvant with the immunogens. In some instances, annual, biannual or other long interval (5-10 years) booster immunization can supplement the initial immunization protocol.

The present methods also include a variety of prime-boost regimens, for example DNA prime-Adenovirus boost regimens. In these methods, one or more priming immunizations are followed by one or more boosting immunizations. The actual immunogenic composition can be the same or different for each immunization and the type of immunogenic composition (e.g., containing protein or expression vector), the route, and formulation of the immunogens can also be varied. For example, if an expression vector is used for the priming and boosting steps, it can either be of the same or different type (e.g., DNA or bacterial or viral expression vector). One useful prime-boost regimen provides for two priming immunizations, four weeks apart, followed by two boosting immunizations at 4 and 8 weeks after the last priming immunization. It should also be readily apparent to one of skill in the art that there are several permutations and combinations that are encompassed using the DNA, bacterial and viral expression vectors of the invention to provide priming and boosting regimens.

A specific embodiment of the invention provides methods of inducing an immune response against HIV in a subject by administering an immunogenic composition of the invention, preferably comprising an adenovirus vector containing DNA encoding one or more of the epitopes of the invention, one or more times to a subject wherein the epitopes are expressed at a level sufficient to induce a specific immune response in the subject. Such immunizations can be repeated multiple times at time intervals of at least 2, 4 or 6 weeks (or more) in accordance with a desired immunization regime.

The immunogenic compositions of the invention can be administered alone, or can be co-administered, or sequentially administered, with other HIV immunogens and/or HIV immunogenic compositions, e.g., with "other" immunological, antigenic or vaccine or therapeutic compositions thereby providing multivalent or "cocktail" or combination compositions of the invention and methods of employing them. Again, the ingredients and manner (sequential or co-administration) of administration, as well as dosages can be determined taking into consideration such factors as the age, sex, weight, species and condition of the particular subject, and the route of administration.

When used in combination, the other HIV immunogens can be administered at the same time or at different times as part of an overall immunization regime, e.g., as part of a prime-boost regimen or other immunization protocol. In an advantageous embodiment, the other HIV immunogen is env, preferably the HIV env trimer.

Many other HIV immunogens are known in the art, one such preferred immunogen is HIVA (described in WO 01/47955), which can be administered as a protein, on a plasmid (e.g., pTHr. HIVA) or in a viral vector (e.g., MVA.RIVA). Another such HIV immunogen is RENTA (described in PCT/US2004/037699), which can also be administered as a protein, on a plasmid (e.g., pTHr. RENTA) or in a viral vector (e.g., MVA.RENTA).

For example, one method of inducing an immune response against HIV in a human subject comprises administering at least one priming dose of an HIV immunogen and at least one boosting dose of an HIV immunogen, wherein the immunogen in each dose can be the same or different, provided that at least one of the immunogens is an epitope

of the present invention, a nucleic acid encoding an epitope of the invention or an expression vector, preferably a VSV vector, encoding an epitope of the invention, and wherein the immunogens are administered in an amount or expressed at a level sufficient to induce an HIV-specific immune response in the subject. The HIV-specific immune response can include an HIV-specific T-cell immune response or an HIV-specific B-cell immune response. Such immunizations can be done at intervals, preferably of at least 2-6 or more weeks.

The present invention will be further illustrated in the following Examples which are given for illustration purposes only and are not intended to limit the invention in any way.

Although the present invention and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the invention as defined in the appended claims.

EXAMPLES

Example 1: Experimental Procedures

SIV Envelope Trimer Design, its Expression and Purification.

SOSIP.664 HIV-1 Env trimer design modifications were incorporated into envelope encoding sequences corresponding to four Chimpanzee (SIVcpzPtt) isolates (GAB1 [GenBank: P17281]; MB897 [GenBank: ABU53023]; EK505 [GenBank: ABD19499]; and MT145 [GenBank: ABD19508]) to express as soluble native trimers (FIG. 8A). These isolates are sensitive to the V2 apex bnAbs, PG9, PG16, and PGT145 (Barbian et al., 2015). Further characterization showed that one of these SIVcpzPtt Env sequences, MT145 SOSIP.664, could be expressed as a soluble Env trimer protein (FIG. 8B). PGT145 Ab affinity-purified MT 145 trimer was efficiently cleaved into gp120 and gp41 subunits, and revealed well-ordered native-like trimer configurations that were highly thermostable, which are all properties displayed by natively folded HIV-1 soluble trimers (FIG. 9A-D) (Pugach et al., 2015; Sanders et al., 2013; Sharma et al., 2015).

Briefly, the following modifications were incorporated into these Envs for soluble trimer expression: a) the Env leader sequence was replaced by Tissue Plasminogen Activator (TPA) signal sequence for higher protein expression; b) a disulfide bond was introduced between gp120 and gp41 subunits by substituting residues A501-C and T605-C respectively in gp120 and gp41; c) the gp120 REKR (SEQ ID NO:358) cleavage site was replaced by Furin inducible R6 site (RRRRRR) (SEQ ID NO:359) for enhancing cleavage efficiency between gp120 and gp41; and d) an I559P substitution in gp41 to stabilize the soluble trimer protein. In addition, a GS-linker and a His-tag were added to the gp41_{ECTO} C-terminus at HXB2 residue 664 position. The codon-optimized SOSIP.664 gp140 gene constructs were synthesized (Geneart, Life Technologies) and cloned into the pCMV3 vector (Genlantis). Recombinant envelope proteins were expressed in HEK293F cells as described elsewhere (Sanders et al., 2013). Briefly, HIV-1 Env trimers WITO, C108, ZM197-ZM233V1V2 and the 4 chimpanzee SIV SOSIP.664 Env-encoding trimer plasmids were cotransfected with a plasmid encoding for Furin (3:1 ratio) into HEK293F cells using PEI-MAX 4000 transfection reagent (Polysciences, Inc.). The secreted soluble trimers proteins were purified from cell supernatants after 5 days using

agarose-bound Gallanthus Nivalis Lectin (GNL) (Vector Labs) or CNBr-activated Sepharose 4B bead (GE Healthcare) bound PGT145 bnAb antibody affinity columns as described previously (Pugach et al., 2015). The affinity-purified proteins were size exclusion chromatography (SEC)-purified with a Superdex 200 10/300 GL column (GE Healthcare) in PBS/TBS. The purified trimers for the immunization experiments were quality control tested for antigenicity with a range of HIV-1 Env-specific neutralizing and non-neutralizing mAbs.

Example 2: Binding to bnAB Precursors

MT145K Trimer Binds Prototype V2 Apex bnAb Precursors.

One property thought to be critical for vaccine immunogens to select rare bnAb precursors is the ability to effectively bind to UCA B cell receptors (Dosenovic et al., 2015; Escolano et al., 2016; Jardine et al., 2015; McGuire et al., 2016; Steichen et al., 2016a). Therefore, to gain or improve binding of the V2 apex bnAb inferred precursor Abs to MT145 Env trimer, glutamine (Q) was substituted with lysine (K) at residue (HXB2 position 171) in strand C of the V2 apex bnAb core epitope (FIG. 1A-B). This substitution is based on the presence of a positively charged motif (KKKK) (SEQ ID NO:5) in CRF250 and CP256. SU strand C V2 Env sequences, both of which bind V2 apex bnAb prototype precursors (Andrabi et al., 2015; Doria-Rose et al., 2014; Gorman et al., 2016). ELISA binding revealed strong binding of the mature V2 apex bnAb prototypes with the MT145-WT trimer and weak but detectable binding with one of the UCA Abs, CAP256 UCA (FIG. 1C). Strikingly, binding with a V2-engineered MT145 trimer (henceforth referred to as "MT145K") not only improved binding to CAP256 UCA Ab but also conferred binding on both PG9 and CH01 iGL Abs (FIG. 1C). The PG9 and CH01 iGL Abs used here had diversity (D; heavy chain) and joining (J; both heavy and light chains) genes reverted to their corresponding germline gene families in the CDRH3s, in addition to the VH and VL regions reported previously (FIG. 10) (Andrabi et al., 2015; Gorman et al., 2016).

Previous mapping studies have examined the HIV core epitope recognized by mature V2 apex bnAbs (Andrabi et al., 2015; Gorman et al., 2016; Landais et al., 2017; Lee et al., 2017; McLellan et al., 2011; Pancera et al., 2013; Walker et al., 2009). To evaluate the contributions of V2 apex core epitope glycan and protein residues to binding by V2 apex bnAb inferred germline-reverted (iGL) Ab versions, MT145K strand C peptide and glycan trimer variants known to eliminate interactions of V2 apex bnAbs with the Env trimer were generated (see Andrabi et al., 2015; McLellan et al., 2011; Pancera et al., 2013). Bio-Layer Interferometry (BLI or octet) binding analyses of the iGL Abs with these trimer variants showed that glycan/peptide epitope requirements of precursor Abs were largely similar to the requirements of corresponding mature Abs (FIG. 1D), suggesting that most contacts with the MT145K V2 apex core epitope are already encoded in the germline configuration for this class of bnAbs. Notably, the mature Abs showed slightly more tolerance to changes within the core protein epitope, particularly for the CAP256.09 bnAb, suggesting that part of the affinity maturation in this class of Abs may be to accommodate variation within the bnAb V2 apex core epitope. Overall, the strand C V2-modification in the MT145 SOSIP.664 trimer conferred binding to multiple V2 apex bnAb germline prototypes.

Example 3: Structure

Architecture of the MT145K Trimer.

The structure of the MT145K trimer was solved by cryo-EM to a global resolution of ~ 4.1 Å (FIG. 11). Data collection methods are summarized in the following table.

Data collection	
Microscope	FEI Titan Krios
Voltage (kV)	300
Detector	Gatan K2 Summit
Recording mode	Counting
Magnification (incl. post-magnification)	49,020
Movie micrograph pixelsize (Å)	1.02
Dose rate ($e^-/[(\text{camera pixel}) \cdot \text{s}]$)	10
Number of frames per movie micrograph	50
Frame exposure time (ms)	200
Movie micrograph exposure time (s)	10
Total dose ($e^-/\text{Å}^2$)	94
Defocus range (μm)	1.0-3.6
EM data processing	
Number of movie micrographs	1,281
Number of molecular projection images in map	44,301
Symmetry	C3
Map resolution (FSC 0.143; Å)	4.1
Mapsharpening B-factor (Å^2)	-130
Structure Building and Validation	
Number of atoms in deposited model	15,192
gp120	3,547
gp41	964
glycans	553
MolProbity score	1.72 (88%)
Clashscore	3.55
EMRinger score	2.26
Deviations from ideal	
Bond length outliers	0 (0%)
Bond angles outliers	0 (0%)
Ramachandran plot	
Favored (%)	88.9
Allowed (%)	10.5
Outliers (%)	0.5

The structure represents the first atomic level structure of an SIV Env trimer. Like other class I fusion proteins, protomers (gp120 and gp41) of MT 145K trimerize to form a meta-stable pre-fusion Env trimer (FIGS. 2A-B, 12). The trimer architecture exhibits a mushroom-like shape with subunits gp120 and gp41 constituting the envelope-distal and proximal entities, respectively (FIG. 2B). Overall, the MT 145K trimer configuration closely resembles that of the trimeric HIV-1 Env spike, with an overall C α root mean square deviation (rmsd) of 1.9 Å (Kwon et al., 2015). Arrangement of V-loops in the MT145K Env trimer is reminiscent of the V-loop arrangement in the HIV Env trimer and is suggestive of a similar role in immune evasion by steric occlusion of underlying conserved epitopes (Julien et al., 2013a; Pancera et al., 2014). Notably, the V1 and V2 loops are largely solvent-exposed and occlude access to the underlying V3 loop (FIG. 2C). Inaccessibility of the V3 loop is mediated by intra-protomer interactions of V1V2 to V3 and by extensive inter-protomer V1V2 trimer interactions at the apex of the spike. The SIV Env trimer exhibits well-ordered V2-V5 loops, while V1 is somewhat disordered (FIG. 2D).

Proximal to the viral membrane is the gp41 subunit that forms the base of the trimer spike and is arranged into heptad repeat-1 (HR1), HR2 and the fusion peptide (FP) (FIG. 2B). Similar to the HIV Env trimer, the three C-terminal helices of HR1 are centrally positioned along the trimer axis per-

pendicular to the viral membrane (Julien et al., 2013a; Lyumkis et al., 2013; Pancera et al., 2014). Intriguingly, the FP region, which has been observed solvent-exposed on the outside of the HIV-1 Env, is positioned in a pocket inside the MT145K trimer and remains sequestered in all three protomers (FIG. 2E).

Example 4: Glycan Analysis

10 Conservation of the Glycan Shield on HIV and SIV Env Trimers.

To compare the nature of the glycan shield on SIVcpzPtt Env and HIV Env, site-specific glycan analysis of the MT145K trimer was performed. The overall oligomannose content of the MT145K trimer is similar to HIV Env (FIG. 3A-B) (Panico et al., 2016; Pritchard et al., 2015). However, although the distributions differed from the HIV-1 clade A strain BG505, which is dominated by Man₅GlcNAc₂ oligomannose-type glycans, MT145K is predominantly Man₈GlcNAc₂ (Behrens et al., 2016). In addition, further processing was evident in the MT145K trimer which showed elevated Man₆₋₇GlcNAc₂ structures (FIG. 3B, 13). The outer domain of gp120 presents a high density of oligomannose glycans that form the intrinsic mannose patch (Bonomelli et al., 2011), which was a highly conserved feature across the two viral species. The apex of the MT145K trimer possesses oligomannose-type glycans at N160 that correspond to the trimer associated mannose patch (TAMP) also observed on HIV-1 Env (Behrens et al., 2017). As for HIV-1, glycans at the base of the trimer at N88 and on gp41 of the MT145K trimer are extensively processed (FIGS. 3A, C, 13).

In the MT145K trimer, glycans at N156 and N262 are predominantly complex-type, whereas the corresponding glycans are oligomannose-type in HIV Env (Behrens et al., 2016). These differences may arise due to the proximity of neighbouring glycans. For instance, the HIV Env glycans at positions N295 and N332, adjacent to the N262 glycan, are absent on MT145K Env, which may lead to increased processing of N262 (FIG. 3A, C). The remarkable conservation in the overall architecture of the SIV and HIV Env glycan shield, despite sharing only $\sim 62\%$ of the amino-acid sequence identity, suggests that the glycan shield has an indispensable role in immune evasion and potentially maintaining functional integrity of the trimer spike. Indeed, the glycan shield is integral to all lentiviral envelopes and appears to have evolved somewhat specifically to mammalian host (FIG. 14). Over the course of lentiviral evolution, the Env glycan density shows an overall gradual progression, and likely peaked in retroviruses infecting non-human primates and plateaued in HIV Envs (FIG. 14) (Zhang et al., 2004). Therefore, the high-density Env glycan shield on HIV must have been established well before chimpanzee SIV crossed into humans. Nevertheless, several glycan positions on HIV-1 Env appear to have subtly shifted after the species cross-over that presumably resulted as an adaptation to the human immune system (FIG. 15).

Example 5: Binding Specificity

60 MT145K Binds V2 Apex bnAbs Almost Exclusively.

To define the overall antigenicity of the MT145K trimer, neutralization sensitivity of MT145K virus (MT145-Q171K) to a broad panel of HIV-1 Env-specific neutralizing and non-neutralizing (nnAbs) mAbs was assessed and the profiles compared to the clade A BG505 HIV virus (FIG. 4A, 16) (Sanders et al., 2013; Voss et al., 2017). Remarkably, the

V2 apex bnAbs, but essentially no other bnAbs or nnAbs (except 35022 gp120-41 interface mAb), exhibited potent neutralizing activities against MT145K virus (FIG. 4A, 16). As previously observed, the BG505 isolate was sensitive to neutralization by all of the bnAbs in the panel, but none of the nnAbs (FIG. 4A FIG. 16).

Next, binding of MT145K trimer and monomeric gp120 to a panel of mAbs was evaluated by ELISA. Consistent with the neutralization results above, bnAbs to the V2 apex site showed robust binding to the MT145K trimer (FIGS. 4B, 16), but other bnAbs and nnAbs did not bind, except for a few mAbs that displayed very weak binding (FIGS. 4B, 16). PG9, 17b and some of the linear V3-loop directed mAbs (2557, 3074, 3904 and 14e) (FIGS. 4B, 16) that bound to the MT145K gp120 monomer. The results suggest that the sequence-dependent epitopes for some of the non-neutralizing V3-loop mAbs are present on monomeric MT145K gp120, but are obscured on the MT145K trimer, as indicated by the MT145K structure. Virus neutralization and trimer binding by mAbs is strongly correlated ($p=0.003$), consistent with the notion that the MT145K soluble trimer adopts a native-like trimeric Env configuration and displays antigenic properties optimal for a vaccine immunogen.

Example 6: Epitope Analysis

HIV bnAb Epitopes on SIV Env.

To gain insight into the differences in the HIV-1 Env bnAb epitopes on MT145K SIV Env that may potentially explain the reactivity of V2 apex bnAbs and non-reactivity of HIV bnAbs targeting other Env epitopes, we took advantage of the previously determined structures of human HIV bnAbs in complex with various HIV Env forms and compared the corresponding epitope regions with the MT145K Env (Garces et al., 2014; Lee et al., 2017; Lee et al., 2016; Ozorowski et al., 2017; Pejchal et al., 2011; Wu et al., 2010). A lysine-rich patch in strand C of the V2 loop (¹⁶⁶RDKKQK¹⁷¹ (SEQ ID NO:355) on BG505 Env) and two nearby glycans N160 and N156 form the core epitope for V2 apex bnAbs on HIV Envs (FIG. 5A, 17) (Gorman et al., 2016; Julien et al., 2013b; Lee et al., 2017; McLellan et al., 2011; Pancera et al., 2013). Both of these features are conserved on the MT145K trimer, thus enabling the human V2 apex bnAbs to be highly effective against the SIV Envs (FIGS. 5A, 15, 17) (Barbian et al., 2015).

Binding of one of the N332-V3 epitope specific bnAbs, PGT128, predominantly relies on the N332 glycan and a neighboring peptide motif³²⁴GDIR³²⁷ (SEQ ID NO:356) at the base of the V3 loop (FIGS. 5B, 17) (Garces et al., 2014; Pejchal et al., 2011; Sok et al., 2016b). The lack of binding to the MT145K trimer by PGT128 and other bnAbs in this class can be explained by the absence of the N332 glycan on this Env. In contrast, 3 of the 4 core protein epitope residues 324G-325D-327R are conserved on MT145K trimer and, in fact, on other chimpanzee SIV Envs (FIGS. 5B, 15, 17). For the PGT128 class bnAbs, the interaction with glycan N332 can be substituted by the N295 glycan observed in some HIV isolates, but not by glycan N334 that is present on the MT145K trimer (Sok et al., 2014a). In fact, the MT145K N334 glycan points in a different direction away from the N332-V3 epitope site making it impossible to facilitate bnAb binding to this epitope. Strikingly, the majority of known SIVcpz Env sequences possess an N334 glycan in place of the more common N332 glycan on the HIV Env, which appears to be a significant glycan shift upon species cross-over as the virus established itself in humans (FIG. 15). In addition, the glycan at N412 in the gp120-V4 region

of MT145K Env may obstructively interfere with bnAb binding, and, particularly, the glycan at N442, unique to the MT145K Env trimer and several other SIVcpz Envs, would clash with CDRH2 of PGT128 and other bnAbs in this class and may prevent them from accessing the epitope (FIG. 5B, 15).

PGT151 represents another glycan-targeting bnAb class (Blattner et al., 2014; Falkowska et al., 2014; Lee et al., 2016) that recognizes several glycans on gp120 (N88, N448) and gp41 (N611 and N637) as well as the fusion peptide. All glycans and fusion peptide residues that contribute to the PGT151 epitope are conserved between HIV and SIVcpz Envs (FIG. 15). Therefore, the lack of PGT151 binding to MT145K is most likely attributable to inaccessibility of the FP on MT145K (FIG. 5C).

The CD4bs is conserved between HIV and SIV to the extent that there is cross-species reactivity with sCD4. Human CD4-IgG2 immunoadhesin binds well to the MT145K trimer, indicating a strong cross-species conservation of the Env CD4bs. Phe43 in domain-1 of human sCD4 would fit well inside the Trp427 Env cavity on the MT145K trimer reminiscent of its interaction with the HIV Env BG505 trimer (FIG. 5D) (Ozorowski et al., 2017). However, the MT145K trimer is nonreactive with CD4bs bnAbs. VRC01, one of the bnAbs in this class, binds to the HIV Env CD4bs bnAb epitope formed by discontinuous protein backbone elements including loop D of the gp120-C2 region and bordered by a glycan at N276 (FIG. 5E-F, 17) (Wu et al., 2010). MT145K lacks the N276 glycan and the proximal N234 glycan, present in most HIV-1 Envs, but instead has a glycan at position 236. Differences in the loop D sequence (FIG. 15) and the glycan at N236, which would clash with VRC01 CDRL1 and CDRL3 loops (FIG. 5F) on the MT145K trimer likely impose the biggest impediment to VRC01 binding. Further, the MT145K gp120-V5 loop has a 6-amino acid insertion at HXB2 position 456 compared to HIV-1 Envs that would clash with the VRC01 LC (FIG. 5F, 15).

Overall, the non-reactivity of HIV Env bnAbs with the MT145K trimer can be largely ascribed to subtle glycan shifts that have occurred in HIV-1 from chimpanzee SIV Env as the virus established itself in humans.

Example 7: Activation of B Cell Precursors

The Engineered MT145K but not the MT145-WT Trimer Activates V2 Apex UCA-Expressing B Cell Precursors In Vivo.

To determine whether the engineered chimpanzee MT145K trimer could efficiently activate HIV V2 apex Ab germline-encoding precursor B cells in vivo and how it compares with the MT145-WT trimer, immunization experiments were conducted in the CH01 unmutated common ancestor (UCA) "HC only" knock-in (KI) mouse model. This KI-mouse model expresses the pre-rearranged heavy chain (V_H DDJ_H) of the CH01 V2 apex bnAb UCA paired with WT mouse light chains. Two groups of 5 CH01 UCA "HC only" KI mice were immunized, each with two repeated doses (at week-0 and week -4) of MT145-WT or MT145K trimer (FIG. 6A). To track the development of Ab responses, ELISA assays were performed on the pre-bleed, 2-week (day 14) post prime (Bleed #1) and 2-week post boost-1 (day 42) (Bleed #2) serum samples with MT145K SOSIP trimer protein and its N160 glycan knock-out variant (MT145K N160K) (FIG. 6B). The pre-bleed serum samples in both immunization groups exhibited weak binding activity with the MT145K trimer that was dependent on the N160

glycan, consistent with the presence of CH01 UCA Abs that do show some binding to MT145K trimer as described above (FIG. 6B).

The immunogen-specific titers of the serum Ab responses post prime immunizations (Bleed #1 samples) marginally increased in the MT145K group but remained largely unchanged in the MT145-WT trimer immunized group. The serum Ab titers post boost-1 immunization (Bleed #2) increased in both the groups and were orders of magnitude higher as compared to the pre-bleed or the post prime Ab binding responses (FIG. 6B). At this immunization step, the serum Ab responses in the MT145K trimer immunized group were solely dependent on the N160 glycan while the MT145-WT trimer immunization group responses targeted the MT145 Env that were mostly independent of the N160 glycan, which forms part of the core V2 apex bnAb epitope (FIG. 6B). Therefore, we conclude that the engineered MT145K trimer but not the MT145-WT, efficiently triggers the epitope specific V2-apex bnAb UCA encoding B cell precursors in vivo. Remarkably, immunizations with Q171K substituted engineered MT145K trimer also appeared to eliminate the non-V2 apex bnAb site Env specific off-target B cell responses that were elicited in the MT145-WT trimer immunization group (FIG. 6B). The results demonstrate that the activation of the HIV Env bnAb-encoding unmutated B cell precursor by immunogens that display binding to their UCA Ab versions is critical for eliciting epitope-specific Ab responses and the findings are consistent with studies that specifically use germline-targeting immunogen molecules to kick-off the bnAb precursor encoding B cell responses in vivo (Dosenovic et al., 2015; Escolano et al., 2016; Jardine et al., 2015; McGuire et al., 2013; Sok et al., 2016a; Steichen et al., 2016b; Tian et al., 2016).

Immune sera were evaluated for neutralization of autologous and heterologous viruses. Reproducible MT145K autologous virus-specific neutralizing Ab responses were induced in the MT145K immunization group but not in the MT145-WT immunization group (FIG. 6C-D). As for the ELISA binding responses, the nAb titers in the MT145K trimer immunized group increased at 2 weeks post prime, as indicated by nAb titers against a CH01 sensitive HIV Env-encoding virus (Q23_17), and further significantly increased after the boost-1 immunization (FIG. 6C). At this point, all animals in the MT145K group developed autologous virus specific nAb responses (FIG. 6C). The nAb responses in MT145K trimer-immunized animals mapped to the glycan N160 and strand C K171 residue, both of which form part of the core epitope for V2-apex bnAbs, suggesting that the MT145K trimer successfully primed V2 apex UCA B cells in an epitope-specific manner in vivo.

Overall, we conclude that the engineered MT145K but not the MT145-WT trimer activated the V2-apex specific bnAb precursor B cells in a UCA-expressing mouse model and further drove maturation along favorable B cell pathways.

Example 8: Immunization Strategies

Combining Chimpanzee SIV MT145K Trimer with HIV Env Trimer Immunizations in the CH01 UCA Model.

Evaluation of the utility of the MT145K trimer in a sequential HIV immunization regime will be best carried out in humans. Nevertheless, we were interested to investigate the effects of combining the chimpanzee SIV MT145K and HIV Env trimers in a prime-boost immunization in the CH01 UCA "HC only" KI mice. We immunized a group of CH01 UCA KI mice twice with chimpanzee SIV MT145K (as

above) and boosted with an HIV Env derived 3-trimer cocktail (C108, WITO and ZM197-ZM233V1V2) (FIG. 7A). NAb titers developed against a few viruses from a global HIV panel that were very sensitive to the mature CH01 bnAb (FIG. 7B). However, viruses that were only somewhat sensitive to CH01 bnAb were not neutralized, suggesting that further affinity maturation would be required for a broad response. The cross-neutralizing responses mapped to the V2 apex bnAb epitope that is shared between HIV and SIV Envs, including to the N160 glycan and the stand C K171 residue.

Overall, the analysis of the immune responses revealed that, due to the extraordinary conservation of the V2 apex bnAb epitope region between HIV and chimpanzee SIV, the MT145K trimer successfully primed human V2 apex bnAb UCA-encoding mouse B cells and induced a V2-focused cross-neutralizing HIV Env specific response that could be further boosted by HIV Env derived trimers.

Example 9: Materials and Methods

Antibodies, expression and purification. HIV-1 envelope specific mAbs to a broad range of epitopes were used, including those that target V2-apex, V3-N332, linear V3, CD4bs, CD4i and gp120-41 Env sites. A dengue antibody (DEN3) was used as control Ab for binding experiments. For PG9 and CH01 V2-apex bnAb inferred germline antibody designs, the heavy and the light chain V-gene of the mature Abs were reverted to their corresponding closest inferred germline gene sequence as determined using the ImMunoGeneTics (IMGT) website (imgt.cines.fr/) (Brochet et al., 2008). The reverted variable heavy and light chain nucleotide sequences were synthesized by Geneart (Life Technologies) and cloned into corresponding IgY1, IgK, and IgL expression vectors as previously described (Tiller et al., 2008), using the Gibson cloning method (NEB, USA). The antibodies were expressed and purified using methods described previously (Sok et al., 2014b). Briefly, the heavy and light chain encoding plasmids were reconstituted (1:1 ratio) in Opti-MEM (Life Technologies), and cotransfected HEK293F cells (Invitrogen) using 293fectin (Invitrogen). The suspension cells were cultured for 4-5 days in a shaker incubator at 8% CO₂, 37.0° C., and 125 rpm. The antibody containing supernatants were harvested, filtered through a 0.22 mm Steriflip units (EMD Millipore) and passed over a protein A or protein G affinity column (GE Healthcare). The bound antibody was eluted from the columns in 0.1 M citric acid, pH 3.0. Column fractions containing IgG were neutralized (2M Tris-base), pooled, and dialyzed against phosphate-buffered saline (PBS), pH 7.4. IgG purity was determined by sodium dodecyl sulfate-polyacrylamide gel electrophoresis, and the concentration was determined by measuring the relative absorbance at 280 nm.

Site-directed mutagenesis. The amino-acid point mutations in Env-encoding plasmids were incorporated by using a QuikChange site-directed mutagenesis kit (Agilent Technologies, USA), according to the manufacturer's instructions. All of the mutations were confirmed by DNA sequence analysis (Eton Bioscience, San Diego, CA).

Differential Scanning Calorimetry. Thermal denaturation was analyzed with a differential scanning calorimetry (DSC) using a MicroCal VP-Capillary DSC instrument (Malvern), at a scanning rate of 1 K/min under 3.0 atmospheres of pressure. Samples were dialyzed in PBS pH 7.4 overnight and protein concentration was adjusted to 0.5 mg/mL prior to measurement. DSC data were analyzed after buffer cor-

rection, normalization, and baseline subtraction using MicroCal VP-Capillary DSC analysis software provided by the manufacturer.

Negative stain electron microscopy and data treatment. Purified M145K sample was deposited on thin-carbon-coated (Edwards Auto 306 carbon evaporator) a C-flat EM grid (Cu400 mesh, 2 μ m hole diameter, 2 m hole spacing) (Protochips, Morrisville, NC, USA) and embedded in 2% (w/v) uranyl formate. The carbon-coated grids were Ar/O₂-plasma-cleaned (Gatan Solarus Model 950 Advanced Plasma System; Gatan Inc., Pleasanton, CA, USA) prior to sample deposition. The uranyl-stained EM sample was then inserted into an FEI Tecnai 12 microscope (Thermo Fisher Scientific, Waltham, MA, USA) equipped with a US4000 CMOS detector (Gatan Inc., Pleasanton, CA, USA). The data was collected at 52,000 $\times$ nominal magnification resulting in a pixel size of 2.05 $\text{\AA}$ at the object level. Data was binned by a factor of 2 prior to data treatment. Projection image identification in the micrographs was performed with a difference-of-Gaussians implementation (Voss et al., 2009). Projection images subsequently underwent 2D alignment and classification by iterative multi-reference alignment/multivariate statistical analysis (Ogura et al., 2003).

CryoEM sample preparation, data collection, processing and analysis. Purified MT145K sample was deposited on a C-flat EM grid (Cu400 mesh, 2 m hole diameter, 2 m hole spacing) (Protochips, Morrisville, NC, USA) that had been Ar/O₂-plasma-cleaned (Gatan Solarus Model 950 Advanced Plasma System; Gatan Inc., Pleasanton, CA, USA) prior to sample deposition. Excess buffer was then blotted away from the grid followed by plunging into and vitrification in liquid ethane cooled by liquid nitrogen using a vitrobot (Thermo Fisher Scientific, Waltham, MA, USA). The resulting cryo-EM specimen was transferred into an FEI Titan Krios microscope (Thermo Fisher Scientific, Waltham, MA, USA) equipped with a Gatan K2 Summit direct electron detector (Gatan Inc., Pleasanton, CA, USA). Dose-fractionated data was collected in electron counting mode at a nominal magnification of 29,000 $\times$ resulting in a pixel size of 1.02 $\text{\AA}$ at the object level. Micrograph movie frame exposure time was 200 ms and each movie micrograph was recorded over 10s (50 movie frames) corresponding to a total dose of 94e⁻⁷/Å². Movie micrograph frames were subsequently aligned (MotionCor2; (Zheng et al., 2017)), dose-weighted and signal-integrated resulting in 1,281 micrographs for further data processing. CTF models were determined using GCTF (Zhang, 2016). Candidate projection images of MT145K were identified using a difference-of-Gaussians implementation (Voss et al., 2009). The resulting set of candidate projection images subsequently underwent 2D alignment and classification by use of Relion 2.1b1 (Scheres, 2012). ~95,000 projection images corresponding to well-formed class averages of MT145K were selected for further data processing. This data class was iteratively angularly refined and reconstructed using a B41 unliganded Env trimer map rendered at 60 $\text{\AA}$ resolution as an initial reference (Ozorowski et al., 2017). The data class then underwent 3D classification into six classes with the initial reconstruction rendered at 60 $\text{\AA}$ resolution as reference. From 3D classification, a subset of 44,301 projection images was selected for final data processing comprising CTF model adjustment at the projection-image level (Zhang, 2016) and angular refinement and reconstruction (Scheres, 2012).

Model building and refinement. A homology model (Modeller; (Webb and Sali, 2016)) was generated from sequence alignment of MT145K and BG505 and the struc-

ture of the latter (PDB: 4TVP). Significant manual rebuilding followed in Coot (Emsley and Cowtan, 2004). A fragment library was then created from the MT145K sequence containing 200 homologous, non-redundant sequences at each MT145K 7-mer position. Library fragment-based, density-guided, real-space rebuilding was then performed (DiMaio et al., 2015) with 319 decoys. The resulting models were evaluated geometrically (MolProbity; (Chen et al., 2010)) and by fit-to-map (EMRinger; (Barad et al., 2015)). The overall best model was selected for further iterations of manual rebuilding and multi-decoy, density-guided, real-space, all-atom Rosetta FastRelax refinement. Finally, glycans were manually built in Coot and restricted, density-guided real-space refinement performed in Phenix 1.12 (Adams et al., 2002) followed by model evaluation by MolProbity, EMRinger and Privateer (Agirre et al., 2015).

Global N-linked glycan analysis. The quantifications and structural characterization of the total glycan pool was achieved by cleaving the N-linked glycans from the surface of the glycoprotein using an in-gel digestion with peptide N-glycosidase F (PNGaseF). The resultant glycans were separated into two aliquots. The first was derivatized with 2-aminobenzoic acid (2-AA) and subjected to HILIC-UPLC analysis using an Acquity UPLC (Waters). To quantify the oligomannose content of the released glycans, the labelled samples were treated with endoglycosidase H (endoH), which selectively cleaves oligomannose glycans. Data analysis and interpretation were performed using Empower software (Waters). The second aliquot of released glycans was subjected to negative ion electrospray ion mobility mass spectrometry using a Synapt G2Si mass spectrometer (Waters). Glycan compositions were determined using collision induced dissociation (CID) fragmentation. Data analysis was performed using Waters Driftscope (version 2.8) software and MassLynx™ (version 4.1). Spectra were interpreted as described previously (Harvey et al., 2009). The glycan compositions were used to generate a sample-specific glycan library that was used to search the glycopeptide data to minimize the number of false-positive assignments in site-specific analysis.

LC-MS glycopeptide analysis. Site-specific N-glycosylation analysis was performed using proteolytic digestion followed by tandem LC-MS. Prior to digestion, trimers were denatured, reduced and alkylated by incubation for 1 h at room temperature (RT) in a 50 mM Tris/HCl, pH 8.0 buffer containing 6 M urea and 5 mM dithiothreitol (DTT), followed by the addition of 20 mM iodoacetamide (IAA) for a further 1 h at RT in the dark, and then additional DTT (20 mM) for another 1 h, to eliminate any residual IAA. The alkylated trimers were buffer-exchanged into 50 mM Tris/HCl, pH 8.0 using Vivaspin columns (GE healthcare) and digested separately with trypsin, elastase and chymotrypsin (Mass Spectrometry Grade, Promega) at a ratio of 1:30 (w/w). Glycopeptides were selected from the protease-digested samples using the ProteoExtract Glycopeptide Enrichment Kit (Merck Millipore) following the manufacturer's protocol. Enriched glycopeptides were analyzed by LC-ESI MS on an Orbitrap fusion mass spectrometer (Thermo Fisher Scientific), as previously described (Behrens et al., 2016), using higher energy collisional dissociation (HCD) fragmentation. Data analysis and glycopeptide identification were performed using Byonic™ (Version 2.7) and Byologic™ software (Version 2.3; Protein Metrics Inc.), as previously described (Behrens et al., 2016).

Glycan modeling. Man₉GlcNAc₂ oligomannose-type glycans were docked and rigid-body fitted at each of the

corresponding Env glycan positions using the MT145K structure presented here or an unliganded BG505 SOSIP.664 structure (PDB: 4ZMJ).

Pseudovirus production. To produce pseudoviruses, Env-encoding plasmids were cotransfected with an Env-deficient backbone plasmid (pSG3ΔEnv) (1:2 ratio) using X-tremeGENE™ 9 (Sigma-Aldrich) DNA transfection reagent. Briefly, 1×10^6 cells in 10 ml of Dulbecco's Modified Eagle Medium (DMEM) were seeded in a 100 mm $\times$ 20 mm cell culture dish (Corning) one day prior to transfection. For transfection, 40 μ l of X-tremeGENE™ 9 was added to 700 μ l of Opti-MEM I reduced serum medium (Thermo Fisher) in tube 1. The Env-encoding plasmid (5 μ g) and pSG3ΔEnv (10 μ g) were added to tube 2 in 700 μ l of Opti-MEM. The tube 1 and tube 2 solutions were mixed together and incubated for 25 min at room temperature. Next, the transfection mixture was added to the media with 293T cells seeded previously and then distributed uniformly. All pseudoviruses were harvested 48-72 h posttransfection, filtered through 0.22 mm Steriflip units (EMD Millipore) and aliquoted for use in neutralization assays.

Neutralization assay. Neutralization was measured by using single-round replication-defective HIV Env-pseudoviruses and TZM-bl target cells (Montefiori, 2005; Seaman et al., 2010). 25 μ l of 3-fold serially diluted mAbs or serum samples were pre-incubated at 37° C. for 1 h with 25 μ l of tissue culture infective dose-50 (TCID₅₀) Env-pseudotyped virus in a half-area 96-well tissue culture plate. TZM-bl target cells (5,000 cells/well) in 50 μ l of DMEM were added and the plates were allowed to grow in humidified incubator at 37° C. and 5% CO₂. The luciferase activity of the lysed cells was read on instrument (Biotek) after 2-3 days, by adding lysis buffer followed by Brightglow (Promega). The 50% inhibitory concentration (IC₅₀) or 50% inhibitory doses (ID₅₀) was reported as the antibody concentration or serum dilution required to reduce infection by half.

ELISA binding assay. ELISA binding experiments were performed as described previously with minor modification (Sanders et al., 2013). ELISA binding with SOSIP.664 trimer proteins with mAbs was carried out by either capturing the trimer proteins onto the anti-His capture antibodies or on the streptavidin coated plates through biotinylated trimers. For trimer biotinylation, the SOSIP.664 proteins were randomly biotinylated using a 2:1 molar ratio of biotin reagent to trimer using the EZ-link-NHS-PEG4-Biotin kit (Thermo Fisher Scientific, 21324). MaxiSorp plates (Thermo Fisher Scientific) were coated overnight at 4C with 2 μ g/mL of anti-His Ab (Thermo Fisher Scientific) or 2 μ g/mL streptavidin (Thermo Fisher Scientific). Plates were blocked for 1 hr with 3% BSA and washed three times with 0.05% Tween 20-PBS (PBS-T) (pH 7.4). Anti-His or Streptavidin-coated plates were incubated with biotinylated trimers in 1% BSA plus PBST for 1.5 hr and washed three times with PBST. 3-fold serially diluted mAbs or sera were added starting at a maximum concentration of 10 μ g/mL (100 μ g/ml for iGL Abs) (sera at 1:100 dilution) in 1% BSA plus PBST, and incubated at room temperature (RT) for 1.5 hr. Plates were washed three times with PBST. Alkaline-phosphatase-conjugated goat anti-human IgG Fc secondary antibody (Jackson ImmunoResearch Laboratories) was diluted 1:1000 in 1% BSA PBST and added to plates for 1 hr at RT. Plates were washed three times with PBST and incubated with phosphatase substrate (Sigma) for 15 mins and the absorbance at 405 nm recorded. The 50% binding (EC₅₀) was recorded as the half of the maximum binding activity and was calculated by linear regression method using Prism 6 Software.

Bio Layer Interferometry (BLI) binding assay. The binding experiments of Abs to the affinity purified trimers were performed with an Octet K2 system (ForteBio, Pall Life Sciences). Briefly, the mAbs or IgGs (10 μ g/mL in PBST) were immobilizing onto hydrated anti-human IgG-Fc biosensors (AHC: ForteBio) for 60 seconds to achieve a binding response of at least 1.0. After Ab capture, the sensor was placed in a PBST wash buffer to remove the unbound Ab to establish a baseline signal. Next, the IgG immobilized sensor was dipped into a solution containing SOSIP.664 trimer protein as analyte and incubated for 120 seconds at 1000 rpm. Following this, the trimer bound to IgG immobilized sensor was removed from the analyte solution and placed into the PBST buffer for 240 seconds at 1000 rpm. The 2 and 4 minute binding intervals respectively denote the association and dissociation binding curves reported in this study. The sensograms were corrected with the blank reference and fit (1:1 binding kinetics model) with the ForteBio Data Analysis version.9 software using the global fitting function. The data are represented as maximum binding response or the association and dissociation curve fits.

Trimer protein immunizations in CH01 UCA HC-only KI-mice. For the immunization experiments, groups of 5 CH01 UCA HC-only knock-in B cell expressing mice were immunized with 25 μ g of the individual trimer protein or 25 μ g total protein of the 3-trimer cocktail (Prime, week-0; Boost-1, week-4 and Boost-2, week-8) along with Glucopyranosyl Lipid Adjuvant in stable emulsion (GLA-SE) as adjuvant. Immunizations were administered intramuscular in the leg of each animal with 25 g of total trimer immunogens. Blood samples were collected at pre-bleed, 2-weeks each, post-prime (Bleed #1), post boost-1 (Bleed #2) and post boost-2 (Bleed #3) immunization time-point for the isolation of sera that were tested for presence of neutralizing antibodies in TZM-bl cell based assay. Serum samples were heat inactivated for potential complement activity at 56° C. for 0.5 h. Mice used in this study were approved by Duke University Institutional Animal Care and Use Committee-approved animal protocols.

Data availability. Cryo-EM reconstructions have been deposited in the Electron Microscopy Data Bank.

Example 10: Immunization Strategies and Responses

Sequential MT145K Trimer Boost Following HIV Trimer Prime Recalls "on Target" V2-Apex Specific Memory B Cell Responses and Disfavors Off-Target Responses.

In addition to priming the V2-apex specific B cell responses, the chimpanzee MT145K trimer can be used as a boost to recall HIV trimer induced V2-apex directed memory B cell responses.

CH01 iGL B cell activation by HIV GT-trimer, CRF250, under rare B cell conditions to mimic their natural circulating frequencies (CH01 iGL B cell diluted up to 1 in 10⁷ cells) was investigated by adoptively transferring CH01 iGL B cells (CD45.2) into WT mice (CD45.1) (FIG. 18A). CRF250 GT-trimer binds CH01 UCA BCR with 100 nM KD affinity and efficiently activated the on-target V2-apex CH01 B cells even at 1 in 10⁷ precursor frequency as determined by ELISA binding of the week-2 (2 wk) post prime Ab responses with CRF250 SOSIP trimer and its N160 glycan-eliminated V2-epitope specific knock-out variant (FIG. 18B). Further boosting with CRF250 trimer recalled the V2-specific Ab responses (6 wk), but also resulted in the development of a strong CRF250 Env-backbone specific off-target B cell responses targeting epitopes outside of the

V2-apex bnAb site, as probed by N160 glycan-dependent binding (FIG. 18B). The development of CRF250-specific off-target B cell responses coincided with reduction of CH01-specific class-switched memory B cell responses, as determined by next generation sequencing of B cells from immunized mice (FIG. 18C). Both the CRF250 trimer-elicited post-prime and the post-boost Ab responses displayed strong binding to the chimpanzee SIV MT145K trimer in a V2-apex bnAb epitope-specific manner but largely no binding to the HIV Env backbone-specific off-target Ab responses (FIG. 18B). The results suggest that sequential prime boosting with HIV and SIV trimers that share V2-apex bnAb site can be advantageous, especially in that off-target responses should be disfavored.

MT145KdV5 SOSIP Trimer Version Induces Superior Cross-Neutralizing Ab Responses Compared to the Parent MT145K SOSIP Trimer in CH01 iGL HC-Only KI Mouse Model.

Infection of non-human primates (NHPs) with Simian-Human Immunodeficiency Virus (SHIV)-encoding MT145K envelope induces a rapid and strong autologous neutralizing antibody response largely directed to the immunodominant hypervariable V5-loop of gp120. In response to this early wave of autologous nAb responses, the circulating MT145K SHIV viruses develop multiple escape solutions, including a 9 amino acid deletion in the V5-loop (George Shaw et al, unpublished). Multiple animals developed this common viral escape solution of 9AA V5-loop deletion. Structure of MT145K SOSIP trimer revealed a peripheral positioning of the flexible V5 loop, consistent with its in vivo B cell immunodominance effects.

The MT145K SOSIP trimer was engineered to eliminate this 9AA immunodominant V5-loop (FIG. 19) and the resulting SOSIP trimer, MT145KdV5 with a 9AA V5-deletion showed native-like trimer antigenic and biophysical properties similar to the parent MT 145K SOSIP trimer (FIG. 20). A head-to-head immunogenicity comparison of MT 145dV5 trimer version with MT 145K in CH01 iGL HC-only KI mice showed induction of a more reproducible and superior cross-neutralizing serum antibody response by MT145KdV5 trimer. Deletion of the immunodominant V5-loop in the MT 145KdV5 can reduce the development of non-desired off-target B cell responses and therefore, the modified version of the trimer can be more optimal for human vaccinations to induce V2-apex bnAbs.

Chimpanzee SIV MT145KdV5 SOSIP Trimer Induces Autologous Neutralizing Antibodies in NHPs

The immunogenicity of MT145KdV5 trimer was investigated in NHPs, whose B cell repertoires closely resemble to that of humans. A group of 6 NHPs was immunized twice (prime, week 0; boost-1, week 8) with MT145KdV5 SOSIP trimer and monitored the development of antibody responses by ELISA and neutralization assay (FIG. 21). ELISA binding of the post-prime serum antibodies revealed a strong development of autologous trimer immunogen specific binding responses (FIG. 21B). The titers of the Ab binding responses substantially increased post boost immunization indicating a B cell recall response. In addition to stronger Ab binding titers, the post boost (10 wk) serum antibody responses exhibited neutralizing activities against the autologous MT145KdV5 virus (FIG. 21C).

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Having thus described in detail preferred embodiments of the present invention, it is to be understood that the invention defined by the above paragraphs is not to be limited to particular details set forth in the above description as many apparent variations thereof are possible without departing from the spirit or scope of the present invention.

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Lys Xaa Lys Lys
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Lys Lys Xaa Lys
 1

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<223> OTHER INFORMATION: Xaa comprises any amino acid.

<400> SEQUENCE: 6

Asn Xaa Asn Xaa Xaa Xaa
1 5

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<223> OTHER INFORMATION: Xaa comprises I, V, T, Q, or M.

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<223> OTHER INFORMATION: Xaa comprises S or T.

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Xaa Xaa Xaa Xaa
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Xaa Xaa Xaa Xaa
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1          5          10          15

Xaa Xaa Xaa Xaa
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<400> SEQUENCE: 10

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Xaa Xaa Xaa Xaa
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Val Tyr Ser Leu
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Val Tyr Ser Leu
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Glu Tyr Ser Phe
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Glu Tyr Ala Leu
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Val Tyr Ser Leu Phe Phe Tyr
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Val Tyr Ser Leu Phe Tyr
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Val Tyr Ser Leu Phe Phe Tyr
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Val Tyr Ser Leu Phe Tyr
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Glu Tyr Ser Phe Phe Phe Tyr
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1 5 10 15

Glu Tyr Ser Phe Phe Tyr
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Glu Tyr Ala Leu Phe Phe Tyr
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Glu Tyr Ala Leu Phe Tyr

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Glu Tyr Ala Phe Phe Tyr
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Trp Gly Thr Met Leu Leu Gly Met Leu Met Ile Cys Ser Ala Thr Glu
 20 25 30

Lys Leu Trp Val Thr Val Tyr Tyr Gly Val Pro Val Trp Lys Glu Ala
 35 40 45

Thr Thr Thr Leu Phe Cys Ala Ser Asp Ala Lys Ala Tyr Asp Thr Glu
 50 55 60

Val His Asn Val Trp Ala Thr His Ala Cys Val Pro Thr Asp Pro Asn
 65 70 75 80

Pro Gln Glu Val Val Leu Val Asn Val Thr Glu Asn Phe Asn Met Trp
 85 90 95

Lys Asn Asp Met Val Glu Gln Met His Glu Asp Ile Ile Ser Leu Trp
 100 105 110

Asp Gln Ser Leu Lys Pro Cys Val Lys Leu Thr Pro Leu Cys Val Ser
 115 120 125

Leu Lys Cys Thr Asp Leu Lys Asn Asp Thr Asn Thr Asn Ser Ser Ser
 130 135 140

Gly Arg Met Ile Met Glu Lys Gly Glu Ile Lys Asn Cys Ser Phe Asn
 145 150 155 160

Ile Ser Thr Ser Ile Arg Gly Lys Val Gln Lys Glu Tyr Ala Phe Phe
 165 170 175

Tyr Lys Leu Asp Ile Ile Pro Ile Asp Asn Asp Thr Thr Ser Tyr Lys
 180 185 190

Leu Thr Ser Cys Asn Thr Ser Val Ile Thr Gln Ala Cys Pro Lys Val
 195 200 205

Ser Phe Glu Pro Ile Pro Ile His Tyr Cys Ala Pro Ala Gly Phe Ala
 210 215 220

Ile Leu Lys Cys Asn Asn Lys Thr Phe Asn Gly Thr Gly Pro Cys Thr
 225 230 235 240

Asn Val Ser Thr Val Gln Cys Thr His Gly Ile Arg Pro Val Val Ser
 245 250 255

Thr Gln Leu Leu Leu Asn Gly Ser Leu Ala Glu Glu Glu Val Val Ile
 260 265 270

Arg Ser Val Asn Phe Thr Asp Asn Ala Lys Thr Ile Ile Val Gln Leu
 275 280 285

Asn Thr Ser Val Glu Ile Asn Cys Thr Arg Pro Asn Asn Asn Thr Arg

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290	295	300
Lys Arg Ile Arg Ile Gln Arg Gly Pro Gly Arg Ala Phe Val Thr Ile 305 310 315 320		
Gly Lys Ile Gly Asn Met Arg Gln Ala His Cys Asn Ile Ser Arg Ala 325 330 335		
Lys Trp Asn Asn Thr Leu Lys Gln Ile Ala Ser Lys Leu Arg Glu Gln 340 345 350		
Phe Gly Asn Asn Lys Thr Ile Ile Phe Lys Gln Ser Ser Gly Gly Asp 355 360 365		
Pro Glu Ile Val Thr His Ser Phe Asn Cys Gly Gly Glu Phe Phe Tyr 370 375 380		
Cys Asn Ser Thr Gln Leu Phe Asn Ser Thr Trp Phe Asn Ser Thr Trp 385 390 395 400		
Ser Thr Glu Gly Ser Asn Asn Thr Glu Gly Ser Asp Thr Ile Thr Leu 405 410 415		
Pro Cys Arg Ile Lys Gln Ile Ile Asn Met Trp Gln Lys Val Gly Lys 420 425 430		
Ala Met Tyr Ala Pro Pro Ile Ser Gly Gln Ile Arg Cys Ser Ser Asn 435 440 445		
Ile Thr Gly Leu Leu Leu Thr Arg Asp Gly Gly Asn Ser Asn Asn Glu 450 455 460		
Ser Glu Ile Phe Arg Pro Gly Gly Gly Asp Met Arg Asp Asn Trp Arg 465 470 475 480		
Ser Glu Leu Tyr Lys Tyr Lys Val Val Lys Ile Glu Pro Leu Gly Val 485 490 495		
Ala Pro Thr Lys Ala Lys Arg Arg Val Val Gln Arg Glu Lys Arg Ala 500 505 510		
Val Gly Ile Gly Ala Leu Phe Leu Gly Phe Leu Gly Ala Ala Gly Ser 515 520 525		
Thr Met Gly Ala Ala Ser Met Thr Leu Thr Val Gln Ala Arg Gln Leu 530 535 540		
Leu Ser Gly Ile Val Gln Gln Gln Asn Asn Leu Leu Arg Ala Ile Glu 545 550 555 560		
Ala Gln Gln His Leu Leu Gln Leu Thr Val Trp Gly Ile Lys Gln Leu 565 570 575		
Gln Ala Arg Ile Leu Ala Val Glu Arg Tyr Leu Lys Asp Gln Gln Leu 580 585 590		
Leu Gly Ile Trp Gly Cys Ser Gly Lys Leu Ile Cys Thr Thr Ala Val 595 600 605		
Pro Trp Asn Ala Ser Trp Ser Asn Lys Ser Leu Glu Gln Ile Trp Asn 610 615 620		
His Thr Thr Trp Met Glu Trp Asp Arg Glu Ile Asn Asn Tyr Thr Ser 625 630 635 640		
Leu Ile His Ser Leu Ile Glu Glu Ser Gln Asn Gln Gln Glu Lys Asn 645 650 655		
Glu Gln Glu Leu Leu Glu Leu Asp Lys Trp Ala Ser Leu Trp Asn Trp 660 665 670		
Phe Asn Ile Thr Asn Trp Leu Trp Tyr Ile Lys Leu Phe Ile Met Ile 675 680 685		
Val Gly Gly Leu Val Gly Leu Arg Ile Val Phe Ala Val Leu Ser Ile 690 695 700		
Val Asn Arg Val Arg Gln Gly Tyr Ser Pro Leu Ser Phe Gln Thr His 705 710 715 720		

[illegible]

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<210> SEQ ID NO 25
<211> LENGTH: 678
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus
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<400> SEQUENCE: 25

Met 1	Asp	Ala	Met	Lys 5	Arg	Gly	Leu	Cys	Cys 10	Val	Leu	Leu	Leu	Cys 15	Gly
Ala	Val	Phe	Val 20	Ser	Pro	Ser	Gln	Glu 25	Ile	His	Ala	Arg	Phe 30	Arg	Arg
Gly	Ala	Arg	Ser 35	Glu	Leu	Trp	Val 40	Thr	Val	Tyr	Tyr	Gly 45	Val	Pro	Val
Trp	His 50	Asp	Ala	Asp	Pro	Val 55	Leu	Phe	Cys	Ala	Ser 60	Asp	Ala	Lys	Ala
His 65	Ser	Thr	Glu	Ala	His 70	Asn	Ile	Trp	Ala	Thr 75	Gln	Ala	Cys	Val	Pro 80
Thr	Asp	Pro	Ser	Pro 85	Gln	Glu	Val	Phe	Leu 90	Pro	Asn	Val	Ile	Glu 95	Ser
Phe	Asn	Met	Trp 100	Lys	Asn	Asn	Met	Val 105	Asp	Gln	Met	His	Glu 110	Asp	Ile
Ile	Ser	Leu 115	Trp	Asp	Gln	Ser	Leu 120	Lys	Pro	Cys	Val	Lys 125	Leu	Thr	Pro
Leu	Cys 130	Val	Thr	Leu	Gln	Cys 135	Ser	Lys	Ala	Asn	Phe 140	Ser	Gln	Ala	Lys
Asn 145	Leu	Thr	Asn	Gln	Thr 150	Ser	Ser	Pro	Pro	Leu 155	Glu	Met	Lys	Asn	Cys 160
Ser	Phe	Asn	Val	Thr 165	Thr	Glu	Leu	Arg	Asp 170	Lys	Lys	Lys	Gln	Val 175	Tyr
Ser	Leu	Phe	Tyr 180	Val	Glu	Asp	Val	Val 185	Asn	Leu	Gly	Asn	Glu 190	Asn	Asn
Thr	Tyr 195	Arg	Ile	Ile	Asn	Cys	Asn 200	Thr	Thr	Ala	Ile	Thr 205	Gln	Ala	Cys
Pro	Lys 210	Thr	Ser	Phe	Glu	Pro 215	Ile	Pro	Ile	His	Tyr 220	Cys	Ala	Pro	Ala
Gly	Phe	Ala	Ile	Leu	Lys	Cys	Asn	Asp	Lys	Asp	Phe	Ser	Gly	Lys	Gly

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225				230					235					240			
Lys	Cys	Thr	Asn	Val	Ser	Thr	Val	His	Cys	Thr	His	Gly	Ile	Lys	Pro		
245				250					255				260				
Val	Val	Thr	Thr	Gln	Leu	Leu	Ile	Asn	Gly	Ser	Leu	Ala	Glu	Gly	Asn		
265				270					275				280				
Ile	Thr	Val	Arg	Val	Glu	Asn	Lys	Ser	Lys	Asn	Thr	Asp	Val	Trp	Ile		
285				290					295				300				
Val	Gln	Leu	Val	Glu	Ala	Val	Ser	Leu	Asn	Cys	His	Arg	Pro	Gly	Asn		
305				310					315				320				
Asn	Thr	Gly	Arg	Glu	Val	Gln	Ile	Gly	Pro	Gly	Met	Thr	Phe	Tyr	Asn		
325				330					335				340				
Ile	Glu	Asn	Val	Val	Gly	Asp	Thr	Arg	Ser	Ala	Tyr	Cys	Lys	Ile	Asn		
345				350					355				360				
Gly	Thr	Thr	Trp	Asn	Arg	Thr	Val	Glu	Glu	Val	Lys	Lys	Ala	Leu	Ala		
365				370					375				380				
Thr	Ser	Ser	Asn	Arg	Thr	Ala	Ala	Asn	Ile	Thr	Leu	Asn	Arg	Ala	Ser		
385				390					395				400				
Gly	Gly	Asp	Pro	Glu	Val	Thr	His	His	Met	Phe	Asn	Cys	Gly	Gly	Glu		
405				410					415				420				
Phe	Phe	Tyr	Cys	Asn	Thr	Ser	Gln	Ile	Phe	Thr	Asp	Asn	Ile	Thr	Asn		
425				430					435				440				
Gly	Ile	Ile	Ile	Leu	Pro	Cys	Arg	Ile	Arg	Gln	Ile	Val	Ser	Ser	Trp		
445				450					455				460				
Met	Arg	Val	Gly	Arg	Gly	Ile	Tyr	Ala	Pro	Pro	Ile	Arg	Gly	Asn	Ile		
465				470					475				480				
Thr	Cys	Asn	Ser	Asn	Ile	Thr	Gly	Leu	Leu	Leu	Thr	Ser	Asp	Thr	Pro		
485				490					495				500				
Val	Thr	Asn	Asn	Ser	Gly	Asn	Leu	Thr	Phe	Arg	Pro	Thr	Gly	Gly	Asn		
505				510					515				520				
Met	Lys	Asp	Ile	Trp	Arg	Ser	Glu	Leu	Tyr	Lys	Tyr	Lys	Val	Val	Arg		
525				530					535				540				
Ile	Glu	Pro	Leu	Ser	Val	Ala	Pro	Thr	Lys	Cys	Arg	Arg	His	Thr	Val		
545				550					555				560				
Ala	Arg	Gln	Lys	Asp	Arg	Arg	Arg	Arg	Arg	Arg	Ala	Ala	Phe	Gly	Leu		
565				570					575				580				
Gly	Ala	Leu	Phe	Leu	Gly	Phe	Leu	Gly	Ala	Ala	Gly	Ser	Thr	Met	Gly		
585				590					595				600				
Ala	Ala	Ala	Val	Thr	Leu	Thr	Val	Gln	Ala	Arg	Gln	Leu	Leu	Ser	Gly		
605				610					615				620				
Ile	Val	Gln	Gln	Gln	Asn	Asn	Leu	Leu	Lys	Ala	Pro	Glu	Ala	Gln	Gln		
625				630					635				640				
His	Leu	Leu	Gln	Leu	Ser	Ile	Trp	Gly	Val	Lys	Gln	Leu	Gln	Ala	Arg		
645				650					655				660				
Leu	Leu	Ala	Val	Glu	Arg	Tyr	Leu	Gln	Asp	Gln	Gln	Ile	Leu	Gly	Leu		
665				670					675				680				
Trp	Gly	Cys	Ser	Gly	Lys	Ala	Val	Cys	Cys	Thr	Thr	Val	Pro	Trp	Asn		
685				690					695				700				
Asn	Ser	Trp	Pro	Gly	Ser	Asn	Ser	Thr	Asp	Asp	Ile	Trp	Gly	Asn	Leu		
705				710					715				720				
Thr	Trp	Gln	Gln	Trp	Asp	Lys	Leu	Val	Ser	Asn	Tyr	Thr	Gly	Lys	Ile		
725				730					735				740				
Phe	Gly	Leu	Leu	Glu	Glu	Ala	Gln	Ser	Gln	Gln	Glu	Lys	Asn	Glu	Arg		
745				750					755				760				

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Asp Leu Leu Glu Leu Asp Gly Ser Gly Ser Gly Gly Ser Gly His His
 660 665 670

His His His His His His
 675

<210> SEQ ID NO 26

<211> LENGTH: 669

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 26

Met Asp Ala Met Lys Arg Gly Leu Cys Cys Val Leu Leu Leu Cys Gly
 1 5 10 15

Ala Val Phe Val Ser Pro Ser Gln Glu Ile His Ala Arg Phe Arg Arg
 20 25 30

Gly Ala Arg Ser Glu Asn Asn Leu Trp Val Thr Val Tyr Tyr Gly Val
 35 40 45

Pro Val Trp Lys Glu Ala Lys Thr Thr Leu Phe Cys Ala Ser Asp Ala
 50 55 60

Lys Ala Gln Ile Ala Glu Ala His Asn Ile Trp Ala Ser Gln Ala Cys
 65 70 75 80

Val Pro Thr Asp Pro Asn Pro Glu Glu Ile Thr Leu Glu Asn Val Thr
 85 90 95

Glu Glu Phe Asp Ala Trp Asn Asn Asn Met Val Asp Gln Met Gln Glu
 100 105 110

Asp Leu Ile Ser Leu Trp Asp Gln Ser Leu Lys Pro Cys Val Lys Leu
 115 120 125

Thr Pro Leu Cys Val Thr Leu Asn Cys Ser Thr Asn Thr Thr Thr Glu
 130 135 140

Ala Ser Gln Val Gln Tyr Asn Cys Ser Phe Asn Val Thr Thr Glu Leu
 145 150 155 160

Arg Asp Lys Lys Lys Gln Val Tyr Ser Leu Phe Tyr Arg Glu Asp Ile
 165 170 175

Thr Ser Leu Asp Ser Asn Lys Thr Val Lys Asn Gly Thr Tyr Arg Leu
 180 185 190

Ile Asn Cys Asn Thr Thr Ala Ile Thr Gln Ala Cys Pro Lys Thr Ser
 195 200 205

Phe Glu Pro Ile Pro Ile Tyr Tyr Cys Ala Pro Ala Gly Phe Ala Leu
 210 215 220

Leu Lys Cys Asn Asp Gln Asn Phe Lys Gly Lys Gly Thr Cys Arg Asn
 225 230 235 240

Val Ser Thr Val His Cys Thr His Gly Ile Lys Pro Val Val Ser Thr
 245 250 255

Gln Phe Leu Leu Asn Gly Ser Leu Ala Glu Gly Asn Lys Thr Val Val
 260 265 270

Arg Val Arg Ser Lys Ser Asn Thr Glu Thr Ile Ile Val Gln Leu Ala
 275 280 285

Thr Ala Ile Tyr Ile Asn Cys Thr Arg Leu Gly Asn Lys Thr Ile Glu
 290 295 300

Gly Ile Pro Ile Gly Pro Gly Gln Ile Phe Tyr Arg Thr Lys Thr Val
 305 310 315 320

Val Gly Asp Thr Arg Gly Ala Glu Cys Arg Ile Asn Gly Thr Ala Trp
 325 330 335

Asn Glu Thr Leu Arg Gln Val Lys Glu Ala Leu Asn Asn Thr Tyr Arg

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340					345					350					
Asn	Leu	Asn	Leu	Ser	Leu	Thr	Glu	Ile	Asn	Phe	Glu	Gly	Ala	Ser	Gly
	355						360					365			
Gly	Asp	Leu	Glu	Val	Thr	Thr	His	Tyr	Phe	Asn	Cys	Gly	Gly	Glu	Phe
	370					375					380				
Phe	Tyr	Cys	Asn	Thr	Ser	Asn	Leu	Phe	Asn	Gln	Ser	Thr	Ile	Asn	Asn
	385					390					395				400
Ile	Thr	His	Ile	Pro	Cys	Arg	Ile	Arg	Gln	Ile	Val	Asn	Gln	Trp	Gln
				405					410					415	
Gly	Val	Gly	Lys	Gly	Ile	Phe	Ala	Pro	Pro	Ile	Arg	Gly	Thr	Ile	Gln
			420					425					430		
Cys	Asn	Ser	Thr	Ile	Thr	Gly	Leu	Leu	Leu	Thr	Arg	Asp	Gly	Lys	Asn
		435					440					445			
Glu	Thr	Glu	Thr	Phe	Arg	Pro	Thr	Gly	Gly	Asp	Met	Arg	Asn	Asn	Trp
	450					455					460				
Arg	Ser	Glu	Leu	Tyr	Lys	Tyr	Lys	Val	Val	Lys	Ile	Glu	Pro	Leu	Gly
	465					470					475				480
Leu	Ala	Pro	Thr	Lys	Cys	Lys	Arg	Arg	Thr	Val	Gln	Arg	Arg	Arg	Arg
				485					490					495	
Arg	Arg	Ala	Val	Gly	Leu	Gly	Val	Met	Phe	Leu	Gly	Phe	Met	Gly	Ala
			500					505					510		
Ala	Gly	Ser	Thr	Met	Gly	Ala	Ala	Ala	Leu	Thr	Leu	Thr	Val	Gln	Ala
		515					520						525		
Arg	Asn	Leu	Leu	Ser	Gly	Ile	Val	Gln	Gln	Gln	Ser	Asn	Leu	Leu	Lys
	530					535					540				
Ala	Pro	Glu	Ala	Gln	Gln	His	Leu	Leu	Gln	Leu	Ser	Val	Trp	Gly	Ile
	545					550					555				560
Lys	Gln	Leu	Gln	Ala	Arg	Val	Leu	Ala	Val	Glu	Arg	Tyr	Leu	Lys	Asp
				565					570					575	
Gln	Gln	Leu	Leu	Gly	Leu	Trp	Gly	Cys	Ser	Gly	Lys	Leu	Ile	Cys	Cys
		580						585					590		
Thr	Asn	Val	Pro	Trp	Asn	Asn	Thr	Trp	Ser	Lys	Lys	Leu	Gly	Thr	Asn
	595						600						605		
Leu	Ser	Phe	Trp	Asp	Asn	Met	Thr	Trp	Met	Gln	Trp	Glu	Lys	Glu	Ile
	610					615					620				
Asp	Asn	Tyr	Thr	Glu	Thr	Ile	Tyr	Glu	Leu	Leu	Thr	Arg	Ser	Gln	Asn
	625					630					635				640
Gln	Gln	Glu	Val	Asn	Glu	Gln	Glu	Leu	Leu	Ala	Leu	Asp	Gly	Ser	Gly
				645					650					655	
Ser	Gly	Gly	Ser	Gly	His	His	His	His	His	His	His	His	His		
			660					665							

<210> SEQ ID NO 27

<211> LENGTH: 665

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 27

Met	Asp	Ala	Met	Lys	Arg	Gly	Leu	Cys	Cys	Val	Leu	Leu	Leu	Cys	Gly
1			5							10				15	

Ala	Val	Phe	Val	Ser	Pro	Ser	Gln	Glu	Ile	His	Ala	Arg	Phe	Arg	Arg
			20					25					30		

Gly	Ala	Arg	Ser	Glu	Leu	Trp	Val	Thr	Val	Tyr	Tyr	Gly	Val	Pro	Val
		35					40					45			

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Trp	His	Asp	Ala	Glu	Thr	Val	Leu	Phe	Cys	Ala	Ser	Asp	Ala	Lys	Ala
	50					55					60				
His	Ser	Thr	Glu	Ala	His	Asn	Ile	Trp	Ala	Thr	Gln	Ala	Cys	Val	Pro
65					70					75					80
Thr	Asp	Pro	Asn	Pro	Gln	Glu	Val	Leu	Ile	Pro	Asn	Val	Thr	Glu	Arg
				85					90					95	
Phe	Asp	Met	Trp	Lys	Asn	Asn	Met	Val	Asp	Gln	Met	Gln	Glu	Asp	Ile
		100						105					110		
Ile	Ser	Leu	Trp	Glu	Gln	Ser	Leu	Lys	Pro	Cys	Val	Lys	Leu	Thr	Pro
		115					120					125			
Leu	Cys	Val	Thr	Leu	Ser	Cys	Ser	Ser	Trp	Arg	Ser	Val	Asn	Asn	Ser
	130					135					140				
Val	Asn	Gln	Thr	Asn	His	Val	Gln	Met	Gln	Asn	Cys	Ser	Phe	Asn	Val
145					150					155					160
Thr	Thr	Glu	Leu	Arg	Asp	Lys	Lys	Lys	Gln	Val	Tyr	Ser	Leu	Phe	Tyr
			165						170					175	
Met	Gly	Asp	Ile	Ile	Pro	Leu	Asp	Thr	Asn	Asn	Ser	Ser	Gly	Asn	Asn
		180						185					190		
Ser	Gln	Tyr	Arg	Leu	Ile	Asn	Cys	Asn	Thr	Thr	Ala	Val	Thr	Gln	Ala
		195					200					205			
Cys	Pro	Lys	Ile	Ser	Phe	Glu	Pro	Ile	Pro	Ile	Tyr	Tyr	Cys	Ala	Pro
	210					215					220				
Pro	Gly	Phe	Ala	Ile	Ile	Lys	Cys	Asn	Asp	Gln	Asp	Phe	Asn	Gly	Thr
225					230					235					240
Gly	Glu	Cys	Asn	Asn	Val	Ser	Thr	Val	Gln	Cys	Thr	His	Gly	Ile	Lys
			245						250					255	
Pro	Val	Ile	Ser	Thr	Gln	Leu	Ile	Leu	Asn	Gly	Ser	Leu	Ala	Thr	Ser
		260						265					270		
Asn	Ile	Val	Ile	Arg	Asn	Asn	Ser	Lys	Asp	Thr	Leu	Leu	Val	Gln	Leu
	275						280					285			
Asn	Glu	Ser	Ile	Pro	Ile	Asn	Cys	Thr	Arg	Pro	Gly	Asn	Lys	Thr	Arg
	290					295					300				
Gly	Gln	Val	Gln	Ile	Gly	Pro	Gly	Met	Thr	Phe	Tyr	Asn	Ile	Glu	Asn
305					310					315					320
Ile	Ile	Gly	Asp	Thr	Arg	Gln	Ala	Tyr	Cys	Glu	Val	Asn	Arg	Thr	Trp
			325						330					335	
Glu	Gln	Ile	Trp	Asn	Thr	Thr	Lys	Gln	Ile	Ile	Ile	Asn	Asn	Arg	Lys
		340						345					350		
Asn	Ile	Thr	Phe	Ile	Pro	Asn	Pro	Gly	Gly	Asp	Leu	Glu	Val	Thr	Asn
	355						360					365			
Leu	Met	Ile	Asn	Cys	Gly	Gly	Glu	Phe	Phe	Tyr	Cys	Asn	Thr	Ser	Gln
	370					375					380				
Leu	Phe	Thr	Lys	Gln	Asn	Gly	Asn	Thr	Thr	Gly	Asn	Ile	Thr	Leu	Gln
385					390					395					400
Cys	Arg	Ile	Arg	Gln	Ile	Val	Asn	Leu	Trp	Thr	Arg	Val	Gly	Lys	Gly
			405						410					415	
Ile	Tyr	Ala	Pro	Pro	Ile	Lys	Gly	Pro	Ile	Asn	Cys	Leu	Ser	Asn	Ile
		420						425					430		
Thr	Gly	Ile	Ile	Leu	Asp	Tyr	Thr	Lys	Ser	Gly	Thr	Glu	Lys	Tyr	Thr
	435						440					445			
Ile	Tyr	Pro	Thr	Gly	Gly	Asp	Met	Thr	Asn	Leu	Trp	Arg	Gln	Glu	Leu
	450					455					460				
Tyr	Lys	Tyr	Lys	Val	Val	Ser	Ile	Glu	Pro	Ile	Gly	Val	Ala	Pro	Gly

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465	470	475	480
Lys Cys Lys Arg His Thr Val Thr Arg Arg Arg Arg Arg Arg Ala Ala			
	485	490	495
Phe Gly Leu Gly Ala Leu Phe Leu Gly Phe Leu Gly Ala Ala Gly Ser			
	500	505	510
Thr Met Gly Ala Ala Ser Ile Thr Leu Thr Val Gln Ala Arg Lys Leu			
	515	520	525
Leu Ser Gly Ile Val Gln Gln Gln Asn Asn Leu Leu Arg Ala Pro Glu			
	530	535	540
Ala Gln Gln His Leu Leu Gln Leu Ser Val Trp Gly Ile Lys Gln Leu			
545	550	555	560
Gln Ala Arg Val Leu Ala Ile Glu Arg Tyr Leu Arg Asp Gln Gln Ile			
	565	570	575
Leu Gly Leu Trp Gly Cys Ser Gly Lys Ser Val Cys Cys Thr Asn Val			
	580	585	590
Pro Trp Asn Thr Thr Trp Ser Asn Asn Asn Ser Tyr Asp Thr Ile Trp			
	595	600	605
Gly Asn Met Thr Trp Gln Asn Trp Asp Glu Gln Val Arg Asn Tyr Ser			
	610	615	620
Gly Val Ile Phe Gly Leu Leu Glu Gln Ala Gln Glu Gln Gln Ser Ile			
625	630	635	640
Asn Glu Lys Ser Leu Leu Glu Leu Asp Gly Ser Gly Ser Gly Gly Ser			
	645	650	655
Gly His His His His His His His			
	660	665	

<210> SEQ ID NO 28
 <211> LENGTH: 675
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 28

Met Asp Ala Met Lys Arg Gly Leu Cys Cys Val Leu Leu Leu Cys Gly			
1	5	10	15
Ala Val Phe Val Ser Pro Ser Gln Glu Ile His Ala Arg Phe Arg Arg			
	20	25	30
Gly Ala Arg Ser Glu Trp Trp Val Thr Val Tyr Tyr Gly Val Pro Val			
	35	40	45
Trp Arg Glu Ala Lys Thr Thr Leu Phe Cys Ala Ser Asp Ala Lys Ser			
	50	55	60
Tyr Ser Thr Glu Ala His Asn Ile Trp Ala Thr Gln Ala Cys Val Pro			
65	70	75	80
Thr Asp Pro Thr Pro Gln Glu Val Leu Leu Pro Asn Val Thr Glu Glu			
	85	90	95
Phe Asn Met Trp Glu Asn Tyr Met Val Asp Gln Met Gln Glu Asp Ile			
	100	105	110
Ile Ser Leu Trp Glu Gln Ser Leu Lys Pro Cys Val Lys Leu Thr Pro			
	115	120	125
Leu Cys Val Thr Leu Thr Cys Asn Asn Pro Thr Asn Thr Ser Cys Thr			
	130	135	140
Asn Ser Thr Asp Asp Arg Leu Gly Asp Arg Met Asn Cys Ser Phe Asn			
145	150	155	160
Val Thr Thr Glu Leu Arg Asp Lys Lys Lys Gln Val Tyr Ser Leu Phe			
	165	170	175

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Tyr	Val	Glu	Asp	Ile	Thr	Ala	Ile	Gly	Asn	Asn	Ser	Thr	Tyr	Arg	Leu	180	185	190	
Ile	Asn	Cys	Asn	Thr	Thr	Ala	Ile	Thr	Gln	Ala	Cys	Pro	Lys	Thr	Ser	195	200	205	
Phe	Glu	Pro	Ile	Pro	Ile	His	Tyr	Cys	Ala	Pro	Ala	Gly	Phe	Ala	Leu	210	215	220	
Leu	Lys	Cys	Asn	Asp	Ile	Asp	Tyr	Lys	Gly	Asn	Glu	Thr	Cys	Lys	Asn	225	230	235	240
Val	Ser	Thr	Val	His	Cys	Thr	His	Gly	Ile	Lys	Pro	Val	Ala	Thr	Thr	245	250	255	
Gln	Leu	Ile	Leu	Asn	Gly	Ser	Thr	Ala	Asp	Asn	Gln	Thr	Val	Ala	Arg	260	265	270	
Ile	Asp	Pro	Ser	Glu	Asn	Leu	Ala	Ile	Ile	Gln	Leu	Lys	Asp	Pro	Val	275	280	285	
Lys	Ile	Thr	Cys	Arg	Arg	Pro	Gly	Asn	Asn	Thr	Arg	Gly	Gln	Ile	Gln	290	295	300	
Ile	Gly	Pro	Ala	Met	Thr	Phe	Tyr	Asn	Ile	Glu	Asn	Val	Val	Gly	Asp	305	310	315	320
Thr	Arg	Lys	Ala	Tyr	Cys	Glu	Ile	Asn	Gly	Thr	Gln	Trp	Ala	Lys	Ala	325	330	335	
Leu	Asn	Glu	Thr	Lys	Glu	Val	Leu	Arg	Asn	Ile	Leu	Arg	Lys	Asn	Ile	340	345	350	
Ser	Phe	Met	Val	Pro	Ser	Gly	Gly	Asp	Pro	Glu	Val	Thr	Asn	His	His	355	360	365	
Phe	Asn	Cys	Gly	Gly	Glu	Phe	Phe	Tyr	Cys	Asn	Thr	Ser	Asp	Ile	Ile	370	375	380	
Asn	Ile	Thr	Lys	Ile	Asn	Lys	Thr	Glu	Asn	Met	Thr	Ile	Ile	Pro	Cys	385	390	395	400
Arg	Ile	Arg	Gln	Ile	Val	Asn	Ser	Trp	Met	Arg	Val	Gly	Lys	Gly	Ile	405	410	415	
Phe	Ala	Pro	Pro	Ile	Arg	Gly	Asn	Ile	Thr	Cys	Thr	Ser	Asn	Ile	Thr	420	425	430	
Gly	Met	Leu	Leu	Glu	Ile	His	Lys	Asn	Arg	Glu	Asp	Gln	Gly	Glu	Asp	435	440	445	
Gln	Asp	Gln	Asn	Asn	Thr	Tyr	Val	Cys	Leu	Thr	Gly	Gly	Asn	Met	Lys	450	455	460	
Asp	Ile	Trp	Arg	Ser	Glu	Leu	Tyr	Lys	Tyr	Lys	Ile	Val	Glu	Ile	Gln	465	470	475	480
Pro	Leu	Gly	Val	Ala	Pro	Thr	Lys	Cys	Arg	Arg	Tyr	Ala	Val	Glu	Lys	485	490	495	
Gln	His	His	Arg	Arg	Arg	Arg	Arg	Ala	Leu	Gly	Leu	Gly	Ala	Leu		500	505	510	
Phe	Leu	Gly	Phe	Leu	Gly	Ala	Ala	Gly	Ser	Thr	Met	Gly	Ala	Ala	Ser	515	520	525	
Val	Val	Leu	Thr	Val	Gln	Ala	Arg	Gln	Leu	Leu	Thr	Gly	Ile	Val	Gln	530	535	540	
Gln	Gln	Asn	Asn	Leu	Leu	Arg	Ala	Pro	Glu	Ala	Gln	Gln	His	Leu	Leu	545	550	555	560
Gln	Leu	Ser	Val	Trp	Gly	Ile	Lys	Gln	Leu	Gln	Ala	Arg	Val	Leu	Ala	565	570	575	
Val	Glu	Arg	Tyr	Leu	Arg	Asp	Gln	Gln	Leu	Leu	Gly	Leu	Trp	Gly	Cys	580	585	590	
Thr	Gly	Lys	Thr	Ile	Cys	Cys	Thr	Ala	Val	Arg	Trp	Asn	Lys	Thr	Trp				


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<210> SEQ ID NO 31
<211> LENGTH: 137
<212> TYPE: PRT
<213> ORGANISM: unknown
<220> FEATURE:
<223> OTHER INFORMATION: synthetic sequence

<400> SEQUENCE: 31

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
1             5             10             15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
             20             25             30
Gly Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
             35             40             45
Ala Val Ile Ser Tyr Asp Gly Ser Asn Lys Tyr Tyr Ala Asp Ser Val
             50             55             60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65             70             75             80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
             85             90             95
Ala Arg Glu Ala Gly Gly Pro Asp Tyr Arg Asn Gly Tyr Asn Tyr Tyr
             100            105            110
Asp Phe Tyr Asp Gly Tyr Tyr Asn Tyr His Tyr Met Asp Val Trp Gly
             115            120            125
Lys Gly Thr Thr Val Thr Val Ser Ser
             130            135

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<210> SEQ ID NO 32
<211> LENGTH: 137
<212> TYPE: PRT
<213> ORGANISM: unknown
<220> FEATURE:
<223> OTHER INFORMATION: synthetic sequence

<400> SEQUENCE: 32

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
          20          25          30
Gly Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
          35          40          45
Ala Val Ile Ser Tyr Asp Gly Ser Asn Lys Tyr Tyr Ala Asp Ser Val
          50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
          85          90          95
Ala Arg Glu Ala Gly Gly Pro Asp Tyr Arg Asn Gly Tyr Asn Tyr Tyr
          100          105          110
Asp Phe Trp Ser Gly Tyr Tyr Tyr Tyr Tyr Tyr Met Asp Val Trp Gly
          115          120          125

Lys Gly Thr Thr Val Thr Val Ser Ser
          130          135

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<210> SEQ ID NO 33
<211> LENGTH: 5
<212> TYPE: PRT
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<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 33

Ser Tyr Gly Met His
1 5

<210> SEQ ID NO 34

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 34

Val Ile Ser Tyr Asp Gly Ser Asn Lys Tyr Tyr Ala Asp Ser Val Lys
1 5 10 15

Gly

<210> SEQ ID NO 35

<211> LENGTH: 28

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 35

Glu Ala Gly Gly Pro Asp Tyr Arg Asn Gly Tyr Asn Tyr Tyr Asp Phe
1 5 10 15

Trp Ser Gly Tyr Tyr Tyr Tyr Tyr Tyr Met Asp Val
20 25

<210> SEQ ID NO 36

<211> LENGTH: 20

<212> TYPE: PRT

<213> ORGANISM: unknown

<220> FEATURE:

<223> OTHER INFORMATION: synthetic sequence

<400> SEQUENCE: 36

Tyr Tyr Tyr Tyr Tyr Tyr Met Asp Val Trp Gly Lys Gly Thr Thr Val
1 5 10 15

Thr Val Ser Ser
20

<210> SEQ ID NO 37

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: unknown

<220> FEATURE:

<223> OTHER INFORMATION: synthetic sequence

<400> SEQUENCE: 37

Tyr Tyr Asp Phe Trp Ser Gly Tyr Tyr Thr
1 5 10

<210> SEQ ID NO 38

<211> LENGTH: 99

<212> TYPE: PRT

<213> ORGANISM: homo sapiens

<400> SEQUENCE: 38

Gln Ser Ala Leu Thr Gln Pro Ala Ser Val Ser Gly Ser Pro Gly Gln
1 5 10 15

Ser Ile Thr Ile Ser Cys Thr Gly Thr Ser Ser Asp Val Gly Gly Tyr
20 25 30

Asn Tyr Val Ser Trp Tyr Gln Gln His Pro Gly Lys Ala Pro Lys Leu
35 40 45

-continued

Met Ile Tyr Glu Val Ser Asn Arg Pro Ser Gly Val Ser Asn Arg Phe
50 55 60

Ser Gly Ser Lys Ser Gly Asn Thr Ala Ser Leu Thr Ile Ser Gly Leu
65 70 75 80

Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Ser Ser Tyr Thr Ser Ser
85 90 95

Ser Thr Leu

<210> SEQ ID NO 39
<211> LENGTH: 110
<212> TYPE: PRT
<213> ORGANISM: unknown
<220> FEATURE:
<223> OTHER INFORMATION: synthetic sequence

<400> SEQUENCE: 39

Gln Ser Ala Leu Thr Gln Pro Ala Ser Val Ser Gly Ser Pro Gly Gln
1 5 10 15

Ser Ile Thr Ile Ser Cys Asn Gly Thr Ser Asn Asp Val Gly Gly Tyr
20 25 30

Glu Ser Val Ser Trp Tyr Gln Gln His Pro Gly Lys Ala Pro Lys Val
35 40 45

Val Ile Tyr Asp Val Ser Lys Arg Pro Ser Gly Val Ser Asn Arg Phe
50 55 60

Ser Gly Ser Lys Ser Gly Asn Thr Ala Ser Leu Thr Ile Ser Gly Leu
65 70 75 80

Gln Ala Glu Asp Glu Gly Asp Tyr Tyr Cys Lys Ser Leu Thr Ser Thr
85 90 95

Arg Arg Arg Val Phe Gly Thr Gly Thr Lys Leu Thr Val Leu
100 105 110

<210> SEQ ID NO 40
<211> LENGTH: 110
<212> TYPE: PRT
<213> ORGANISM: unknown
<220> FEATURE:
<223> OTHER INFORMATION: synthetic sequence

<400> SEQUENCE: 40

Gln Ser Ala Leu Thr Gln Pro Ala Ser Val Ser Gly Ser Pro Gly Gln
1 5 10 15

Ser Ile Thr Ile Ser Cys Thr Gly Thr Ser Ser Asp Val Gly Gly Tyr
20 25 30

Asn Tyr Val Ser Trp Tyr Gln Gln His Pro Gly Lys Ala Pro Lys Leu
35 40 45

Met Ile Tyr Glu Val Ser Asn Arg Pro Ser Gly Val Ser Asn Arg Phe
50 55 60

Ser Gly Ser Lys Ser Gly Asn Thr Ala Ser Leu Thr Ile Ser Gly Leu
65 70 75 80

Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Ser Ser Tyr Thr Ser Ser
85 90 95

Ser Thr Leu Val Phe Gly Thr Gly Thr Lys Leu Thr Val Leu
100 105 110

<210> SEQ ID NO 41
<211> LENGTH: 110
<212> TYPE: PRT
<213> ORGANISM: unknown

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<220> FEATURE:

<223> OTHER INFORMATION: synthetic sequence

<400> SEQUENCE: 41

Gln Ser Ala Leu Thr Gln Pro Ala Ser Val Ser Gly Ser Pro Gly Gln
1 5 10 15
Ser Ile Thr Ile Ser Cys Thr Gly Thr Ser Ser Asp Val Gly Gly Tyr
20 25 30
Asn Tyr Val Ser Trp Tyr Gln Gln His Pro Gly Lys Ala Pro Lys Leu
35 40 45
Met Ile Tyr Glu Val Ser Asn Arg Pro Ser Gly Val Ser Asn Arg Phe
50 55 60
Ser Gly Ser Lys Ser Gly Asn Thr Ala Ser Leu Thr Ile Ser Gly Leu
65 70 75 80
Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Ser Ser Tyr Thr Ser Ser
85 90 95
Ser Thr Trp Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105 110

<210> SEQ ID NO 42

<211> LENGTH: 14

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 42

Thr Gly Thr Ser Ser Asp Val Gly Gly Tyr Asn Tyr Val Ser
1 5 10

<210> SEQ ID NO 43

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 43

Glu Val Ser Asn Arg Pro Ser
1 5

<210> SEQ ID NO 44

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 44

Ser Ser Tyr Thr Ser Ser Ser Thr Trp Val
1 5 10

<210> SEQ ID NO 45

<211> LENGTH: 12

<212> TYPE: PRT

<213> ORGANISM: unknown

<220> FEATURE:

<223> OTHER INFORMATION: synthetic sequence

<400> SEQUENCE: 45

Trp Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
1 5 10

<210> SEQ ID NO 46

<211> LENGTH: 98

<212> TYPE: PRT

<213> ORGANISM: homo sapiens

<400> SEQUENCE: 46

-continued

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asp Asp Tyr
 20 25 30
 Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Gly Ile Asn Trp Asn Gly Gly Ser Thr Gly Tyr Ala Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Leu Tyr His Cys
 85 90 95

Ala Arg

<210> SEQ ID NO 47
 <211> LENGTH: 133
 <212> TYPE: PRT
 <213> ORGANISM: homo sapiens

<400> SEQUENCE: 47

Glu Val Gln Leu Val Glu Ser Gly Ala Asn Val Val Arg Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Lys Ala Ser Gly Phe Ile Phe Glu Asn Phe
 20 25 30
 Gly Phe Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Gln Trp Val
 35 40 45
 Ala Gly Leu Asn Trp Asn Gly Gly Asp Thr Arg Tyr Ala Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Arg Met Ser Arg Asp Asn Ser Arg Asn Phe Val Tyr
 65 70 75 80
 Leu Asp Met Asp Lys Val Gly Val Asp Asp Thr Ala Phe Tyr Tyr Cys
 85 90 95
 Ala Arg Gly Thr Asp Tyr Thr Ile Asp Asp Ala Gly Ile His Tyr Gln
 100 105 110
 Gly Ser Gly Thr Phe Trp Tyr Phe Asp Leu Trp Gly Arg Gly Thr Leu
 115 120 125
 Val Ser Val Ser Ser
 130

<210> SEQ ID NO 48
 <211> LENGTH: 133
 <212> TYPE: PRT
 <213> ORGANISM: unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: synthetic sequence

<400> SEQUENCE: 48

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asp Asp Tyr
 20 25 30
 Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Gly Ile Asn Trp Asn Gly Gly Ser Thr Gly Tyr Ala Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Tyr

-continued

65	70	75	80
Leu Gln Met Asn Ser	Leu Arg Ala Glu Asp	Thr Ala Leu Tyr His	Cys
	85	90	95
Ala Arg Gly Thr Asp Tyr Thr Ile Asp Asp Ala Gly Ile His Tyr Gln			
	100	105	110
Gly Ser Gly Thr Phe Trp Tyr Phe Asp Leu Trp Gly Arg Gly Thr Leu			
	115	120	125
Val Ser Val Ser Ser			
130			

<210> SEQ ID NO 49
 <211> LENGTH: 133
 <212> TYPE: PRT
 <213> ORGANISM: unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: synthetic sequence

<400> SEQUENCE: 49

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Arg Pro Gly Gly			
1	5	10	15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asp Asp Tyr			
	20	25	30
Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val			
	35	40	45
Ser Gly Ile Asn Trp Asn Gly Gly Ser Thr Gly Tyr Ala Asp Ser Val			
	50	55	60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Tyr			
65	70	75	80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Leu Tyr His Cys			
	85	90	95
Ala Arg Gly Thr Asp Tyr Thr Ile Asp Asp Ala Gly Ile His Tyr Tyr			
	100	105	110
Gly Ser Gly Ser Tyr Trp Tyr Phe Asp Leu Trp Gly Arg Gly Thr Leu			
	115	120	125
Val Thr Val Ser Ser			
130			

<210> SEQ ID NO 50
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 50

Asp Tyr Gly Met Ser
1 5

<210> SEQ ID NO 51
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 51

Gly Ile Asn Trp Asn Gly Gly Ser Thr Gly Tyr Ala Asp Ser Val Lys
1 5 10 15

Gly

<210> SEQ ID NO 52
 <211> LENGTH: 24
 <212> TYPE: PRT

-continued

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 52

Gly Thr Asp Tyr Thr Ile Asp Asp Ala Gly Ile His Tyr Tyr Gly Ser
 1 5 10 15

Gly Ser Tyr Trp Tyr Phe Asp Leu
 20

<210> SEQ ID NO 53

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: unknown

<220> FEATURE:

<223> OTHER INFORMATION: synthetic sequence

<400> SEQUENCE: 53

Tyr Trp Tyr Phe Asp Leu Trp Gly Arg Gly Thr Leu Val Thr Val Ser
 1 5 10 15

Ser

<210> SEQ ID NO 54

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: unknown

<220> FEATURE:

<223> OTHER INFORMATION: synthetic sequence

<400> SEQUENCE: 54

Tyr Tyr Tyr Gly Ser Gly Ser Tyr Tyr Asn
 1 5 10

<210> SEQ ID NO 55

<211> LENGTH: 97

<212> TYPE: PRT

<213> ORGANISM: homo sapiens

<400> SEQUENCE: 55

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Ser Ser Ser
 20 25 30

Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu
 35 40 45

Ile Tyr Gly Ala Ser Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser
 50 55 60

Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu
 65 70 75 80

Pro Glu Asp Phe Ala Val Tyr Tyr Cys Cys Gln Gln Tyr Gly Ser Ser
 85 90 95

Pro

<210> SEQ ID NO 56

<211> LENGTH: 109

<212> TYPE: PRT

<213> ORGANISM: homo sapiens

<400> SEQUENCE: 56

Glu Ile Val Leu Ala Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser His Asn Val His Pro Lys
 20 25 30

-continued

Tyr Phe Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ser Pro Arg Leu Leu
 35 40 45

Ile Tyr Gly Gly Ser Thr Arg Ala Ala Gly Ile Pro Gly Lys Phe Ser
 50 55 60

Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Val Asp
 65 70 75 80

Pro Glu Asp Phe Ala Val Tyr Tyr Cys Cys Gln Gln Tyr Gly Gly Ser
 85 90 95

Pro Tyr Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 57
 <211> LENGTH: 109
 <212> TYPE: PRT
 <213> ORGANISM: unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: synthetic sequence

<400> SEQUENCE: 57

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Ser Ser Ser
 20 25 30

Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu
 35 40 45

Ile Tyr Gly Ala Ser Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser
 50 55 60

Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu
 65 70 75 80

Pro Glu Asp Phe Ala Val Tyr Tyr Cys Cys Gln Gln Tyr Gly Ser Ser
 85 90 95

Pro Tyr Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 58
 <211> LENGTH: 109
 <212> TYPE: PRT
 <213> ORGANISM: unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: synthetic sequence

<400> SEQUENCE: 58

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Ser Ser Ser
 20 25 30

Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu
 35 40 45

Ile Tyr Gly Ala Ser Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser
 50 55 60

Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu
 65 70 75 80

Pro Glu Asp Phe Ala Val Tyr Tyr Cys Cys Gln Gln Tyr Gly Ser Ser
 85 90 95

Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
 100 105

-continued

<210> SEQ ID NO 59
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 59

Arg Ala Ser Gln Ser Val Ser Ser Ser Tyr Leu Ala
1 5 10

<210> SEQ ID NO 60
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 60

Gly Ala Ser Ser Arg Ala Thr
1 5

<210> SEQ ID NO 61
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 61

Gln Gln Tyr Gly Ser Ser Pro Trp Thr
1 5

<210> SEQ ID NO 62
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: unknown
<220> FEATURE:
<223> OTHER INFORMATION: synthetic sequence

<400> SEQUENCE: 62

Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
1 5 10

<210> SEQ ID NO 63
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Human immunodeficiency virus type 1

<400> SEQUENCE: 63

Asn Cys Ser Phe Asn Ile Ser Thr Ser Ile Arg Gly Lys Val Gln Lys
1 5 10 15
Glu Tyr Ala Phe
20

<210> SEQ ID NO 64
<211> LENGTH: 33
<212> TYPE: PRT
<213> ORGANISM: Human immunodeficiency virus type 1

<400> SEQUENCE: 64

Gly Lys Ile Gly Asn Met Arg Gln Ala His Cys Asn Ile Ser Arg Ala
1 5 10 15

Lys Trp Asn Asn Thr Leu Lys Gln Ile Ala Ser Lys Leu Arg Glu Gln
20 25 30

Phe

<210> SEQ ID NO 65
<211> LENGTH: 39
<212> TYPE: PRT

-continued

<213> ORGANISM: Human immunodeficiency virus type 1

<400> SEQUENCE: 65

Glu Gly Ser Asp Thr Ile Thr Leu Pro Cys Arg Ile Lys Gln Ile Ile
1 5 10 15

Asn Met Trp Gln Lys Val Gly Lys Ala Met Tyr Ala Pro Pro Ile Ser
20 25 30

Gly Gln Ile Arg Cys Ser Ser
35

<210> SEQ ID NO 66

<211> LENGTH: 36

<212> TYPE: PRT

<213> ORGANISM: Human immunodeficiency virus type 1

<400> SEQUENCE: 66

Ser Thr Gln Leu Leu Leu Asn Gly Ser Leu Ala Glu Glu Glu Val Val
1 5 10 15

Ile Arg Ser Val Asn Phe Thr Asp Asn Ala Lys Thr Ile Ile Val Gln
20 25 30

Leu Asn Thr Ser
35

<210> SEQ ID NO 67

<211> LENGTH: 18

<212> TYPE: PRT

<213> ORGANISM: Human immunodeficiency virus type 1

<400> SEQUENCE: 67

Cys Asn Asn Lys Thr Phe Asn Gly Thr Gly Pro Cys Thr Asn Val Ser
1 5 10 15

Thr Val

<210> SEQ ID NO 68

<211> LENGTH: 12

<212> TYPE: PRT

<213> ORGANISM: Human immunodeficiency virus type 1

<400> SEQUENCE: 68

Asn Ile Thr Gly Leu Leu Leu Thr Arg Asp Gly Gly
1 5 10

<210> SEQ ID NO 69

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Human immunodeficiency virus type 1

<400> SEQUENCE: 69

Val Val Leu Val Asn Val Thr Glu Asn Phe
1 5 10

<210> SEQ ID NO 70

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Human immunodeficiency virus type 1

<400> SEQUENCE: 70

Asn Ile Thr Gly Leu Leu Leu Thr
1 5

<210> SEQ ID NO 71

<211> LENGTH: 28

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<212> TYPE: PRT
<213> ORGANISM: Human immunodeficiency virus type 1

<400> SEQUENCE: 71

Gln Arg Glu Lys Arg Ala Val Gly Ile Gly Ala Leu Phe Leu Gly Phe
1 5 10 15

Leu Gly Ala Ala Gly Ser Thr Met Gly Ala Ala Ser
20 25

<210> SEQ ID NO 72
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Human immunodeficiency virus type 1

<400> SEQUENCE: 72

Thr Thr Ala Val Pro Trp Asn Ala Ser Trp
1 5 10

<210> SEQ ID NO 73
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Human immunodeficiency virus type 1

<400> SEQUENCE: 73

Gln Ile Trp Asn His Thr Thr Trp Met Glu Trp Asp Arg Glu Ile Asn
1 5 10 15

Asn Tyr Thr Ser Leu Ile His Ser Leu Ile
20 25

<210> SEQ ID NO 74
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 74

Asn Cys Ser Phe Asn Val Thr Thr Glu Leu Arg Asp Lys Lys Arg Gln
1 5 10 15

Val Tyr Ser Leu
20

<210> SEQ ID NO 75
<211> LENGTH: 34
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 75

Glu Asn Val Val Gly Asp Thr Arg Lys Ala Tyr Cys Glu Ile Asn Gly
1 5 10 15

Thr Gln Trp Ala Lys Ala Leu Asn Glu Thr Lys Glu Val Leu Arg Asn
20 25 30

Ile Leu

<210> SEQ ID NO 76
<211> LENGTH: 39
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 76

Lys Thr Glu Asn Met Thr Ile Ile Pro Cys Arg Ile Arg Gln Ile Val
1 5 10 15

Asn Ser Trp Met Arg Val Gly Lys Gly Ile Phe Ala Pro Pro Ile Arg
20 25 30

-continued

Gly Asn Ile Thr Cys Thr Ser
35

<210> SEQ ID NO 77
<211> LENGTH: 33
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 77

Thr Thr Gln Leu Ile Leu Asn Gly Ser Thr Ala Asp Asn Gln Thr Val
1 5 10 15

Ala Arg Ile Asp Pro Ser Glu Asn Leu Ala Ile Ile Gln Leu Lys Asp
20 25 30

Pro

<210> SEQ ID NO 78
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 78

Cys Asn Asp Ile Asp Tyr Lys Gly Asn Glu Thr Cys Lys Asn Val Ser
1 5 10 15

Thr Val

<210> SEQ ID NO 79
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 79

Asn Ile Thr Gly Met Leu Leu Glu Ile His Lys Asn Arg Glu Asp Gln
1 5 10 15

Gly

<210> SEQ ID NO 80
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 80

Val Leu Leu Pro Asn Val Thr Glu Glu Phe
1 5 10

<210> SEQ ID NO 81
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 81

Asn Ile Thr Gly Met Leu Leu Glu
1 5

<210> SEQ ID NO 82
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 82

His Arg Glu Lys Arg Ala Leu Gly Leu Gly Ala Leu Phe Leu Gly Phe
1 5 10 15

-continued

Leu Gly Ala Ala Gly Ser Thr Met Gly Ala Ala Ser
20 25

<210> SEQ ID NO 83
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus
 <400> SEQUENCE: 83

Pro Thr Ala Val Arg Trp Asn Lys Thr Trp
1 5 10

<210> SEQ ID NO 84
 <211> LENGTH: 26
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus
 <400> SEQUENCE: 84

Val Ile Trp Asn Asn Tyr Thr Trp Gln Gln Trp Asp Arg Glu Val Asn
1 5 10 15

Asn Tyr Thr Gly Leu Ile Tyr Thr Leu Leu
20 25

<210> SEQ ID NO 85
 <211> LENGTH: 20
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus
 <400> SEQUENCE: 85

Asn Cys Ser Phe Asn Val Thr Thr Glu Leu Arg Asp Lys Lys Lys Gln
1 5 10 15

Val Tyr Ser Leu
20

<210> SEQ ID NO 86
 <211> LENGTH: 34
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus
 <400> SEQUENCE: 86

Lys Thr Val Val Gly Asp Thr Arg Gly Ala Glu Cys Arg Ile Asn Gly
1 5 10 15

Thr Ala Trp Asn Glu Thr Leu Arg Gln Val Lys Glu Ala Leu Gln His
20 25 30

Ile Ser

<210> SEQ ID NO 87
 <211> LENGTH: 39
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus
 <400> SEQUENCE: 87

Thr Ile Asn Asn Ile Thr His Ile Pro Cys Arg Ile Arg Gln Ile Val
1 5 10 15

Asn Gln Trp Gln Gly Val Gly Lys Gly Ile Phe Ala Pro Pro Ile Arg
20 25 30

Gly Thr Ile Gln Cys Asn Ser
35

<210> SEQ ID NO 88
 <211> LENGTH: 36
 <212> TYPE: PRT

-continued

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 88

Ser Thr Gln Phe Leu Leu Asn Gly Ser Leu Ala Glu Gly Asn Lys Thr
1 5 10 15

Val Val Arg Val Arg Ser Lys Ser Asn Thr Glu Thr Ile Ile Val Gln
20 25 30

Leu Ala Thr Ala
35

<210> SEQ ID NO 89

<211> LENGTH: 18

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 89

Cys Asn Asp Gln Asn Phe Lys Gly Lys Gly Thr Cys Arg Asn Val Ser
1 5 10 15

Thr Val

<210> SEQ ID NO 90

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 90

Thr Ile Thr Gly Leu Leu Leu Thr Arg Asp Gly
1 5 10

<210> SEQ ID NO 91

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 91

Ile Thr Leu Glu Asn Val Thr Glu Glu Phe
1 5 10

<210> SEQ ID NO 92

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 92

Thr Ile Thr Gly Leu Leu Leu Thr
1 5

<210> SEQ ID NO 93

<211> LENGTH: 28

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 93

Gln Arg Glu Lys Arg Ala Val Gly Leu Gly Val Met Phe Leu Gly Phe
1 5 10 15

Met Gly Ala Ala Gly Ser Thr Met Gly Ala Ala Ala
20 25

<210> SEQ ID NO 94

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

-continued

<400> SEQUENCE: 94

Thr Thr Asn Val Pro Trp Asn Asn Thr Trp
 1 5 10

<210> SEQ ID NO 95

<211> LENGTH: 25

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 95

Phe Trp Asp Asn Met Thr Trp Met Gln Trp Glu Lys Glu Ile Asp Asn
 1 5 10 15

Tyr Thr Glu Thr Ile Tyr Glu Leu Leu
 20 25

<210> SEQ ID NO 96

<211> LENGTH: 20

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 96

Asn Cys Ser Phe Asn Thr Thr Thr Glu Leu Arg Asp Lys Lys Lys Gln
 1 5 10 15

Thr Tyr Ser Leu
 20

<210> SEQ ID NO 97

<211> LENGTH: 34

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 97

Glu Thr Val Val Gly Asp Thr Arg Glu Ala His Cys Glu Ile Asn Gln
 1 5 10 15

Thr His Trp Tyr Lys Ile Leu Asn Gln Val Lys Arg Glu Leu Thr Thr
 20 25 30

Val Phe

<210> SEQ ID NO 98

<211> LENGTH: 39

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 98

Thr Leu His Asn Asn Leu Thr Leu Pro Cys Arg Ile Arg Gln Ile Val
 1 5 10 15

Asn Leu Trp Leu Arg Val Gly Lys Gly Ile Phe Ala Pro Pro Ile Arg
 20 25 30

Gly Asn Ile Arg Cys Asn Ser
 35

<210> SEQ ID NO 99

<211> LENGTH: 31

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 99

Ser Thr Gln Leu Ile Leu Asn Gly Ser Leu Ala Glu Glu Lys Ile Met
 1 5 10 15

Ile Arg Tyr Lys Ala Asp Lys Val Leu Val Gln Leu Asn Thr Ser
 20 25 30

-continued

<210> SEQ ID NO 100
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 100

Cys Asn Asn Lys Thr Tyr Asn Gly Ser Gly Pro Cys Asn Asn Val Ser
1 5 10 15

Thr Val

<210> SEQ ID NO 101
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 101

Thr Ile Thr Gly Leu Ile Leu Glu
1 5

<210> SEQ ID NO 102
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 102

Ile Glu Leu Ser Asn Val Thr Glu Asn Phe
1 5 10

<210> SEQ ID NO 103
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 103

Thr Ile Thr Gly Leu Ile Leu Glu
1 5

<210> SEQ ID NO 104
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 104

Asp Arg Glu Lys Arg Ala Ala Gly Leu Gly Val Phe Phe Leu Gly Phe
1 5 10 15

Leu Gly Ala Ala Gly Ser Thr Met Gly Ala Ala Ser
 20 25

<210> SEQ ID NO 105
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 105

Thr Thr Ser Val Pro Trp Asn Arg Thr Trp
1 5 10

<210> SEQ ID NO 106
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 106

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Asp Ile Trp Asn Asn Leu Thr Trp Gln Gln Trp Asp Lys Leu Ile Thr
1 5 10 15

Asn Tyr Thr Gly Thr Ile Phe Gly Leu Leu
20 25

<210> SEQ ID NO 107
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 107

Asn Cys Thr Phe Asn Thr Thr Thr Glu Leu Arg Asp Lys Lys Lys Gln
1 5 10 15

Val Tyr Ser Leu
20

<210> SEQ ID NO 108
<211> LENGTH: 34
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 108

Asp His Ile Phe Gly Asp Thr Arg Lys Ala Phe Cys Glu Leu Asn Gly
1 5 10 15

Thr Thr Trp Asn Glu Thr Leu Gln Lys Val Arg Glu Ser Leu Ile Lys
20 25 30

Glu Ile

<210> SEQ ID NO 109
<211> LENGTH: 39
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 109

Pro Phe His Glu Asn Met Thr Ile Pro Cys Arg Ile Arg Gln Ile Val
1 5 10 15

Asn Ser Trp Met Arg Val Gly Arg Gly Ile Tyr Ala Pro Pro Ile Pro
20 25 30

Gly His Ile Thr Cys Asn Ser
35

<210> SEQ ID NO 110
<211> LENGTH: 36
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 110

Thr Thr Gln Leu Ile Ile Asn Gly Ser Leu Ala Thr Glu Asn Ile Thr
1 5 10 15

Val Arg Val Asn Asn Ala Ser Lys Asn Thr His Asp Trp Ile Val Gln
20 25 30

Leu Ser Thr Ala
35

<210> SEQ ID NO 111
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 111

-continued

Cys	Asn	Asp	Val	Asn	Phe	Ser	Gly	Lys	Gly	Lys	Cys	Arg	Asn	Val	Ser
1				5					10					15	

Thr Val

<210> SEQ ID NO 112
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 112

Leu	Ile	Thr	Gly	Leu	Ile	Leu	Thr	Arg	Asp	His	Val
1				5					10		

<210> SEQ ID NO 113
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 113

Val	Val	Leu	His	Asn	Val	Thr	Glu	Asp	Phe
1				5					10

<210> SEQ ID NO 114
 <211> LENGTH: 8
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 114

Leu	Ile	Thr	Gly	Leu	Ile	Leu	Thr
1				5			

<210> SEQ ID NO 115
 <211> LENGTH: 29
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 115

Arg	Arg	Gln	Lys	Arg	Ala	Ala	Phe	Gly	Leu	Gly	Ala	Leu	Phe	Leu	Gly
1				5					10					15	

Phe	Leu	Gly	Ala	Ala	Gly	Ser	Thr	Met	Gly	Ala	Ala	Ala
			20					25				

<210> SEQ ID NO 116
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 116

Tyr	Thr	Thr	Val	Pro	Trp	Asn	Lys	Thr	Trp
1				5					10

<210> SEQ ID NO 117
 <211> LENGTH: 26
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 117

Asp	Ile	Trp	Asn	Asn	Leu	Thr	Trp	Gln	Gln	Trp	Asp	Lys	Leu	Ile	Thr
1				5					10					15	

Asn	Tyr	Thr	Gly	Thr	Ile	Phe	Gly	Leu	Leu
			20					25	

<210> SEQ ID NO 118

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<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 118

Asn Cys Ser Phe Asn Val Thr Thr Glu Leu Arg Asp Lys Lys Lys Gln
1             5             10             15

Val Tyr Ser Leu
20

<210> SEQ ID NO 119
<211> LENGTH: 33
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 119

Glu Asn Val Val Gly Asp Thr Arg Ser Ala Tyr Cys Lys Ile Asn Gly
1             5             10             15

Thr Thr Trp Asn Arg Thr Val Glu Glu Val Lys Gly Phe Ser Asn Leu
20             25             30

Ile

<210> SEQ ID NO 120
<211> LENGTH: 39
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 120

Ile Thr Asn Gly Ile Ile Ile Leu Pro Cys Arg Ile Arg Gln Ile Val
1             5             10             15

Ser Ser Trp Met Arg Val Gly Arg Gly Ile Tyr Ala Pro Pro Ile Arg
20             25             30

Gly Asn Ile Thr Cys Asn Ser
35

<210> SEQ ID NO 121
<211> LENGTH: 36
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 121

Thr Thr Gln Leu Leu Ile Asn Gly Ser Leu Ala Glu Gly Asn Ile Thr
1             5             10             15

Val Arg Val Glu Asn Lys Ser Lys Asn Thr Asp Val Trp Ile Val Gln
20             25             30

Leu Val Glu Ala
35

<210> SEQ ID NO 122
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 122

Cys Asn Asp Lys Asp Phe Ser Gly Lys Gly Lys Cys Thr Asn Val Ser
1             5             10             15

Thr Val

<210> SEQ ID NO 123
<211> LENGTH: 12
<212> TYPE: PRT

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<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 123

Asn Ile Thr Gly Leu Leu Leu Thr Ser Asp Thr Pro
1 5 10

<210> SEQ ID NO 124

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 124

Val Phe Leu Pro Asn Val Ile Glu Ser Phe
1 5 10

<210> SEQ ID NO 125

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 125

Asn Ile Thr Gly Leu Leu Thr
1 5

<210> SEQ ID NO 126

<211> LENGTH: 29

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 126

Asp Arg Gln Lys Arg Ala Ala Phe Gly Leu Gly Ala Leu Phe Leu Gly
1 5 10 15

Phe Leu Gly Ala Ala Gly Ser Thr Met Gly Ala Ala Ala
20 25

<210> SEQ ID NO 127

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 127

Tyr Thr Thr Val Pro Trp Asn Asn Ser Trp
1 5 10

<210> SEQ ID NO 128

<211> LENGTH: 26

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 128

Asp Ile Trp Gly Asn Leu Thr Trp Gln Gln Trp Asp Lys Leu Val Ser
1 5 10 15

Asn Tyr Thr Gly Lys Ile Phe Gly Leu Leu
20 25

<210> SEQ ID NO 129

<211> LENGTH: 20

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 129

Asn Cys Ser Phe Asn Ala Thr Thr Glu Leu Arg Asp Arg Lys Lys Gln
1 5 10 15

-continued

Val Tyr Ser Leu
20

<210> SEQ ID NO 130
 <211> LENGTH: 34
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 130

Glu Thr Val Ile Gly Asp Thr Arg Gln Ala Phe Cys Gln Leu Asn Lys
 1 5 10 15

Thr Val Trp Thr Asn Thr Phe Lys Lys Val Arg His Ala Leu Asn Glu
 20 25 30

Thr Tyr

<210> SEQ ID NO 131
 <211> LENGTH: 39
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 131

Thr Lys Asp Asp Asn Ile Thr Ile Pro Cys Arg Ile Arg Gln Ile Val
 1 5 10 15

Arg Leu Trp Gln Arg Val Gly Arg Gly Ile Phe Leu Pro Pro Ile Arg
 20 25 30

Gly Thr Ile Asn Cys Ile Ser
 35

<210> SEQ ID NO 132
 <211> LENGTH: 35
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 132

Ser Thr Gln Phe Leu Leu Asn Gly Thr Leu Glu Glu Lys Val Thr Val
 1 5 10 15

Leu Asp Arg Asn Val Ser Asn Asp Met Asp Thr Ile Ile Val Lys Leu
 20 25 30

Asn Glu Thr
 35

<210> SEQ ID NO 133
 <211> LENGTH: 18
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 133

Cys Asn Asp Lys Asn Phe Thr Gly Ile Gly Gln Cys Lys Asn Val Ser
 1 5 10 15

Thr Val

<210> SEQ ID NO 134
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 134

Asn Ile Thr Gly Ile Leu Phe Ala Gln Gln
 1 5 10

<210> SEQ ID NO 135

-continued

<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 135

Ile Glu Leu Thr Asn Val Thr Glu Tyr Phe
1 5 10

<210> SEQ ID NO 136
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 136

Asn Ile Thr Gly Ile Leu Phe Ala
1 5

<210> SEQ ID NO 137
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 137

His Arg Glu Lys Arg Ala Ala Val Gly Leu Gly Ala Leu Phe Leu Gly
1 5 10 15

Phe Leu Gly Ala Ala Gly Ser Thr Met Gly Ala Ala Ser
20 25

<210> SEQ ID NO 138
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 138

Thr Thr Ser Val Pro Trp Asn Thr Thr Trp
1 5 10

<210> SEQ ID NO 139
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 139

Asp Ile Trp Tyr Asn Met Thr Trp Met Gln Trp Asp Lys Glu Val Ser
1 5 10 15

Asn Tyr Thr Asp Val Ile Tyr Asn Leu Leu
20 25

<210> SEQ ID NO 140
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 140

Asn Cys Thr Phe Asn Val Thr Thr Glu Leu Arg Gly
1 5 10

<210> SEQ ID NO 141
<211> LENGTH: 34
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 141

Glu Asn Ile Val Gly Asn Thr Arg Lys Ala Phe Cys Lys Val Asn Gly

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1	5	10	15
Ser	Gln	Trp	Trp
Asn	Met	Lys	Gln
Asn	Ile	Ile	Gln
Arg	Phe	Lys	Ala
20	25	30	

Glu His

<210> SEQ ID NO 142
 <211> LENGTH: 39
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 142

Lys	Thr	Asn	Ile	Thr	Ile	Ile	Leu	Pro	Cys	Lys	Ile	Arg	Gln	Ile	Val
1		5					10						15		

Asn	Ser	Trp	Met	Arg	Val	Gly	Lys	Gly	Ile	Tyr	Ala	Pro	Pro	Ile	Arg
20						25							30		

Gly	Asn	Leu	Ser	Cys	Asn	Ser
35						

<210> SEQ ID NO 143
 <211> LENGTH: 35
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 143

Thr	Thr	Gln	Leu	Ile	Leu	Asn	Gly	Ser	Ile	Thr	Gln	Gln	Ile	Met	Ile
1		5					10						15		

Arg	Ser	Lys	Asn	Ile	Ser	Ser	Asn	Ser	Phe	Asn	Ile	Ile	Val	Gln	Phe
20							25						30		

Asn	Glu	Thr
35		

<210> SEQ ID NO 144
 <211> LENGTH: 18
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 144

Cys	Asn	Glu	Glu	Asn	Ser	Gln	Glu	Met	Gly	Tyr	Cys	Glu	Asn	Val	Ser
1		5					10						15		

Thr Val

<210> SEQ ID NO 145
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 145

Asn	Ile	Thr	Gly	Leu	Leu	Leu	Thr	Arg	Asp	Gly	Pro
1		5					10				

<210> SEQ ID NO 146
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 146

Val	Phe	Leu	Lys	Asn	Val	Lys	Glu	Asn	Phe
1		5					10		

<210> SEQ ID NO 147
 <211> LENGTH: 8

-continued

<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 147

Asn Ile Thr Gly Leu Leu Thr
1 5

<210> SEQ ID NO 148
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 148

His Arg Ala Lys Arg Ala Ala Phe Ala Ala Gly Ser Thr Met Gly Ala
1 5 10 15

Ala Ala

<210> SEQ ID NO 149
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 149

Tyr Thr Asn Val Pro Trp Asn Arg Asn Trp
1 5 10

<210> SEQ ID NO 150
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 150

Glu Ile Trp Asn Asn Leu Thr Trp Asn Glu Trp Asp Lys Gln Val Ser
1 5 10 15

Asn Tyr Thr Thr Glu Ile Phe Lys Ala Leu
20 25

<210> SEQ ID NO 151
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 151

Asn Cys Ser Phe Asn Val Thr Thr Glu Leu Arg Asp Lys Lys Lys Gln
1 5 10 15

Val Tyr Ser Leu
20

<210> SEQ ID NO 152
<211> LENGTH: 34
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 152

Glu Asn Val Val Gly Asp Thr Arg Lys Ala Tyr Cys Thr Leu Asn Gly
1 5 10 15

Ser Asn Trp Glu Glu Ala Met Ser Lys Val Lys Lys Glu Leu Glu Lys
20 25 30

Ile Thr

<210> SEQ ID NO 153
<211> LENGTH: 39

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<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 153

Asn Lys Ser Glu Glu Ile Ile Leu Pro Cys Arg Ile Lys Gln Ile Val
1 5 10 15

Asn Leu Trp Thr Arg Val Gly Lys Gly Ile Tyr Ala Pro Pro Ile Arg
20 25 30

Gly Asn Ile Ser Cys Lys Ser
35

<210> SEQ ID NO 154
<211> LENGTH: 36
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 154

Thr Thr Gln Leu Ile Leu Asn Gly Ser Leu Ala Glu Gly Asn Val Thr
1 5 10 15

Ile Arg Val Lys Asn Ala Ser Ser Asn Ser Gly Asn Tyr Ile Val Gln
20 25 30

Leu Ala Arg Ser
35

<210> SEQ ID NO 155
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 155

Cys Asn Asp Thr Asn Tyr Leu Gly Thr Gly Lys Cys His Asn Val Ser
1 5 10 15

Thr Val

<210> SEQ ID NO 156
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (10)..(10)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<400> SEQUENCE: 156

Asn Ile Thr Gly Ile Leu Leu Ala Arg Xaa Glu Gly
1 5 10

<210> SEQ ID NO 157
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 157

Val Phe Leu Pro Asn Val Thr Glu Asn Phe
1 5 10

<210> SEQ ID NO 158
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 158

Asn Ile Thr Gly Ile Leu Leu Ala

-continued

1 5

<210> SEQ ID NO 159
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 159

Lys Arg Gln Lys Arg Ala Ala Phe Gly Leu Gly Ala Leu Phe Leu Gly
1 5 10 15

Phe Leu Gly Ala Ala Gly Ser Thr Met Gly Ala Ala Ala
 20 25

<210> SEQ ID NO 160
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 160

Tyr Thr Thr Val Pro Trp Asn Lys Thr Trp
1 5 10

<210> SEQ ID NO 161
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 161

Tyr Ile Trp Asn Asn Leu Thr Trp Glu Gln Trp Asp Val His Val Ser
1 5 10 15

Asn Tyr Thr Gly Val Ile Phe Gly Leu Leu
 20 25

<210> SEQ ID NO 162
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 162

Asn Cys Thr Phe Asn Met Thr Thr Glu Val Arg Asp Lys Glu Lys Lys
1 5 10 15

Val Tyr Ser Leu
 20

<210> SEQ ID NO 163
<211> LENGTH: 34
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 163

Glu Asn Ile Val Gly Asp Thr Arg Arg Ala Tyr Cys Gln Ile Asn Arg
1 5 10 15

Thr Val Trp Asp Glu Arg Leu Asn Glu Thr Gly Gln Ala Leu Arg Glu
 20 25 30

Leu Phe

<210> SEQ ID NO 164
<211> LENGTH: 39
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 164

-continued

Lys Gly Asp Asn Thr Thr Phe Phe Pro Cys Arg Ile Arg Gln Ile Val
1 5 10 15

Asn Ser Trp Met Arg Val Gly Lys Gly Ile Tyr Ala Pro Pro Ile Arg
20 25 30

Gly Val Ile Ser Cys Thr Ser
35

<210> SEQ ID NO 165
<211> LENGTH: 31
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (23)..(23)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (26)..(26)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (30)..(30)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (31)..(31)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<400> SEQUENCE: 165

Thr Thr Gln Leu Ile Leu Asn Gly Ser Val Ala Glu Asn Lys Thr Ile
1 5 10 15

Ala Arg Arg Asn Gly Tyr Xaa Phe Leu Xaa Gln Phe Gln Xaa Xaa
20 25 30

<210> SEQ ID NO 166
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 166

Cys Asn Met Pro Asn Phe Asn Gly Thr Gly Thr Gly Arg Cys Asn Asn
1 5 10 15

Ile Ser Thr Val
20

<210> SEQ ID NO 167
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 167

Asn Val Thr Gly Ile Ile Leu Glu Thr Gly His Gly
1 5 10

<210> SEQ ID NO 168
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 168

Val Gln Ile Pro Asn Val Thr Glu Asn Phe
1 5 10

<210> SEQ ID NO 169
<211> LENGTH: 8
<212> TYPE: PRT

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<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 169

Asn Val Thr Gly Ile Ile Leu Glu
1 5

<210> SEQ ID NO 170

<211> LENGTH: 29

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 170

Ser Arg Glu Lys Arg Ala Ala Phe Gly Leu Gly Ala Leu Phe Leu Gly
1 5 10 15

Phe Leu Gly Ala Ala Gly Ser Thr Met Gly Ala Ala Ser
20 25

<210> SEQ ID NO 171

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 171

Tyr Thr Thr Val Pro Trp Asn Asn Thr Trp
1 5 10

<210> SEQ ID NO 172

<211> LENGTH: 26

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 172

Glu Ile Trp Asn Asn Leu Thr Trp Gln Asp Trp Asp Lys Arg Val Lys
1 5 10 15

Asn Tyr Ser Gly Val Ile Phe Ser Leu Ile
20 25

<210> SEQ ID NO 173

<211> LENGTH: 20

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 173

Asn Cys Ser Phe Asn Gln Thr Thr Glu Phe Arg Asp Lys Lys Lys Gln
1 5 10 15

Ile Tyr Ser Leu
20

<210> SEQ ID NO 174

<211> LENGTH: 34

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 174

Glu Asn Val Val Gly Asp Thr Arg Lys Ala Tyr Cys His Val Asn Ala
1 5 10 15

Thr Leu Trp Asp Gln Ser Leu Ser Arg Ala Met Glu Ala Ile Asn Lys
20 25 30

Thr Leu

<210> SEQ ID NO 175

<211> LENGTH: 39

-continued

<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 175

His Arg Asn Gly Thr Leu Val Ala Pro Cys Arg Leu Arg Gln Ile Val
1 5 10 15

Asn His Trp Gly Ile Val Ser Lys Gly Ile Tyr Leu Pro Pro Arg Arg
20 25 30

Gly Thr Leu Lys Cys Asn Ser
35

<210> SEQ ID NO 176
<211> LENGTH: 35
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 176

Ala Thr Ala Leu His Leu Asn Gly Thr Leu Glu Glu Asn Glu Thr Lys
1 5 10 15

Ala Tyr Phe Ala Asp Val Asn Val Asn Pro Pro Leu Leu Val Lys Phe
20 25 30

Asn Glu Ser
35

<210> SEQ ID NO 177
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 177

Cys Asn Asp Gln Gly Tyr Thr Gly Asn Gly Thr Cys Ser Asn Val Thr
1 5 10 15

Val Val

<210> SEQ ID NO 178
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 178

Asn Ile Thr Gly Leu Ile Met Thr Ala Glu Lys Gly
1 5 10

<210> SEQ ID NO 179
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 179

Val Pro Leu Ala Asn Val Ser Ile Asp Phe
1 5 10

<210> SEQ ID NO 180
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 180

Asn Ile Thr Gly Leu Ile Met Thr
1 5

<210> SEQ ID NO 181

-continued

<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 181

Thr Arg Ser Arg Arg Asp Val Gly Ile Gly Leu Leu Phe Leu Gly Phe
1 5 10 15

Leu Ser Ala Ala Gly Ser Thr Met Gly Ala Ala Ser
 20 25

<210> SEQ ID NO 182
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 182

His Ser Ser Val Pro Trp Asn Leu Ser Trp
1 5 10

<210> SEQ ID NO 183
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 183

Cys Ile Trp Asn Asn Leu Thr Trp Gln Glu Trp Asp Arg Leu Val Gln
1 5 10 15

Asn Ala Ser Glu Thr Ile Tyr Ser Leu Leu
 20 25

<210> SEQ ID NO 184
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 184

Asn Cys Ser Phe Gln Gln Thr Thr Glu Phe Arg Asp Lys Lys Lys Gln
1 5 10 15

Ile Tyr Ser Leu
 20

<210> SEQ ID NO 185
<211> LENGTH: 33
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 185

Glu Asn Val Val Gly Asp Thr Arg Lys Ala Tyr Cys Trp Val Asn Ala
1 5 10 15

Thr Leu Trp Tyr Lys Ser Leu Asp Asn Ala Trp Arg Leu Pro Lys His
 20 25 30

Arg

<210> SEQ ID NO 186
<211> LENGTH: 40
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 186

Ala Leu Asn Lys Thr His Leu Leu Ala Gln Cys Arg Leu Arg Gln Ile
1 5 10 15

Val Asn His Trp Gly Ile Val Ser Lys Gly Ile Tyr Leu Pro Pro Arg

-continued

20	25	30
Lys Gly His Ile Lys Cys Val Ser		
35	40	

<210> SEQ ID NO 187
 <211> LENGTH: 35
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 187

Ala Thr Ala Leu His Leu Asn Gly Thr Leu Glu Glu Lys Glu Thr Lys
1 5 10 15

Ala Tyr Phe Val Asn Ala Thr Asn Asn Pro Pro Leu Leu Val Lys Phe
20 25 30

Asn Glu Ser
35

<210> SEQ ID NO 188
 <211> LENGTH: 18
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 188

Cys Lys Asp Gln Asn Tyr Thr Gly Asn Gly Ile Cys Tyr Asn Val Thr
1 5 10 15

Val Val

<210> SEQ ID NO 189
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 189

Asn Ile Thr Gly Leu Ile Met Thr Ala Glu Gly Gly
1 5 10

<210> SEQ ID NO 190
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 190

Val Pro Ile Ala Asn Val Ser Ile Lys Phe
1 5 10

<210> SEQ ID NO 191
 <211> LENGTH: 8
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 191

Asn Ile Thr Gly Leu Ile Met Thr
1 5

<210> SEQ ID NO 192
 <211> LENGTH: 28
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 192

Thr Arg Ser Arg Arg Asp Val Gly Ile Gly Leu Leu Phe Leu Gly Phe
1 5 10 15

-continued

Leu Ser Ala Ala Gly Ser Thr Met Gly Ala Ala Ser
20 25

<210> SEQ ID NO 193
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 193

His Ser Ser Val Pro Trp Asn Leu Ser Trp
1 5 10

<210> SEQ ID NO 194
 <211> LENGTH: 26
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 194

Cys Ile Trp Asn Asn Leu Thr Trp Gln Glu Trp Asp Lys Leu Val Gln
1 5 10 15

Asn Ser Ser Glu Thr Ile Tyr Ser Leu Leu
20 25

<210> SEQ ID NO 195
 <211> LENGTH: 20
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 195

Asn Cys Ser Phe Asn Gln Thr Thr Glu Phe Arg Asp Lys Lys Lys Gln
1 5 10 15

Ile Tyr Ser Leu
20

<210> SEQ ID NO 196
 <211> LENGTH: 34
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 196

Glu Asn Val Val Gly Asp Thr Arg Lys Ala Tyr Cys Ser Val Asn Ala
1 5 10 15

Thr Thr Trp Tyr Arg Asn Leu Asp Trp Ala Met Ala Ala Ile Asn Thr
20 25 30

Thr Met

<210> SEQ ID NO 197
 <211> LENGTH: 39
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 197

Asn Thr His Gly Thr Leu Val Ala Pro Cys Arg Leu Arg Gln Ile Val
1 5 10 15

Asn His Trp Gly Ile Val Ser Lys Gly Val Tyr Leu Pro Pro Arg Arg
20 25 30

Gly Thr Val Lys Cys His Ser
35

<210> SEQ ID NO 198
 <211> LENGTH: 35
 <212> TYPE: PRT

-continued

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 198

Ala Thr Ala Leu His Leu Asn Gly Ser Leu Glu Glu Glu Glu Thr Lys
1 5 10 15

Ala Tyr Phe Val Asn Thr Ser Val Asn Thr Pro Leu Leu Val Lys Phe
20 25 30

Asn Val Ser
35

<210> SEQ ID NO 199

<211> LENGTH: 18

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 199

Cys Arg Asp Gln Asn Phe Thr Gly Lys Gly Gln Cys Ser Asn Val Ser
1 5 10 15

Val Val

<210> SEQ ID NO 200

<211> LENGTH: 12

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 200

Asn Ile Thr Gly Leu Ile Met Thr Ala Glu Lys Asp
1 5 10

<210> SEQ ID NO 201

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 201

Val Thr Leu Ala Asn Val Ser Ile Arg Phe
1 5 10

<210> SEQ ID NO 202

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 202

Asn Ile Thr Gly Leu Ile Met Thr
1 5

<210> SEQ ID NO 203

<211> LENGTH: 28

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 203

Thr Arg Ser Arg Arg Asp Val Gly Ile Gly Leu Leu Phe Leu Gly Phe
1 5 10 15

Leu Ser Ala Ala Gly Ser Thr Met Gly Ala Ala Ser
20 25

<210> SEQ ID NO 204

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

-continued

<400> SEQUENCE: 204

His Ser Ser Val Pro Trp Asn Leu Thr Trp
 1 5 10

<210> SEQ ID NO 205

<211> LENGTH: 26

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 205

Cys Ile Trp Asn Asn Leu Thr Trp Gln Glu Trp Asp Arg Leu Val Glu
 1 5 10 15

Asn Ser Thr Gly Thr Ile Tyr Ser Leu Leu
 20 25

<210> SEQ ID NO 206

<211> LENGTH: 20

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 206

Lys Cys Ser Phe Asn Gln Thr Thr Glu Phe Arg Asp Lys Lys Lys Gln
 1 5 10 15

Ile Tyr Ser Leu
 20

<210> SEQ ID NO 207

<211> LENGTH: 34

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 207

Glu Asn Val Val Gly Asp Thr Arg Lys Ala Phe Cys Ile Val Asn Ala
 1 5 10 15

Thr Glu Trp Ser Asn Ala Leu Asn Glu Ser Lys Ile Ala Ile Asn Glu
 20 25 30

Thr Leu

<210> SEQ ID NO 208

<211> LENGTH: 39

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 208

Gly Lys Asn Gly Ser Ile Val Ala Gln Cys Arg Leu Arg Gln Ile Val
 1 5 10 15

Asn His Trp Gly Ile Val Ser Lys Gly Ile Tyr Leu Pro Pro Arg Arg
 20 25 30

Gly Thr Ile Lys Cys Thr Ser
 35

<210> SEQ ID NO 209

<211> LENGTH: 35

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 209

Ala Thr Ala Leu His Leu Asn Gly Thr Leu Glu Glu Asn Lys Thr Thr
 1 5 10 15

Ala Tyr Phe Ala Asp Ser Thr His Asn Gln Pro Leu Leu Val Lys Phe
 20 25 30

-continued

His Lys Ala
35

<210> SEQ ID NO 210
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 210

Cys Lys Asp Gln Asn Phe Thr Gly Lys Gly Pro Cys Ser Asn Val Ser
1 5 10 15

Val Val

<210> SEQ ID NO 211
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 211

Asn Ile Thr Gly Leu Ile Leu Thr Ala Glu Gly Gly
1 5 10

<210> SEQ ID NO 212
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 212

Val Pro Leu Ala Asn Met Thr Val Asp Phe
1 5 10

<210> SEQ ID NO 213
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 213

Asn Ile Thr Gly Leu Ile Leu Thr
1 5

<210> SEQ ID NO 214
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 214

Thr Arg Ser Lys Arg Asp Val Gly Ile Gly Leu Leu Phe Leu Gly Phe
1 5 10 15

Leu Ser Ala Ala Gly Ser Thr Met Gly Ala Ala Ser
20 25

<210> SEQ ID NO 215
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 215

His Ser Ser Val Pro Trp Asn Ile Thr Trp
1 5 10

<210> SEQ ID NO 216
<211> LENGTH: 26
<212> TYPE: PRT

-continued

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 216

Cys Ile Trp Asn Asn Leu Thr Trp Gln Glu Trp Asp Lys Leu Val Lys
1 5 10 15

Asn Ser Ser Glu Thr Ile Tyr Ser Leu Leu
20 25

<210> SEQ ID NO 217

<211> LENGTH: 20

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 217

Asn Cys Ser Phe Asn Gln Thr Thr Glu Phe Arg Asp Lys Lys Lys Gln
1 5 10 15

Ile Tyr Ser Leu
20

<210> SEQ ID NO 218

<211> LENGTH: 34

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 218

Glu Asn Ile Val Gly Asn Thr Arg Lys Ala Tyr Cys Thr Val Asn Tyr
1 5 10 15

Gln Glu Trp Ser His Ala Ile Gly Glu Ala Lys Lys Val Ala Glu Glu
20 25 30

Ala Leu

<210> SEQ ID NO 219

<211> LENGTH: 39

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 219

Asn Ser Ser Arg Pro Leu Val Ala Pro Cys Lys Leu Arg Gln Ile Val
1 5 10 15

Asn His Trp Gly Ile Val Ser Lys Gly Ile Tyr Leu Pro Pro Arg Arg
20 25 30

Gly Asn Leu Thr Cys Arg Ser
35

<210> SEQ ID NO 220

<211> LENGTH: 35

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 220

Ala Thr Ala Leu His Leu Asn Gly Thr Leu Glu Lys His Asn Thr Thr
1 5 10 15

Ala Tyr Phe Ala Thr Pro Thr His Asn Lys Pro Leu Leu Ile Lys Phe
20 25 30

Asn Lys Ser
35

<210> SEQ ID NO 221

<211> LENGTH: 18

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

-continued

<400> SEQUENCE: 221

Cys Asn Ser Ala Asn Phe Thr Gly Val Gly Thr Cys Asn Asn Val Ser
1 5 10 15

Val Val

<210> SEQ ID NO 222

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 222

Asn Ile Thr Gly Phe Ile Thr Thr Trp Asp
1 5 10

<210> SEQ ID NO 223

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 223

Val Pro Leu Ser Asn Val Ser Val Asp Phe
1 5 10

<210> SEQ ID NO 224

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 224

Asn Ile Thr Gly Phe Ile Thr Thr
1 5

<210> SEQ ID NO 225

<211> LENGTH: 28

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 225

Thr Arg Ser Lys Arg Asp Val Gly Ile Gly Leu Leu Phe Leu Gly Phe
1 5 10 15

Leu Ser Ala Ala Gly Ser Thr Met Gly Ala Ala Ser
20 25

<210> SEQ ID NO 226

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 226

His Ser Asn Val Val Trp Asn Met Thr Trp
1 5 10

<210> SEQ ID NO 227

<211> LENGTH: 26

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 227

Cys Ile Trp Asn Asn Leu Thr Trp Gln Gln Trp Glu Lys Leu Val Ala
1 5 10 15

Asn Ser Thr Glu Glu Ile Tyr Thr Leu Leu
20 25

-continued

<210> SEQ ID NO 228
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 228

Asn Cys Ser Phe Asn Gln Thr Thr Glu Phe Arg Asp Lys Lys Lys Asn
1 5 10 15

Ile Tyr Ser Leu
20

<210> SEQ ID NO 229
<211> LENGTH: 34
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 229

Glu Asn Ile Phe Gly Asp Thr Arg Lys Ala Phe Cys Tyr Val Asn Ala
1 5 10 15

Thr Glu Trp Arg Arg Thr Leu Glu Met Ala Lys Thr Ala Leu Arg Glu
20 25 30

Ala Thr

<210> SEQ ID NO 230
<211> LENGTH: 39
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 230

Thr Asp Gln Thr Asn Leu Val Ala Pro Cys Arg Leu Arg Gln Ile Val
1 5 10 15

Asn His Trp Gly Ile Val Ala Lys Ala Ile Tyr Leu Pro Pro Arg Arg
20 25 30

Gly Thr Val Lys Cys Val Ser
35

<210> SEQ ID NO 231
<211> LENGTH: 35
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 231

Ala Ser Trp Leu His Leu Asn Gly Thr Tyr Glu Lys Glu Lys Thr Lys
1 5 10 15

Val Tyr His Thr Asn Val Thr Asn Asn Pro Pro Leu Leu Val Lys Phe
20 25 30

Asn Glu Thr
35

<210> SEQ ID NO 232
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 232

Cys Asn Asp Ala Asn Phe Thr Gly Ala Gly Glu Cys Thr Asn Val Ser
1 5 10 15

Val Val

-continued

<210> SEQ ID NO 233
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 233

Asn Ile Thr Gly Phe Leu Met Thr
1 5

<210> SEQ ID NO 234
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 234

Val Pro Leu Pro Asn Val Ser Ile Glu Phe
1 5 10

<210> SEQ ID NO 235
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 235

Asn Ile Thr Gly Phe Leu Met Thr
1 5

<210> SEQ ID NO 236
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 236

Thr Arg Ser Lys Arg Asp Val Gly Ile Gly Leu Leu Phe Leu Gly Phe
1 5 10 15

Leu Ser Ala Ala Gly Ser Thr Met Gly Ala Ala Ser
20 25

<210> SEQ ID NO 237
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 237

His Ser Thr Val Val Trp Asn Asp Thr Trp
1 5 10

<210> SEQ ID NO 238
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 238

Cys Ile Trp Gln Asn Leu Thr Trp Gln Gln Trp Ser Glu Leu Val Asp
1 5 10 15

Asn Ser Thr Asp Thr Ile Tyr Thr Leu Leu
20 25

<210> SEQ ID NO 239
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 239

-continued

Glu Cys Phe Phe Asn Gln Thr Thr Glu Phe Arg Asp Lys Lys Lys Gln
1 5 10 15

Ile Tyr Ser Leu
20

<210> SEQ ID NO 240
<211> LENGTH: 35
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 240

Arg Asn Val Ile Gly Asp Thr Arg Lys Ala Phe Cys Tyr Val Asn Lys
1 5 10 15

Thr Leu Trp Glu Asn Ala Leu Arg Gln Gln Lys Gln Thr Lys Leu Leu
20 25 30

Lys Thr Pro
35

<210> SEQ ID NO 241
<211> LENGTH: 39
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 241

Glu Ile Asp Gly Met Leu Ile Ala Pro Cys Arg Leu Arg Gln Ile Val
1 5 10 15

Asn His Trp Gly Ile Val Ser Arg Gly Ile Tyr Leu Pro Pro Arg Glu
20 25 30

Gly Gln Val Lys Cys Val Ser
35

<210> SEQ ID NO 242
<211> LENGTH: 35
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 242

Ala Thr Ala Leu His Leu Asn Gly Ser Val Glu Glu Gly Glu Thr Lys
1 5 10 15

Ala Tyr Tyr Val Glu Ala Lys His Asn Ala Pro Leu Leu Ile Lys Phe
20 25 30

Asn Lys Ala
35

<210> SEQ ID NO 243
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 243

Cys Lys Asp Glu Asn Phe Arg Gly Lys Gly Asn Cys Thr Thr Val Arg
1 5 10 15

Ile Val

<210> SEQ ID NO 244
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 244

Ser Ile Thr Gly Phe Ile Met Thr

-continued

1 5

<210> SEQ ID NO 245
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 245

Ile Pro Ile Asn Ile Thr Thr Glu Phe
 1 5

<210> SEQ ID NO 246
 <211> LENGTH: 8
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 246

Ser Ile Thr Gly Phe Ile Met Thr
 1 5

<210> SEQ ID NO 247
 <211> LENGTH: 28
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 247

Ser Arg Ser Lys Arg Glu Ile Gly Met Gly Met Ile Phe Met Gly Phe
 1 5 10 15

Leu Ser Ala Ala Gly Ser Thr Met Gly Ala Ala Ser
 20 25

<210> SEQ ID NO 248
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 248

His Ser Ser Leu Glu Trp Asn Ser Ser Trp
 1 5 10

<210> SEQ ID NO 249
 <211> LENGTH: 26
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 249

Cys Ile Trp Ser Gln Tyr Thr Trp Gln Gln Trp Asp Glu Glu Ile Arg
 1 5 10 15

Asn Ile Ser Asp Val Ile Tyr Leu Ser Leu
 20 25

<210> SEQ ID NO 250
 <211> LENGTH: 20
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 250

Glu Cys Asn Phe Asn Gln Ser Thr Gly Phe Lys Asp Lys Lys Gln Lys
 1 5 10 15

Met Lys Ala Ile
 20

<210> SEQ ID NO 251

-continued

<211> LENGTH: 34
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 251

Glu Ile Ala Thr Gly Asp Thr Arg Lys Ala Phe Cys Thr Val Asn Lys
1 5 10 15

Thr Leu Trp Glu Gln Ala Arg Asn Lys Thr Glu His Val Leu Ala Glu
20 25 30

His Trp

<210> SEQ ID NO 252
<211> LENGTH: 39
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 252

Ile Thr Asn Gly Ala Leu Ile Ala His Cys Arg Ile Lys Gln Ile Val
1 5 10 15

Asn His Trp Gly Ile Val Ser Lys Gly Ile Tyr Leu Ala Pro Arg Arg
20 25 30

Gly Asn Val Ser Cys Thr Ser
35

<210> SEQ ID NO 253
<211> LENGTH: 32
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 253

Ala Thr Trp Leu Leu Leu Asn Gly Thr Tyr Gln Thr Asn Thr Ser Val
1 5 10 15

Val Met Asn Gly Ala Lys Met Asn Leu Cys Leu Asp Leu Glu Lys Asn
20 25 30

<210> SEQ ID NO 254
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 254

Cys Glu Asp Glu Asp Phe Thr Gly Val Gly Met Cys Lys Asn Val Ser
1 5 10 15

Val Val

<210> SEQ ID NO 255
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 255

Ser Ile Thr Gly Ile Met Leu Glu Gly
1 5

<210> SEQ ID NO 256
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 256

Val Arg Leu Asn Thr Ser Val Trp Phe
1 5

-continued

<210> SEQ ID NO 257
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 257

Ser Ile Thr Gly Ile Met Leu Glu
1 5

<210> SEQ ID NO 258
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 258

Ser Arg Gln Lys Arg Gly Ile Gly Ile Gly Leu Phe Phe Leu Gly Leu
1 5 10 15
Leu Ser Ala Ala Gly Ser Thr Met Gly Ala Ala Ser
20 25

<210> SEQ ID NO 259
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 259

His Thr Thr Val Pro Trp Asn Asn Ser Trp
1 5 10

<210> SEQ ID NO 260
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 260

Cys Ile Trp Glu Asn Met Thr Trp Gln Glu Trp Asp Arg Leu Val Gln
1 5 10 15
Asn Ser Thr Gly Gln Ile Tyr Asn Ile Leu
20 25

<210> SEQ ID NO 261
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 261

Asn Cys Thr Phe Asn Thr Thr Thr Glu Leu Arg Asp Lys Lys Gln Gln
1 5 10 15
Ile Tyr Ser Leu
20

<210> SEQ ID NO 262
<211> LENGTH: 34
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 262

Glu Ala Leu Ile Gly Asn Pro Arg Lys Ala Ser Cys His Leu Asn Gly
1 5 10 15
Thr Leu Trp Asn Asn Ile Leu Asn Arg Ile Lys Gln Lys Ile Lys Asn
20 25 30

-continued

Ser Thr

<210> SEQ ID NO 263
<211> LENGTH: 39
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 263

Ser Asp Thr Ser Glu Tyr Ile Leu Pro Cys Lys Ile Arg Gln Val Val
1 5 10 15

Asn Ser Trp Met Arg Val Gly Lys Gly Ile Phe Ala Pro Pro Arg Arg
20 25 30

Gly Thr Ile Thr Cys Asn Ser
35

<210> SEQ ID NO 264
<211> LENGTH: 39
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 264

Thr Thr Gln Leu Ile Leu Asn Gly Ser Leu Ala Glu Glu Glu Glu Val
1 5 10 15

Val Phe Arg Thr Lys Asn Met Thr Ala Pro Gly Leu Ser Asp Thr Val
20 25 30

Ile Val Gln Leu Lys Arg Ala
35

<210> SEQ ID NO 265
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 265

Cys Asn Asp Lys Asp Tyr Pro Gly Lys Gly Lys Cys Lys Asn Val Ser
1 5 10 15

Thr Val

<210> SEQ ID NO 266
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 266

Thr Ile Thr Gly Leu Leu Leu Glu Val Gln Asn Gly
1 5 10

<210> SEQ ID NO 267
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 267

Val Ile Leu Ile Asn Val Thr Glu Glu Phe
1 5 10

<210> SEQ ID NO 268
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 268

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Thr Ile Thr Gly Leu Leu Glu
1 5

<210> SEQ ID NO 269
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 269

Asp Arg Ser Lys Arg Ala Ala Phe Gly Leu Gly Ala Leu Phe Leu Gly
1 5 10 15

Phe Leu Gly Ala Ala Gly Ser Thr Met Gly Ala Ala Ser
20 25

<210> SEQ ID NO 270
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 270

Tyr Ser Thr Val Pro Trp Asn Thr Ser Trp
1 5 10

<210> SEQ ID NO 271
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 271

Asp Ile Trp Ser Asn Leu Thr Trp Gln Gln Trp Asp Lys Leu Val Glu
1 5 10 15

Asn His Thr Gly Thr Ile Phe Gln Leu Leu
20 25

<210> SEQ ID NO 272
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 272

Asn Cys Ser Phe Asn Val Thr Thr Glu Leu Arg Asp Lys Lys Lys Gln
1 5 10 15

Val Tyr Ser Leu
20

<210> SEQ ID NO 273
<211> LENGTH: 31
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 273

Glu Asn Ile Ile Gly Asp Thr Arg Gln Ala Tyr Cys Glu Val Asn Arg
1 5 10 15

Thr Trp Glu Gln Ile Trp Asn Thr Thr Lys Gln Ile Ile Ile Asn
20 25 30

<210> SEQ ID NO 274
<211> LENGTH: 39
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 274

Asn Thr Thr Gly Asn Ile Thr Leu Gln Cys Arg Ile Arg Gln Ile Val

-continued

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1           5           10           15
Asn Leu Trp Thr Arg Val Gly Lys Gly Ile Tyr Ala Pro Pro Ile Lys
      20           25           30

Gly Pro Ile Asn Cys Leu Ser
      35

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<210> SEQ ID NO 275
<211> LENGTH: 32
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

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<400> SEQUENCE: 275

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Ser Thr Gln Leu Ile Leu Asn Gly Ser Leu Ala Thr Ser Asn Ile Val
1           5           10           15

Ile Arg Asn Asn Ser Lys Asp Thr Leu Leu Val Gln Leu Asn Glu Ser
      20           25           30

```

```

<210> SEQ ID NO 276
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

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<400> SEQUENCE: 276

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Cys Asn Asp Gln Asp Phe Asn Gly Thr Gly Glu Cys Asn Asn Val Ser
1           5           10           15

Thr Val

```

```

<210> SEQ ID NO 277
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

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<400> SEQUENCE: 277

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```

Asn Ile Thr Gly Ile Ile Leu Asp Tyr Thr
1           5           10

```

```

<210> SEQ ID NO 278
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

```

```

<400> SEQUENCE: 278

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```

Val Leu Ile Pro Asn Val Thr Glu Arg Phe
1           5           10

```

```

<210> SEQ ID NO 279
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

```

```

<400> SEQUENCE: 279

```

```

Asn Ile Thr Gly Ile Ile Leu Asp
1           5

```

```

<210> SEQ ID NO 280
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

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<400> SEQUENCE: 280

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```

Thr Arg Gln Lys Arg Ala Ala Phe Gly Leu Gly Ala Leu Phe Leu Gly
1           5           10           15

```

-continued

Phe Leu Gly Ala Ala Gly Ser Thr Met Gly Ala Ala Ser
20 25

<210> SEQ ID NO 281
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 281

Tyr Thr Asn Val Pro Trp Asn Thr Thr Trp
1 5 10

<210> SEQ ID NO 282
 <211> LENGTH: 26
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 282

Thr Ile Trp Gly Asn Met Thr Trp Gln Asn Trp Asp Glu Gln Val Arg
1 5 10 15

Asn Tyr Ser Gly Val Ile Phe Gly Leu Leu
20 25

<210> SEQ ID NO 283
 <211> LENGTH: 20
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 283

Asn Cys Thr Phe Asn Met Thr Thr Glu Val Arg Asp Lys Lys Lys Glu
1 5 10 15

Val Tyr Ser Leu
20

<210> SEQ ID NO 284
 <211> LENGTH: 34
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 284

Glu Asn Ile Val Gly Asp Pro Arg Gln Ala His Cys Asn Val Ser Lys
1 5 10 15

Ala Gln Trp Met Lys Lys Leu Asn Glu Thr Ala Leu Ala Ile Ala Lys
20 25 30

Tyr Asp

<210> SEQ ID NO 285
 <211> LENGTH: 39
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 285

Ser Asp Asn Ser Thr Leu Thr Ile Ser Cys Arg Ile Arg Gln Ile Val
1 5 10 15

Asn Ser Trp Met Arg Val Gly Lys Gly Ile Tyr Ala Pro Pro Ile Arg
20 25 30

Gly Asn Ile Thr Cys Ile Ser
35

<210> SEQ ID NO 286
 <211> LENGTH: 35
 <212> TYPE: PRT

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<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 286

Thr Thr Gln Val Ile Leu Asn Gly Ser Leu Ala Glu Glu Gly Pro Ala
1 5 10 15

Val Ala Arg Ser Glu Asp Leu Arg His Lys Ile Ile Ile Ile Gln Phe
20 25 30

Lys Glu Gly
35

<210> SEQ ID NO 287

<211> LENGTH: 18

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 287

Cys Asn Asp Gln Lys Phe Asn Gly Thr Gly Pro Cys Thr Asn Ile Ser
1 5 10 15

Thr Val

<210> SEQ ID NO 288

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 288

Asn Ile Thr Gly Leu Leu Leu Glu Ala Gly Gly
1 5 10

<210> SEQ ID NO 289

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 289

Val Tyr Met Pro Asn Val Ser Glu Ser Phe
1 5 10

<210> SEQ ID NO 290

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 290

Asn Ile Thr Gly Leu Leu Leu Glu
1 5

<210> SEQ ID NO 291

<211> LENGTH: 29

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 291

Ala Arg Glu Lys Arg Ala Ala Phe Gly Leu Gly Ala Leu Phe Leu Gly
1 5 10 15

Phe Leu Gly Ala Ala Gly Ser Thr Met Gly Ala Ala Ser
20 25

<210> SEQ ID NO 292

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

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<400> SEQUENCE: 292

Tyr Thr Thr Val Pro Trp Asn Ala Thr Trp
1 5 10

<210> SEQ ID NO 293

<211> LENGTH: 26

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 293

Glu Ile Trp Asn Asn Leu Thr Trp Gln Asp Trp Asp Lys Lys Val Lys
1 5 10 15

Asn Tyr Ser Gly Val Ile Phe Ser Leu Ile
20 25

<210> SEQ ID NO 294

<211> LENGTH: 20

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 294

Asn Cys Ser Phe Asn Ile Thr Thr Glu Leu Arg Asp Lys Lys Lys Gln
1 5 10 15

Val Tyr Ser Leu
20

<210> SEQ ID NO 295

<211> LENGTH: 34

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 295

Pro Lys Ile Val Gly Asp Val Arg Glu Ala His Cys Asn Ile Ser Lys
1 5 10 15

Leu Thr Trp Glu Lys Gln Arg Lys Tyr Thr Leu Glu Ile Ile Lys Lys
20 25 30

Glu Ala

<210> SEQ ID NO 296

<211> LENGTH: 39

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 296

Thr Asp Asn Thr Thr Ile Thr Leu Lys Cys Arg Ile Arg Gln Ile Val
1 5 10 15

Asn Gln Trp Met Arg Val Gly Lys Gly Ile Phe Ala Pro Pro Ile Lys
20 25 30

Gly Val Leu Ser Cys Asn Ser
35

<210> SEQ ID NO 297

<211> LENGTH: 33

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 297

Thr Thr Gln Leu Ile Ile Asn Gly Ser Leu Ala Thr Lys Asn Val Thr
1 5 10 15

Val Arg Ser Lys Asn Phe Ala Asp Ile Ile Leu Val Gln Phe Ser Glu
20 25 30

-continued

Gly

<210> SEQ ID NO 298
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 298

Cys Asn Glu Lys Asp Phe Lys Gly Lys Gly Glu Cys Lys Asn Val Ser
1 5 10 15

Thr Val

<210> SEQ ID NO 299
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 299

Asn Ile Thr Gly Met Phe Leu Thr Ala Ala Gln
1 5 10

<210> SEQ ID NO 300
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 300

Val His Leu Pro Asn Val Thr Glu Lys Phe
1 5 10

<210> SEQ ID NO 301
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 301

Asn Ile Thr Gly Met Phe Leu Thr
1 5

<210> SEQ ID NO 302
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 302

Lys Arg Glu Lys Arg Ala Ala Phe Gly Leu Gly Ala Leu Phe Leu Gly
1 5 10 15

Phe Leu Gly Ala Ala Gly Ser Thr Met Gly Ala Ala Ser
20 25

<210> SEQ ID NO 303
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 303

Tyr Thr Thr Val Pro Trp Asn Asp Thr Trp
1 5 10

<210> SEQ ID NO 304
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

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<400> SEQUENCE: 304

Ala Ile Trp Gly Asn Leu Thr Trp Gln Glu Trp Asp Arg Lys Val Arg
1 5 10 15

Asn Tyr Ser Gly Thr Ile Phe Ser Leu Ile
20 25

<210> SEQ ID NO 305

<211> LENGTH: 20

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 305

Lys Cys Phe Phe Asn Met Thr Thr Glu Leu Arg Asp Lys Gln Lys Gln
1 5 10 15

Val Tyr Ser Leu
20

<210> SEQ ID NO 306

<211> LENGTH: 34

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<220> FEATURE:

<221> NAME/KEY: MISC_FEATURE

<222> LOCATION: (14)..(14)

<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<220> FEATURE:

<221> NAME/KEY: MISC_FEATURE

<222> LOCATION: (23)..(23)

<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<220> FEATURE:

<221> NAME/KEY: MISC_FEATURE

<222> LOCATION: (24)..(24)

<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<220> FEATURE:

<221> NAME/KEY: MISC_FEATURE

<222> LOCATION: (27)..(27)

<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<400> SEQUENCE: 306

Glu Asn Ile Ile Gly Asp Thr Arg Gln Ala Tyr Cys Asn Xaa Ser Ala
1 5 10 15

Ala Glu Trp Glu Gln Arg Xaa Xaa Asp Thr Xaa Arg Ala Ile Lys Ser
20 25 30

Leu Lys

<210> SEQ ID NO 307

<211> LENGTH: 39

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 307

Ile Asn Ser Ser Asn Ile Ile Ile Asn Cys Arg Ile Arg Gln Ile Val
1 5 10 15

Asn Ser Trp Met Arg Val Gly Lys Gly Ile Tyr Ala Pro Pro Ile Arg
20 25 30

Gly Thr Ile Ser Cys Thr Ser
35

<210> SEQ ID NO 308

<211> LENGTH: 36

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<220> FEATURE:

<221> NAME/KEY: MISC_FEATURE

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<222> LOCATION: (16)..(16)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (17)..(17)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (21)..(21)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (23)..(23)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (24)..(24)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (27)..(27)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<400> SEQUENCE: 308

Thr Thr Gln Leu Ile Leu Asn Gly Ser Leu Ala Glu Asn Ser Thr Xaa
1 5 10 15

Xaa Arg Thr Glu Xaa Met Xaa Xaa Met Asp Xaa Thr Ile Ile Val Gln
20 25 30

Phe His Lys Arg
35

<210> SEQ ID NO 309
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 309

Cys Asn Glu Pro Asp Phe Asn Gly Thr Gly Lys Cys Arg Asn Ile Ser
1 5 10 15

Thr Val

<210> SEQ ID NO 310
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 310

Asn Ile Thr Gly Leu Leu Leu Glu Leu Asp Arg Gly Leu Pro Arg
1 5 10 15

<210> SEQ ID NO 311
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 311

Val His Val Pro Asn Ile Thr Glu Thr Phe
1 5 10

<210> SEQ ID NO 312
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 312

Asn Ile Thr Gly Leu Leu Leu Glu
1 5

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<210> SEQ ID NO 313
<211> LENGTH: 29
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 313

Thr Arg Glu Lys Arg Ala Ala Phe Gly Leu Gly Ala Leu Phe Leu Gly
1 5 10 15

Phe Leu Gly Ala Ala Gly Ser Thr Met Gly Ala Ala Ser
 20 25

<210> SEQ ID NO 314
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 314

Tyr Thr Thr Val Pro Trp Asn Ser Thr Trp
1 5 10

<210> SEQ ID NO 315
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 315

Glu Ile Trp Asn Asn Leu Thr Trp Gln Asp Trp Asp Lys Arg Val Lys
1 5 10 15

Asn Tyr Ser Gly Val Ile Phe Asp Leu Ile
 20 25

<210> SEQ ID NO 316
<211> LENGTH: 650
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 316

Met Asp Ala Met Lys Arg Gly Leu Cys Cys Val Leu Leu Leu Cys Gly
1 5 10 15

Ala Val Phe Val Ser Pro Ser Gln Glu Ile His Ala Arg Phe Arg Arg
 20 25 30

Gly Ala Arg Glu Asn Trp Trp Val Thr Val Tyr Tyr Gly Val Pro Val
 35 40 45

Trp Arg Glu Ala Lys Thr Thr Leu Phe Cys Ala Ser Asp Ala Lys Ser
 50 55 60

Tyr Ser Thr Glu Ala His Asn Ile Trp Ala Thr Gln Ala Cys Val Pro
65 70 75 80

Thr Asp Pro Thr Pro Gln Glu Val Leu Leu Pro Asn Val Thr Glu Glu
 85 90 95

Phe Asn Met Trp Glu Asn Tyr Met Val Asp Gln Met Gln Glu Asp Ile
 100 105 110

Ile Ser Leu Trp Glu Gln Ser Leu Lys Pro Cys Val Lys Leu Thr Pro
 115 120 125

Leu Cys Val Thr Leu Thr Cys Asn Asn Pro Thr Asn Thr Ser Cys Thr
 130 135 140

Asn Ser Thr Asp Asp Arg Leu Gly Asp Met Arg Asn Cys Ser Phe Asn
145 150 155 160

Val Thr Thr Glu Leu Arg Asp Lys Lys Arg Lys Val Tyr Ser Leu Phe

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165								170					175				
Tyr	Val	Glu	Asp	Ile	Thr	Ala	Ile	Gly	Asn	Asn	Ser	Thr	Tyr	Arg	Leu		
			180					185					190				
Ile	Asn	Cys	Asn	Thr	Thr	Ala	Ile	Thr	Gln	Ala	Cys	Pro	Lys	Thr	Ser		
		195					200					205					
Phe	Glu	Pro	Ile	Pro	Ile	His	Tyr	Cys	Ala	Pro	Ala	Gly	Phe	Ala	Leu		
	210					215					220						
Leu	Lys	Cys	Asn	Asp	Ile	Asp	Tyr	Lys	Gly	Asn	Glu	Thr	Cys	Lys	Asn		
225					230					235					240		
Val	Ser	Thr	Val	His	Cys	Thr	His	Gly	Ile	Lys	Pro	Val	Ala	Thr	Thr		
				245					250					255			
Gln	Leu	Ile	Leu	Asn	Gly	Ser	Thr	Ala	Asp	Asn	Gln	Thr	Val	Ala	Arg		
			260					265					270				
Ile	Asp	Pro	Ser	Glu	Asn	Leu	Ala	Ile	Ile	Gln	Leu	Lys	Asp	Pro	Val		
		275					280					285					
Lys	Ile	Thr	Cys	Arg	Arg	Pro	Gly	Asn	Asn	Thr	Arg	Gly	Gln	Ile	Gln		
	290					295					300						
Ile	Gly	Pro	Ala	Met	Thr	Phe	Tyr	Asn	Ile	Glu	Asn	Val	Val	Gly	Asp		
305					310					315					320		
Thr	Arg	Lys	Ala	Tyr	Cys	Glu	Ile	Asn	Gly	Thr	Gln	Trp	Ala	Lys	Ala		
				325					330					335			
Leu	Asn	Glu	Thr	Lys	Glu	Val	Leu	Arg	Asn	Ile	Leu	Arg	Lys	Asn	Ile		
			340					345					350				
Ser	Phe	Met	Val	Pro	Ser	Gly	Gly	Asp	Pro	Glu	Val	Thr	Asn	His	His		
		355					360					365					
Phe	Asn	Cys	Gly	Gly	Glu	Phe	Phe	Tyr	Cys	Asn	Thr	Ser	Glu	Ile	Ile		
	370					375					380						
Asn	Ile	Thr	Lys	Ile	Asn	Lys	Thr	Glu	Asn	Met	Thr	Ile	Ile	Pro	Cys		
385					390					395					400		
Arg	Ile	Arg	Gln	Ile	Val	Asn	Ser	Trp	Met	Arg	Val	Gly	Lys	Gly	Ile		
				405					410					415			
Phe	Ala	Pro	Pro	Ile	Arg	Gly	Asn	Ile	Thr	Cys	Thr	Ser	Asn	Ile	Thr		
			420					425					430				
Gly	Met	Leu	Leu	Glu	Ile	His	Lys	Asp	Gln	Asn	Asn	Thr	Tyr	Val	Cys		
		435					440					445					
Leu	Thr	Gly	Gly	Asn	Met	Lys	Asp	Ile	Trp	Arg	Ser	Glu	Leu	Tyr	Lys		
	450					455					460						
Tyr	Lys	Ile	Val	Glu	Ile	Gln	Pro	Leu	Gly	Val	Ala	Pro	Thr	Lys	Cys		
465					470					475					480		
Arg	Arg	Tyr	Ala	Val	Glu	Lys	Gln	His	His	Arg	Arg	Arg	Arg	Arg	Arg		
				485					490					495			
Ala	Leu	Gly	Leu	Gly	Ala	Leu	Phe	Leu	Gly	Phe	Leu	Gly	Ala	Ala	Gly		
			500					505					510				
Ser	Thr	Met	Gly	Ala	Ala	Ser	Val	Val	Leu	Thr	Val	Gln	Ala	Arg	Gln		
		515					520					525					
Leu	Leu	Thr	Gly	Ile	Val	Gln	Gln	Gln	Asn	Asn	Leu	Leu	Arg	Ala	Pro		
	530					535					540						
Glu	Ala	Gln	Gln	His	Leu	Leu	Gln	Leu	Ser	Val	Trp	Gly	Ile	Lys	Gln		
545					550					555					560		
Leu	Gln	Ala	Arg	Val	Leu	Ala	Val	Glu	Arg	Tyr	Leu	Arg	Asp	Gln	Gln		
				565					570					575			
Leu	Leu	Gly	Leu	Trp	Gly	Cys	Thr	Gly	Lys	Thr	Ile	Cys	Cys	Thr	Ala		
			580					585					590				

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Val Arg Trp Asn Lys Thr Trp Gly Asn Ile Ser Asp Tyr Gln Val Ile
595 600 605

Trp Asn Asn Tyr Thr Trp Gln Gln Trp Asp Arg Glu Val Asn Asn Tyr
610 615 620

Thr Gly Leu Ile Tyr Thr Leu Leu Glu Glu Ala Asn Thr Gln Gln Glu
625 630 635 640

Lys Asn Glu Lys Glu Leu Leu Glu Leu Asp
645 650

<210> SEQ ID NO 317
 <211> LENGTH: 836
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 317

Met Lys Val Met Glu Lys Lys Lys Arg Leu Trp Leu Ser Tyr Cys Leu
1 5 10 15

Leu Ser Ser Leu Ile Ile Pro Gly Leu Ser Ser Leu Trp Ala Thr Val
20 25 30

Tyr Tyr Gly Val Pro Val Trp Arg Asp Val Glu Thr Thr Leu Phe Cys
35 40 45

Ala Ser Asp Ala Lys Ala Tyr Lys Gln Glu Ala His Asn Ile Trp Ala
50 55 60

Thr Gln Ala Cys Val Pro Thr Asp Pro Asn Pro Gln Glu Val His Leu
65 70 75 80

Pro Asn Val Thr Glu Lys Phe Asp Met Trp Glu Asn Asn Met Ala Glu
85 90 95

Gln Met Gln Glu Asp Ile Ile Ser Leu Trp Asp Gln Ser Leu Lys Pro
100 105 110

Cys Ile Lys Leu Thr Pro Leu Cys Val Thr Met Thr Cys Leu Asn Pro
115 120 125

Asp Ser Asn Ser Ser Ala Val Asn Thr Thr Asp Ile Met Arg Asn Cys
130 135 140

Ser Phe Asn Ile Thr Thr Glu Leu Arg Asp Lys Lys Lys Gln Val Tyr
145 150 155 160

Ser Leu Phe Tyr Val Asp Asp Leu Ala His Ile Asn Asn Asn Thr Tyr
165 170 175

Arg Leu Ile Asn Cys Asn Thr Thr Ala Ile Thr Gln Ala Cys Pro Lys
180 185 190

Thr Ser Phe Glu Pro Ile Pro Ile His Tyr Cys Ala Pro Pro Gly Phe
195 200 205

Ala Ile Leu Lys Cys Asn Glu Lys Asp Phe Lys Gly Lys Gly Glu Cys
210 215 220

Lys Asn Val Ser Thr Val Gln Cys Thr His Gly Ile Lys Pro Val Val
225 230 235 240

Thr Thr Gln Leu Ile Ile Asn Gly Ser Leu Ala Thr Lys Asn Val Thr
245 250 255

Val Arg Ser Lys Asn Phe Ala Asp Ile Ile Leu Val Gln Phe Ser Glu
260 265 270

Gly Val Asn Met Thr Cys Ile Arg Pro Gly Asn Asn Thr Val Gly Asn
275 280 285

Val Gln Leu Gly Pro Gly Met Thr Phe Tyr Asn Ile Pro Lys Ile Val
290 295 300

Gly Asp Val Arg Glu Ala His Cys Asn Ile Ser Lys Leu Thr Trp Glu

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305	310	315	320
Lys Gln Arg Lys Tyr Thr Leu Glu Ile Ile Lys Lys Glu Ala Asn Leu	325	330	335
Thr Lys Val Glu Leu Ile Pro Asn Ala Gly Gly Asp Pro Glu Val Val	340	345	350
Asn Met Met Leu Asn Cys Gly Gly Glu Phe Phe Tyr Cys Asn Thr Ile	355	360	365
Pro Leu Phe Asn Met Thr Tyr Asn Asn Thr Asp Asn Thr Thr Ile Thr	370	375	380
Leu Lys Cys Arg Ile Arg Gln Ile Val Asn Gln Trp Met Arg Val Gly	385	390	400
Lys Gly Ile Phe Ala Pro Pro Ile Lys Gly Val Leu Ser Cys Asn Ser	405	410	415
Asn Ile Thr Gly Met Ile Leu Asp Ile Ser Ile Ser Ala Val Asn Asn	420	425	430
Asp Ser Arg Asn Ile Thr Val Met Pro Thr Gly Gly Asp Met Thr Ala	435	440	445
Leu Trp Lys Asn Glu Leu His Lys Tyr Lys Val Val Ser Ile Glu Pro	450	455	460
Ile Gly Val Ala Pro Gly Lys Ala Lys Arg His Thr Val Lys Arg Glu	465	470	480
Lys Arg Ala Ala Phe Gly Leu Gly Ala Leu Phe Leu Gly Phe Leu Gly	485	490	495
Ala Ala Gly Ser Thr Met Gly Ala Ala Ser Val Val Leu Thr Val Gln	500	505	510
Ala Arg Gln Leu Leu Ser Gly Ile Val Gln Gln Gln Asn Asn Leu Leu	515	520	525
Arg Ala Ile Glu Ala Gln Gln His Leu Leu Gln Leu Ser Val Trp Gly	530	535	540
Ile Lys Gln Leu Gln Ala Arg Val Leu Ala Val Glu Arg Tyr Leu Lys	545	550	555
Asp Gln Gln Ile Leu Gly Leu Trp Gly Cys Ser Gly Lys Thr Ile Cys	565	570	575
Tyr Thr Thr Val Pro Trp Asn Asp Thr Trp Ser Asn Asn Leu Ser Tyr	580	585	590
Asp Ala Ile Trp Gly Asn Leu Thr Trp Gln Glu Trp Asp Arg Lys Val	595	600	605
Arg Asn Tyr Ser Gly Thr Ile Phe Ser Leu Ile Glu Gln Ala Gln Glu	610	615	620
Gln Gln Asn Thr Asn Glu Lys Ser Leu Leu Glu Leu Asp Gln Trp Ser	625	630	635
Ser Leu Trp Asn Trp Phe Asp Ile Thr Asn Trp Leu Trp Tyr Ile Lys	645	650	655
Ile Phe Leu Ile Val Val Ala Ser Leu Val Gly Ile Arg Ile Val Gly	660	665	670
Val Ile Phe Ser Leu Val Ala Lys Val Arg Gln Gly Tyr Ser Pro Leu	675	680	685
Ser Leu Gln Thr Leu Phe Pro Thr Thr Arg Glu Pro Asp Arg Pro Glu	690	695	700
Gly Thr Glu Glu Gly Ala Gly Lys Thr Asp Asn Val Arg Ser Thr Arg	705	710	715
Leu Val Ser Gly Phe Leu Ala Leu Val Trp Glu Asp Leu Arg Asn Leu	725	730	735

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Leu Ile Phe Leu Tyr His Arg Leu Glu Asp Leu Leu Leu Ile Leu Arg
 740 745 750
 Arg Thr Val Gln Ile Leu Gly Gln Asn Ile Asn Lys Gly Leu Gln Leu
 755 760 765
 Leu Asn Glu Leu Arg Ala Arg Cys Trp Gly Val Ile Ala Tyr Trp Ala
 770 775 780
 Arg Glu Leu Lys Val Ser Ala Thr Ser Leu Leu Asp Thr Thr Ala Ile
 785 790 795 800
 Ala Val Ala Glu Gly Thr Asp Arg Ile Ile Glu Leu Thr Arg Arg Leu
 805 810 815
 Phe Leu Gly Ile Ile His Ile Pro Arg Arg Ile Arg Gln Gly Leu Glu
 820 825 830
 Arg Ser Leu Leu
 835

<210> SEQ ID NO 318
 <211> LENGTH: 859
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (154)..(154)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid.
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (284)..(284)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid.
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (287)..(287)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid.
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (291)..(291)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid.
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (292)..(292)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (303)..(303)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid.
 <400> SEQUENCE: 318

Met Lys Val Met Glu Lys Ile Lys Gly Ser Leu Ile Glu Lys Tyr Met
 1 5 10 15
 Phe Met Gly Leu Ile Ile Pro Cys Leu Thr Gly Ser Asp Gln Leu Trp
 20 25 30
 Ala Thr Val Tyr Tyr Gly Val Pro Val Trp Lys Glu Ala Asp Thr Thr
 35 40 45
 Leu Phe Cys Ala Ser Glu Ala Ser Ala Leu Asn Lys Glu Ala His Asn
 50 55 60
 Ile Trp Ala Ser Gln Ala Cys Val Pro Thr Asp Pro Asn Pro Gln Glu
 65 70 75 80
 Val Gln Ile Pro Asn Val Thr Glu Asn Phe Asn Met Trp Asn Asn Thr
 85 90 95
 Met Ala Glu Gln Met Asn Thr Asp Ile Ile Ser Leu Trp Asp Gln Ser
 100 105 110
 Leu Lys Pro Cys Val Lys Leu Thr Pro Ile Cys Val Thr Met Glu Cys
 115 120 125
 Arg Lys Val Thr Phe Asn Ser Thr Ser Asn Arg Asn Lys Thr Ser Thr

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130					135					140					
Met 145	Thr	Thr	Asn	Ser	Pro 150	Asn	Glu	Lys	Xaa	Asp 155	Ser	Thr	Val	Lys	Asn 160
Cys	Thr	Phe	Asn	Met 165	Thr	Thr	Glu	Val	Arg 170	Asp	Lys	Glu	Lys	Lys 175	Val
Tyr	Ser	Leu	Phe	Tyr 180	Val	Asp	Asp	Leu	Val 185	Leu	Ile	Asp	Asn	Asp	Thr
Asp	Thr	Tyr	Arg	Leu 195	Ile	Asn	Cys	Asn	Thr	Thr	Ala	Ile	Thr	Gln	Ala
Cys	Pro 210	Lys	Thr	Ser	Phe	Glu 215	Pro	Ile	Pro	Ile	His 220	Tyr	Cys	Ala	Thr
Pro 225	Gly	Tyr	Ala	Ile	Met 230	Lys	Cys	Asn	Met	Pro 235	Asn	Phe	Asn	Gly	Thr 240
Gly	Thr	Gly	Arg	Cys 245	Asn	Asn	Ile	Ser	Thr 250	Val	Gln	Cys	Thr	His 255	Gly
Ile	Arg	Pro	Val	Val 260	Thr	Thr	Gln	Leu	Ile 265	Leu	Asn	Gly	Ser	Val	Ala
Glu	Asn	Lys	Thr	Ile	Ala	Arg	Arg	Asn	Gly	Tyr	Xaa	Phe	Leu	Xaa	Gln
Phe	Gln	Xaa	Xaa	Val	Ser	Ile 295	Asn	Cys	Thr	Arg	Pro 300	Gly	Asn	Xaa	Ser
Arg 305	Gly	Gln	Ile	Gln	Ile 310	Gly	Pro	Gly	Met	Thr 315	Phe	Tyr	Asn	Ile	Glu 320
Asn	Ile	Val	Gly	Asp 325	Thr	Arg	Arg	Ala	Tyr 330	Cys	Gln	Ile	Asn	Arg 335	Thr
Val	Trp	Asp	Glu	Arg 340	Leu	Asn	Glu	Thr	Gly 345	Gln	Ala	Leu	Arg	Glu	Leu
Phe	Thr	Asn	Leu	Thr	Gln	Val	Asn	Phe	Thr	Val	Ser	Pro	Gly	Gly	Asp
Pro 370	Glu	Val	Thr	Asn	Met	Met 375	Phe	Asn	Cys	Gly	Gly 380	Glu	Phe	Phe	Tyr
Cys 385	Asn	Thr	Thr	Thr	Leu	Phe	Asn	Tyr	Thr	Trp 395	Lys	Asn	Asn	Asn	Ile 400
Thr	Lys	Gly	Asp	Asn 405	Thr	Thr	Phe	Phe	Pro 410	Cys	Arg	Ile	Arg	Gln	Ile 415
Val	Asn	Ser	Trp	Met 420	Arg	Val	Gly	Lys	Gly 425	Ile	Tyr	Ala	Pro	Pro	Ile
Arg	Gly	Val	Ile	Ser	Cys	Thr	Ser	Asn	Val	Thr	Gly	Ile	Ile	Leu	Glu
Thr	Gly	His	Gly	Ile	Asn	Asn 455	Ser	Ile	Thr	Asn	Ile 460	Thr	Leu	Tyr	Pro
Thr 465	Gly	Gly	Asn	Met	Val	Asp 470	Leu	Trp	Arg	Leu	Glu 475	Leu	His	Lys	Tyr 480
Lys	Val	Val	Ser	Ile 485	Glu	Pro	Ile	Gly	Val 490	Ala	Pro	Ser	Lys	Ala	Lys 495
Arg	His	Thr	Val	Ser	Arg	Glu	Lys	Arg	Ala 505	Ala	Phe	Gly	Leu	Gly	Ala
Leu	Phe	Leu	Gly	Phe	Leu	Gly	Ala	Ala	Gly 520	Ser	Thr	Met	Gly	Ala	Ala
Ser	Val	Val	Leu	Thr	Val	Gln 535	Ala	Arg	Gln	Leu	Leu 540	Ser	Gly	Ile	Val
Gln 545	Gln	Gln	Asn	Asn	Leu	Leu	Arg	Ala	Ile	Glu 555	Ala	Gln	Gln	His	Leu 560

Leu	Gln	Leu	Ser	Val	Trp	Gly	Ile	Lys	Gln	Leu	Gln	Ala	Arg	Val	Leu
				565					570					575	
Ala	Val	Glu	Arg	Tyr	Leu	Arg	Asp	Gln	Gln	Ile	Leu	Gly	Leu	Trp	Gly
			580					585					590		
Cys	Ser	Gly	Lys	Ala	Ile	Cys	Tyr	Thr	Thr	Val	Pro	Trp	Asn	Asn	Thr
		595					600					605			
Trp	Ser	Ala	Asn	Thr	Ser	Phe	Asp	Glu	Ile	Trp	Asn	Asn	Leu	Thr	Trp
	610					615					620				
Gln	Asp	Trp	Asp	Lys	Arg	Val	Lys	Asn	Tyr	Ser	Gly	Val	Ile	Phe	Ser
625					630					635					640
Leu	Ile	Glu	Gln	Ala	Gln	Glu	Gln	Gln	Asn	Thr	Asn	Glu	Lys	Ser	Leu
				645					650					655	
Leu	Glu	Leu	Asp	Gln	Trp	Ser	Ser	Leu	Trp	Asn	Trp	Phe	Asp	Ile	Thr
			660					665					670		
Arg	Trp	Leu	Trp	Tyr	Ile	Lys	Leu	Phe	Ile	Met	Ile	Val	Ala	Gly	Leu
		675					680					685			
Val	Gly	Ile	Arg	Ile	Val	Gly	Ala	Ile	Ile	Ser	Phe	Val	Ala	Lys	Val
	690					695					700				
Arg	Gln	Gly	Tyr	Ser	Pro	Leu	Ser	Leu	Gln	Thr	Leu	Ile	Pro	Thr	Thr
705					710					715					720
Arg	Glu	Pro	Asp	Arg	Pro	Gly	Gly	Thr	Glu	Glu	Asp	Val	Gly	Glu	Pro
				725					730					735	
Gly	Lys	Gly	Arg	Ser	Ile	Arg	Leu	Val	Ser	Gly	Phe	Leu	Ala	Leu	Ile
			740					745					750		
Trp	Glu	Asp	Leu	Arg	Asn	Leu	Leu	Leu	Phe	Cys	Tyr	His	Arg	Leu	Arg
		755					760					765			
Asp	Leu	Leu	Leu	Ile	Leu	Gly	Arg	Thr	Leu	Glu	Asn	Leu	Gly	Gln	Ser
		770				775						780			
Leu	Asn	Lys	Gly	Leu	Gln	Gln	Leu	Arg	Asn	Phe	Ser	Arg	Tyr	Leu	Trp
785					790					795					800
Gly	Val	Ile	Thr	Tyr	Trp	Gly	Arg	Glu	Leu	Gln	Thr	Ser	Ala	Ile	Ser
				805					810					815	
Leu	Leu	Asp	Ala	Thr	Ala	Ile	Ala	Val	Ala	Glu	Gly	Thr	Asp	Arg	Ile
			820					825					830		
Leu	Glu	Val	Ala	Gln	Ile	Ile	Gly	Arg	Gly	Ile	Leu	His	Ile	Pro	Arg
		835					840					845			
Arg	Ile	Arg	Gln	Gly	Leu	Glu	Arg	Ser	Leu	Leu					
	850					855									

<400> SEQUENCE: 319

Met	Lys	Val	Met	Gly	Lys	Thr	Arg	Arg	Arg	Trp	Gln	Pro	Tyr	Cys	Ile
1				5					10					15	
Ile	Met	Ala	Leu	Ile	Ile	Pro	Cys	Leu	Lys	Thr	Glu	Ser	Asp	Gln	Trp
			20					25					30		
Trp	Val	Thr	Val	Tyr	Tyr	Gly	Val	Pro	Val	Trp	Lys	Glu	Ala	Thr	Thr
		35				40						45			
Pro	Leu	Phe	Cys	Ala	Ala	Asn	Ala	Thr	Ala	Leu	Lys	Glu	Glu	Pro	His
	50					55					60				
Asn	Ile	Trp	Ala	Thr	Gln	Ala	Cys	Val	Pro	Thr	Asp	Pro	Ser	Pro	Glu

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65					70					75					80
Glu	Val	Ile	Leu	Ile	Asn	Val	Thr	Glu	Glu	Phe	Asn	Val	Trp	Asp	Asn
				85					90					95	
Ala	Met	Val	Glu	Gln	Met	Gln	Glu	Asp	Ile	Thr	Ser	Leu	Trp	Asp	Gln
			100					105					110		
Ser	Leu	Arg	Pro	Cys	Val	Lys	Leu	Asn	Pro	Leu	Cys	Val	Gln	Leu	Thr
		115					120					125			
Cys	Thr	Ser	Ile	Asn	Ala	Thr	Gly	Asn	Glu	Thr	Asn	Ala	Thr	Gly	Asn
	130					135					140				
Gly	Ile	Glu	Lys	Asp	Gly	Leu	Ala	Gln	Asp	Met	Arg	Asn	Cys	Thr	Phe
145					150					155					160
Asn	Thr	Thr	Thr	Glu	Leu	Arg	Asp	Lys	Lys	Gln	Gln	Ile	Tyr	Ser	Leu
				165				170							175
Phe	Trp	Lys	Asn	Asp	Leu	Val	Gly	Thr	Asn	Asn	Thr	Phe	Arg	Leu	Ile
			180					185					190		
Asn	Cys	Asn	Thr	Thr	Ala	Ile	Thr	Gln	Ala	Cys	Pro	Lys	Thr	Ser	Phe
		195					200					205			
Glu	Pro	Ile	Pro	Ile	His	Tyr	Cys	Ala	Pro	Ala	Gly	Phe	Ala	Leu	Leu
	210					215					220				
Lys	Cys	Asn	Asp	Lys	Asp	Tyr	Pro	Gly	Lys	Gly	Lys	Cys	Lys	Asn	Val
225					230					235					240
Ser	Thr	Val	His	Cys	Thr	His	Gly	Ile	Lys	Pro	Thr	Val	Thr	Thr	Gln
				245					250						255
Leu	Ile	Leu	Asn	Gly	Ser	Leu	Ala	Glu	Glu	Glu	Glu	Val	Val	Phe	Arg
			260					265					270		
Thr	Lys	Asn	Met	Thr	Ala	Pro	Gly	Leu	Ser	Asp	Thr	Val	Ile	Val	Gln
		275					280					285			
Leu	Lys	Arg	Ala	Ile	Pro	Ile	Asn	Cys	Ser	Arg	Pro	Gly	Asn	Asn	Thr
	290					295					300				
Gly	Arg	Ala	Ile	Asn	Leu	Ser	Pro	Gly	Thr	Thr	Phe	Phe	Asn	Thr	Glu
305				310							315				320
Ala	Leu	Ile	Gly	Asn	Pro	Arg	Lys	Ala	Ser	Cys	His	Leu	Asn	Gly	Thr
			325					330						335	
Leu	Trp	Asn	Asn	Ile	Leu	Asn	Arg	Ile	Lys	Gln	Lys	Ile	Lys	Asn	Ser
			340					345					350		
Thr	Thr	Trp	His	Arg	Gly	Asp	Ile	Thr	Phe	Thr	Lys	His	Pro	Gly	Gly
		355				360						365			
Asp	Pro	Glu	Val	Val	Asn	Phe	Met	Phe	Asn	Cys	Gly	Gly	Glu	Phe	Phe
	370				375						380				
Tyr	Cys	Asn	Thr	Ser	Arg	Leu	Ile	Thr	Cys	Asn	Ser	Ser	Asp	Thr	Ser
385					390					395					400
Glu	Tyr	Ile	Leu	Pro	Cys	Lys	Ile	Arg	Gln	Val	Val	Asn	Ser	Trp	Met
			405					410						415	
Arg	Val	Gly	Lys	Gly	Ile	Phe	Ala	Pro	Pro	Arg	Arg	Gly	Thr	Ile	Thr
			420					425					430		
Cys	Asn	Ser	Thr	Ile	Thr	Gly	Leu	Leu	Leu	Glu	Val	Gln	Asn	Gly	Thr
		435					440					445			
Gly	Asn	Asn	Thr	Glu	Val	Tyr	Leu	Ser	Gly	Gly	Asp	Met	Arg	Asp	Ile
	450					455					460				
Trp	Arg	Ser	Glu	Leu	Tyr	Lys	Tyr	Lys	Ile	Val	Lys	Ile	Glu	Pro	Leu
465					470					475					480
Gly	Val	Ala	Pro	Thr	Lys	Ala	Lys	Arg	Tyr	Thr	Val	Ala	Lys	Ala	Leu
				485					490						495

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Asp Arg Ser Lys Arg Ala Ala Phe Gly Leu Gly Ala Leu Phe Leu Gly
 500 505 510
 Phe Leu Gly Ala Ala Gly Ser Thr Met Gly Ala Ala Ser Val Met Leu
 515 520 525
 Thr Val Gln Ala Arg Asn Leu Leu Ser Gly Ile Val Gln Gln Gln Asn
 530 535 540
 Asn Leu Leu Arg Ala Ile Glu Ala Gln Gln His Leu Leu Gln Leu Ser
 545 550 555 560
 Val Trp Gly Ile Lys Gln Leu Gln Ala Arg Val Leu Ala Val Glu Arg
 565 570 575
 Tyr Leu Lys Asp Gln Gln Val Leu Ala Leu Trp Gly Cys Ser Gly Lys
 580 585 590
 Thr Val Cys Tyr Ser Thr Val Pro Trp Asn Thr Ser Trp Asn Ser Asn
 595 600 605
 Lys Ser Tyr Glu Asp Ile Trp Ser Asn Leu Thr Trp Gln Gln Trp Asp
 610 615 620
 Lys Leu Val Glu Asn His Thr Gly Thr Ile Phe Gln Leu Leu Gln Arg
 625 630 635 640
 Ala His Glu Gln Gln Asn Ser Asn Glu Lys Glu Leu Leu Glu Leu Asp
 645 650 655
 Gln Trp Ser Ser Leu Trp Asn Trp Phe Asp Ile Thr Asn Trp Leu Trp
 660 665 670
 Tyr Ile Lys Met Phe Ile Ile Leu Val Gly Gly Val Ile Gly Leu Arg
 675 680 685
 Ile Val Ile Gly Val Val Asn Ile Ile Arg Arg Ser Arg Gln Gly Tyr
 690 695 700
 Ser Pro Leu Ser Phe Gln Thr Leu Ile Pro Ala Thr Gly Gly Pro Glu
 705 710 715 720
 Val Pro Glu Gly Ile Gly Gly Gly Gly Gly Glu Gln Asp Arg Gly Arg
 725 730 735
 Ser Val Arg Leu Val Ser Gly Phe Leu Ala Leu Cys Trp Asp Asp Ile
 740 745 750
 Arg Asn Leu Thr Ile Phe Leu Tyr Arg Leu Leu Arg Asp Cys Leu Ser
 755 760 765
 Val Leu Trp Ser Leu Leu Glu Gln Leu Gly Gln Gln Val Leu Arg Gly
 770 775 780
 Leu Arg Leu Leu Arg Glu Leu Leu Ser Gln Leu Lys Gly Ile Gly Gln
 785 790 795 800
 Tyr Trp Leu Gln Glu Leu Arg Thr Ser Ala Ile Ser Leu Leu Asp Thr
 805 810 815
 Thr Ala Ile Ile Val Ala Glu Arg Thr Asp Thr Ile Ile Glu Val Ala
 820 825 830
 Thr Arg Ile Gly Arg Gly Ile Leu His Ile Pro Arg Arg Ile Arg Gln
 835 840 845
 Gly Leu Glu Arg Ala Leu Leu Glu
 850 855

<210> SEQ ID NO 320

<211> LENGTH: 871

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 320

Met Lys Asn Leu Ile Gly Ile Thr Leu Ile Leu Ile Ile Thr Ile Leu

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1	5	10	15
Gly Ile Gly Phe Ser Thr Tyr Tyr Thr Thr Val Phe Tyr Gly Val Pro	20	25	30
Val Trp Lys Glu Ala Gln Pro Thr Leu Phe Cys Ala Ser Asp Ala Asp	35	40	45
Ile Thr Ser Arg Asp Lys His Asn Ile Trp Ala Thr His Asn Cys Val	50	55	60
Pro Leu Asp Pro Asn Pro Tyr Glu Val Thr Leu Ala Asn Val Ser Ile	65	70	75
Arg Phe Asn Met Glu Glu Asn Tyr Met Val Gln Glu Met Lys Glu Asp	85	90	95
Ile Leu Ser Leu Phe Gln Gln Ser Phe Lys Pro Cys Val Lys Leu Thr	100	105	110
Pro Phe Cys Ile Lys Met Thr Cys Thr Met Thr Asn Thr Thr Asn Lys	115	120	125
Thr Leu Asn Ser Ala Thr Thr Thr Leu Thr Pro Thr Val Asn Leu Ser	130	135	140
Ser Ile Pro Asn Tyr Glu Val Tyr Asn Cys Ser Phe Asn Gln Thr Thr	145	150	155
Glu Phe Arg Asp Lys Lys Lys Gln Ile Tyr Ser Leu Phe Tyr Arg Glu	165	170	175
Asp Ile Val Lys Glu Asp Gly Asn Asn Asn Ser Tyr Tyr Leu His Asn	180	185	190
Cys Asn Thr Ser Val Ile Thr Gln Glu Cys Asp Lys Ser Thr Phe Glu	195	200	205
Pro Ile Pro Ile Arg Tyr Cys Ala Pro Ala Gly Phe Ala Leu Leu Lys	210	215	220
Cys Arg Asp Gln Asn Phe Thr Gly Lys Gly Gln Cys Ser Asn Val Ser	225	230	235
Val Val His Cys Thr His Gly Ile Tyr Pro Met Ile Ala Thr Ala Leu	245	250	255
His Leu Asn Gly Ser Leu Glu Glu Glu Glu Thr Lys Ala Tyr Phe Val	260	265	270
Asn Thr Ser Val Asn Thr Pro Leu Leu Val Lys Phe Asn Val Ser Ile	275	280	285
Asn Leu Thr Cys Glu Arg Thr Gly Asn Asn Thr Arg Gly Gln Val Gln	290	295	300
Ile Gly Pro Gly Met Thr Phe Tyr Asn Ile Glu Asn Val Val Gly Asp	305	310	315
Thr Arg Lys Ala Tyr Cys Ser Val Asn Ala Thr Thr Trp Tyr Arg Asn	325	330	335
Leu Asp Trp Ala Met Ala Ala Ile Asn Thr Thr Met Arg Ala Arg Asn	340	345	350
Glu Thr Val Gln Gln Thr Phe Gln Trp Gln Arg Asp Gly Asp Pro Glu	355	360	365
Val Thr Ser Phe Trp Phe Asn Cys Gln Gly Glu Phe Phe Tyr Cys Asn	370	375	380
Leu Thr Asn Trp Thr Asn Thr Trp Thr Ala Asn Arg Thr Asn Asn Thr	385	390	395
His Gly Thr Leu Val Ala Pro Cys Arg Leu Arg Gln Ile Val Asn His	405	410	415
Trp Gly Ile Val Ser Lys Gly Val Tyr Leu Pro Pro Arg Arg Gly Thr	420	425	430

Val	Lys	Cys	His	Ser	Asn	Ile	Thr	Gly	Leu	Ile	Met	Thr	Ala	Glu	Lys
435						440						445			
Asp	Asn	Asn	Asn	Ser	Tyr	Thr	Pro	Gln	Phe	Ser	Ala	Val	Val	Glu	Asp
450						455			460						
Tyr	Trp	Lys	Val	Glu	Leu	Ala	Arg	Tyr	Lys	Val	Val	Glu	Ile	Gln	Pro
465			470						475			480			
Leu	Ser	Val	Ala	Pro	Arg	Pro	Gly	Lys	Arg	Pro	Glu	Ile	Lys	Ala	Asn
			485						490			495			
His	Thr	Arg	Ser	Arg	Arg	Asp	Val	Gly	Ile	Gly	Leu	Leu	Phe	Leu	Gly
			500						505			510			
Phe	Leu	Ser	Ala	Ala	Gly	Ser	Thr	Met	Gly	Ala	Ala	Ser	Ile	Ala	Leu
515						520						525			
Thr	Ala	Gln	Ala	Arg	Gly	Leu	Leu	Ser	Gly	Ile	Val	Gln	Gln	Gln	Gln
530						535			540						
Asn	Leu	Leu	Gln	Ala	Ile	Glu	Ala	Gln	Gln	His	Leu	Leu	Gln	Leu	Ser
545			550						555			560			
Val	Trp	Gly	Ile	Lys	Gln	Leu	Gln	Ala	Arg	Met	Leu	Ala	Val	Glu	Lys
			565						570			575			
Tyr	Ile	Arg	Asp	Gln	Gln	Leu	Leu	Ser	Leu	Trp	Gly	Cys	Ala	Asn	Lys
			580			585						590			
Leu	Val	Cys	His	Ser	Ser	Val	Pro	Trp	Asn	Leu	Thr	Trp	Ala	Glu	Asp
595						600						605			
Ser	Thr	Lys	Cys	Asn	His	Ser	Asp	Ala	Lys	Tyr	Tyr	Asp	Cys	Ile	Trp
610						615			620						
Asn	Asn	Leu	Thr	Trp	Gln	Glu	Trp	Asp	Arg	Leu	Val	Glu	Asn	Ser	Thr
625			630						635			640			
Gly	Thr	Ile	Tyr	Ser	Leu	Leu	Glu	Lys	Ala	Gln	Thr	Gln	Gln	Glu	Lys
			645						650			655			
Asn	Lys	Gln	Glu	Leu	Leu	Glu	Leu	Asp	Lys	Trp	Ser	Ser	Leu	Trp	Asp
			660			665						670			
Trp	Phe	Asp	Ile	Thr	Gln	Trp	Leu	Trp	Tyr	Ile	Lys	Ile	Ala	Ile	Ile
675						680			685						
Ile	Val	Ala	Gly	Leu	Val	Gly	Leu	Arg	Ile	Leu	Met	Phe	Ile	Val	Asn
690						695			700						
Val	Val	Lys	Gln	Val	Arg	Gln	Gly	Tyr	Thr	Pro	Leu	Phe	Ser	Gln	Ile
705			710						715			720			
Pro	Thr	Gln	Ala	Glu	Gln	Asp	Pro	Glu	Gln	Pro	Gly	Gly	Ile	Ala	Gly
			725						730			735			
Gly	Gly	Gly	Gly	Arg	Asp	Asn	Ile	Arg	Trp	Thr	Pro	Ser	Pro	Ala	Gly
			740			745						750			
Phe	Phe	Ser	Ile	Val	Trp	Glu	Asp	Leu	Arg	Asn	Leu	Leu	Ile	Trp	Ile
755						760			765						
Tyr	Gln	Thr	Phe	Gln	Asn	Phe	Ile	Trp	Ile	Leu	Trp	Ile	Ser	Leu	Gln
770						775			780						
Ala	Leu	Lys	Gln	Gly	Ile	Ile	Ser	Leu	Ala	His	Ser	Leu	Val	Ile	Val
785			790						795			800			
His	Arg	Thr	Ile	Ile	Val	Gly	Val	Arg	Gln	Ile	Ile	Glu	Trp	Ser	Ser
			805			810						815			
Asn	Thr	Tyr	Ala	Ser	Leu	Arg	Val	Leu	Leu	Ile	Gln	Ala	Ile	Asp	Arg
			820			825						830			
Leu	Ala	Asn	Phe	Thr	Gly	Trp	Trp	Thr	Asp	Leu	Ile	Ile	Glu	Gly	Val
835						840			845						

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Val Tyr Ile Ala Arg Gly Ile Arg Asn Ile Pro Arg Arg Ile Arg Gln
850 855 860

Gly Leu Glu Leu Ala Leu Asn
865 870

<210> SEQ ID NO 321

<211> LENGTH: 865

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 321

Met Lys Ala Met Glu Arg Lys Lys Leu Asn Trp Ile Ile Cys Tyr Met
1 5 10 15

Leu Met Gly Leu Ile Thr Pro Cys Leu Thr Gly His Glu Trp Trp Ala
20 25 30

Thr Val Tyr Tyr Gly Val Pro Val Trp Lys Glu Ala Glu Thr Thr Leu
35 40 45

Phe Cys Ala Ser Glu Ala Ser Ala Leu Asn Lys Glu Ala His Asn Ile
50 55 60

Trp Ala Ser Gln Ala Cys Val Pro Val Asp Pro Asn Pro Gln Glu Val
65 70 75 80

Tyr Met Pro Asn Val Ser Glu Ser Phe Asn Met Trp Lys Asn Asn Met
85 90 95

Ala Glu Gln Met Gln Glu Asp Ile Ser Ser Leu Trp Asp Gln Ser Leu
100 105 110

Lys Pro Cys Val Lys Leu Thr Pro Ile Cys Val Thr Met Gln Cys Thr
115 120 125

Thr Leu Lys Asn Ile Ser Thr Thr Asn Thr Asn Ser Thr Glu Leu Lys
130 135 140

Asn Ile Lys Glu Ile Met Lys Asn Cys Thr Phe Asn Met Thr Thr Glu
145 150 155 160

Val Arg Asp Lys Lys Lys Glu Val Tyr Ser Leu Phe Tyr Val Asp Asp
165 170 175

Ile Val Pro Ile Gly Gly Asn Lys Ser Asn Asp Thr Tyr Arg Leu Ile
180 185 190

Asn Cys Asn Thr Thr Thr Val Thr Gln Ala Cys Pro Lys Thr Ser Phe
195 200 205

Glu Pro Ile Pro Ile His Tyr Cys Ala Pro Pro Gly Phe Ala Ile Met
210 215 220

Lys Cys Asn Asp Gln Lys Phe Asn Gly Thr Gly Pro Cys Thr Asn Ile
225 230 235 240

Ser Thr Val Gln Cys Thr His Gly Ile Arg Pro Val Val Thr Thr Gln
245 250 255

Val Ile Leu Asn Gly Ser Leu Ala Glu Glu Gly Pro Ala Val Ala Arg
260 265 270

Ser Glu Asp Leu Arg His Lys Ile Ile Ile Ile Gln Phe Lys Glu Gly
275 280 285

Ile Asn Ile Ile Cys Met Arg Pro Gly Asn Asn Ser Arg Gly Gln Leu
290 295 300

Gln Ile Gly Pro Gly Val Ser Phe Tyr Asn Ile Glu Asn Ile Val Gly
305 310 315 320

Asp Pro Arg Gln Ala His Cys Asn Val Ser Lys Ala Gln Trp Met Lys
325 330 335

Lys Leu Asn Glu Thr Ala Leu Ala Ile Ala Lys Tyr Asp Lys Ser Ile
340 345 350

Ala	Lys	Val	Thr	Phe	Thr	Pro	Asn	Pro	Gly	Gly	Asp	Pro	Glu	Ile	Thr
355						360						365			
Asn	Met	Met	Phe	Asn	Cys	Gly	Gly	Glu	Phe	Phe	Tyr	Cys	Asn	Thr	Thr
370						375				380					
Pro	Leu	Phe	Asn	Lys	Phe	Asn	Tyr	Thr	Cys	Asp	Asn	Gln	Asn	Asn	Cys
385						390				395				400	
Thr	Arg	Asn	Asn	Thr	Asn	Asp	Thr	Asn	Ser	Asp	Asn	Ser	Thr	Leu	Thr
				405				410						415	
Ile	Ser	Cys	Arg	Ile	Arg	Gln	Ile	Val	Asn	Ser	Trp	Met	Arg	Val	Gly
		420						425				430			
Lys	Gly	Ile	Tyr	Ala	Pro	Pro	Ile	Arg	Gly	Asn	Ile	Thr	Cys	Ile	Ser
435						440						445			
Asn	Ile	Thr	Gly	Leu	Leu	Leu	Glu	Ala	Gly	Gly	Lys	Trp	Glu	Asn	Asn
450						455						460			
Thr	Ile	Thr	Val	Tyr	Pro	Ser	Gly	Gly	Glu	Met	Arg	Asp	Leu	Trp	Arg
465				470						475				480	
Leu	Glu	Leu	Asn	Arg	Tyr	Lys	Val	Val	Ser	Ile	Glu	Pro	Ile	Gly	Val
				485				490						495	
Ala	Pro	Thr	Lys	Ala	Lys	Arg	His	Thr	Val	Ala	Arg	Glu	Lys	Arg	Ala
		500						505				510			
Ala	Phe	Gly	Leu	Gly	Ala	Leu	Phe	Leu	Gly	Phe	Leu	Gly	Ala	Ala	Gly
515						520						525			
Ser	Thr	Met	Gly	Ala	Ala	Ser	Val	Val	Leu	Thr	Val	Gln	Ala	Arg	His
530						535						540			
Leu	Leu	Ser	Gly	Ile	Val	Gln	Gln	Gln	Asn	Asn	Leu	Leu	Arg	Ala	Ile
545				550						555				560	
Glu	Ala	Gln	Gln	His	Leu	Leu	Gln	Leu	Ser	Val	Trp	Gly	Ile	Lys	Gln
				565				570						575	
Leu	Gln	Ala	Arg	Val	Leu	Ala	Val	Glu	Arg	Tyr	Leu	Lys	Asp	Gln	Gln
		580						585				590			
Ile	Leu	Ser	Leu	Trp	Gly	Cys	Ser	Gly	Lys	Ala	Ile	Cys	Tyr	Thr	Thr
595						600						605			
Val	Pro	Trp	Asn	Ala	Thr	Trp	Ser	Ala	Asn	Thr	Ser	Tyr	Asp	Glu	Ile
610						615						620			
Trp	Asn	Asn	Leu	Thr	Trp	Gln	Asp	Trp	Asp	Lys	Lys	Val	Lys	Asn	Tyr
625				630						635				640	
Ser	Gly	Val	Ile	Phe	Ser	Leu	Ile	Glu	Gln	Ala	Gln	Glu	Gln	Gln	Asn
				645				650						655	
Thr	Asn	Glu	Lys	Asp	Leu	Leu	Glu	Leu	Asp	Gln	Trp	Ser	Ser	Leu	Trp
		660				665						670			
Ser	Trp	Phe	Asn	Ile	Thr	Gln	Trp	Leu	Trp	Tyr	Ile	Lys	Ile	Phe	Leu
675						680						685			
Ile	Val	Val	Ala	Gly	Leu	Ile	Gly	Phe	Arg	Leu	Ile	Gly	Ile	Val	Met
690						695				700					
Ser	Val	Ile	Ala	Lys	Val	Arg	Gln	Gly	Tyr	Ser	Pro	Leu	Ser	Leu	Gln
705				710						715				720	

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Phe Tyr His Arg Leu Ile Asp Leu Leu Ser Ile Leu Trp Arg Thr Val
 770          775          780

Gln Ile Leu Gly Gln Ser Ile Asn Lys Gly Leu Gln Gln Leu Lys Asp
 785          790          795          800

Phe Ser Thr Tyr Leu Lys Gly Val Leu Thr Tyr Trp Gly Arg Glu Ile
          805          810          815

Gln Ile Ser Ala Thr Ser Leu Leu Asp Thr Thr Ala Ile Ala Val Ala
          820          825          830

Asn Gly Thr Asp Arg Ile Ile Glu Phe Val Gln Ile Val Gly Arg Gly
 835          840          845

Ile Leu His Ile Pro Arg Arg Ile Arg Gln Gly Leu Glu Arg Ser Leu
 850          855          860

Leu
865

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<210> SEQ ID NO 322
<211> LENGTH: 864
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

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<400> SEQUENCE: 322

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Met Lys Val Met Glu Glu Arg Lys Arg Ser Phe Asn Gly Phe Tyr Met
 1          5          10          15

Thr Met Thr Ile Cys Met Ala Leu Leu Ile Pro Cys Leu Thr Ser Asp
 20          25          30

Glu Trp Val Thr Val Tyr Tyr Gly Val Pro Val Trp Arg Glu Val Asn
 35          40          45

Thr Val Leu Phe Cys Ala Ser Asp Ala Lys Ala His Ser Thr Glu Ala
 50          55          60

His Asn Ile Trp Ala Thr His Ala Cys Val Pro Thr Asp Pro Asn Pro
 65          70          75          80

Gln Glu Val Val Leu His Asn Val Thr Glu Asp Phe Asn Met Trp Asn
 85          90          95

Asn Gln Met Val Glu Gln Met Gln Glu Asp Ile Ser Ser Leu Trp Asp
100          105          110

Gln Ser Leu Lys Pro Cys Val Lys Leu Thr Pro Leu Cys Val Thr Met
115          120          125

Lys Cys Ser Asn Val Thr Arg Gln Asn Ser Thr Ser Asn Ser Asn Asn
130          135          140

Asn Pro Thr Leu Lys Asn Asn Thr Ser Gly Lys Asn Glu Thr Gly Val
145          150          155          160

Leu Gln Met Lys Asn Cys Thr Phe Asn Thr Thr Thr Glu Leu Arg Asp
165          170          175

Lys Lys Lys Gln Val Tyr Ser Leu Phe Tyr Val Asp Asp Leu Gln Ser
180          185          190

Leu Gly Ser Gly Thr Gly Asp Thr Tyr Thr Met Ile Asn Cys Asn Thr
195          200          205

Thr Ala Ile Thr Gln Ala Cys Pro Lys Val Ser Phe Glu Pro Ile Pro
210          215          220

Ile His Tyr Cys Ala Pro Ala Gly Phe Ala Ile Leu Lys Cys Asn Asp
225          230          235          240

Val Asn Phe Ser Gly Lys Gly Lys Cys Arg Asn Val Ser Thr Val His
245          250          255

Cys Thr His Gly Ile Lys Pro Val Val Thr Thr Gln Leu Ile Ile Asn
260          265          270

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Gly	Ser	Leu	Ala	Thr	Glu	Asn	Ile	Thr	Val	Arg	Val	Asn	Asn	Ala	Ser
	275						280					285			
Lys	Asn	Thr	His	Asp	Trp	Ile	Val	Gln	Leu	Ser	Thr	Ala	Val	Asn	Leu
	290					295					300				
Thr	Cys	Lys	Arg	Val	Gly	Asn	Asn	Thr	Arg	Gly	Lys	Val	Gln	Ile	Gly
305					310					315					320
Pro	Gly	Met	Thr	Phe	Tyr	Asn	Met	Asp	His	Ile	Phe	Gly	Asp	Thr	Arg
				325					330					335	
Lys	Ala	Phe	Cys	Glu	Leu	Asn	Gly	Thr	Thr	Trp	Asn	Glu	Thr	Leu	Gln
			340					345					350		
Lys	Val	Arg	Glu	Ser	Leu	Ile	Lys	Glu	Ile	Lys	Ala	Asn	Ala	Asn	Gly
	355						360					365			
Thr	Tyr	Asn	Ile	Thr	Phe	Glu	Pro	Ser	Ser	Gly	Gly	Asp	Pro	Glu	Ile
	370					375					380				
Ala	Asn	His	Met	Phe	Asn	Cys	Gly	Gly	Glu	Phe	Phe	Tyr	Cys	Asp	Thr
385					390					395					400
Arg	Lys	Met	Phe	Asn	Glu	Ser	Glu	Pro	Phe	His	Glu	Asn	Met	Thr	Ile
				405					410					415	
Pro	Cys	Arg	Ile	Arg	Gln	Ile	Val	Asn	Ser	Trp	Met	Arg	Val	Gly	Arg
			420					425					430		
Gly	Ile	Tyr	Ala	Pro	Pro	Ile	Pro	Gly	His	Ile	Thr	Cys	Asn	Ser	Leu
	435						440					445			
Ile	Thr	Gly	Leu	Ile	Leu	Thr	Arg	Asp	His	Val	Asn	Asn	Thr	Asn	Asn
	450					455					460				
Thr	Phe	Arg	Pro	Ile	Gly	Gly	Asp	Met	Lys	Asn	Ile	Trp	Arg	Ser	Glu
465					470					475					480
Leu	Tyr	Lys	Tyr	Lys	Val	Val	Arg	Ile	Glu	Pro	Leu	Ser	Val	Ala	Pro
				485					490					495	
Thr	Lys	Ala	Lys	Arg	His	Thr	Val	Gly	Glu	Arg	Arg	Gln	Lys	Arg	Ala
			500					505					510		
Ala	Phe	Gly	Leu	Gly	Ala	Leu	Phe	Leu	Gly	Phe	Leu	Gly	Ala	Ala	Gly
	515					520						525			
Ser	Thr	Met	Gly	Ala	Ala	Ala	Val	Thr	Leu	Thr	Val	Gln	Ala	Arg	Gln
	530					535					540				
Leu	Leu	Ser	Gly	Ile	Val	Gln	Gln	Gln	Asn	Asn	Leu	Leu	Arg	Ala	Ile
545					550					555					560
Glu	Ala	Gln	Gln	His	Leu	Leu	Gln	Leu	Thr	Val	Trp	Gly	Val	Lys	Gln
				565					570					575	
Leu	Gln	Ala	Arg	Leu	Leu	Ala	Val	Glu	Arg	Tyr	Leu	Gln	Asp	Gln	Gln
			580					585					590		
Ile	Leu	Gly	Leu	Trp	Gly	Cys	Ser	Gly	Lys	Ser	Ile	Cys	Tyr	Thr	Thr
	595					600						605			
Val	Pro	Trp	Asn	Lys	Thr	Trp	Ser	Gly	Lys	Ser	Met	Ser	Asp	Ile	Trp
	610					615					620				
Asn	Asn	Leu	Thr	Trp	Gln	Gln	Trp	Asp	Lys	Leu	Ile	Thr	Asn	Tyr	Thr
625					630					635					640
Gly	Thr	Ile	Phe	Gly	Leu	Leu	Glu	Glu	Ala	Gln	Ser	Gln	Gln	Glu	Lys
				645					650					655	
Asn	Glu	Lys	Asp	Leu	Leu	Glu	Leu	Asp	Gln	Trp	Ala	Ser	Leu	Trp	Asn
			660					665					670		
Trp	Phe	Asp	Ile	Thr	Asn	Trp	Leu	Trp	Tyr	Ile	Lys	Ile	Phe	Leu	Met
	675						680					685			

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Ala	Val	Gly	Gly	Ile	Ile	Gly	Leu	Arg	Ile	Ile	Met	Ser	Val	Val	Ser
690						695					700				
Val	Ile	Arg	Arg	Val	Arg	Gln	Gly	Tyr	Ser	Pro	Leu	Ser	Leu	Gln	Thr
705					710					715					720
Leu	Ile	Pro	Ala	Gln	Arg	Glu	Pro	Gly	Thr	Leu	Gly	Glu	Thr	Gly	Gly
				725					730					735	
Glu	Gly	Gly	Glu	Pro	Gly	Ser	Gly	Arg	Ser	Val	Arg	Leu	Val	Thr	Gly
			740					745					750		
Cys	Trp	Pro	Leu	Ile	Trp	Asp	Asp	Leu	Arg	Asn	Leu	Val	Ile	Trp	Ser
		755					760					765			
Tyr	Arg	Ser	Leu	Thr	Ser	Leu	Ala	Cys	Ile	Val	Trp	Arg	Gln	Leu	Glu
	770					775					780				
Lys	Leu	Gly	Gln	Leu	Leu	Ile	Asn	Ile	Leu	Arg	Leu	Ile	Gln	Glu	Lys
785					790					795					800
Leu	Thr	Leu	Leu	Arg	Gly	Ile	Ile	Gln	Tyr	Trp	Gly	Arg	Glu	Leu	Arg
				805					810					815	
Thr	Ser	Ala	Thr	Ser	Leu	Leu	Asp	Ala	Thr	Ala	Ile	Ala	Val	Gly	Glu
			820					825					830		
Gly	Thr	Asp	Arg	Ile	Ile	Arg	Ala	Val	Gln	Ile	Val	Phe	Arg	Ile	Ile
		835					840					845			
Gly	Asn	Ile	Pro	Arg	Arg	Ile	Arg	Gln	Gly	Leu	Glu	Arg	Thr	Leu	Leu
	850					855					860				

<210> SEQ ID NO 323

<211> LENGTH: 880

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 323

Met	Lys	Ala	Met	Glu	Thr	Gln	Arg	Asn	Cys	Arg	Thr	Leu	Ser	Leu	Lys
1			5					10						15	
Glu	Ile	Ile	Leu	Cys	Thr	Leu	Val	Leu	Gly	Ile	Ile	Gly	Ile	Ile	Lys
			20					25					30		
Cys	Glu	Asp	Asn	Met	Trp	Val	Thr	Val	Tyr	Tyr	Gly	Val	Pro	Val	Trp
	35					40					45				
Arg	Glu	Ala	Asp	Thr	Thr	Leu	Phe	Cys	Ala	Ser	Asp	Ala	Lys	Ala	Gln
	50					55					60				
Asn	Pro	Glu	Val	His	Asn	Val	Trp	Ala	Ser	Gln	Ala	Cys	Val	Ser	Thr
65				70				75						80	
Asn	Pro	Asn	Pro	Glu	Ile	Glu	Leu	Thr	Asn	Val	Thr	Glu	Tyr	Phe	
			85					90					95		
Asn	Ala	Trp	Glu	Asn	Asn	Met	Val	Glu	Gln	Met	His	Glu	Asp	Ile	Val
	100							105					110		
Asn	Leu	Trp	Asp	Gln	Ser	Val	Lys	Pro	Cys	Val	Lys	Leu	Ile	Pro	Leu
	115						120					125			
Cys	Val	Thr	Leu	Asn	Cys	Ser	Leu	Phe	Lys	Cys	Ile	Lys	Glu	Asn	Gly
	130					135					140				
Asn	Thr	Thr	Asn	Cys	Thr	Val	Gln	Ile	Ser	Thr	Gly	Asn	Asp	Ser	Thr
145					150					155					160
Ala	Asn	Asn	Ile	Thr	Val	Gly	Thr	Ile	Asp	Met	Tyr	Asn	Cys	Ser	Phe
			165					170						175	
Asn	Ala	Thr	Thr	Glu	Leu	Arg	Asp	Arg	Lys	Lys	Gln	Val	Tyr	Ser	Leu
			180					185					190		
Phe	Tyr	Arg	Gln	Asp	Leu	Glu	Pro	Leu	Glu	Gly	Asn	Lys	Pro	Pro	Glu
	195						200					205			

Gly 210	Asp	Lys	Asn	Ala		Leu	Tyr 215	Arg	Leu	Tyr		Asn	Cys 220	Asn	Thr	Thr	Ala
Met 225	Thr	Gln	Ala	Cys	Ser 230	Lys	Val	Ser	Leu	Glu	Pro 235	Ile	Pro	Ile		His 240	
Tyr	Cys	Ala	Pro	Ala 245	Gly	Phe	Ala	Leu	Leu 250	Lys	Cys	Asn	Asp	Lys 255	Asn		
Phe	Thr	Gly	Ile 260	Gly	Gln	Cys	Lys	Asn 265	Val	Ser	Thr	Val	His 270	Cys	Thr		
His	Gly 275	Ile	Arg	Pro	Val	Val	Ser 280	Thr	Gln	Phe	Leu 285	Leu	Asn	Gly	Thr		
Leu 290	Glu	Glu	Lys	Val	Thr	Val 295	Leu	Asp	Arg	Asn	Val 300	Ser	Asn	Asp	Met		
Asp 305	Thr	Ile	Ile	Val	Lys 310	Leu	Asn	Glu	Thr	Val 315	Arg	Leu	Asn	Cys	Thr 320		
Arg	Thr	Gly	Asn 325	Asn	Thr	Ile	Lys	Gly	Ile 330	Pro	Ile	Gly	Pro	Ser 335	Gln		
Ile	Phe	Tyr	Gly 340	Ile	Glu	Thr	Val	Ile	Gly 345	Asp	Thr	Arg	Gln	Ala	Phe		
Cys	Gln 355	Leu	Asn	Lys	Thr	Val	Trp 360	Thr	Asn	Thr	Phe	Lys 365	Lys	Val	Arg		
His 370	Ala	Leu	Asn	Glu	Thr	Tyr 375	Lys	Gly	Tyr	Leu	Gly 380	Asn	Glu	Thr	Ile		
Thr 385	Phe	Gly	Pro	Ser	Thr 390	Gly	Gly	Asp	Leu	Glu	Val 395	Thr	Asn	Leu	His 400		
Leu	Ile	Cys	Gly 405	Gly	Glu	Phe	Phe	Tyr	Cys 410	Asn	Thr	Ser	Ile	Leu 415	Phe		
Asn	Thr	Ser	Ile 420	Ile	Phe	Asn	Glu	Thr	Lys 425	Asp	Asp	Asn	Ile	Thr	Ile		
Pro	Cys	Arg 435	Ile	Arg	Gln	Ile	Val 440	Arg	Leu	Trp	Gln	Arg 445	Val	Gly	Arg		
Gly 450	Ile	Phe	Leu	Pro	Pro	Ile 455	Arg	Gly	Thr	Ile	Asn	Cys 460	Ile	Ser	Asn		
Ile 465	Thr	Gly	Ile	Leu	Phe	Ala 470	Gln	Gln	Lys	Thr	Asp	Arg	Met	Asn	Lys 480		
Ser	Ala	Met	Phe 485	Thr	Pro	Val	Gly	Gly	Glu 490	Met	Arg	Asn	Asn	Trp	Arg 495		
Ser	Glu	Leu	Tyr 500	Lys	Tyr	Lys	Val	Val	Arg 505	Ile	Glu	Pro	Leu	Gly	Val 510		
Ala	Pro	Thr 515	Lys	Ala	Lys	Arg	Arg 520	Thr	Val	His	Arg	Glu	Lys	Arg	Ala		
Ala	Val	Gly 530	Leu	Gly	Ala	Leu 535	Phe	Leu	Gly	Phe	Leu	Gly 540	Ala	Ala	Gly		
Ser 545	Thr	Met	Gly	Ala	Ala 550	Ser	Leu	Thr	Leu	Thr	Val	Gln	Ala	Arg	Gln 560		
Leu	Leu	Ser	Gly 565	Ile	Val	Gln	Gln	Gln	Ser 570	Asn	Leu	Leu	Arg	Ala	Ile 575		
Glu	Ala	Gln	Gln 580	His	Leu	Leu	Gln	Leu	Ser 585	Val	Trp	Gly	Ile	Lys	Gln 590		
Leu	Gln	Ala 595	Arg	Val	Leu	Ala	Val 600	Glu	Arg	Tyr	Leu	Lys 605	Asp	Gln	Gln		
Leu	Leu	Gly 610	Leu	Trp	Gly	Cys 615	Ser	Gly	Lys	Leu	Ile 620	Cys	Thr	Thr	Ser		

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Val	Pro	Trp	Asn	Thr	Thr	Trp	Thr	Asn	Lys	Ser	Tyr	Asp	Asp	Ile	Trp	625	630	635	640
Tyr	Asn	Met	Thr	Trp	Met	Gln	Trp	Asp	Lys	Glu	Val	Ser	Asn	Tyr	Thr	645	650	655	
Asp	Val	Ile	Tyr	Asn	Leu	Leu	Glu	Lys	Ala	Gln	Thr	Gln	Gln	Glu	Asn	660	665	670	
Asn	Glu	Lys	Glu	Leu	Leu	Glu	Leu	Asp	Lys	Trp	Ala	Ser	Leu	Trp	Asn	675	680	685	
Trp	Phe	Asp	Ile	Thr	Ser	Trp	Leu	Trp	Tyr	Ile	Lys	Ile	Phe	Ile	Ile	690	695	700	
Ile	Val	Gly	Gly	Leu	Ile	Gly	Leu	Arg	Ile	Val	Phe	Ala	Leu	Leu	Ser	705	710	715	720
Ile	Val	Asn	Arg	Val	Arg	Gln	Gly	Tyr	Ser	Pro	Leu	Ser	Phe	Gln	Thr	725	730	735	
Leu	Ile	Pro	Ala	Arg	Arg	Asp	Arg	Asp	Arg	Pro	Glu	Glu	Ile	Glu	Glu	740	745	750	
Gly	Gly	Gly	Glu	Pro	Asp	Asn	Val	Arg	Ser	Ile	Arg	Leu	Val	Ser	Gly	755	760	765	
Phe	Leu	Ala	Leu	Ala	Trp	Asn	Asp	Leu	Arg	Asp	Leu	Cys	Leu	Phe	Leu	770	775	780	
Tyr	His	Arg	Leu	Arg	Asp	Leu	Leu	Leu	Ile	Val	Leu	Arg	Thr	Leu	Glu	785	790	795	800
Leu	Val	Gly	Gln	Thr	Leu	Leu	Lys	Gly	Leu	Arg	Arg	Gly	Arg	Glu	Ala	805	810	815	
Leu	Ile	His	Leu	Arg	Gly	Ile	Leu	Gln	Tyr	Trp	Gly	Gln	Glu	Leu	Lys	820	825	830	
Thr	Ser	Ala	Ile	Ser	Leu	Leu	Asp	Thr	Thr	Ala	Ile	Ala	Val	Ala	Glu	835	840	845	
Gly	Thr	Asp	Arg	Ile	Ile	Glu	Ile	Ala	Gln	Arg	Phe	Gly	Arg	Gly	Ile	850	855	860	
Leu	Asn	Ile	Pro	Arg	Arg	Ile	Arg	Gln	Gly	Leu	Glu	Arg	Ala	Leu	Leu	865	870	875	880

<210> SEQ ID NO 324

<211> LENGTH: 845

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 324

Met	Arg	Val	Lys	Gly	Thr	Trp	Thr	Ser	Cys	Met	Pro	Tyr	Tyr	Val	Val	1	5	10	15
Gly	Ile	Leu	Ser	Leu	Cys	Ser	Leu	Ile	Ile	Ser	Ser	Thr	Lys	Asn	Glu	20	25	30	
Asn	Leu	Trp	Val	Thr	Val	Tyr	Tyr	Gly	Val	Pro	Val	Trp	Lys	Glu	Ala	35	40	45	
Glu	Ala	Thr	Leu	Phe	Cys	Ala	Ser	Asp	Ala	Lys	Ala	Gln	Lys	Ala	Glu	50	55	60	
Ala	His	Asn	Ile	Trp	Ala	Ser	Gln	Ala	Cys	Val	Pro	Thr	Asp	Pro	Asp	65	70	75	80
Pro	Lys	Ala	Ile	Glu	Leu	Ser	Asn	Val	Thr	Glu	Asn	Phe	Asn	Met	Trp	85	90	95	
Thr	Asn	Ala	Met	Val	Asn	Gln	Met	Gln	Gln	Asp	Val	Ile	Ser	Leu	Trp	100	105	110	
Asp	Gln	Ser	Leu	Lys	Pro	Cys	Ile	Lys	Leu	Thr	Pro	Leu	Cys	Val	Thr	115	120	125	

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Leu	His	Cys	Ser	Ile	Pro	Lys	Phe	Asp	Asn	Ser	Ser	Ile	Asn	Ser	Ser		
130						135					140						
Asn	Glu	Thr	Met	Lys	Asn	Cys	Ser	Phe	Asn	Thr	Thr	Thr	Glu	Leu	Arg		
145					150					155					160		
Asp	Lys	Lys	Lys	Gln	Thr	Tyr	Ser	Leu	Phe	Tyr	Val	Asp	Asp	Leu	Thr		
				165					170					175			
Gln	Ile	Asn	Lys	Thr	Thr	Glu	Ser	Tyr	Lys	Leu	Ile	Asn	Cys	Asn	Thr		
				180					185					190			
Thr	Ala	Ile	Thr	Gln	Ala	Cys	Pro	Lys	Ile	Ser	Phe	Glu	Pro	Ile	Pro		
				195				200					205				
Ile	His	Tyr	Cys	Ala	Pro	Ala	Gly	Phe	Ala	Leu	Leu	Arg	Cys	Asn	Asn		
	210					215						220					
Lys	Thr	Tyr	Asn	Gly	Ser	Gly	Pro	Cys	Asn	Asn	Val	Ser	Thr	Val	His		
225					230					235					240		
Cys	Thr	His	Gly	Ile	Lys	Pro	Val	Ile	Ser	Thr	Gln	Leu	Ile	Leu	Asn		
				245					250					255			
Gly	Ser	Leu	Ala	Glu	Glu	Lys	Ile	Met	Ile	Arg	Tyr	Lys	Ala	Asp	Lys		
			260					265					270				
Val	Leu	Val	Gln	Leu	Asn	Thr	Ser	Ile	His	Ile	Asn	Cys	Thr	Arg	Val		
			275					280				285					
Gly	Asn	Lys	Thr	Ile	Lys	Gly	Ile	Pro	Leu	Gly	Pro	Gly	Gln	Leu	Phe		
	290					295					300						
Tyr	Gly	Thr	Glu	Thr	Val	Val	Gly	Asp	Thr	Arg	Glu	Ala	His	Cys	Glu		
305					310					315					320		
Ile	Asn	Gln	Thr	His	Trp	Tyr	Lys	Ile	Leu	Asn	Gln	Val	Lys	Arg	Glu		
				325					330					335			
Leu	Thr	Thr	Val	Phe	Asn	Glu	Ser	Asn	Lys	Thr	Val	His	Phe	Ala	Asn		
			340					345					350				
Ser	Ser	Gly	Gly	Asp	Pro	Glu	Val	Ala	Asn	Leu	His	Phe	Asn	Cys	Gly		
		355					360					365					
Gly	Glu	Phe	Phe	Tyr	Cys	Asn	Thr	Ser	Ala	Leu	Phe	Asn	Asp	Thr	Leu		
	370					375					380						
Leu	Arg	Asn	Lys	Thr	Leu	Leu	Asn	Glu	Thr	Leu	His	Asn	Asn	Leu	Thr		
385					390					395					400		
Leu	Pro	Cys	Arg	Ile	Arg	Gln	Ile	Val	Asn	Leu	Trp	Leu	Arg	Val	Gly		
			405					410						415			
Lys	Gly	Ile	Phe	Ala	Pro	Pro	Ile	Arg	Gly	Asn	Ile	Arg	Cys	Asn	Ser		
			420					425					430				
Thr	Ile	Thr	Gly	Leu	Ile	Leu	Glu	Lys	His	Thr	Asn	Thr	Asp	Asn	Ile		
			435				440						445				
Thr	Phe	Arg	Pro	Ile	Gly	Gly	Asp	Met	Thr	Asp	Ile	Trp	Arg	Ser	Glu		
	450					455					460						
Leu	Tyr	Asn	Tyr	Lys	Ile	Val	Lys	Ile	Glu	Pro	Leu	Gly	Val	Ala	Pro		
465					470					475					480		
Thr	Arg	Ala	Arg	Arg	Arg	Thr	Val	Asp	Arg	Glu	Lys	Arg	Ala	Ala	Gly		
				485					490					495			
Leu	Gly	Val	Phe	Phe	Leu	Gly	Phe	Leu	Gly	Ala	Ala	Gly	Ser	Thr	Met		
			500				505						510				
Gly	Ala	Ala	Ser	Leu	Thr	Leu	Thr	Val	Gln	Ala	Arg	Gln	Leu	Leu	Thr		
		515					520					525					
Gly	Ile	Val	Gln	Gln	Gln	Ser	Asn	Leu	Leu	Arg	Ala	Ile	Glu	Ala	Gln		
	530					535					540						

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Gln His Leu Leu Gln Leu Ser Val Trp Gly Ile Lys Gln Leu Gln Ala
 545 550 555 560
 Arg Val Leu Ala Ile Glu Arg Tyr Leu Lys Asp Gln Gln Leu Leu Gly
 565 570 575
 Ile Trp Gly Cys Ser Gly Lys Leu Ile Cys Thr Thr Ser Val Pro Trp
 580 585 590
 Asn Arg Thr Trp Ser Asn Lys Thr Tyr Asn Glu Ile Trp Asp Asn Met
 595 600 605
 Thr Trp Met Glu Trp Asp Arg Glu Val Arg Asn Tyr Thr Glu Ile Ile
 610 615 620
 Tyr Gly Leu Ile Glu Gln Ala Gln Asp Gln Gln Glu Asn Asn Glu Lys
 625 630 635 640
 Lys Leu Leu Glu Leu Asp His Trp Thr Ser Leu Trp Asn Trp Phe Asp
 645 650 655
 Ile Ser His Trp Leu Trp Tyr Ile Lys Ile Phe Ile Met Ile Ile Gly
 660 665 670
 Gly Leu Ile Val Cys Arg Ile Ile Phe Ala Val Leu Ala Ile Val Asn
 675 680 685
 Arg Val Arg Gln Gly Tyr Ser Pro Leu Ser Phe Gln Thr Leu Ser Pro
 690 695 700
 Ile Pro Arg Gly Leu Ala Gln Pro Gly Gly Thr Glu Glu Glu Gly Gly
 705 710 715 720
 Glu Gln Gly Thr Asp Arg Ser Val Arg Leu Leu Asp Gly Phe Leu Ala
 725 730 735
 Leu Ala Trp Asn Asp Leu Arg Asp Leu Cys Leu Phe Leu Tyr His Arg
 740 745 750
 Leu Thr Asp Leu Val Leu Ile Ala Arg Arg Thr Leu Glu Ile Val Gly
 755 760 765
 Gln Tyr Thr Leu Lys Gly Leu Arg Leu Val Trp Glu Ala Leu Leu Tyr
 770 775 780
 Leu Gln Gly Ile Leu Gln Tyr Trp Gly Lys Glu Leu Lys Thr Ser Ala
 785 790 795 800
 Thr Ser Leu Leu Asp Thr Thr Ala Ile Ala Val Ala Glu Gly Thr Asp
 805 810 815
 Arg Ile Ile Glu Ile Val Gln Arg Ile Gly Arg Gly Ile Leu His Ile
 820 825 830
 Pro Arg Arg Ile Arg Gln Gly Phe Glu Arg Ala Leu Leu
 835 840 845

<210> SEQ ID NO 325

<211> LENGTH: 835

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 325

Met Lys Val Thr Glu Met Gln Lys Asn Trp Leu Ile Cys Cys Leu Leu
 1 5 10 15
 Ile Gly Leu Ile Lys Ile Ile Gly Ser Glu Leu Trp Val Thr Val Tyr
 20 25 30
 Tyr Gly Val Pro Val Trp Arg Asp Ala Glu Thr Val Leu Phe Cys Ala
 35 40 45
 Ser Asp Ala Lys Ala His Ser Thr Glu Ala His Asn Ile Trp Ala Thr
 50 55 60
 Gln Ala Cys Val Pro Thr Asp Pro Asn Pro Gln Glu Val Leu Ile Pro
 65 70 75 80

Asn	Val	Thr	Glu	Arg	Phe	Asp	Met	Trp	Lys	Asn	Asn	Met	Val	Asp	Gln	
				85					90					95		
Met	Gln	Glu	Asp	Ile	Ile	Ser	Leu	Trp	Glu	Gln	Ser	Leu	Lys	Pro	Cys	
				100					105					110		
Val	Lys	Leu	Thr	Pro	Leu	Cys	Val	Thr	Leu	Ser	Cys	Ser	Ser	Trp	Arg	
				115					120					125		
Ser	Val	Asn	Asn	Ser	Val	Asn	Gln	Thr	Asn	His	Val	Gln	Met	Gln	Asn	
				130					135					140		
Cys	Ser	Phe	Asn	Val	Thr	Thr	Glu	Leu	Arg	Asp	Lys	Lys	Lys	Gln	Val	
				145					150					155		
Tyr	Ser	Leu	Phe	Tyr	Met	Gly	Asp	Ile	Ile	Pro	Leu	Asp	Thr	Asn	Asn	
				165					170					175		
Ser	Ser	Gly	Asn	Asn	Ser	Gln	Tyr	Arg	Leu	Ile	Asn	Cys	Asn	Thr	Thr	
				180					185					190		
Ala	Val	Thr	Gln	Ala	Cys	Pro	Lys	Ile	Ser	Phe	Glu	Pro	Ile	Pro	Ile	
				195					200					205		
Tyr	Tyr	Cys	Ala	Pro	Pro	Gly	Phe	Ala	Ile	Ile	Lys	Cys	Asn	Asp	Gln	
				210					215					220		
Asp	Phe	Asn	Gly	Thr	Gly	Glu	Cys	Asn	Asn	Val	Ser	Thr	Val	Gln	Cys	
				225					230					235		
Thr	His	Gly	Ile	Lys	Pro	Val	Ile	Ser	Thr	Gln	Leu	Ile	Leu	Asn	Gly	
				245					250					255		
Ser	Leu	Ala	Thr	Ser	Asn	Ile	Val	Ile	Arg	Asn	Asn	Ser	Lys	Asp	Thr	
				260					265					270		
Leu	Leu	Val	Gln	Leu	Asn	Glu	Ser	Ile	Pro	Ile	Asn	Cys	Thr	Arg	Pro	
				275					280					285		
Gly	Asn	Lys	Thr	Arg	Gly	Gln	Val	Gln	Ile	Gly	Pro	Gly	Met	Thr	Phe	
				290					295					300		
Tyr	Asn	Ile	Glu	Asn	Ile	Ile	Gly	Asp	Thr	Arg	Gln	Ala	Tyr	Cys	Glu	
				305					310					315		
Val	Asn	Arg	Thr	Trp	Glu	Gln	Ile	Trp	Asn	Thr	Thr	Lys	Gln	Ile	Ile	
				325					330					335		
Ile	Asn	Asn	Arg	Lys	Asn	Ile	Thr	Phe	Ile	Pro	Asn	Pro	Gly	Gly	Asp	
				340					345					350		
Leu	Glu	Val	Thr	Asn	Leu	Met	Ile	Asn	Cys	Gly	Gly	Glu	Phe	Phe	Tyr	
				355					360					365		
Cys	Asn	Thr	Ser	Gln	Leu	Phe	Thr	Asn	Gln	Asn	Gly	Asn	Thr	Thr	Gly	
				370					375					380		
Asn	Ile	Thr	Leu	Gln	Cys	Arg	Ile	Arg	Gln	Ile	Val	Asn	Leu	Trp	Thr	
				385					390					395		
Arg	Val	Gly	Lys	Gly	Ile	Tyr	Ala	Pro	Pro	Ile	Lys	Gly	Pro	Ile	Asn	
				405					410					415		
Cys	Leu	Ser	Asn	Ile	Thr	Gly	Ile	Ile	Leu	Asp	Tyr	Thr	Lys	Ser	Gly	
				420					425					430		
Thr	Glu	Lys	Tyr	Thr	Ile	Tyr	Pro	Thr	Gly	Gly	Asp	Met	Thr	Asn	Leu	
				435					440					445		
Trp	Arg	Gln	Glu	Leu	Tyr	Lys	Tyr	Lys	Val	Val	Ser	Ile	Glu	Pro	Ile	
				450					455					460		
Gly	Val	Ala	Pro	Gly	Lys	Ala	Lys	Arg	His	Thr	Val	Thr	Arg	Gln	Lys	
				465					470					475		
Arg	Ala	Ala	Phe	Gly	Leu	Gly	Ala	Leu	Phe	Leu	Gly	Phe	Leu	Gly		

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Ala	Gly	Ser	Thr	Met	Gly	Ala	Ala	Ser	Ile	Thr	Leu	Thr	Val	Gln	Ala	
				500				505					510			
Arg	Lys	Leu	Leu	Ser	Gly	Ile	Val	Gln	Gln	Gln	Asn	Asn	Leu	Leu	Arg	
		515					520					525				
Ala	Ile	Glu	Ala	Gln	Gln	His	Leu	Leu	Gln	Leu	Ser	Val	Trp	Gly	Ile	
	530					535					540					
Lys	Gln	Leu	Gln	Ala	Arg	Val	Leu	Ala	Ile	Glu	Arg	Tyr	Leu	Arg	Asp	
545					550					555					560	
Gln	Gln	Ile	Leu	Gly	Leu	Trp	Gly	Cys	Ser	Gly	Lys	Ser	Val	Cys	Tyr	
			565					570						575		
Thr	Asn	Val	Pro	Trp	Asn	Thr	Thr	Trp	Ser	Asn	Asn	Asn	Ser	Tyr	Asp	
			580					585					590			
Thr	Ile	Trp	Gly	Asn	Met	Thr	Trp	Gln	Asn	Trp	Asp	Glu	Gln	Val	Arg	
		595					600					605				
Asn	Tyr	Ser	Gly	Val	Ile	Phe	Gly	Leu	Leu	Glu	Gln	Ala	Gln	Glu	Gln	
	610					615					620					
Gln	Ser	Ile	Asn	Glu	Lys	Ser	Leu	Leu	Glu	Leu	Asp	Gln	Trp	Ser	Ser	
625					630					635					640	
Leu	Trp	Asn	Trp	Phe	Asp	Ile	Thr	Lys	Trp	Leu	Trp	Tyr	Ile	Lys	Ile	
				645					650					655		
Phe	Ile	Met	Val	Val	Ala	Gly	Ile	Val	Gly	Ile	Arg	Ile	Ile	Ser	Ile	
			660					665					670			
Ile	Met	Ser	Met	Val	Ala	Arg	Val	Arg	Gln	Gly	Tyr	Ser	Pro	Leu	Ser	
		675					680					685				
Leu	Gln	Thr	Leu	Ile	Pro	Thr	Thr	Arg	Gly	Pro	Asp	Arg	Pro	Glu	Arg	
	690					695					700					
Thr	Glu	Glu	Asp	Ala	Gly	Glu	Leu	Asp	Asn	Gly	Arg	Ser	Val	Arg	Leu	
705					710					715					720	
Val	Ser	Gly	Phe	Leu	Ala	Leu	Ala	Trp	Glu	Asp	Phe	Arg	Asn	Leu	Leu	
				725					730					735		
Leu	Phe	Leu	Tyr	His	Arg	Leu	Thr	Asp	Cys	Leu	Ser	Ile	Leu	Arg	Arg	
			740					745					750			
Thr	Leu	Glu	Leu	Leu	Arg	Gln	Asn	Ile	His	Lys	Gly	Leu	Gln	Leu	Leu	
		755					760					765				
Asn	Glu	Leu	Arg	Ile	Tyr	Leu	Trp	Gly	Ile	Ile	Ala	Tyr	Trp	Gly	Arg	
	770					775					780					
Glu	Leu	Lys	Ile	Ser	Ala	Ile	Asn	Leu	Leu	Asp	Thr	Thr	Ala	Val	Ala	
785					790					795					800	
Val	Ala	Glu	Gly	Thr	Asp	Arg	Ile	Ile	Glu	Leu	Val	Gln	Arg	Ile	Gly	
				805					810					815		
Arg	Gly	Ile	Leu	His	Ile	Pro	Arg	Arg	Ile	Arg	Gln	Gly	Leu	Glu	Arg	
			820					825					830			
Ala	Leu	Leu														
		835														

<210> SEQ ID NO 326

<211> LENGTH: 855

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 326

Met	Lys	Val	Met	Gly	Met	Lys	Tyr	Tyr	Trp	Trp	Arg	Asn	Ser	Cys	Arg
1				5					10					15	

Ser	Ile	Ser	Ile	Lys	Leu	Ile	Leu	Ile	Gly	Trp	Ile	Ala	Ser	Cys	Phe
			20					25					30		

Gly 35	Glu 35	Asn 35	Trp 35	Trp 35	Val 40	Thr 40	Val 40	Tyr 40	Tyr 40	Gly 45	Val 45	Pro 45	Val 45	Trp 45	
Arg 50	Glu 50	Ala 50	Lys 50	Thr 50	Thr 55	Leu 55	Phe 55	Cys 55	Ala 55	Ser 60	Asp 60	Ala 60	Lys 60	Ser 60	Tyr 60
Ser 65	Thr 65	Glu 65	Ala 65	His 65	Asn 70	Ile 70	Trp 70	Ala 70	Thr 70	Gln 75	Ala 75	Cys 75	Val 75	Pro 75	Thr 80
Asp 85	Pro 85	Thr 85	Pro 85	Gln 85	Glu 85	Val 85	Leu 85	Leu 90	Pro 90	Asn 90	Val 90	Thr 95	Glu 95	Glu 95	Phe 95
Asn 100	Met 100	Trp 100	Glu 100	Asn 100	Tyr 100	Met 105	Val 105	Asp 105	Gln 105	Met 110	Gln 110	Glu 110	Asp 110	Ile 115	Ile 115
Ser 115	Leu 115	Trp 115	Glu 115	Gln 115	Ser 115	Leu 120	Lys 120	Pro 120	Cys 120	Val 125	Lys 125	Leu 125	Thr 125	Pro 130	Leu 130
Cys 130	Val 130	Thr 130	Leu 130	Thr 130	Cys 135	Asn 135	Asn 135	Pro 140	Thr 140	Asn 140	Thr 140	Ser 145	Cys 145	Thr 150	Asn 150
Ser 145	Thr 145	Asp 145	Asp 145	Arg 145	Leu 150	Gly 150	Asp 150	Met 155	Arg 155	Asn 155	Cys 155	Ser 160	Phe 160	Asn 165	Val 160
Thr 165	Thr 165	Glu 165	Leu 165	Arg 165	Asp 165	Lys 170	Lys 170	Arg 170	Gln 170	Val 175	Tyr 175	Ser 175	Leu 175	Phe 175	Tyr 175
Val 180	Glu 180	Asp 180	Ile 180	Thr 180	Ala 180	Ile 185	Gly 185	Asn 185	Asn 185	Ser 190	Thr 190	Tyr 190	Arg 190	Leu 195	Ile 195
Asn 195	Cys 195	Asn 195	Thr 195	Thr 195	Ala 195	Ile 200	Thr 200	Gln 200	Ala 200	Cys 205	Pro 205	Lys 205	Thr 205	Ser 210	Phe 210
Glu 210	Pro 210	Ile 210	Pro 210	Ile 210	His 215	Tyr 215	Cys 215	Ala 220	Pro 220	Ala 220	Gly 220	Phe 225	Ala 225	Leu 230	Leu 230
Lys 225	Cys 225	Asn 225	Asp 225	Ile 225	Asp 230	Tyr 230	Lys 230	Gly 235	Asn 235	Glu 235	Thr 235	Cys 240	Lys 240	Asn 245	Val 240
Ser 245	Thr 245	Val 245	His 245	Cys 245	Thr 245	His 250	Gly 250	Ile 250	Lys 250	Pro 255	Val 255	Ala 255	Thr 255	Thr 255	Gln 255
Leu 260	Ile 260	Leu 260	Asn 260	Gly 260	Ser 260	Thr 265	Ala 265	Asp 265	Asn 265	Gln 270	Thr 270	Val 270	Ala 270	Arg 275	Ile 275
Asp 275	Pro 275	Ser 275	Glu 275	Asn 275	Leu 275	Ala 280	Ile 280	Ile 280	Gln 280	Leu 285	Lys 285	Asp 285	Pro 285	Val 290	Lys 290
Ile 290	Thr 290	Cys 290	Arg 290	Arg 290	Pro 295	Gly 295	Asn 295	Asn 295	Thr 295	Arg 300	Gly 300	Gln 300	Ile 300	Gln 305	Ile 305
Gly 305	Pro 305	Ala 305	Met 305	Thr 305	Phe 310	Tyr 310	Asn 310	Ile 310	Glu 315	Asn 315	Val 315	Val 315	Gly 320	Asp 320	Thr 320
Arg 325	Lys 325	Ala 325	Tyr 325	Cys 325	Glu 325	Ile 330	Asn 330	Gly 330	Thr 330	Gln 330	Trp 335	Ala 335	Lys 335	Ala 340	Leu 340
Asn 340	Glu 340	Thr 340	Lys 340	Glu 340	Val 340	Leu 345	Arg 345	Asn 345	Ile 345	Leu 345	Arg 350	Lys 350	Asn 350	Ile 350	Ser 350
Phe 355	Met 355	Val 355	Pro 355	Ser 355	Gly 355	Gly 360	Asp 360	Pro 360	Glu 360	Val 365	Thr 365	Asn 365	His 365	His 370	Phe 370
Asn 370	Cys 370	Gly 370	Gly 370	Glu 370	Phe 375	Phe 375	Tyr 375	Cys 375	Asn 375	Thr 380	Ser 380	Glu 380	Ile 380	Ile 385	Asn 385
Ile 385	Thr 385	Lys 385	Ile 385	Asn 385	Lys 390	Thr 390	Glu 390	Asn 390	Met 395	Thr 395	Ile 395	Ile 395	Pro 395	Cys 400	Arg 400
Ile 405	Arg 405	Gln 405	Ile 405	Val 405	Asn 405	Ser 410	Trp 410	Met 410	Arg 410	Val 410	Gly 415	Lys 415	Gly 415	Ile 415	Phe 415
Ala 420	Pro 420	Pro 420	Ile 420	Arg 420	Gly 420	Asn 425	Ile 425	Thr 425	Cys 425	Thr 430	Ser 430	Asn 4			

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<210> SEQ ID NO 327
<211> LENGTH: 876
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 327

Met Arg Asn Leu Ile Gly Thr Thr Leu Thr Leu Ile Ile Ile Ile Leu
 1           5           10           15
Gly Ile Gly Ser Lys Glu Tyr Tyr Thr Thr Val Phe Tyr Gly Val Pro
          20           25           30
Ile Trp Lys Glu Ala Thr Pro Ser Leu Phe Cys Ala Ser Asn Ala Asp
          35           40           45
Ile Ala Ser Arg Asp Lys His Asn Ile Trp Ala Thr His Asn Cys Val
          50           55           60
Pro Leu Asp Pro Asn Pro Tyr Glu Val Pro Leu Ala Asn Val Ser Ile
          65           70           75           80
Asp Phe Asp Met Glu Lys Asn Tyr Met Val Glu Glu Met Lys Glu Asp
          85           90           95
Leu Leu Ser Leu Phe Gln Gln Ser Phe Lys Pro Cys Val Lys Leu Thr
          100          105          110
Pro Phe Cys Val Thr Met Lys Cys Thr Val Val Asn Thr Thr Val Val
          115          120          125
Asn Ala Thr Ala Pro Thr Pro Lys Val Ser Ala Thr Val Ala Thr Thr
          130          135          140
Pro Thr Ile Asn Leu Thr Asn Ile Lys Glu Tyr Glu Ile Phe Asn Cys
          145          150          155          160
Ser Phe Asn Gln Thr Thr Glu Phe Arg Asp Lys Lys Lys Gln Ile Tyr
          165          170          175
Ser Leu Phe Tyr Arg Glu Asp Ile Met Lys Val Gly Ser Glu Ser Thr
          180          185          190
Lys Asn Thr Asp Tyr Tyr Leu His Asn Cys Asn Thr Ser Ala Ile Thr
          195          200          205
Gln Glu Cys Asp Lys Ser Thr Phe Glu Pro Ile Pro Ile Arg Tyr Cys
          210          215          220
Ala Pro Ala Gly Phe Ala Met Leu Lys Cys Asn Asp Gln Gly Tyr Thr
          225          230          235          240
Gly Asn Gly Thr Cys Ser Asn Val Thr Val Val His Cys Thr His Gly
          245          250          255
Ile Phe Pro Met Ile Ala Thr Ala Leu His Leu Asn Gly Thr Leu Glu
          260          265          270
Glu Asn Glu Thr Lys Ala Tyr Phe Ala Asp Val Asn Val Asn Pro Pro
          275          280          285
Leu Leu Val Lys Phe Asn Glu Ser Val Ser Leu Thr Cys Glu Arg Thr
          290          295          300
Gly Asn Asn Thr Arg Gly Gln Val Gln Ile Gly Pro Gly Met Thr Phe
          305          310          315          320
Tyr Asn Ile Glu Asn Val Val Gly Asp Thr Arg Lys Ala Tyr Cys His
          325          330          335
Val Asn Ala Thr Leu Trp Asp Gln Ser Leu Ser Arg Ala Met Glu Ala
          340          345          350
Ile Asn Lys Thr Leu Ala Leu Tyr Asn Lys Ala Val Asn Glu Lys Phe
          355          360          365
Glu Trp Ser Lys Gly Asp Pro Glu Val Ser Ser Phe Trp Phe Asn Cys
          370          375          380

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Gln	Gly	Glu	Phe	Phe	Tyr	Cys	Asn	Leu	Thr	Gly	Trp	Thr	Asn	Pro	Trp	385	390	395	400
Asn	Asn	Thr	Asn	His	Arg	Asn	Gly	Thr	Leu	Val	Ala	Pro	Cys	Arg	Leu	405	410	415	
Arg	Gln	Ile	Val	Asn	His	Trp	Gly	Ile	Val	Ser	Lys	Gly	Ile	Tyr	Leu	420	425	430	
Pro	Pro	Arg	Arg	Gly	Thr	Leu	Lys	Cys	Asn	Ser	Asn	Ile	Thr	Gly	Leu	435	440	445	
Ile	Met	Thr	Ala	Glu	Lys	Gly	Gly	Asn	Asn	Thr	Val	Pro	Thr	Phe	Ser	450	455	460	
Gly	Lys	Val	Glu	Asp	Tyr	Trp	Lys	Val	Glu	Leu	Ala	Arg	Tyr	Lys	Val	465	470	475	480
Val	Glu	Ile	Gln	Pro	Leu	Ser	Val	Ala	Pro	Arg	Pro	Gly	Lys	Arg	Pro	485	490	495	
Glu	Ile	Lys	Ala	Asn	His	Thr	Arg	Ser	Arg	Arg	Asp	Val	Gly	Ile	Gly	500	505	510	
Leu	Leu	Phe	Leu	Gly	Phe	Leu	Ser	Ala	Ala	Gly	Ser	Thr	Met	Gly	Ala	515	520	525	
Ala	Ser	Ile	Ala	Leu	Thr	Ala	Gln	Ala	Arg	Gly	Leu	Leu	Ser	Asp	Ile	530	535	540	
Val	Gln	Gln	Gln	Gln	Asn	Leu	Leu	Gln	Ala	Ile	Glu	Ala	Gln	Gln	His	545	550	555	560
Leu	Leu	Gln	Leu	Ser	Val	Trp	Gly	Ile	Lys	Gln	Leu	Gln	Ala	Arg	Met	565	570	575	
Leu	Ala	Val	Glu	Lys	Tyr	Ile	Arg	Asp	Gln	Gln	Tyr	Leu	Ser	Ile	Trp	580	585	590	
Gly	Cys	Ala	Asn	Lys	Leu	Val	Cys	His	Ser	Ser	Val	Pro	Trp	Asn	Leu	595	600	605	
Ser	Trp	Ala	Glu	Asp	Ser	Gln	Lys	Cys	Asn	Ser	Thr	Asn	Thr	Lys	Tyr	610	615	620	
Tyr	Glu	Cys	Ile	Trp	Asn	Asn	Leu	Thr	Trp	Gln	Glu	Trp	Asp	Arg	Leu	625	630	635	640
Val	Gln	Asn	Ala	Ser	Glu	Thr	Ile	Tyr	Ser	Leu	Leu	Glu	Ile	Ala	Gln	645	650	655	
Thr	Gln	Gln	Glu	Lys	Asn	Lys	Gln	Glu	Leu	Leu	Glu	Leu	Asp	Lys	Trp	660	665	670	
Ser	Ser	Leu	Trp	Asp	Trp	Phe	Asp	Ile	Thr	Gln	Trp	Leu	Trp	Tyr	Ile	675	680	685	
Arg	Leu	Ala	Ile	Ile	Ile	Val	Ala	Gly	Leu	Val	Gly	Leu	Arg	Ile	Leu	690	695	700	
Met	Phe	Ile	Val	Asn	Val	Val	Lys	Lys	Val	Arg	Gln	Gly	Tyr	Met	Pro	705	710	715	720
Leu	Phe	Ser	Gln	Thr	Pro	Thr	Gln	Ala	Gln	Gln	Asp	Pro	Glu	Gln	Pro	725	730	735	
Gly	Gly	Ile	Ala	Glu	Gly	Gly	Gly	Arg	Gly	Asn	Ile	Arg	Trp	Thr		740	745	750	
Pro	Ser	Pro	Thr	Gly	Phe	Phe	Ser	Ile	Val	Trp	Glu	Asp	Leu	Arg	Asn	755	760	765	
Leu	Leu	Ile	Trp	Leu	Tyr	Gln	Thr	Cys	Arg	Asn	Phe	Ile	Trp	Val	Leu	770	775	780	
Trp	Thr	Ile	Leu	Gln	Ala	Leu	Lys	Gln	Gly	Thr	Ile	Ser	Leu	Ala	Asn	785	790	795	800
Asn	Leu	Val	Ile	Val	His	Arg	Tyr	Ile	Val	Val	Lys	Ile	Arg	Gln	Ile				

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805	810	815
Ile Glu Trp Cys His Asn Thr Tyr Ala Ser Leu Arg Ala Ser Leu Ile		
820	825	830
His Ala Ile Asp Arg Leu Ala Asp Phe Thr Gly Trp Trp Thr Asp Leu		
835	840	845
Ile Ile Glu Gly Ile Thr Tyr Ile Gly Arg Gly Ile Arg Asn Ile Pro		
850	855	860
Arg Arg Ile Arg Gln Gly Leu Glu Ile Ala Leu Asn		
865	870	875
 <210> SEQ ID NO 328		
<211> LENGTH: 896		
<212> TYPE: PRT		
<213> ORGANISM: Simian immunodeficiency virus		
 <400> SEQUENCE: 328		
Met Lys Met Arg Asn Leu Ile Gly Ile Thr Leu Ile Leu Ile Ile Leu		
1	5	10
Thr Leu Gly Ile Gly Ser Arg Asp Tyr Tyr Thr Thr Val Phe Tyr Gly		
20	25	30
Val Pro Ile Trp Lys Asp Ala Thr Pro Ser Leu Phe Cys Ala Ser Asp		
35	40	45
Ala Asp Val Thr Ser Arg Glu Lys His Asn Ile Trp Ala Thr His Asn		
50	55	60
Cys Val Pro Leu Asp Pro Asn Pro Tyr Glu Val Pro Ile Ala Asn Val		
65	70	75
Ser Ile Lys Phe Asn Met Glu Glu Asn Tyr Met Val Lys Glu Met Lys		
85	90	95
Glu Asp Leu Leu Ser Leu Phe Gln Gln Ser Phe Lys Pro Cys Val Lys		
100	105	110
Leu Thr Pro Phe Cys Val Val Met Asn Cys Thr Lys Ser Gln Asn Val		
115	120	125
Thr Thr Pro Ala Ser Thr Ser Thr Ala Ala Pro Val Asn Ala Thr Thr		
130	135	140
Ile Pro Val Asn Leu Thr Asp Ile Pro Asn Tyr Glu Leu Phe Asn Cys		
145	150	155
Ser Phe Gln Gln Thr Thr Glu Phe Arg Asp Lys Lys Lys Gln Ile Tyr		
165	170	175
Ser Leu Phe Tyr Arg Glu Asp Ile Met Lys Asp Asn Arg Ser Met Gln		
180	185	190
Asp Asn Arg Ser Asn Ser Ser Lys Lys Ser Asn Asn Ser Asn Asn Ser		
195	200	205
Gly Tyr Tyr Leu His Asn Cys Asn Thr Ser Ala Ile Thr Gln Glu Cys		
210	215	220
Asp Lys Ser Thr Phe Glu Pro Ile Pro Ile Arg Tyr Cys Ala Pro Ala		
225	230	235
Gly Tyr Ala Met Leu Lys Cys Lys Asp Gln Asn Tyr Thr Gly Asn Gly		
245	250	255
Ile Cys Tyr Asn Val Thr Val Val His Cys Thr His Gly Ile Phe Pro		
260	265	270
Met Ile Ala Thr Ala Leu His Leu Asn Gly Thr Leu Glu Glu Lys Glu		
275	280	285
Thr Lys Ala Tyr Phe Val Asn Ala Thr Asn Asn Pro Pro Leu Leu Val		
290	295	300

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Lys	Phe	Asn	Glu	Ser	Val	Pro	Leu	Thr	Cys	Gly	Arg	Thr	Gly	Asn	Asn	
305					310					315					320	
Val	Arg	Gly	Gln	Val	Gln	Ile	Gly	Pro	Gly	Met	Thr	Phe	Tyr	Asn	Thr	
			325						330					335		
Glu	Asn	Val	Val	Gly	Asp	Thr	Arg	Lys	Ala	Tyr	Cys	Trp	Val	Asn	Ala	
		340						345					350			
Thr	Leu	Trp	Tyr	Lys	Ser	Leu	Asp	Asn	Ala	Met	Glu	Ala	Ile	Thr	Lys	
	355					360						365				
Thr	Leu	Lys	Lys	Ser	Gly	Leu	Asn	Val	Thr	Arg	Arg	Phe	Asp	Trp	His	
	370					375					380					
Lys	Gly	Gly	Asp	Leu	Glu	Val	Gln	Asn	Phe	Trp	Phe	Asn	Cys	Gln	Gly	
385					390					395					400	
Glu	Phe	Phe	Tyr	Cys	Asn	Leu	Thr	Thr	Trp	Thr	Gly	Asn	Trp	Thr	Arg	
			405						410					415		
Asn	Arg	Thr	His	Gln	Thr	Ser	Ala	Leu	Asn	Lys	Thr	His	Leu	Leu	Ala	
			420					425					430			
Gln	Cys	Arg	Leu	Arg	Gln	Ile	Val	Asn	His	Trp	Gly	Ile	Val	Ser	Lys	
	435						440					445				
Gly	Ile	Tyr	Leu	Pro	Pro	Arg	Lys	Gly	His	Ile	Lys	Cys	Val	Ser	Asn	
	450					455					460					
Ile	Thr	Gly	Leu	Ile	Met	Thr	Ala	Glu	Gly	Gly	Asn	Ser	Asn	Arg	Thr	
465					470					475					480	
Pro	Thr	Phe	Ser	Ser	Lys	Val	Glu	Asp	Tyr	Trp	Lys	Val	Glu	Leu	Ala	
			485						490					495		
Arg	Tyr	Lys	Val	Val	Glu	Ile	Gln	Pro	Leu	Ser	Val	Ala	Pro	Arg	Pro	
			500					505					510			
Gly	Lys	Arg	Pro	Glu	Ile	Lys	Ala	Asn	His	Thr	Arg	Ser	Arg	Arg	Asp	
	515						520					525				
Val	Gly	Ile	Gly	Leu	Leu	Phe	Leu	Gly	Phe	Leu	Ser	Ala	Ala	Gly	Ser	
	530					535					540					
Thr	Met	Gly	Ala	Ala	Ser	Leu	Ala	Leu	Thr	Ala	Gln	Ala	Arg	Gly	Leu	
545					550					555					560	
Leu	Ser	Gly	Ile	Val	Gln	Gln	Gln	Gln	Asn	Leu	Leu	Gln	Ala	Ile	Glu	
			565						570					575		
Ala	Gln	Gln	His	Leu	Leu	Gln	Leu	Ser	Val	Trp	Gly	Ile	Lys	Gln	Leu	
			580					585					590			
Gln	Ala	Arg	Met	Leu	Ala	Val	Glu	Lys	Tyr	Ile	Arg	Asp	Gln	Gln	Tyr	
	595						600					605				
Leu	Ser	Ile	Trp	Gly	Cys	Ala	Asn	Lys	Leu	Val	Cys	His	Ser	Ser	Val	
	610					615					620					
Pro	Trp	Asn	Leu	Ser	Trp	Ala	Gly	Thr	Ser	Ser	Lys	Cys	Asn	Tyr	Thr	
625					630						635				640	
Gly	Ala	Lys	Tyr	Tyr	Glu	Cys	Ile	Trp	Asn	Asn	Leu	Thr	Trp	Gln	Glu	
			645						650					655		
Trp	Asp	Lys	Leu	Val	Gln	Asn	Ser	Ser	Glu	Thr	Ile	Tyr	Ser	Leu	Leu	
			660					665					670			
Glu	Thr	Ala	Gln	Thr	Gln	Gln	Glu	Arg	Asn	Lys	Gln	Glu	Leu	Leu	Glu	
	675						680					685				
Leu	Asp	Lys	Trp	Ser	Ser	Leu	Trp	Asp	Trp	Phe	Asp	Ile	Thr	Gln	Trp	
	690					695					700					
Leu	Trp	Tyr	Ile	Lys	Ile	Ala	Ile	Ile	Ile	Val	Ala	Gly	Leu	Val	Gly	
705					710					715					720	
Leu	Arg	Ile	Leu	Met	Phe	Ile	Val	Asn	Val	Val	Lys	Lys	Val	Arg	Gln	

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725					730					735					
Gly	Tyr	Met	Pro	Leu	Phe	Ser	Gln	Thr	Pro	Thr	Gln	Ala	Glu	Gln	Asp
			740					745					750		
Pro	Glu	Gln	Pro	Gly	Gly	Thr	Ala	Gly	Gly	Gly	Gly	Gly	Gly	Gly	Asn
		755					760					765			
Phe	Arg	Trp	Thr	Pro	Ser	Pro	Thr	Gly	Phe	Phe	Ser	Ile	Val	Trp	Glu
		770					775					780			
Asp	Leu	Arg	Asn	Leu	Leu	Ile	Trp	Ile	Tyr	Gln	Thr	Cys	Gln	Asn	Phe
				790								795			800
Ile	Trp	Val	Leu	Trp	Thr	Ile	Leu	Gln	Ala	Leu	Lys	Gln	Gly	Thr	Ile
			805						810					815	
Ser	Leu	Ala	His	Asn	Leu	Val	Ile	Val	His	Arg	Tyr	Ile	Ile	Val	Arg
			820						825					830	
Val	Arg	Gln	Thr	Ile	Glu	Trp	Cys	Gly	Asn	Thr	Tyr	Ala	Ser	Leu	Arg
			835					840					845		
Ala	Ser	Leu	Ile	His	Ala	Ile	Asp	Arg	Leu	Ala	Asp	Phe	Thr	Gly	Trp
		850					855					860			
Trp	Thr	Asp	Leu	Leu	Ile	Glu	Gly	Val	Val	Tyr	Ile	Ala	Arg	Gly	Ile
				870								875			880
Arg	Asn	Ile	Pro	Arg	Arg	Ile	Arg	Gln	Gly	Leu	Glu	Ile	Ala	Leu	Asn
				885					890					895	

<210> SEQ ID NO 329

<211> LENGTH: 871

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 329

Met	Lys	Asn	Leu	Ile	Gly	Ile	Thr	Leu	Ile	Leu	Ile	Ile	Thr	Ile	Leu
1			5					10					15		
Gly	Ile	Gly	Phe	Ser	Thr	Tyr	Tyr	Thr	Thr	Val	Phe	Tyr	Gly	Val	Pro
			20					25					30		
Val	Trp	Lys	Glu	Ala	Gln	Pro	Thr	Leu	Phe	Cys	Ala	Ser	Asp	Ala	Asp
		35					40					45			
Ile	Thr	Ser	Arg	Asp	Lys	His	Asn	Ile	Trp	Ala	Thr	His	Asn	Cys	Val
		50				55						60			
Pro	Leu	Asp	Pro	Asn	Pro	Tyr	Glu	Val	Thr	Leu	Ala	Asn	Val	Ser	Ile
		65			70							75			80
Arg	Phe	Asn	Met	Glu	Glu	Asn	Tyr	Met	Val	Gln	Glu	Met	Lys	Glu	Asp
			85					90					95		
Ile	Leu	Ser	Leu	Phe	Gln	Gln	Ser	Phe	Lys	Pro	Cys	Val	Lys	Leu	Thr
			100					105					110		
Pro	Phe	Cys	Ile	Lys	Met	Thr	Cys	Thr	Met	Thr	Asn	Thr	Thr	Asn	Lys
		115					120						125		
Thr	Leu	Asn	Ser	Ala	Thr	Thr	Thr	Leu	Thr	Pro	Thr	Val	Asn	Leu	Ser
		130					135					140			
Ser	Ile	Pro	Asn	Tyr	Glu	Val	Tyr	Asn	Cys	Ser	Phe	Asn	Gln	Thr	Thr
		145			150							155			160
Glu	Phe	Arg	Asp	Lys	Lys	Lys	Gln	Ile	Tyr	Ser	Leu	Phe	Tyr	Arg	Glu
				165					170					175	
Asp	Ile	Val	Lys	Glu	Asp	Gly	Asn	Asn	Asn	Ser	Tyr	Tyr	Leu	His	Asn
			180					185					190		
Cys	Asn	Thr	Ser	Val	Ile	Thr	Gln	Glu	Cys	Asp	Lys	Ser	Thr	Phe	Glu
		195					200						205		

Pro 210	Ile	Pro	Ile	Arg	Tyr	Cys 215	Ala	Pro	Ala	Gly	Phe 220	Ala	Leu	Leu	Lys
Cys 225	Arg	Asp	Gln	Asn	Phe 230	Thr	Gly	Lys	Gly	Gln 235	Cys	Ser	Asn	Val	Ser 240
Val	Val	His	Cys	Thr 245	His	Gly	Ile	Tyr	Pro 250	Met	Ile	Ala	Thr	Ala 255	Leu
His	Leu	Asn	Gly 260	Ser	Leu	Glu	Glu	Glu 265	Glu	Thr	Lys	Ala	Tyr 270	Phe	Val
Asn	Thr	Ser 275	Val	Asn	Thr	Pro	Leu 280	Leu	Val	Lys	Phe	Asn 285	Val	Ser	Ile
Asn 290	Leu	Thr	Cys	Glu	Arg	Thr 295	Gly	Asn	Asn	Thr	Arg 300	Gly	Gln	Val	Gln
Ile 305	Gly	Pro	Gly	Met 310	Thr	Phe	Tyr	Asn	Ile	Glu 315	Asn	Val	Val	Gly	Asp 320
Thr	Arg	Lys	Ala	Tyr 325	Cys	Ser	Val	Asn	Ala 330	Thr	Thr	Trp	Tyr 335	Arg	Asn
Leu	Asp	Trp	Ala 340	Met	Ala	Ala	Ile	Asn 345	Thr	Thr	Met	Arg 350	Ala	Arg	Asn
Glu	Thr 355	Val	Gln	Gln	Thr	Phe	Gln 360	Trp	Gln	Arg	Asp 365	Gly	Asp	Pro	Glu
Val 370	Thr	Ser	Phe	Trp	Phe	Asn 375	Cys	Gln	Gly	Glu	Phe 380	Phe	Tyr	Cys	Asn
Leu 385	Thr	Asn	Trp	Thr	Asn 390	Thr	Trp	Thr	Ala 395	Asn	Arg	Thr	Asn	Asn	Thr 400
His	Gly	Thr	Leu	Val 405	Ala	Pro	Cys	Arg	Leu 410	Arg	Gln	Ile	Val	Asn 415	His
Trp	Gly	Ile	Val 420	Ser	Lys	Gly	Val	Tyr 425	Leu	Pro	Pro	Arg 430	Arg	Gly	Thr
Val	Lys 435	Cys	His	Ser	Asn	Ile	Thr 440	Gly	Leu	Ile	Met 445	Thr	Ala	Glu	Lys
Asp 450	Asn	Asn	Asn	Ser	Tyr	Thr 455	Pro	Gln	Phe	Ser	Ala 460	Val	Val	Glu	Asp
Tyr 465	Trp	Lys	Val	Glu	Leu 470	Ala	Arg	Tyr	Lys 475	Val	Val	Glu	Ile	Gln	Pro 480
Leu	Ser	Val	Ala	Pro 485	Arg	Pro	Gly	Lys	Arg 490	Pro	Glu	Ile	Lys	Ala 495	Asn
His	Thr	Arg	Ser 500	Arg	Arg	Asp	Val	Gly 505	Ile	Gly	Leu	Leu	Phe 510	Leu	Gly
Phe	Leu 515	Ser	Ala	Ala	Gly	Ser	Thr 520	Met	Gly	Ala	Ala 525	Ser	Ile	Ala	Leu
Thr 530	Ala	Gln	Ala	Arg	Gly	Leu 535	Leu	Ser	Gly	Ile	Val 540	Gln	Gln	Gln	Gln
Asn 545	Leu	Leu	Gln	Ala	Ile 550	Glu	Ala	Gln	Gln	His 555	Leu	Leu	Gln	Leu	Ser 560
Val	Trp	Gly	Ile	Lys 565	Gln	Leu	Gln	Ala	Arg 570	Met	Leu	Ala	Val	Glu	Lys 575
Tyr	Ile	Arg	Asp 580	Gln	Gln	Leu	Leu	Ser 585	Leu	Trp	Gly	Cys 590	Ala	Asn	Lys
Leu	Val 595	Cys	His	Ser	Ser	Val	Pro 600	Trp	Asn	Leu	Thr	Trp 605	Ala	Glu	Asp
Ser 610	Thr	Lys	Cys	Asn	His	Ser 615	Asp	Ala	Lys	Tyr	Tyr 620	Asp	Cys	Ile	Trp
Asn	Asn	Leu	Thr	Trp	Gln	Glu	Trp	Asp	Arg	Leu	Val	Glu	Asn	Ser	Thr

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625	630	635	640
Gly Thr Ile Tyr Ser Leu Leu Glu Lys Ala Gln Thr Gln Gln Glu Lys	645	650	655
Asn Lys Gln Glu Leu Leu Glu Leu Asp Lys Trp Ser Ser Leu Trp Asp	660	665	670
Trp Phe Asp Ile Thr Gln Trp Leu Trp Tyr Ile Lys Ile Ala Ile Ile	675	680	685
Ile Val Ala Gly Leu Val Gly Leu Arg Ile Leu Met Phe Ile Val Asn	690	695	700
Val Val Lys Gln Val Arg Gln Gly Tyr Thr Pro Leu Phe Ser Gln Ile	705	710	715
Pro Thr Gln Ala Glu Gln Asp Pro Glu Gln Pro Gly Gly Ile Ala Gly	725	730	735
Gly Gly Gly Gly Arg Asp Asn Ile Arg Trp Thr Pro Ser Pro Ala Gly	740	745	750
Phe Phe Ser Ile Val Trp Glu Asp Leu Arg Asn Leu Leu Ile Trp Ile	755	760	765
Tyr Gln Thr Phe Gln Asn Phe Ile Trp Ile Leu Trp Ile Ser Leu Gln	770	775	780
Ala Leu Lys Gln Gly Ile Ile Ser Leu Ala His Ser Leu Val Ile Val	785	790	795
His Arg Thr Ile Ile Val Gly Val Arg Gln Ile Ile Glu Trp Ser Ser	805	810	815
Asn Thr Tyr Ala Ser Leu Arg Val Leu Leu Ile Gln Ala Ile Asp Arg	820	825	830
Leu Ala Asn Phe Thr Gly Trp Trp Thr Asp Leu Ile Ile Glu Gly Val	835	840	845
Val Tyr Ile Ala Arg Gly Ile Arg Asn Ile Pro Arg Arg Ile Arg Gln	850	855	860
Gly Leu Glu Leu Ala Leu Asn	865	870	

<210> SEQ ID NO 330
 <211> LENGTH: 876
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (81)..(81)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (91)..(91)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (179)..(179)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (428)..(428)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
 <400> SEQUENCE: 330

Met Arg Asn Leu Ile Gly Thr Thr Leu Thr Leu Ile Ile Ile Leu	1	5	10	15
Gly Ile Gly Ser Lys Glu Tyr Tyr Thr Thr Val Phe Tyr Gly Val Pro	20	25	30	
Ile Trp Lys Glu Ala Thr Pro Ser Leu Phe Cys Ala Ser Asn Ala Asp	35	40	45	

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Ile	Ala	Ser	Arg	Asp	Lys	His	Asn	Ile	Trp	Ala	Thr	His	Asn	Cys	Val
50						55					60				
Pro	Leu	Asp	Pro	Asn	Pro	Tyr	Glu	Val	Pro	Leu	Ala	Asn	Val	Ser	Ile
65				70					75					80	
Xaa	Phe	Asp	Met	Glu	Lys	Asn	Tyr	Met	Val	Xaa	Glu	Met	Lys	Glu	Asp
			85					90					95		
Leu	Leu	Ser	Leu	Phe	Gln	Gln	Ser	Phe	Lys	Pro	Cys	Val	Lys	Leu	Thr
			100				105						110		
Pro	Phe	Cys	Val	Thr	Met	Lys	Cys	Thr	Val	Val	Asn	Thr	Thr	Val	Val
	115					120					125				
Asn	Ala	Thr	Ala	Pro	Thr	Pro	Lys	Val	Ser	Ala	Thr	Val	Ala	Thr	Thr
130						135					140				
Pro	Thr	Ile	Asn	Leu	Thr	Asn	Ile	Lys	Glu	Tyr	Glu	Ile	Phe	Asn	Cys
145				150					155					160	
Ser	Phe	Asn	Gln	Thr	Thr	Glu	Phe	Arg	Asp	Lys	Lys	Lys	Gln	Ile	Tyr
			165					170					175		
Ser	Leu	Xaa	Tyr	Arg	Glu	Asp	Ile	Met	Lys	Val	Gly	Ser	Glu	Ser	Thr
		180					185						190		
Lys	Asn	Thr	Asp	Tyr	Tyr	Leu	His	Asn	Cys	Asn	Thr	Ser	Ala	Ile	Thr
	195					200					205				
Gln	Glu	Cys	Asp	Lys	Ser	Thr	Phe	Glu	Pro	Ile	Pro	Ile	Arg	Tyr	Cys
210						215					220				
Ala	Pro	Ala	Gly	Phe	Ala	Met	Leu	Lys	Cys	Asn	Asp	Gln	Gly	Tyr	Thr
225				230					235					240	
Gly	Asn	Gly	Thr	Cys	Ser	Asn	Val	Thr	Val	Val	His	Cys	Thr	His	Gly
			245					250					255		
Ile	Phe	Pro	Met	Ile	Ala	Thr	Ala	Leu	His	Leu	Asn	Gly	Thr	Leu	Glu
		260					265						270		
Glu	Asn	Glu	Thr	Lys	Ala	Tyr	Phe	Ala	Asp	Val	Asn	Val	Asn	Pro	Pro
	275					280					285				
Leu	Leu	Val	Lys	Phe	Asn	Glu	Ser	Val	Ser	Leu	Thr	Cys	Glu	Arg	Thr
290					295						300				
Gly	Asn	Asn	Thr	Arg	Gly	Gln	Val	Gln	Ile	Gly	Pro	Gly	Met	Thr	Phe
305				310					315					320	
Tyr	Asn	Ile	Glu	Asn	Val	Val	Gly	Asp	Thr	Arg	Lys	Ala	Tyr	Cys	His
			325				330						335		
Val	Asn	Ala	Thr	Leu	Trp	Asp	Gln	Ser	Leu	Ser	Arg	Ala	Met	Glu	Ala
		340					345						350		
Ile	Asn	Lys	Thr	Leu	Ala	Leu	Tyr	Asn	Lys	Ala	Val	Asn	Glu	Lys	Phe
	355					360					365				
Glu	Trp	Ser	Lys	Gly	Asp	Pro	Glu	Val	Ser	Ser	Phe	Trp	Phe	Asn	Cys
370					375						380				
Gln	Gly	Glu	Phe	Phe	Tyr	Cys	Asn	Leu	Thr	Gly	Trp	Thr	Asn	Pro	Trp
385				390					395					400	
Asn	Asn	Thr	Asn	His	Arg	Asn	Gly	Thr	Leu	Val	Ala	Pro	Cys	Arg	Leu
			405					410					415		
Arg	Gln	Ile	Val	Asn	His	Trp	Gly	Ile	Val	Ser	Xaa	Gly	Ile	Tyr	Leu
		420					425						430		
Pro	Pro	Arg	Arg	Gly	Thr	Leu	Lys	Cys	Asn	Ser	Asn	Ile	Thr	Gly	Leu
		435				440						445			
Ile	Met	Thr	Ala	Glu	Lys	Gly	Gly	Asn	Asn	Thr	Val	Pro	Thr	Phe	Ser
450					455						460				

Gly 465	Lys	Val	Glu	Asp	Tyr 470	Trp	Lys	Val	Glu	Leu	Ala	Arg	Tyr	Lys	Val 480
Val	Glu	Ile	Gln	Pro 485	Leu	Ser	Val	Ala	Pro 490	Arg	Pro	Gly	Lys	Arg	Pro 495
Glu	Ile	Lys	Ala	Asn 500	His	Thr	Arg	Ser	Arg 505	Arg	Asp	Val	Gly	Ile	Gly
Leu	Leu	Phe	Leu	Gly	Phe	Leu	Ser	Ala	Ala 520	Gly	Ser	Thr	Met	Gly	Ala
Ala	Ser	Ile	Ala	Leu	Thr	Ala	Gln	Ala	Arg 535	Gly	Leu	Leu	Ser	Asp	Ile
Val	Gln	Gln	Gln	Gln	Asn 550	Leu	Leu	Gln	Ala 555	Ile	Glu	Ala	Gln	Gln	His 560
Leu	Leu	Gln	Leu	Ser 565	Val	Trp	Gly	Ile	Lys 570	Gln	Leu	Gln	Ala	Arg	Met 575
Leu	Ala	Val	Glu	Lys	Tyr	Ile	Arg	Asp 585	Gln	Gln	Tyr	Leu	Ser	Ile	Trp
Gly	Cys	Ala	Asn	Lys	Leu	Val	Cys	His 600	Ser	Ser	Val	Pro	Trp	Asn	Leu
Ser	Trp	Ala	Glu	Asp	Ser	Gln	Lys	Cys 615	Asn	Ser	Thr	Asn	Thr	Lys	Tyr
Tyr 625	Glu	Cys	Ile	Trp	Asn 630	Asn	Leu	Thr	Trp 635	Gln	Glu	Trp	Asp	Arg	Leu 640
Val	Gln	Asn	Ala	Ser 645	Glu	Thr	Ile	Tyr	Ser 650	Leu	Leu	Glu	Ile	Ala	Gln 655
Thr	Gln	Gln	Glu	Lys 660	Asn	Lys	Gln	Glu	Leu 665	Leu	Glu	Leu	Asp	Lys	Trp
Ser	Ser	Leu	Trp	Asp	Trp	Phe	Asp 680	Ile	Thr	Gln	Trp	Leu	Trp	Tyr	Ile
Arg 690	Leu	Ala	Ile	Ile	Ile	Val	Ala 695	Gly	Leu	Val	Gly 700	Leu	Arg	Ile	Leu
Met 705	Phe	Ile	Val	Asn 710	Val	Val	Lys	Lys	Val 715	Arg	Gln	Gly	Tyr	Met	Pro 720
Leu	Phe	Ser	Gln	Thr 725	Pro	Thr	Gln	Ala	Gln 730	Gln	Asp	Pro	Glu	Gln	Pro 735
Gly	Gly	Ile	Ala	Glu 740	Gly	Gly	Gly	Gly	Arg 745	Gly	Asn	Ile	Arg	Trp	Thr
Pro	Ser	Pro	Thr	Gly	Phe	Phe	Ser 760	Ile	Val	Trp	Glu	Asp 765	Leu	Arg	Asn
Leu 770	Leu	Ile	Trp	Leu	Tyr	Gln	Thr 775	Cys	Arg	Asn	Phe	Ile	Trp	Val	Leu
Trp 785	Thr	Ile	Leu	Gln	Ala 790	Leu	Lys	Gln	Gly 795	Thr	Ile	Ser	Leu	Ala	Asn 800
Asn	Leu	Val	Ile	Val 805	His	Arg	Tyr	Ile	Val 810	Val	Lys	Ile	Arg	Gln	Ile 815
Ile	Glu	Trp	Cys	His 820	Asn	Thr	Tyr	Ala	Ser 825	Leu	Arg	Ala	Ser	Leu	Ile 830
His	Ala	Ile	Asp	Arg	Leu	Ala	Asp 840	Phe	Thr	Gly	Trp	Trp 845	Thr	Asp	Leu
Ile 850	Ile	Glu	Gly	Ile	Thr	Tyr	Ile 855	Gly	Arg	Gly	Ile	Arg	Asn	Ile	Pro
Arg 865	Arg	Ile	Arg	Gln	Gly	Leu	Glu 870	Ile	Ala	Leu	Asn				

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<210> SEQ ID NO 331
<211> LENGTH: 896
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (259)..(259)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (287)..(287)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (322)..(322)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (336)..(336)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
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<222> LOCATION: (338)..(338)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (340)..(340)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (361)..(361)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (371)..(371)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (379)..(379)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (380)..(380)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (382)..(382)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<400> SEQUENCE: 331

Met Lys Met Arg Asn Leu Ile Gly Ile Thr Leu Ile Leu Ile Ile Leu
1             5             10             15

Thr Leu Gly Ile Gly Ser Arg Asp Tyr Tyr Thr Thr Val Phe Tyr Gly
20             25             30

Val Pro Ile Trp Lys Asp Ala Thr Pro Ser Leu Phe Cys Ala Ser Asp
35             40             45

Ala Asp Val Thr Ser Arg Glu Lys His Asn Ile Trp Ala Thr His Asn
50             55             60

Cys Val Pro Leu Asp Pro Asn Pro Tyr Glu Val Pro Ile Ala Asn Val
65             70             75             80

Ser Ile Lys Phe Asn Met Glu Glu Asn Tyr Met Val Lys Glu Met Lys
85             90             95

Glu Asp Leu Leu Ser Leu Phe Gln Gln Ser Phe Lys Pro Cys Val Lys
100            105            110

Leu Thr Pro Phe Cys Val Val Met Asn Cys Thr Lys Ser Gln Asn Val
115            120            125

Thr Thr Pro Ala Ser Thr Ser Thr Ala Ala Pro Val Asn Ala Thr Thr
130            135            140

Ile Pro Val Asn Leu Thr Asp Ile Pro Asn Tyr Glu Leu Phe Asn Cys

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145					150						155					160
Ser	Phe	Gln	Gln	Thr	Thr	Glu	Phe	Arg	Asp	Lys	Lys	Lys	Gln	Ile	Tyr	
				165					170						175	
Ser	Leu	Phe	Tyr	Arg	Glu	Asp	Ile	Met	Lys	Asp	Asn	Arg	Ser	Met	Gln	
			180					185					190			
Asp	Asn	Arg	Ser	Asn	Ser	Ser	Lys	Lys	Ser	Asn	Asn	Ser	Asn	Asn	Ser	
		195					200					205				
Gly	Tyr	Tyr	Leu	His	Asn	Cys	Asn	Thr	Ser	Ala	Ile	Thr	Gln	Glu	Cys	
	210				215						220					
Asp	Lys	Ser	Thr	Phe	Glu	Pro	Ile	Pro	Ile	Arg	Tyr	Cys	Ala	Pro	Ala	
225					230					235					240	
Gly	Tyr	Ala	Met	Leu	Lys	Cys	Lys	Asp	Gln	Asn	Tyr	Thr	Gly	Asn	Gly	
			245						250					255		
Ile	Cys	Xaa	Asn	Val	Thr	Val	Val	His	Cys	Thr	His	Gly	Ile	Phe	Pro	
			260					265					270			
Met	Ile	Ala	Thr	Ala	Leu	His	Leu	Asn	Gly	Thr	Leu	Glu	Glu	Xaa	Glu	
		275					280					285				
Thr	Lys	Ala	Tyr	Phe	Val	Asn	Ala	Thr	Asn	Asn	Pro	Pro	Leu	Leu	Val	
	290					295					300					
Lys	Phe	Asn	Glu	Ser	Val	Pro	Leu	Thr	Cys	Gly	Arg	Thr	Gly	Asx	Asn	
305					310					315					320	
Val	Xaa	Gly	Gln	Val	Gln	Ile	Gly	Pro	Gly	Met	Thr	Phe	Tyr	Asn	Xaa	
			325						330					335		
Glu	Xaa	Val	Xaa	Gly	Asp	Thr	Arg	Lys	Ala	Tyr	Cys	Trp	Val	Asn	Ala	
			340					345					350			
Thr	Leu	Trp	Tyr	Lys	Ser	Leu	Asp	Xaa	Ala	Met	Glu	Ala	Ile	Thr	Lys	
	355						360					365				
Thr	Leu	Xaa	Lys	Ser	Gly	Leu	Asx	Val	Thr	Xaa	Xaa	Phe	Xaa	Trp	His	
	370					375				380						
Lys	Gly	Gly	Asp	Leu	Glu	Val	Gln	Asn	Phe	Trp	Phe	Asn	Cys	Gln	Gly	
385				390						395					400	
Glu	Phe	Phe	Tyr	Cys	Asn	Leu	Thr	Thr	Trp	Thr	Gly	Asn	Trp	Thr	Arg	
			405						410						415	
Asn	Arg	Thr	His	Gln	Thr	Ser	Ala	Leu	Asn	Lys	Thr	His	Leu	Leu	Ala	
			420					425					430			
Gln	Cys	Arg	Leu	Arg	Gln	Ile	Val	Asn	His	Trp	Gly	Ile	Val	Ser	Lys	
		435					440					445				
Gly	Ile	Tyr	Leu	Pro	Pro	Arg	Lys	Gly	His	Ile	Lys	Cys	Val	Ser	Asn	
	450					455					460					
Ile	Thr	Gly	Leu	Ile	Met	Thr	Ala	Glu	Gly	Gly	Asn	Ser	Asn	Arg	Thr	
465				470						475					480	
Pro	Thr	Phe	Ser	Ser	Lys	Val	Glu	Asp	Tyr	Trp	Lys	Val	Glu	Leu	Ala	
			485						490					495		
Arg	Tyr	Lys	Val	Val	Glu	Ile	Gln	Pro	Leu	Ser	Val	Ala	Pro	Arg	Pro	
			500					505					510			
Gly	Lys	Arg	Pro	Glu	Ile	Lys	Ala	Asn	His	Thr	Arg	Ser	Arg	Arg	Asp	
		515					520					525				
Val	Gly	Ile	Gly	Leu	Leu	Phe	Leu	Gly	Phe	Leu	Ser	Ala	Ala	Gly	Ser	
	530					535						540				
Thr	Met	Gly	Ala	Ala	Ser	Leu	Ala	Leu	Thr	Ala	Gln	Ala	Arg	Gly	Leu	
545					550					555					560	
Leu	Ser	Gly	Ile	Val	Gln	Gln	Gln	Gln	Asn	Leu	Leu	Gln	Ala	Ile	Glu	
			565						570					575		

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Ala Gln Gln His Leu Leu Gln Leu Ser Val Trp Gly Ile Lys Gln Leu
 580 585 590
 Gln Ala Arg Met Leu Ala Val Glu Lys Tyr Ile Arg Asp Gln Gln Tyr
 595 600 605
 Leu Ser Ile Trp Gly Cys Ala Asn Lys Leu Val Cys His Ser Ser Val
 610 615 620
 Pro Trp Asn Leu Ser Trp Ala Gly Thr Ser Ser Lys Cys Asn Tyr Thr
 625 630 635 640
 Gly Ala Lys Tyr Tyr Glu Cys Ile Trp Asn Asn Leu Thr Trp Gln Glu
 645 650 655
 Trp Asp Lys Leu Val Gln Asn Ser Ser Glu Thr Ile Tyr Ser Leu Leu
 660 665 670
 Glu Thr Ala Gln Thr Gln Gln Glu Arg Asn Lys Gln Glu Leu Leu Glu
 675 680 685
 Leu Asp Lys Trp Ser Ser Leu Trp Asp Trp Phe Asp Ile Thr Gln Trp
 690 695 700
 Leu Trp Tyr Ile Lys Ile Ala Ile Ile Ile Val Ala Gly Leu Val Gly
 705 710 715 720
 Leu Arg Ile Leu Met Phe Ile Val Asn Val Val Lys Lys Val Arg Gln
 725 730 735
 Gly Tyr Met Pro Leu Phe Ser Gln Thr Pro Thr Gln Ala Glu Gln Asp
 740 745 750
 Pro Glu Gln Pro Gly Gly Thr Ala Gly Gly Gly Gly Gly Gly Gly Asn
 755 760 765
 Phe Arg Trp Thr Pro Ser Pro Thr Gly Phe Phe Ser Ile Val Trp Glu
 770 775 780
 Asp Leu Arg Asn Leu Leu Ile Trp Ile Tyr Gln Thr Cys Gln Asn Phe
 785 790 795 800
 Ile Trp Val Leu Trp Thr Ile Leu Gln Ala Leu Lys Gln Gly Thr Ile
 805 810 815
 Ser Leu Ala His Asn Leu Val Ile Val His Arg Tyr Ile Ile Val Arg
 820 825 830
 Val Arg Gln Thr Ile Glu Trp Cys Gly Asn Thr Tyr Ala Ser Leu Arg
 835 840 845
 Ala Ser Leu Ile His Ala Ile Asp Arg Leu Ala Asp Phe Thr Gly Trp
 850 855 860
 Trp Thr Asp Leu Leu Ile Glu Gly Val Val Tyr Ile Ala Arg Gly Ile
 865 870 875 880
 Arg Asn Ile Pro Arg Arg Ile Arg Gln Gly Leu Glu Ile Ala Leu Asn
 885 890 895

<210> SEQ ID NO 332
 <211> LENGTH: 852
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (174)..(174)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (178)..(178)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (179)..(179)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
 <220> FEATURE:

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<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (261)..(261)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
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<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
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<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
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<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (269)..(269)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
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<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (272)..(272)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
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<222> LOCATION: (322)..(322)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (331)..(331)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (332)..(332)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (335)..(335)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

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<400> SEQUENCE: 332

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Met Arg Val Met Glu Arg Ile Arg Gly Ser Leu Trp Leu Met Gly Leu
1           5           10          15

Ile Ile Pro Cys Leu Thr Gly Ser Asp Gln Leu Trp Ala Thr Val Tyr
20          25          30

Tyr Gly Val Pro Val Trp Lys Asp Ala Glu Thr Thr Leu Phe Cys Ala
35          40          45

Ala Asp Ala Ser Ala Leu Asn Lys Glu Ala His Asn Ile Trp Ala Ser
50          55          60

Gln Ala Cys Val Pro Thr Asp Pro Asn Pro Gln Glu Val His Val Pro
65          70          75          80

Asn Ile Thr Glu Thr Phe Asp Met Trp Glu Asn Asn Met Val Glu Gln
85          90          95

Met Gln Thr Asp Ile Ile Ser Leu Trp Asp Gln Ser Leu Lys Pro Cys
100         105         110

Val Lys Leu Thr Pro Leu Cys Val Thr Met His Cys Ile Pro Tyr Asn
115         120         125

Ile Ser Ser Thr Gly Asn Asp Thr Arg Asn Ser Ser Thr Ser Gln Met
130         135         140

Val Lys Lys Cys Phe Phe Asn Met Thr Thr Glu Leu Arg Asp Lys Gln
145         150         155         160

Lys Gln Val Tyr Ser Leu Phe Tyr Val Asp Asp Ile Val Xaa Ile Asn
165         170         175

Gly Xaa Xaa Ser Tyr Arg Leu Ile Asn Cys Asn Thr Thr Ala Ile Thr
180         185         190

Gln Ala Cys Pro Lys Thr Ser Phe Glu Pro Ile Pro Ile His Tyr Cys

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195					200					205					
Ala	Pro	Pro	Gly	Phe	Ala	Ile	Met	Lys	Cys	Asn	Glu	Pro	Asp	Phe	Asn
210						215					220				
Gly	Thr	Gly	Lys	Cys	Arg	Asn	Ile	Ser	Thr	Val	Gln	Cys	Thr	His	Gly
225					230					235					240
Ile	Lys	Pro	Val	Val	Thr	Thr	Gln	Leu	Ile	Leu	Asn	Gly	Ser	Leu	Ala
				245					250					255	
Glu	Asn	Ser	Thr	Xaa	Xaa	Arg	Thr	Glu	Xaa	Met	Xaa	Xaa	Met	Asp	Xaa
			260					265					270		
Thr	Ile	Ile	Val	Gln	Phe	His	Lys	Arg	Val	Gln	Met	Thr	Cys	Ile	Arg
	275						280					285			
Pro	Gly	Asn	Asn	Ser	Arg	Gly	Gln	Ile	Gln	Ile	Gly	Pro	Ala	Met	Ser
290						295					300				
Phe	Tyr	Asn	Leu	Glu	Asn	Ile	Ile	Gly	Asp	Thr	Arg	Gln	Ala	Tyr	Cys
305					310					315					320
Asn	Xaa	Ser	Ala	Ala	Glu	Trp	Glu	Gln	Arg	Xaa	Xaa	Asp	Thr	Xaa	Arg
			325						330					335	
Ala	Ile	Lys	Ser	Leu	Lys	Pro	Gly	Ser	Asn	Ile	Thr	Phe	Thr	Asp	Gln
			340						345				350		
Gln	Gly	Gly	Asp	Pro	Glu	Ile	Val	Asn	Met	Met	Phe	Asn	Cys	Gly	Gly
	355					360						365			
Glu	Phe	Phe	Tyr	Cys	Asn	Thr	Thr	Pro	Leu	Phe	Asn	Asn	Ser	Trp	Thr
370						375					380				
Asn	Thr	Thr	Ser	Ser	Asp	Ile	Asn	Ser	Ser	Asn	Ile	Ile	Ile	Asn	Cys
385					390					395					400
Arg	Ile	Arg	Gln	Ile	Val	Asn	Ser	Trp	Met	Arg	Val	Gly	Lys	Gly	Ile
			405						410					415	
Tyr	Ala	Pro	Pro	Ile	Arg	Gly	Thr	Ile	Ser	Cys	Thr	Ser	Asn	Ile	Thr
	420							425					430		
Gly	Leu	Leu	Leu	Glu	Leu	Asp	Arg	Gly	Leu	Pro	Arg	Asn	Gly	Thr	Asn
	435						440					445			
Thr	Asn	Asn	Lys	Leu	Thr	Leu	Tyr	Pro	Thr	Gly	Gly	Glu	Met	Arg	Asp
450						455					460				
Leu	Trp	Arg	Leu	Glu	Leu	His	Lys	Tyr	Lys	Val	Val	Ser	Ile	Glu	Pro
465					470					475					480
Ile	Gly	Val	Ala	Pro	Ser	Lys	Ala	Lys	Arg	His	Thr	Val	Thr	Arg	Glu
			485						490					495	
Lys	Arg	Ala	Ala	Phe	Gly	Leu	Gly	Ala	Leu	Phe	Leu	Gly	Phe	Leu	Gly
		500						505					510		
Ala	Ala	Gly	Ser	Thr	Met	Gly	Ala	Ala	Ser	Leu	Thr	Leu	Thr	Val	Gln
	515					520						525			
Ala	Arg	Gln	Leu	Leu	Ser	Gly	Ile	Val	Gln	Gln	Gln	Asn	Asn	Leu	Leu
530						535						540			
Arg	Ala	Ile	Glu	Ala	Gln	Gln	His	Leu	Leu	Gln	Leu	Ser	Val	Trp	Gly
545					550					555					560
Ile	Lys	Gln	Leu	Gln	Ala	Arg	Val	Leu	Ala	Val	Glu	Arg	Tyr	Leu	Lys
			565						570					575	
Asp	Gln	Gln	Ile	Leu	Gly	Leu	Trp	Gly	Cys	Ser	Gly	Lys	Ser	Ile	Cys
			580					585					590		
Tyr	Thr	Thr	Val	Pro	Trp	Asn	Ser	Thr	Trp	Gly	Ala	Asn	Thr	Ser	Tyr
	595					600						605			
Asp	Glu	Ile	Trp	Asn	Asn	Leu	Thr	Trp	Gln	Asp	Trp	Asp	Lys	Arg	Val
610						615					620				

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Lys Asn Tyr Ser Gly Val Ile Phe Asp Leu Ile Glu Gln Ala Gln Glu
 625 630 635 640
 Gln Gln Asn Thr Asn Glu Arg Ser Leu Leu Glu Leu Asp Gln Trp Ser
 645 650 655
 Ser Leu Trp Asn Trp Phe Asp Ile Ser Asn Trp Leu Trp Tyr Ile Lys
 660 665 670
 Ile Phe Leu Met Ala Val Ala Gly Leu Ile Gly Ile Arg Ile Val Gly
 675 680 685
 Val Ile Met Ser Leu Ile Ala Lys Val Arg Gln Gly Tyr Ser Pro Leu
 690 695 700
 Ser Leu Gln Thr Leu Ile Pro Thr Thr Arg Gly Pro Asp Arg Pro Gly
 705 710 715 720
 Gly Thr Glu Glu Glu Gly Gly Glu Pro Gly Arg Gly Thr Ser Thr Arg
 725 730 735
 Leu Val Ser Gly Phe Leu Ala Leu Ala Trp Asp Asp Leu Arg Asn Leu
 740 745 750
 Leu Leu Phe Leu Tyr His Arg Leu Ile Asp Leu Leu Ser Ile Leu Arg
 755 760 765
 Arg Thr Leu Glu Ile Leu Gly Gln Asn Ile Asn Lys Gly Leu Gln Trp
 770 775 780
 Leu Arg Asp Ile Ser Arg His Leu Trp Gly Val Ile Ala Tyr Trp Gly
 785 790 795 800
 Arg Glu Leu Gln Ile Ser Ala Thr Ser Leu Leu Asp Thr Thr Ala Ile
 805 810 815
 Ala Val Ala Glu Gly Thr Asp Arg Ile Ile Glu Val Val Gln Arg Ile
 820 825 830
 Gly Arg Gly Ile Leu His Ile Pro Arg Arg Ile Arg Gln Gly Leu Glu
 835 840 845
 Arg Ser Leu Leu
 850

<210> SEQ ID NO 333

<211> LENGTH: 844

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 333

Met Lys Val Met Gly Thr Gln Arg Ser Trp Lys Gln Phe Leu Ser Leu
 1 5 10 15
 Arg Ile Val Met Cys Leu Trp Leu Leu Gly Val Ile Arg Ser Glu Asn
 20 25 30
 Asn Leu Trp Val Thr Val Tyr Tyr Gly Val Pro Val Trp Lys Glu Ala
 35 40 45
 Lys Thr Thr Leu Phe Cys Ala Ser Asp Ala Lys Ala Gln Ile Ala Glu
 50 55 60
 Ala His Asn Ile Trp Ala Ser Gln Ala Cys Val Pro Thr Asp Pro Asn
 65 70 75 80
 Pro Glu Glu Ile Thr Leu Glu Asn Val Thr Glu Glu Phe Asp Ala Trp
 85 90 95
 Asn Asn Asn Met Val Asp Gln Met Gln Glu Asp Leu Ile Ser Leu Trp
 100 105 110
 Asp Gln Ser Leu Lys Pro Cys Val Lys Leu Thr Pro Leu Cys Val Thr
 115 120 125
 Leu Asn Cys Ser Thr Asn Thr Thr Thr Glu Ala Ser Gln Val Gln Tyr

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130	135	140
Asn Cys Ser Phe Asn Val Thr Thr Glu Leu Arg Asp Lys Lys Lys Gln 145 150 155 160		
Val Tyr Ser Leu Phe Tyr Arg Glu Asp Ile Thr Ser Leu Asp Ser Asn 165 170 175		
Lys Thr Val Lys Asn Gly Thr Tyr Arg Leu Ile Asn Cys Asn Thr Thr 180 185 190		
Ala Ile Thr Gln Ala Cys Pro Lys Thr Ser Phe Glu Pro Ile Pro Ile 195 200 205		
Tyr Tyr Cys Ala Pro Ala Gly Phe Ala Leu Leu Lys Cys Asn Asp Gln 210 215 220		
Asn Phe Lys Gly Lys Gly Thr Cys Arg Asn Val Ser Thr Val His Cys 225 230 235 240		
Thr His Gly Ile Lys Pro Val Val Ser Thr Gln Phe Leu Leu Asn Gly 245 250 255		
Ser Leu Ala Glu Gly Asn Lys Thr Val Val Arg Val Arg Ser Lys Ser 260 265 270		
Asn Thr Glu Thr Ile Ile Val Gln Leu Ala Thr Ala Ile Tyr Ile Asn 275 280 285		
Cys Thr Arg Leu Gly Asn Lys Thr Ile Glu Gly Ile Pro Ile Gly Pro 290 295 300		
Gly Gln Ile Phe Tyr Arg Thr Lys Thr Val Val Gly Asp Thr Arg Gly 305 310 315 320		
Ala Glu Cys Arg Ile Asn Gly Thr Ala Trp Asn Glu Thr Leu Arg Gln 325 330 335		
Val Lys Glu Ala Leu Asn Asn Thr Tyr Arg Asn Leu Asn Leu Ser Leu 340 345 350		
Thr Glu Ile Asn Phe Glu Gly Ala Ser Gly Gly Asp Leu Glu Val Thr 355 360 365		
Thr His Tyr Phe Asn Cys Gly Gly Glu Phe Phe Tyr Cys Asn Thr Ser 370 375 380		
Asn Leu Phe Asn Gln Ser Thr Ile Asn Asn Ile Thr His Ile Pro Cys 385 390 395 400		
Arg Ile Arg Gln Ile Val Asn Gln Trp Gln Gly Val Gly Lys Gly Ile 405 410 415		
Phe Ala Pro Pro Ile Arg Gly Thr Ile Gln Cys Asn Ser Thr Ile Thr 420 425 430		
Gly Leu Leu Leu Thr Arg Asp Gly Lys Asn Glu Thr Glu Thr Phe Arg 435 440 445		
Pro Thr Gly Gly Asp Met Arg Asn Asn Trp Arg Ser Glu Leu Tyr Lys 450 455 460		
Tyr Lys Val Val Lys Ile Glu Pro Leu Gly Leu Ala Pro Thr Lys Ala 465 470 475 480		
Lys Arg Arg Thr Val Gln Arg Glu Lys Arg Ala Val Gly Leu Gly Val 485 490 495		
Met Phe Leu Gly Phe Met Gly Ala Ala Gly Ser Thr Met Gly Ala Ala 500 505 510		
Ala Leu Thr Leu Thr Val Gln Ala Arg Asn Leu Leu Ser Gly Ile Val 515 520 525		
Gln Gln Gln Ser Asn Leu Leu Lys Ala Ile Glu Ala Gln Gln His Leu 530 535 540		
Leu Gln Leu Ser Val Trp Gly Ile Lys Gln Leu Gln Ala Arg Val Leu 545 550 555 560		

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Ala Val Glu Arg Tyr Leu Lys Asp Gln Gln Leu Leu Gly Leu Trp Gly
 565 570 575
 Cys Ser Gly Lys Leu Ile Cys Thr Thr Asn Val Pro Trp Asn Asn Thr
 580 585 590
 Trp Ser Lys Lys Leu Gly Thr Asn Leu Ser Phe Trp Asp Asn Met Thr
 595 600 605
 Trp Met Gln Trp Glu Lys Glu Ile Asp Asn Tyr Thr Glu Thr Ile Tyr
 610 615 620
 Glu Leu Leu Thr Arg Ser Gln Asn Gln Gln Glu Val Asn Glu Gln Glu
 625 630 635 640
 Leu Leu Ala Leu Asp Lys Trp Ser Ser Leu Trp Asn Trp Phe Asp Ile
 645 650 655
 Thr Gln Trp Leu Trp Tyr Ile Lys Leu Phe Val Met Ile Val Gly Gly
 660 665 670
 Leu Ile Gly Ile Arg Ile Val Phe Ala Met Leu Ser Ile Val Asn Arg
 675 680 685
 Val Arg Gln Gly Tyr Ser Pro Leu Ser Phe Gln Thr Pro Phe Pro Val
 690 695 700
 Arg Arg Asp Pro Asp Arg Pro Glu Gly Ile Glu Glu Gly Gly Gly Glu
 705 710 715 720
 Asn Asp Asn Val Arg Ser Val Arg Leu Val Ser Gly Phe Leu Ala Leu
 725 730 735
 Ala Trp Asp Asp Leu Arg Asn Leu Cys Leu Phe Ser Tyr His Arg Leu
 740 745 750
 Arg Asp Leu Leu Trp Ile Val Arg Arg Thr Leu Glu Ile Val Gly Gln
 755 760 765
 Lys Phe Leu Ala Gly Leu Arg Leu Val Trp Glu Thr Leu Thr Tyr Leu
 770 775 780
 Lys Ser Ile Leu Gln Tyr Trp Gly Gln Glu Leu Lys Thr Ser Ala Ile
 785 790 795 800
 Ser Leu Leu Asp Ala Thr Ala Ile Ala Val Ala Glu Gly Thr Asp Arg
 805 810 815
 Ile Ile Glu Ile Ala Gln Arg Val Gly Arg Gly Ile Leu His Ile Pro
 820 825 830
 Arg Arg Ile Arg Gln Gly Leu Glu Arg Ala Leu Leu
 835 840

<210> SEQ ID NO 334
 <211> LENGTH: 867
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (86)..(86)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (141)..(141)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (554)..(554)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
 <400> SEQUENCE: 334

Met Arg Asp Met Arg Ala Met Arg Lys Lys Gln Leu Asn Leu Trp Lys
 1 5 10 15
 Gly Val Met Ser Leu Gly Leu Ile Leu Thr Cys Ile Glu Ala Thr His

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20					25					30					
Tyr	Ala	Thr	Val	Tyr	Tyr	Gly	Val	Pro	Val	Trp	Gln	Glu	Ala	Asn	Val
		35					40					45			
Thr	Leu	Phe	Cys	Ala	Ala	Asp	Ala	Asn	Tyr	Val	Ser	Gln	Glu	Gln	His
		50				55					60				
Asn	Ile	Trp	Ala	Thr	Gln	Ala	Cys	Val	Pro	Thr	Asp	Pro	Ser	Pro	Ile
		65			70				75						80
Glu	Phe	Pro	Leu	Glu	Xaa	Val	Thr	Glu	Thr	Phe	Asp	Ile	Trp	Asn	Asn
			85					90						95	
Asn	Met	Val	Asp	Gln	Met	Gln	Asp	Asp	Ile	Ile	Ser	Leu	Trp	Asp	Gln
			100					105					110		
Ser	Leu	Lys	Pro	Cys	Val	Lys	Leu	Thr	Val	Met	Cys	Val	Thr	Leu	Asn
		115					120					125			
Cys	Thr	Lys	Thr	Thr	Glu	Thr	Thr	Thr	Gln	Thr	Asn	Xaa	Thr	Asn	Thr
		130				135					140				
Thr	Ile	Pro	Pro	Tyr	Ile	Pro	Gln	Leu	Asp	Ile	Tyr	Asn	Cys	Ser	Phe
					150					155					160
Asn	Val	Thr	Thr	Val	Leu	Lys	Asp	Lys	Lys	Thr	Lys	Gln	Gln	Ala	Leu
				165					170					175	
Phe	Tyr	Lys	Gln	Asp	Ile	Ile	Lys	Thr	Asp	Glu	Thr	Asn	Glu	Thr	Lys
			180					185					190		
Glu	Tyr	Ser	Glu	Tyr	Arg	Leu	Ile	Asn	Cys	Asn	Thr	Ser	Thr	Ile	Ser
		195				200						205			
Gln	Ala	Cys	Pro	Lys	Val	Ser	Phe	Glu	Pro	Leu	Pro	Ile	Gln	Tyr	Cys
		210				215					220				
Ala	Pro	Ala	Gly	Tyr	Ala	Leu	Met	Lys	Cys	Asn	Asp	Lys	Gln	Phe	Asn
					230					235					240
Gly	Thr	Gly	Glu	Cys	Lys	Asn	Val	Ser	Ile	Val	His	Cys	Thr	His	Gly
			245						250					255	
Ile	Arg	Pro	Thr	Val	Ser	Thr	Gln	Leu	Ile	Leu	Asn	Gly	Thr	Leu	Ala
			260					265					270		
Lys	Glu	His	Ala	Thr	Ile	Ile	Ser	Lys	Asn	Ser	Thr	Asp	Ser	Gly	Lys
		275					280					285			
Asp	Ile	Ile	Val	Lys	Leu	Ala	Arg	Ser	Val	Glu	Ile	Thr	Cys	Glu	Arg
		290				295					300				
Thr	Ser	Asn	Asn	Thr	Arg	Gly	Gln	Ile	Gln	Val	Gly	Pro	Met	Thr	Ile
					310					315					320
Tyr	Asn	Ser	Glu	Asn	Ile	Val	Gly	Asn	Thr	Arg	Lys	Ala	Phe	Cys	Lys
				325					330					335	
Tyr	Asn	Lys	Thr	Asn	Trp	Gln	Asn	Ala	Leu	Lys	Asp	Thr	Val	Arg	Ala
								345					350		
Leu	Lys	Ala	Asn	Gln	Ile	Arg	Phe	Gln	Asn	Ser	Ser	Gly	Gly	Asp	Pro
		355					360					365			
Glu	Val	Thr	Phe	Leu	His	Phe	Asn	Cys	His	Gly	Glu	Phe	Phe	Tyr	Cys
		370				375					380				
Asp	Thr	Thr	Lys	Met	Phe	Asn	Tyr	Asn	Cys	Thr	Lys	Glu	Ser	Cys	Asp
					390					395					400
Cys	Ile	Thr	Gly	Asn	Cys	Thr	Asn	Tyr	Ile	Pro	Cys	His	Leu	Lys	Gln
				405					410					415	
Val	Val	Met	Ser	Trp	Met	Arg	Val	Gly	Ser	Gly	Leu	Phe	Ala	Pro	Pro
				420				425					430		
Ile	Arg	Gly	Thr	Leu	Arg	Cys	Lys	Ser	Asn	Ile	Thr	Gly	Ile	Ile	Leu
		435					440					445			

Gln	Arg	Asp	Val	Pro	Leu	Asn	Lys	Thr	Asn	Glu	Asn	Ser	Thr	Asp	Tyr
450						455			460						
Asn	Thr	Leu	Arg	Pro	Ile	Gly	Gly	Asp	Met	Thr	Asn	Ile	Trp	Arg	Ser
465			470			475			480						
Glu	Leu	Tyr	Pro	Tyr	Lys	Val	Val	Lys	Val	Lys	Ala	Leu	Ser	Val	Ala
			485			490			495						
Pro	Thr	Lys	Ala	Arg	Arg	Pro	Val	Ile	Ser	His	Asn	Arg	Glu	Lys	Arg
			500			505			510						
Ala	Ala	Gly	Leu	Gly	Met	Leu	Phe	Leu	Gly	Phe	Met	Ser	Ala	Ala	Gly
515						520			525						
Ser	Thr	Met	Gly	Ala	Ala	Ala	Val	Thr	Leu	Thr	Val	Gln	Ala	Arg	Gln
530						535			540						
Val	Leu	His	Gly	Ile	Val	Gln	Gln	Gln	Xaa	Asn	Met	Leu	Arg	Ala	Ile
545			550			555			560						
Ala	Ala	Gln	Gln	Glu	Leu	Leu	Arg	Leu	Ser	Val	Trp	Gly	Ile	Arg	Gln
			565			570			575						
Leu	Arg	Ala	Arg	Leu	Leu	Ala	Ile	Glu	Thr	Tyr	Leu	Arg	Asp	Gln	Gln
			580			585			590						
Leu	Leu	Gly	Leu	Trp	Gly	Cys	Thr	Gly	Arg	Leu	Ile	Cys	Tyr	Thr	Asn
595						600			605						
Val	Pro	Trp	Asn	Lys	Thr	Trp	Thr	Gly	Lys	Asn	Glu	Thr	Glu	Leu	Asp
610						615			620						
Asp	Ile	Trp	Gly	Asn	Met	Thr	Trp	Gln	Gln	Trp	Asp	Lys	Leu	Val	Asp
625			630			635			640						
Asn	Tyr	Thr	Asp	Thr	Ile	Phe	Leu	Glu	Ile	Gln	Lys	Ala	Gln	Glu	Gln
			645			650			655						
Gln	Glu	Val	Asn	Glu	Lys	Ala	Leu	Leu	Glu	Leu	Asp	Lys	Trp	Ala	Asp
			660			665			670						
Leu	Trp	Ser	Trp	Leu	Asp	Ile	Thr	Gln	Trp	Leu	Trp	Tyr	Ile	Lys	Ile
675						680			685						
Phe	Ile	Met	Ile	Ile	Ala	Gly	Leu	Val	Gly	Leu	Arg	Ile	Leu	Met	Ala
690						695			700						
Ile	Ile	Asn	Met	Cys	His	Arg	Val	Arg	Gln	Gly	Tyr	Ser	Pro	Leu	Ser
705			710			715			720						
Phe	Gln	Ile	Leu	Ala	Arg	Asn	Gln	Val	Pro	Ala	Gly	Ile	Val	Pro	Glu
			725			730			735						
Ile	Gly	Glu	Glu	Gly	Gly	Arg	Ala	Asp	Ser	Asp	Arg	Ser	Ile	Arg	Leu
			740			745			750						
Leu	Ala	Gly	Phe	Leu	Pro	Leu	Ile	Trp	Asp	Asp	Leu	Arg	Thr	Leu	Ile
755						760			765						
Val	Trp	Ile	Tyr	Arg	Thr	Leu	Val	Ile	Leu	Leu	Ser	Gly	Ile	Lys	Glu
770			775			780			785						
Ile	Thr	Leu	Ser	Leu	Ile	Glu	His	Leu	Thr	Arg	Leu	Leu	Arg	Arg	Val
785			790			795			800						
Asn	Asn	Leu	Thr	Arg	Asp	Cys	Phe	Ala	Phe	Ile	Ala	Tyr	Trp	Gly	Gln
			805			810			815						
Glu	Leu	Lys	Gln	Ser	Ala	Ile	Ser	Leu	Leu	Asp	Cys	Val	Ala	Val	Trp
			820			825			830						
Thr	Ala	Asp	Trp	Thr	Asp	Gln	Val	Ile	Ala	Ile	Ala	Gln	Arg	Ile	Gly
835						840			845						
Arg	Gly	Ile	Leu	Asn	Ile	Pro	Arg	Arg	Ile	Arg	Gln	Gly	Leu	Glu	Arg
850			855			860			865						

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Ser Leu Leu
865

<210> SEQ ID NO 335
 <211> LENGTH: 872
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus
 <400> SEQUENCE: 335

Met	Arg	Asp	Met	Arg	Ala	Met	Arg	Lys	Lys	Gln	Leu	Asn	Leu	Trp	Lys
1			5					10						15	
Trp	Val	Met	Ser	Leu	Gly	Leu	Ile	Leu	Thr	Cys	Ile	Glu	Ala	Thr	His
		20					25						30		
Tyr	Ala	Thr	Val	Tyr	Tyr	Gly	Val	Pro	Val	Trp	Arg	Glu	Ala	Asn	Val
		35				40					45				
Thr	Leu	Phe	Cys	Ala	Ala	Asp	Ala	Asn	Tyr	Val	Ser	Lys	Glu	Gln	His
	50				55					60					
Asn	Ile	Trp	Ala	Thr	Gln	Ala	Cys	Val	Pro	Thr	Asp	Pro	Arg	Pro	Thr
65				70					75					80	
Glu	Phe	Pro	Leu	Glu	Asn	Val	Thr	Glu	Thr	Phe	Asn	Ile	Trp	Lys	Asn
			85					90						95	
Tyr	Met	Val	Asp	Gln	Met	Gln	Asp	Asp	Ile	Val	Ser	Leu	Trp	Asp	Gln
		100					105						110		
Ser	Leu	Lys	Pro	Cys	Val	Lys	Leu	Thr	Val	Met	Cys	Val	Thr	Leu	Asn
		115				120						125			
Cys	Thr	Gln	Leu	Asn	Thr	Thr	Asn	Thr	Thr	Gln	Thr	Glu	Met	Thr	Asn
	130				135						140				
Thr	Thr	Ile	Pro	Pro	His	Thr	Thr	Ile	Pro	Pro	His	Ile	Pro	Gln	Leu
145				150					155						160
Asp	Ile	Tyr	Asn	Cys	Ser	Phe	Asn	Val	Thr	Thr	Val	Leu	Lys	Asp	Lys
			165					170						175	
Thr	Thr	Lys	Gln	Gln	Ala	Leu	Phe	Tyr	Arg	Gln	Asp	Ile	Ile	Glu	Thr
		180					185							190	
Gly	Glu	Thr	Gly	Thr	Asn	Gly	Ile	Lys	Gly	Tyr	Arg	Leu	Ile	Asn	Cys
	195				200							205			
Asn	Thr	Ser	Thr	Ile	Ser	Gln	Ala	Cys	Pro	Lys	Val	Ser	Phe	Glu	Pro
	210				215						220				
Leu	Pro	Ile	Gln	Tyr	Cys	Ala	Pro	Ala	Gly	Tyr	Ala	Leu	Met	Lys	Cys
225				230					235					240	
Asn	Asp	Ile	Gln	Phe	Asn	Gly	Thr	Gly	Glu	Cys	Arg	Asn	Val	Ser	Ile
			245					250						255	
Val	His	Cys	Thr	His	Gly	Ile	Arg	Pro	Thr	Val	Ser	Thr	Gln	Leu	Ile
		260					265						270		
Leu	Asn	Gly	Thr	Leu	Ala	Glu	Gly	Asn	Ala	Thr	Ile	Ile	Ser	Lys	Asn
	275					280						285			
Ser	Ser	Asp	Ser	Gly	Lys	Asp	Ile	Ile	Val	Lys	Leu	Ala	Arg	Pro	Val
	290				295						300				
Glu	Ile	Thr	Cys	Glu	Arg	Thr	Gly	Asn	Asn	Thr	Arg	Gly	Gln	Ile	Gln
305				310						315				320	
Ile	Gly	Pro	Met	Thr	Ile	Tyr	Asn	Ser	Glu	Asn	Ile	Val	Gly	Asn	Thr
			325					330						335	
Arg	Lys	Ala	Phe	Cys	Lys	Tyr	Asn	Glu	Thr	Asn	Trp	Gln	Gly	Ala	Leu
		340						345					350		
Lys	Asp	Thr	Val	Gln	Ala	Leu	Asn	Glu	Ser	Leu	Ile	His	Phe	Arg	Asn
		355					360					365			

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Ser	Ser	Gly	Gly	Asp	Ser	Glu	Val	Thr	Phe	Leu	His	Phe	Asn	Cys	His
370						375					380				
Gly	Glu	Phe	Phe	Tyr	Cys	Asn	Thr	Thr	Lys	Met	Phe	Ser	Tyr	Asn	Cys
385					390					395					400
Thr	Thr	Glu	Gly	Cys	Asn	Thr	Thr	Glu	Thr	Ser	Asn	Thr	Thr	Glu	Thr
			405					410						415	
Thr	Lys	Tyr	Ile	Pro	Cys	His	Leu	Lys	Gln	Val	Val	Arg	Ser	Trp	Met
		420						425					430		
Arg	Val	Gly	Ser	Gly	Leu	Phe	Ala	Pro	Pro	Ile	Arg	Gly	Thr	Leu	Arg
		435					440					445			
Cys	Ile	Ser	Asn	Ile	Thr	Gly	Ile	Ile	Leu	Gln	Arg	Asp	Ala	Pro	Leu
450						455					460				
Lys	Thr	Asp	Glu	Asn	Ser	Thr	Leu	Arg	Pro	Ile	Gly	Gly	Asp	Met	Thr
465					470					475					480
Asn	Ile	Trp	Arg	Ser	Glu	Leu	Tyr	Pro	Tyr	Lys	Val	Val	Arg	Val	Lys
			485						490					495	
Ala	Leu	Thr	Val	Ala	Pro	Thr	Lys	Ala	Lys	Arg	Pro	Val	Ile	Gly	His
			500					505					510		
Asn	Arg	Glu	Lys	Arg	Ala	Ala	Gly	Leu	Gly	Met	Leu	Phe	Leu	Gly	Phe
		515					520					525			
Met	Ser	Ala	Ala	Gly	Ser	Thr	Met	Gly	Ala	Ala	Ala	Val	Thr	Leu	Thr
530						535					540				
Val	Gln	Ala	Arg	Gln	Val	Leu	His	Gly	Ile	Val	Gln	Gln	Gln	Asn	Asn
545					550					555					560
Met	Leu	Arg	Ala	Ile	Ala	Ala	Gln	Gln	Glu	Leu	Leu	Arg	Leu	Ser	Val
			565					570					575		
Trp	Gly	Ile	Arg	Gln	Leu	Arg	Ala	Arg	Leu	Leu	Ala	Ile	Glu	Thr	Tyr
			580					585					590		
Leu	Arg	Asp	Gln	Gln	Leu	Leu	Gly	Leu	Trp	Gly	Cys	Thr	Gly	Arg	Leu
		595					600				605				
Ile	Cys	Tyr	Thr	Asn	Val	Pro	Trp	Asn	Lys	Thr	Trp	Thr	Gly	Lys	Asn
610					615						620				
Asp	Thr	Glu	Leu	Asp	Asp	Ile	Trp	Gly	Asn	Met	Thr	Trp	Gln	Gln	Trp
625					630					635					640
Asp	Lys	Leu	Val	Asp	Asn	Tyr	Thr	Asp	Thr	Ile	Phe	Leu	Glu	Ile	Gln
			645					650					655		
Arg	Ala	Gln	Glu	Gln	Gln	Glu	Ala	Asn	Gly	Lys	Ala	Leu	Leu	Glu	Leu
			660					665				670			
Asp	Lys	Trp	Ala	Asp	Leu	Trp	Ser	Trp	Leu	Asp	Ile	Thr	Gln	Trp	Leu
		675					680				685				
Trp	Tyr	Ile	Lys	Ile	Phe	Ile	Ile	Ile	Ile	Ala	Gly	Leu	Val	Gly	Leu
690					695					700					
Arg	Ile	Leu	Met	Ala	Ile	Ile	Asn	Met	Cys	His	Arg	Val	Arg	Gln	Gly
705					710					715					720
Tyr	Ser	Pro	Leu	Ser	Phe	Gln	Ile	Leu	Gly	Arg	Asn	Gln	Val	Pro	Ala
			725					730					735		
Gly	Ile	Val	Pro	Glu	Ile	Gly	Glu	Glu	Gly	Gly	Arg	Ala	Asp	Asn	Asp
		740					745						750		
Arg	Ser	Ile	Arg	Leu	Leu	Ala	Gly	Phe	Leu	Pro	Leu	Ile	Trp	Asp	Asp
		755					760					765			
Ile	Arg	Thr	Leu	Val	Val	Trp	Ile	Tyr	Arg	Thr	Leu	Val	Ile	Leu	Leu
770					775					780					

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Ser Gly Ile Lys Glu Ile Thr Leu Ser Leu Ile Glu Tyr Leu Thr Arg
785 790 795 800

Leu Leu Arg Arg Val Asn Asn Leu Thr Arg Asp Cys Phe Ala Phe Ile
805 810 815

Ala Tyr Trp Gly Gln Glu Leu Lys Gln Ser Ala Ile Ser Leu Leu Asp
820 825 830

Cys Val Ala Val Trp Thr Ala Asn Trp Thr Asp Gln Val Ile Ala Ile
835 840 845

Ala Gln Arg Ile Gly Arg Gly Ile Leu Asn Ile Pro Arg Arg Ile Arg
850 855 860

Gln Gly Leu Glu Arg Ser Leu Leu
865 870

<210> SEQ ID NO 336

<211> LENGTH: 872

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<220> FEATURE:

<221> NAME/KEY: MISC_FEATURE

<222> LOCATION: (630)..(630)

<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (633)..(633)

<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<220> FEATURE:

<221> NAME/KEY: MISC_FEATURE

<222> LOCATION: (634)..(634)

<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<220> FEATURE:

<221> NAME/KEY: MISC_FEATURE

<222> LOCATION: (666)..(666)

<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<220> FEATURE:

<221> NAME/KEY: MISC_FEATURE

<222> LOCATION: (696)..(696)

<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<220> FEATURE:

<221> NAME/KEY: MISC_FEATURE

<222> LOCATION: (697)..(697)

<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<220> FEATURE:

<221> NAME/KEY: MISC_FEATURE

<222> LOCATION: (728)..(728)

<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<220> FEATURE:

<221> NAME/KEY: MISC_FEATURE

<222> LOCATION: (808)..(808)

<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<400> SEQUENCE: 336

Met Arg Asp Met Arg Ala Met Arg Lys Lys Gln Leu Asn Leu Trp Lys
1 5 10 15

Trp Val Met Ser Leu Gly Leu Ile Leu Thr Cys Ile Glu Ala Thr His
20 25 30

Tyr Ala Thr Val Tyr Tyr Gly Val Pro Val Trp Arg Glu Ala Asn Val
35 40 45

Thr Leu Phe Cys Ala Ala Asp Ala Asn Tyr Val Ser Lys Glu Gln His
50 55 60

Asn Ile Trp Val Thr Gln Ala Cys Val Pro Thr Asp Pro Arg Pro Thr
65 70 75 80

Glu Phe Pro Leu Glu Asn Val Thr Glu Thr Phe Asn Ile Trp Lys Asn
85 90 95

Tyr Met Val Asp Gln Met Gln Asp Asp Ile Val Ser Leu Trp Asp Gln
100 105 110

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Ser	Leu	Lys	Pro	Cys	Val	Lys	Leu	Thr	Val	Met	Cys	Val	Thr	Leu	Asn
		115					120					125			
Cys	Thr	Gln	Leu	Asn	Thr	Thr	Asn	Thr	Thr	Gln	Thr	Glu	Met	Thr	Asn
	130					135					140				
Thr	Thr	Ile	Pro	Pro	His	Thr	Thr	Ile	Pro	Pro	His	Ile	Pro	Gln	Leu
	145				150					155					160
Asp	Ile	Tyr	Asn	Cys	Ser	Phe	Asn	Val	Thr	Thr	Val	Leu	Lys	Asp	Lys
				165					170					175	
Thr	Thr	Lys	Gln	Gln	Ala	Leu	Phe	Tyr	Arg	Gln	Asp	Ile	Ile	Glu	Thr
			180					185					190		
Gly	Glu	Thr	Gly	Thr	Asn	Gly	Ile	Lys	Gly	Tyr	Arg	Leu	Ile	Asn	Cys
		195					200					205			
Asn	Thr	Ser	Thr	Ile	Ser	Gln	Ala	Cys	Pro	Lys	Val	Ser	Phe	Glu	Pro
	210					215					220				
Leu	Pro	Ile	Gln	Tyr	Cys	Ala	Pro	Ala	Gly	Tyr	Ala	Leu	Met	Lys	Cys
	225				230					235					240
Asn	Asp	Ile	Gln	Phe	Asn	Gly	Thr	Gly	Glu	Cys	Arg	Asn	Val	Ser	Ile
				245					250					255	
Val	His	Cys	Thr	His	Gly	Ile	Arg	Pro	Thr	Val	Ser	Thr	Gln	Leu	Ile
			260					265					270		
Leu	Asn	Gly	Thr	Leu	Ala	Glu	Gly	Asn	Ala	Thr	Ile	Ile	Ser	Lys	Asn
		275					280					285			
Ser	Ser	Asp	Ser	Gly	Lys	Asp	Ile	Ile	Val	Lys	Leu	Ala	Arg	Pro	Val
	290					295					300				
Glu	Ile	Thr	Cys	Glu	Arg	Thr	Gly	Asn	Asn	Thr	Arg	Gly	Gln	Ile	Gln
	305				310					315					320
Ile	Gly	Pro	Met	Thr	Ile	Tyr	Asn	Ser	Glu	Asn	Ile	Val	Gly	Asn	Thr
				325					330					335	
Arg	Lys	Ala	Phe	Cys	Lys	Tyr	Asn	Glu	Thr	Asn	Trp	Gln	Gly	Ala	Leu
			340					345					350		
Lys	Asp	Thr	Val	Gln	Ala	Leu	Asn	Glu	Ser	Leu	Ile	His	Phe	Arg	Asn
		355					360					365			
Ser	Ser	Gly	Gly	Asp	Pro	Glu	Val	Thr	Phe	Leu	His	Phe	Asn	Cys	His
	370					375					380				
Gly	Glu	Phe	Phe	Tyr	Cys	Asn	Thr	Thr	Lys	Met	Phe	Ser	Tyr	Asn	Cys
	385				390					395					400
Thr	Thr	Glu	Gly	Cys	Asn	Thr	Thr	Glu	Thr	Ser	Asn	Thr	Thr	Glu	Thr
				405				410						415	
Thr	Lys	Tyr	Ile	Pro	Cys	His	Leu	Lys	Gln	Val	Val	Arg	Ser	Trp	Met
			420					425					430		
Arg	Val	Gly	Ser	Gly	Leu	Phe	Ala	Pro	Pro	Ile	Arg	Gly	Thr	Leu	Arg
		435					440					445			
Cys	Ile	Ser	Asn	Ile	Thr	Gly	Ile	Ile	Leu	Gln	Arg	Asp	Ala	Pro	Leu
	450					455					460				
Lys	Thr	Asp	Glu	Asn	Ser	Thr	Leu	Arg	Pro	Ile	Gly	Gly	Asp	Met	Thr
	465				470					475					480
Asn	Ile	Trp	Arg	Ser	Glu	Leu	Tyr	Pro	Tyr	Lys	Val	Val	Arg	Val	Lys
				485					490					495	
Ala	Leu	Thr	Val	Ala	Pro	Thr	Lys	Ala	Lys	Arg	Pro	Val	Ile	Gly	His
			500				505						510		
Asn	Arg	Glu	Lys	Arg	Ala	Ala	Gly	Leu	Gly	Met	Leu	Phe	Leu	Gly	Phe
		515					520					525			
Met	Ser	Ala	Ala	Gly	Ser	Thr	Met	Gly	Ala	Ala	Ala	Val	Thr	Leu	Thr

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530	535	540
Val Gln Ala Arg Gln Val Leu His Gly Ile Val Gln Gln Gln Asn Asn		
545	550	555 560
Met Leu Arg Ala Ile Ala Ala Gln Gln Glu Leu Leu Arg Leu Ser Val		
565	570	575
Trp Gly Ile Arg Gln Leu Arg Ala Arg Leu Leu Ala Ile Glu Thr Tyr		
580	585	590
Leu Arg Asp Gln Gln Leu Leu Gly Leu Trp Gly Cys Thr Gly Arg Leu		
595	600	605
Ile Cys Tyr Thr Asn Val Pro Trp Asn Lys Thr Trp Thr Gly Lys Asn		
610	615	620
Asp Thr Glu Leu Asp Xaa Ile Trp Xaa Xaa Met Thr Trp Gln Gln Trp		
625	630	635 640
Asp Lys Leu Val Asp Asn Tyr Thr Asp Thr Ile Phe Leu Glu Ile Gln		
645	650	655
Arg Ala Gln Glu Gln Gln Glu Ala Asn Xaa Lys Ala Leu Leu Glu Leu		
660	665	670
Asp Lys Trp Ala Asp Leu Trp Ser Trp Leu Asp Ile Thr Gln Trp Leu		
675	680	685
Trp Tyr Ile Lys Ile Phe Ile Xaa Xaa Ile Ala Gly Leu Val Gly Leu		
690	695	700
Arg Ile Leu Met Ala Ile Ile Asn Met Cys His Arg Val Arg Gln Gly		
705	710	715 720
Tyr Ser Pro Leu Ser Phe Gln Xaa Leu Gly Arg Asn Gln Val Pro Ala		
725	730	735
Gly Ile Val Pro Glu Ile Gly Glu Glu Gly Gly Arg Ala Asp Asn Asp		
740	745	750
Arg Ser Ile Arg Leu Leu Ala Gly Phe Leu Pro Leu Ile Trp Asp Asp		
755	760	765
Ile Arg Thr Leu Val Val Trp Ile Tyr Arg Thr Leu Val Ile Leu Leu		
770	775	780
Ser Gly Ile Lys Glu Ile Thr Leu Ser Leu Ile Glx Tyr Leu Thr Arg		
785	790	795 800
Leu Leu Arg Arg Val Asn Asn Xaa Thr Arg Asp Cys Phe Ala Phe Ile		
805	810	815
Ala Tyr Trp Gly Gln Glu Leu Lys Gln Ser Ala Ile Ser Leu Leu Asp		
820	825	830
Cys Val Ala Val Trp Thr Ala Asn Trp Thr Asp Gln Val Ile Ala Ile		
835	840	845
Ala Gln Arg Ile Gly Arg Gly Ile Leu Asn Ile Pro Arg Arg Ile Arg		
850	855	860
Gln Gly Leu Glu Arg Ser Leu Leu		
865	870	

<210> SEQ ID NO 337

<211> LENGTH: 872

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 337

Met Arg Asp Met Arg Ala Met Arg Lys Lys Gln Leu Asn Leu Trp Lys
1 5 10 15
Trp Val Met Ser Leu Gly Leu Ile Leu Thr Cys Ile Glu Ala Thr His
20 25 30

Tyr	Ala	Thr	Val	Tyr	Tyr	Gly	Val	Pro	Val	Trp	Arg	Glu	Ala	Asn	Val
35															
Thr	Leu	Phe	Cys	Ala	Ala	Asp	Ala	Asn	Tyr	Val	Ser	Lys	Glu	Gln	His
50															
Asn	Ile	Trp	Ala	Thr	Gln	Ala	Cys	Val	Pro	Thr	Asp	Pro	Arg	Pro	Thr
65															
Glu	Phe	Pro	Leu	Glu	Asn	Val	Thr	Glu	Thr	Phe	Asn	Ile	Trp	Lys	Asn
85															
Tyr	Met	Val	Asp	Gln	Met	Gln	Asp	Asp	Ile	Val	Ser	Leu	Trp	Asp	Gln
100															
Ser	Leu	Lys	Pro	Cys	Val	Lys	Leu	Thr	Val	Met	Cys	Val	Thr	Leu	Asn
115															
Cys	Thr	Gln	Leu	Asn	Thr	Thr	Asn	Thr	Thr	Gln	Thr	Glu	Met	Thr	Asn
130															
Thr	Thr	Ile	Pro	Pro	His	Thr	Thr	Ile	Pro	Pro	His	Ile	Pro	Gln	Leu
145															
Asp	Ile	Tyr	Asn	Cys	Ser	Phe	Asn	Val	Thr	Thr	Val	Leu	Lys	Asp	Lys
165															
Thr	Thr	Lys	Gln	Gln	Ala	Leu	Phe	Tyr	Arg	Gln	Asp	Ile	Ile	Glu	Thr
180															
Gly	Glu	Thr	Gly	Thr	Asn	Gly	Ile	Lys	Gly	Tyr	Arg	Leu	Ile	Asn	Cys
195															
Asn	Thr	Ser	Thr	Ile	Ser	Gln	Ala	Cys	Pro	Lys	Val	Ser	Phe	Glu	Pro
210															
Leu	Pro	Ile	Gln	Tyr	Cys	Ala	Pro	Ala	Gly	Tyr	Ala	Leu	Met	Lys	Cys
225															
Asn	Asp	Ile	Gln	Phe	Asn	Gly	Thr	Gly	Glu	Cys	Arg	Asn	Val	Ser	Ile
245															
Val	His	Cys	Thr	His	Gly	Ile	Arg	Pro	Thr	Val	Ser	Thr	Gln	Leu	Ile
260															
Leu	Asn	Gly	Thr	Leu	Ala	Glu	Gly	Asn	Ala	Thr	Ile	Ile	Ser	Lys	Asn
275															
Ser	Ser	Asp	Ser	Gly	Lys	Asp	Ile	Ile	Val	Lys	Leu	Ala	Arg	Pro	Val
290															
Glu	Ile	Thr	Cys	Glu	Arg	Thr	Gly	Asn	Asn	Thr	Arg	Gly	Gln	Ile	Gln
305															
Ile	Gly	Pro	Met	Thr	Ile	Tyr	Asn	Ser	Glu	Asn	Ile	Val	Gly	Asn	Thr
325															
Arg	Lys	Ala	Phe	Cys	Lys	Tyr	Asn	Glu	Thr	Asn	Trp	Gln	Gly	Ala	Leu
340															
Lys	Asp	Thr	Val	Gln	Ala	Leu	Asn	Glu	Ser	Leu	Ile	His	Phe	Arg	Asn
355															
Ser	Ser	Gly	Gly	Asp	Pro	Glu	Val	Thr	Phe	Leu	His	Phe	Asn	Cys	His
370															
Gly	Glu	Phe	Phe	Tyr	Cys	Asn	Thr	Thr	Lys	Met	Phe	Ser	Tyr	Asn	Cys
385															
Thr	Thr	Glu	Gly	Cys	Asn	Thr	Thr	Glu	Thr	Ser	Asn	Thr	Thr	Glu	Thr
405															
Thr	Lys	Tyr	Ile	Pro	Cys	His	Leu	Lys	Gln	Val	Val	Arg	Ser	Trp	Met
420															
Arg	Val	Gly	Ser	Gly	Leu	Phe	Ala	Pro	Pro	Ile	Arg	Gly	Thr	Leu	Arg
435															
Cys	Ile	Ser	Asn	Ile	Thr	Gly	Ile	Ile	Leu	Gln	Arg	Asp	Ala	Pro	Leu
440															
445															

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450	455	460
Lys Thr Asp Glu Asn Ser Thr Leu Arg Pro Ile Gly Gly Asp Met Thr		
465	470	475 480
Asn Ile Trp Arg Ser Glu Leu Tyr Pro Tyr Lys Val Val Arg Val Lys		
	485	490 495
Ala Leu Thr Val Ala Pro Thr Lys Ala Lys Arg Pro Val Ile Gly His		
	500	505 510
Asn Arg Glu Lys Arg Ala Ala Gly Leu Gly Met Leu Phe Leu Gly Phe		
	515	520 525
Met Ser Ala Ala Gly Ser Thr Met Gly Ala Ala Val Thr Leu Thr		
	530	535 540
Val Gln Ala Arg Gln Val Leu His Gly Ile Val Gln Gln Gln Asn Asn		
	545	550 555 560
Met Leu Arg Ala Ile Ala Ala Gln Gln Glu Leu Leu Arg Leu Ser Val		
	565	570 575
Trp Gly Ile Arg Gln Leu Arg Ala Arg Leu Leu Ala Ile Glu Thr Tyr		
	580	585 590
Leu Arg Asp Gln Gln Leu Leu Gly Leu Trp Gly Cys Thr Gly Arg Leu		
	595	600 605
Ile Cys Tyr Thr Asn Val Pro Trp Asn Lys Thr Trp Thr Gly Lys Asn		
	610	615 620
Asp Thr Glu Leu Asp Asp Ile Trp Gly Asn Met Thr Trp Gln Gln Trp		
	625	630 635 640
Asp Lys Leu Val Asp Asn Tyr Thr Asp Thr Ile Phe Leu Glu Ile Gln		
	645	650 655
Arg Ala Gln Glu Gln Gln Glu Ala Asn Glu Lys Ala Leu Leu Glu Leu		
	660	665 670
Asp Lys Trp Ala Asp Leu Trp Ser Trp Leu Asp Ile Thr Gln Trp Leu		
	675	680 685
Trp Tyr Ile Lys Ile Phe Ile Ile Ile Ile Ala Gly Leu Val Gly Leu		
	690	695 700
Arg Ile Leu Met Ala Ile Ile Asn Met Cys His Arg Val Arg Gln Gly		
	705	710 715 720
Tyr Ser Pro Leu Ser Phe Gln Ile Leu Gly Arg Asn Gln Val Pro Ala		
	725	730 735
Gly Ile Val Pro Glu Ile Gly Glu Glu Gly Gly Arg Ala Asp Asn Asp		
	740	745 750
Arg Ser Ile Arg Leu Leu Ala Gly Phe Leu Pro Leu Ile Trp Asp Asp		
	755	760 765
Ile Arg Thr Leu Val Val Trp Ile Tyr Arg Thr Leu Val Ile Leu Leu		
	770	775 780
Ser Gly Ile Lys Glu Ile Thr Leu Ser Leu Ile Glu Tyr Leu Thr Arg		
	785	790 795 800
Leu Leu Arg Arg Val Asn Asn Leu Thr Arg Asp Cys Phe Ala Phe Ile		
	805	810 815
Ala Tyr Trp Gly Gln Glu Leu Lys Gln Ser Ala Ile Ser Leu Leu Asp		
	820	825 830
Cys Val Ala Val Trp Thr Ala Asn Trp Thr Asp Gln Val Ile Ala Ile		
	835	840 845
Ala Gln Arg Ile Gly Arg Gly Ile Leu Asn Ile Pro Arg Arg Ile Arg		
	850	855 860
Gln Gly Leu Glu Arg Ser Leu Leu		
865	870	

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<210> SEQ ID NO 338
<211> LENGTH: 876
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (345)..(345)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (446)..(446)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (450)..(450)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (454)..(454)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (455)..(455)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (460)..(460)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (467)..(467)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (801)..(801)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (807)..(807)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<400> SEQUENCE: 338

Met Lys Asp Met Arg Ala Met Arg Lys Lys Gln Leu Asn Leu Trp Lys
1             5             10             15

Gly Val Met Ser Leu Gly Leu Leu Leu Thr Cys Ile Glu Ala Lys Thr
20             25             30

Glu His Tyr Ala Thr Val Tyr Tyr Gly Val Pro Val Trp Arg Glu Ala
35             40             45

Lys Val Thr Leu Phe Cys Ala Ala Asp Ala Ser Phe Val Ser Lys Glu
50             55             60

Gln His Asn Ile Trp Ala Thr Gln Ala Cys Val Pro Thr Asp Pro Ser
65             70             75             80

Pro Thr Glu Phe Leu Leu Glu Asn Val Thr Glu Thr Phe Asp Ile Trp
85             90             95

Lys Asn Tyr Met Val Glu Gln Met Gln Glu Asp Ile Ile Ser Leu Trp
100            105            110

Asp Gln Ser Leu Lys Pro Cys Val Lys Leu Thr Val Met Cys Val Thr
115            120            125

Met Thr Cys Ser Asn Asp Thr Lys Gly Ser Asn Lys Ser Thr Ala Thr
130            135            140

Thr Thr Ser Ala Asn Thr Pro Thr Ser Thr Glu Thr Thr Ile Pro Thr
145            150            155            160

Asn Ile Pro Glu Leu Asp Ile Phe Asn Cys Thr Phe Asn Val Thr Thr
165            170            175

Val Leu Lys Asp Lys Lys Thr Lys Gln Gln Ala Leu Phe Tyr Lys Gln

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180							185					190				
Asp	Ile	Thr	Glu	Thr	Asp	Lys	Asn	Glu	Tyr	Arg	Leu	Ile	Asn	Cys	Asn	
			195				200				205					
Thr	Ser	Thr	Ile	Ser	Gln	Ala	Cys	Pro	Lys	Val	Ser	Phe	Glu	Pro	Leu	
			210				215				220					
Pro	Ile	Gln	Tyr	Cys	Ala	Pro	Ala	Gly	Tyr	Pro	Leu	Leu	Lys	Cys	Asn	
			225				230				235					
Asp	Glu	Gln	Phe	Asn	Gly	Thr	Gly	Glu	Cys	Asn	Asn	Val	Ser	Ile	Val	
			245				250				255					
His	Cys	Thr	His	Gly	Ile	Arg	Pro	Thr	Val	Ser	Thr	Gln	Leu	Ile	Leu	
			260				265				270					
Asn	Gly	Thr	Leu	Ala	Lys	Gly	Asn	Asn	Ala	Thr	Ile	Ile	Ser	Ser	Asn	
			275				280				285					
Ser	Ser	Asp	Ser	Ala	Thr	Asp	Ile	Ile	Val	Lys	Leu	Ala	Asn	Pro	Val	
			290				295				300					
Glu	Ile	Thr	Cys	Glu	Arg	Thr	Gly	Asn	Asn	Thr	Arg	Gly	Gln	Ile	Gln	
			305				310				315					
Val	Gly	Pro	Leu	Thr	Ile	Tyr	Asn	Ser	Glu	Asn	Ile	Ile	Gly	Asn	Thr	
			325				330				335					
Arg	Lys	Ala	Phe	Cys	Lys	Tyr	Asn	Xaa	Thr	Ala	Trp	Gln	Lys	Ala	Leu	
			340				345				350					
Gln	Ala	Thr	Val	Arg	Ala	Leu	Lys	Thr	Asn	Leu	Thr	His	Tyr	Phe	Asn	
			355				360				365					
Lys	Ser	Ser	Gly	Gly	Asp	Pro	Glu	Val	Thr	Ser	Leu	His	Phe	Asn	Cys	
			370				375				380					
His	Gly	Glu	Phe	Phe	Tyr	Cys	Asn	Thr	Thr	Lys	Met	Phe	Asn	Tyr	Lys	
			385				390				395					
Cys	Thr	Gly	Glu	His	Cys	Asn	Cys	Thr	Asp	Lys	Glu	Leu	Cys	Asn	Lys	
			405				410				415					
Thr	Ala	Glu	Met	Glu	Tyr	Ile	Pro	Cys	Arg	Leu	Lys	Gln	Val	Val	Asn	
			420				425				430					
Ser	Trp	Met	Arg	Val	Gly	Ser	Gly	Leu	Phe	Ala	Pro	Pro	Xaa	Arg	Gly	
			435				440				445					
Thr	Xaa	Arg	Cys	Ile	Xaa	Xaa	Ile	Thr	Gly	Ile	Xaa	Leu	Gln	Arg	Asp	
			450				455				460					
Thr	Pro	Xaa	Ala	Glu	Asn	Asn	Asn	Asn	Thr	Thr	Leu	Arg	Pro	Leu	Gly	
			465				470				475					
Gly	Glu	Met	Arg	Asn	Ile	Trp	Arg	Ser	Glu	Leu	Tyr	Pro	Tyr	Lys	Val	
			485				490				495					
Val	Lys	Val	Lys	Ala	Leu	Ser	Val	Ala	Pro	Thr	Lys	Ala	Lys	Arg	Pro	
			500				505				510					
Val	Val	Ser	His	Asn	Arg	Glu	Lys	Arg	Ala	Ala	Gly	Leu	Gly	Met	Leu	
			515				520				525					
Phe	Leu	Gly	Phe	Met	Ser	Ala	Ala	Gly	Ser	Thr	Met	Gly	Ala	Ala	Ala	
			530				535				540					
Ile	Thr	Leu	Thr	Val	Gln	Ala	Arg	Gln	Val	Leu	His	Gly	Ile	Val	Gln	
			545				550				555					
Gln	Gln	Asn	Asn	Met	Leu	Arg	Ala	Ile	Ala	Ala	Gln	Gln	Glu	Leu	Leu	
			565				570				575					
Arg	Leu	Ser	Val	Trp	Gly	Ile	Arg	Gln	Leu	Arg	Ala	Arg	Leu	Leu	Ala	
			580				585				590					
Ile	Glu	Thr	Tyr	Leu	Arg	Asp	Gln	Gln	Leu	Leu	Gly	Leu	Trp	Gly	Cys	
			595				600				605					

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Thr Gly Lys Leu Ile Cys Tyr Thr Asn Val Pro Trp Asn Thr Thr Trp
 610 615 620
 Thr Gly Lys Ser Asp Ser Glu Leu Asp Gly Ile Trp Gly Asn Leu Thr
 625 630 635 640
 Trp Gln Gln Trp Asp Lys Leu Val Asp Asn Tyr Thr Asp Thr Ile Phe
 645 650 655
 Leu Glu Ile Gln Lys Ala Gln Glu Gln Gln Glu Ala Asn Glu Lys Ala
 660 665 670
 Leu Leu Glu Leu Asp Lys Trp Ala Asp Leu Trp Ser Trp Leu Asp Ile
 675 680 685
 Thr Gln Trp Ile Trp Tyr Ile Arg Ile Phe Ile Ile Val Ile Ala Gly
 690 695 700
 Leu Val Gly Leu Lys Ile Leu Met Ala Ile Ile Asn Ile Cys His Arg
 705 710 715 720
 Val Arg Gln Gly Tyr Ser Pro Leu Ser Phe Gln Ile Leu Gly Arg Asn
 725 730 735
 Gln Val Pro Ala Gly Ile Val Pro Glu Ile Gly Glu Glu Gly Gly Arg
 740 745 750
 Ala Asp Ser Asp Arg Ser Ile Arg Leu Leu Ala Gly Phe Leu Pro Leu
 755 760 765
 Ile Trp Asp Asp Leu Arg Thr Leu Val Val Trp Ile Tyr Arg Thr Leu
 770 775 780
 Val Ile Leu Leu Ser Gly Ile Lys Glu Ile Thr Leu Ser Leu Ile Glu
 785 790 795 800
 Xaa Leu Thr Arg Leu Leu Xaa Arg Val Asn Asn Leu Thr Arg Asp Cys
 805 810 815
 Phe Ala Phe Ile Ala Tyr Trp Gly Gln Glu Leu Lys Gln Ser Ala Ile
 820 825 830
 Ser Leu Leu Asp Cys Val Ala Val Trp Thr Ala Asp Trp Thr Asp Gln
 835 840 845
 Val Ile Ala Ile Ala Gln Arg Ile Gly Arg Gly Ile Leu Asn Ile Pro
 850 855 860
 Arg Arg Ile Arg Gln Gly Leu Glu Arg Ser Leu Leu
 865 870 875

<210> SEQ ID NO 339
 <211> LENGTH: 870
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (459)..(459)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (621)..(621)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (682)..(682)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
 <400> SEQUENCE: 339

Met Arg Ala Met Glu Arg Thr Lys Arg Ser Trp Asn Ile Leu Ser Arg
 1 5 10 15
 Ser Met Val Leu Leu Thr Leu Cys Leu Thr Phe Asn Val Ala Thr Asp
 20 25 30
 Tyr Trp Val Thr Val Tyr Tyr Gly Val Pro Val Trp Lys Asp Ala Glu

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35	40	45
Thr Thr Leu Phe Cys Ala Ser Glu Ala Lys Pro His Ser Ser Glu Ala		
50	55	60
His Asn Ile Trp Ala Thr Gln Ala Cys Val Pro Ile Asp Pro Asn Pro		
65	70	75
Glu Glu Val Phe Leu Pro Asn Val Thr Glu Asn Phe Ser Met Trp Asp		
	85	90
Asn Pro Met Val Ser Gln Met Gln Glu Asp Ile Ile Ser Leu Trp Glu		
	100	105
Gln Ser Leu Lys Pro Cys Val Lys Leu Thr Pro Leu Cys Val Thr Leu		
	115	120
Asn Cys Ser Thr Ala Ile Phe Arg Ala Asn Arg Thr Ser Thr Asn Thr		
	130	135
Thr Thr Pro Ser Thr Ile Ala Thr Ser Pro Thr Thr Asp Thr Ile Tyr		
145	150	155
Glu Met Lys Asn Cys Ser Phe Asn Val Thr Thr Glu Leu Arg Asp Lys		
	165	170
Lys Lys Gln Val Tyr Ser Leu Phe Tyr Val Asp Asp Val Val Lys Leu		
	180	185
Asn Asp Gly Asp Ser Thr Asn Ser Ser Thr Tyr Arg Leu Ile Asn Cys		
	195	200
Asn Thr Thr Ala Ile Thr Gln Ala Cys Pro Lys Ile Ser Phe Glu Pro		
	210	215
Ile Pro Ile His Tyr Cys Ala Pro Ala Gly Phe Ala Ile Leu Lys Cys		
225	230	235
Asn Asp Thr Asn Tyr Leu Gly Thr Gly Lys Cys His Asn Val Ser Thr		
	245	250
Val His Cys Thr His Gly Ile Arg Pro Ala Val Thr Thr Gln Leu Ile		
	260	265
Leu Asn Gly Ser Leu Ala Glu Gly Asn Val Thr Ile Arg Val Lys Asn		
	275	280
Ala Ser Ser Asn Ser Gly Asn Tyr Ile Val Gln Leu Ala Arg Ser Val		
	290	295
Glu Ile Leu Cys Arg Arg Glu Gly Asn Asn Thr Arg Gly Gln Val Gln		
305	310	315
Ile Gly Pro Ala Met Thr Phe Tyr Asn Ile Glu Asn Val Val Gly Asp		
	325	330
Thr Arg Lys Ala Tyr Cys Thr Leu Asn Gly Ser Asn Trp Glu Glu Ala		
	340	345
Met Ser Lys Val Lys Lys Glu Leu Glu Lys Ile Thr Asn Val Thr Asn		
	355	360
Ile Thr Phe Ala Asp Gln Arg Gln Gly Ser Asp Pro Glu Val Glu Asn		
	370	375
His Met Phe Asn Cys Gly Gly Glu Phe Phe Tyr Cys Asn Thr Ser Lys		
385	390	395
Leu Phe Asn Asn Lys Thr Gly Arg Gly Gly Asn Lys Ser Glu Glu Ile		
	405	410
Ile Leu Pro Cys Arg Ile Lys Gln Ile Val Asn Leu Trp Thr Arg Val		
	420	425
Gly Lys Gly Ile Tyr Ala Pro Pro Ile Arg Gly Asn Ile Ser Cys Lys		
	435	440
Ser Asn Ile Thr Gly Ile Leu Leu Ala Arg Xaa Glu Gly Lys Asp Asn		
	450	455
		460

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Gln Asn Asn Leu Thr Phe Arg Pro Ile Gly Gly Asn Met Glu Asp Leu
 465 470 475 480
 Trp Arg Ser Glu Leu Tyr Lys Tyr Lys Val Val Arg Ile Glu Pro Leu
 485 490 495
 Ala Val Ala Pro Ser Lys Ala Lys Arg Arg Thr Val Glu Thr Glu Lys
 500 505 510
 Arg Gln Lys Arg Ala Ala Phe Gly Leu Gly Ala Leu Phe Leu Gly Phe
 515 520 525
 Leu Gly Ala Ala Gly Ser Thr Met Gly Ala Ala Ala Val Thr Leu Thr
 530 535 540
 Val Gln Ala Arg Gln Leu Leu Ser Gly Ile Val Gln Gln Gln Asn Asn
 545 550 555 560
 Leu Leu Arg Ala Ile Glu Ala Gln Gln His Leu Leu Gln Leu Ser Ile
 565 570 575
 Trp Gly Val Lys Gln Leu Gln Gly Arg Leu Leu Ala Val Glu Arg Tyr
 580 585 590
 Leu Gln Asp Gln Gln Ile Leu Gly Leu Trp Gly Cys Ser Gly Lys Ala
 595 600 605
 Ile Cys Tyr Thr Thr Val Pro Trp Asn Lys Thr Trp Xaa Gly Asn Tyr
 610 615 620
 Ser Tyr Glu Tyr Ile Trp Asn Asn Leu Thr Trp Glu Gln Trp Asp Val
 625 630 635 640
 His Val Ser Asn Tyr Thr Gly Val Ile Phe Gly Leu Leu Glu Glu Ala
 645 650 655
 Gln Gly Gln Gln Glu Lys Asn Glu Arg Glu Leu Leu Glu Leu Asp Lys
 660 665 670
 Trp Ser Ser Leu Trp Ser Trp Phe Asx Xaa Thr His Trp Leu Trp Tyr
 675 680 685
 Ile Lys Ile Phe Ile Met Val Val Ala Gly Ile Val Gly Leu Arg Ile
 690 695 700
 Cys Met Ile Val Ile Ser Val Ile Arg Arg Val Arg Gln Gly Tyr Ser
 705 710 715 720
 Pro Leu Ser Leu Gln Thr Leu Leu Pro Val Gln Arg Gly Pro Gly Val
 725 730 735
 Leu Asp Gly Thr Asp Glu Glu Gly Gly Glu Gln Gly Arg Gly Arg Ser
 740 745 750
 Val Arg Leu Val Thr Gly Cys Leu Pro Leu Ile Trp Asp Asp Leu Arg
 755 760 765
 Ser Leu Leu Ile Trp Phe Tyr Gln Thr Leu Ser Asn Ser Ala Cys Val
 770 775 780
 Val Gln Arg His Leu Glu Arg Leu Ile Gln Leu Ile Leu Lys Gly Leu
 785 790 795 800
 Arg Leu Leu Trp Glu Lys Ile Arg Leu Cys Trp Gly Val Leu Gln Tyr
 805 810 815
 Trp Lys Arg Glu Leu Lys Thr Ser Ala Ile Ser Leu Leu Asp Ala Thr
 820 825 830
 Ala Ile Arg Val Ala Glu Gly Thr Asp Gln Ile Ile Phe Ala Ala Gln
 835 840 845
 Arg Ile Gly Arg Gly Phe Leu Asn Ile Pro Arg Arg Ile Arg Gln Gly
 850 855 860
 Leu Glu Arg Ala Leu Leu
 865 870

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<210> SEQ ID NO 340
<211> LENGTH: 868
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (461)..(461)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (463)..(463)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (767)..(767)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<400> SEQUENCE: 340

Met Arg Ala Met Glu Arg Thr Lys Arg Ser Trp Asn Ile Leu Ser Arg
1          5          10          15

Ser Met Val Leu Leu Thr Leu Cys Leu Thr Phe Asn Val Ala Thr Glu
20          25          30

Tyr Trp Val Ala Val Tyr Tyr Gly Val Pro Val Trp Lys Asp Ala Glu
35          40          45

Thr Thr Leu Phe Cys Ala Ser Glu Ala Lys Pro His Ser Ser Glu Ala
50          55          60

His Asn Ile Trp Ala Thr Gln Ala Cys Val Pro Ile Asp Pro Asn Pro
65          70          75          80

Glu Glu Val Phe Leu Thr Asn Val Thr Glu Asn Phe Ser Met Trp Asp
85          90          95

Asn Pro Met Val Ser Gln Met Gln Glu Asp Ile Ile Ser Leu Trp Glu
100         105         110

Gln Ser Leu Lys Pro Cys Val Lys Leu Thr Leu Tyr Val Leu Thr Pro
115         120         125

Asn Cys Ser Thr Ala Glu Phe Lys Asn Asn Thr Asn Thr Asn Thr Thr
130         135         140

Thr Thr Gly Pro Ala Asn Val Thr Asn Tyr Glu Met Lys Asn Cys Ser
145         150         155         160

Phe Asn Val Thr Thr Glu Leu Arg Asp Arg Lys Lys Arg Val Tyr Ser
165         170         175

Leu Phe Tyr Val Asp Asp Val Glu Lys Leu Asn Asp Gly Asn Ser Thr
180         185         190

Asn Ser Thr Tyr Arg Leu Ile Asn Cys Asn Thr Thr Ala Ile Thr Gln
195         200         205

Ala Cys Pro Lys Ile Ser Phe Glu Pro Ile Pro Ile His Tyr Cys Ala
210         215         220

Pro Ala Gly Phe Ala Ile Leu Lys Cys Asn Asp Thr Asn Tyr Ser Gly
225         230         235         240

Thr Gly Lys Cys His Asn Val Ser Thr Val His Cys Thr His Gly Ile
245         250         255

Met Pro Ala Val Thr Thr Gln Leu Ile Leu Asn Gly Ser Leu Ala Glu
260         265         270

Gly Asn Val Thr Ile Arg Val Lys Asn Ala Ser Ser Asn Ser Gly Asn
275         280         285

Phe Ile Val Gln Leu Ala Lys Pro Val Glu Ile Leu Cys Arg Arg Glu
290         295         300

Gly Asn Asn Thr Arg Gly Gln Val Gln Ile Gly Pro Gly Met Thr Tyr
305         310         315         320

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Tyr	Asn	Ile	Glu	Asn	Val	Ile	Gly	Asn	Thr	Arg	Lys	Ala	Tyr	Cys	Lys
			325						330			335			
Leu	Asn	Arg	Ser	Asn	Trp	Glu	Asp	Ala	Met	Ser	Lys	Val	Lys	Lys	Glu
			340						345			350			
Leu	Glu	Lys	Ile	Thr	Asn	Val	Thr	Lys	Val	Ile	Phe	Ala	Asn	Gln	Ser
			355						360			365			
His	Gly	Ser	Asp	Pro	Glu	Val	Glu	Asn	His	Met	Phe	Asn	Cys	Gly	Gly
			370						375			380			
Glu	Phe	Phe	Tyr	Cys	Asn	Thr	Ser	Lys	Leu	Phe	Asn	Asn	Lys	Thr	Gly
			385						390			400			
Ile	Asn	Asp	Thr	Thr	Gly	Lys	Gly	Thr	Asn	Glu	Ser	Glu	Ile	Ile	Leu
			405						410			415			
Pro	Cys	Arg	Leu	Arg	Gln	Ile	Val	Asn	Leu	Trp	Thr	Arg	Val	Gly	Lys
			420						425			430			
Gly	Ile	Tyr	Ala	Pro	Pro	Ile	Arg	Gly	Asn	Ile	Ser	Cys	Lys	Ser	Asn
			435						440			445			
Ile	Thr	Gly	Ile	Leu	Leu	Ala	Arg	Glu	Glu	Gly	Asx	Xaa	Asn	Xaa	Thr
			450						455			460			
Asn	Leu	Thr	Phe	Lys	Pro	Ile	Gly	Gly	Asn	Met	Glu	Asp	Leu	Trp	Arg
			465						470			475			
Ser	Glu	Leu	Tyr	Lys	Tyr	Lys	Val	Val	Arg	Ile	Glu	Pro	Leu	Ala	Val
			485						490			495			
Ala	Pro	Ser	Lys	Ala	Lys	Arg	Arg	Thr	Val	Glu	Thr	Glu	Lys	Arg	Gln
			500						505			510			
Lys	Arg	Ala	Ala	Phe	Gly	Leu	Gly	Ala	Leu	Phe	Leu	Gly	Phe	Leu	Gly
			515						520			525			
Ala	Ala	Gly	Ser	Thr	Met	Gly	Ala	Ala	Ala	Val	Thr	Leu	Thr	Val	Gln
			530						535			540			
Ala	Arg	Gln	Leu	Leu	Ser	Gly	Ile	Val	Gln	Gln	Gln	Asn	Asn	Leu	Leu
			545						550			555			
Arg	Ala	Ile	Glu	Ala	Gln	Gln	His	Leu	Leu	Gln	Leu	Ser	Ile	Trp	Gly
			565						570			575			
Val	Lys	Gln	Leu	Gln	Gly	Arg	Leu	Leu	Ala	Val	Glu	Arg	Tyr	Leu	Gln
			580						585			590			
Asp	Gln	Gln	Ile	Leu	Gly	Leu	Trp	Gly	Cys	Ser	Gly	Lys	Ala	Ile	Cys
			595						600			605			
Tyr	Thr	Thr	Val	Pro	Trp	Asn	Lys	Thr	Trp	Phe	Gly	Asn	Tyr	Ser	Tyr
			610						615			620			
Glu	Tyr	Ile	Trp	Asn	Asn	Leu	Thr	Trp	Glu	Gln	Trp	Asp	Val	His	Val
			625						630			635			
Ser	Asn	Tyr	Thr	Gly	Val	Ile	Phe	Gly	Leu	Leu	Glu	Glu	Ala	Gln	Gly
			645						650			655			
Gln	Gln	Glu	Lys	Asn	Glu	Arg	Glu	Leu	Leu	Glu	Leu	Asp	Lys	Trp	Ser
			660						665			670			
Ser	Leu	Trp	Ser	Trp	Phe	Asp	Ile	Thr	His	Trp	Leu	Trp	Tyr	Ile	Lys
			675						680			685			
Ile	Phe	Ile	Met	Val	Val	Ala	Gly	Ile	Val	Gly	Leu	Arg	Ile	Cys	Met
			690						695			700			
Ile	Val	Ile	Ser	Val	Ile	Arg	Arg	Val	Arg	Gln	Gly	Tyr	Ser	Pro	Leu
			705						710			715			
Ser	Leu	Gln	Thr	Leu	Leu	Pro	Val	Gln	Arg	Gly	Pro	Gly	Val	Leu	Asp
			725						730			735			

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Gly	Thr	Asp	Glu	Glu	Gly	Gly	Glu	Gln	Gly	Arg	Gly	Arg	Ser	Val	Arg
			740					745					750		
Leu	Val	Thr	Gly	Cys	Leu	Pro	Leu	Ile	Trp	Asp	Asp	Leu	Arg	Xaa	Leu
		755					760					765			
Leu	Ile	Trp	Phe	Tyr	Gln	Thr	Leu	Ser	Asn	Ser	Ala	Cys	Val	Val	Gln
	770					775					780				
Arg	His	Leu	Glu	Arg	Leu	Ile	Gln	Leu	Ile	Leu	Lys	Gly	Leu	Arg	Leu
	785				790					795					800
Leu	Trp	Glu	Lys	Ile	Arg	Leu	Cys	Trp	Gly	Val	Leu	Gln	Tyr	Trp	Thr
			805						810					815	
Arg	Glu	Leu	Lys	Thr	Ser	Ala	Ile	Ser	Leu	Leu	Asp	Ala	Ile	Ala	Ile
			820					825					830		
Arg	Val	Ala	Glu	Gly	Thr	Asp	Gln	Ile	Ile	Phe	Ala	Ala	Gln	Arg	Ile
		835					840					845			
Gly	Arg	Gly	Phe	Leu	Asn	Ile	Pro	Gly	Arg	Ile	Arg	Gln	Gly	Leu	Leu
	850				855						860				
Arg	Ala	Leu	Leu												
	865														

<210> SEQ ID NO 341

<211> LENGTH: 798

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 341

Met	Arg	Val	Arg	Glu	Met	Lys	Lys	Leu	Trp	Ser	Phe	Trp	Val	Leu	Gly
1				5					10					15	
Leu	Gly	Phe	Leu	Ala	Leu	Ser	Leu	Thr	Ser	Asp	Ser	Asn	Trp	Trp	Val
		20					25						30		
Thr	Val	Tyr	Leu	Gly	Val	Pro	Val	Trp	Lys	Asp	Ala	Glu	Thr	Thr	Leu
		35				40						45			
Phe	Cys	Ala	Ser	Asp	Ala	Lys	Ala	Tyr	Ser	Thr	Glu	Ala	His	Asn	Ile
	50					55					60				
Trp	Ala	Thr	Gln	Ala	Cys	Val	Pro	Ile	Asp	Pro	Asn	Pro	Gln	Glu	Val
	65				70				75					80	
Phe	Leu	Lys	Asn	Val	Lys	Glu	Asn	Phe	Asn	Met	Trp	Asp	Asn	Pro	Met
			85					90						95	
Val	Asp	Gln	Met	Gln	Glu	Asp	Ile	Ile	Ser	Leu	Trp	Asp	Gln	Ser	Leu
		100					105						110		
Lys	Pro	Cys	Val	Lys	Leu	Thr	Pro	Leu	Cys	Val	Thr	Leu	Asn	Cys	Thr
		115					120					125			
Asn	Val	Asn	Ala	Thr	Ser	Pro	Thr	Thr	Lys	Pro	Pro	Thr	Thr	Thr	Pro
	130					135					140				
Thr	Thr	Val	Ser	Thr	Val	Ser	Thr	Thr	Ile	Pro	Leu	Asn	Asp	Ser	Ile
	145				150					155					160
Phe	Glu	Asp	Met	Cys	Asn	Cys	Thr	Phe	Asn	Val	Thr	Thr	Glu	Leu	Arg
			165					170						175	
Gly	Ser	Ser	Asn	Asn	Ser	Tyr	Asn	Ser	Tyr	Arg	Leu	Ile	Asn	Cys	Asn
			180					185					190		
Thr	Thr	Ala	Ile	Thr	Gln	Ala	Cys	Pro	Lys	Thr	Ser	Phe	Glu	Pro	Ile
		195					200					205			
Pro	Ile	His	Tyr	Cys	Ala	Pro	Ala	Gly	Phe	Ala	Ile	Leu	Arg	Cys	Asn
	210					215					220				
Glu	Glu	Asn	Ser	Gln	Glu	Met	Gly	Tyr	Cys	Glu	Asn	Val	Ser	Thr	Val
	225				230					235					240

His 245	Cys	Thr	His	Gly	Ile	Lys	Pro	Val	Val	Thr	Thr	Gln	Leu	Ile	Leu
Asn	Gly	Ser	Ile	Thr	Gln	Gln	Ile	Met	Ile	Arg	Ser	Lys	Asn	Ile	Ser
Ser	Asn	Ser	Phe	Asn	Ile	Ile	Val	Gln	Phe	Asn	Glu	Thr	Ile	Pro	Ile
Lys	Cys	Ile	Arg	Pro	Gly	Asn	Asn	Thr	Arg	Gly	Gln	Ile	Gln	Ile	Gly
Pro 305	Ala	Met	Thr	Phe	Tyr	Asn	Ile	Glu	Asn	Ile	Val	Gly	Asn	Thr	Arg 320
Lys	Ala	Phe	Cys	Lys	Val	Asn	Gly	Ser	Gln	Trp	Trp	Asn	Met	Lys	Gln
Asn	Ile	Ile	Gln	Arg	Phe	Lys	Ala	Glu	His	Lys	Leu	Asn	Val	Thr	Phe
Asn	Ser	Ser	Ala	Gly	Gly	Asp	Pro	Glu	Ile	Thr	Asn	Phe	Met	Val	Asn
Cys	His	Gly	Glu	Phe	Phe	Tyr	Cys	Asn	Thr	Thr	Pro	Leu	Phe	Thr	Gly
Asn 385	Lys	Thr	Asn	Ile	Thr	Ile	Ile	Leu	Pro	Cys	Lys	Ile	Arg	Gln	Ile 400
Val	Asn	Ser	Trp	Met	Arg	Val	Gly	Lys	Gly	Ile	Tyr	Ala	Pro	Pro	Ile
Arg	Gly	Asn	Leu	Ser	Cys	Asn	Ser	Asn	Ile	Thr	Gly	Leu	Leu	Leu	Thr
Arg	Asp	Gly	Pro	Asp	Lys	Asn	Ile	Asn	Glu	Thr	Glu	Thr	Leu	Arg	Pro
Gly	Gly	Gly	Asp	Met	Lys	Asp	Ile	Trp	Arg	Ser	Glu	Leu	Tyr	Lys	Tyr
Lys 465	Val	Val	Lys	Ile	Glu	Pro	Leu	Ala	Val	Ala	Pro	Thr	Lys	Ala	Arg 480
Arg	Tyr	Thr	Ile	Asn	Met	Glu	Lys	His	Arg	Ala	Lys	Arg	Ala	Ala	Phe
Ala	Ala	Gly	Ser	Thr	Met	Gly	Ala	Ala	Ala	Val	Thr	Leu	Thr	Val	Gln
Ala	Arg	Asn	Leu	Leu	Ser	Gly	Ile	Val	Gln	Gln	Gln	Asn	Asn	Leu	Leu
Arg	Ala	Ile	Glu	Ala	Gln	Gln	His	Leu	Leu	Gln	Leu	Ser	Val	Trp	Gly
Val 545	Lys	Gln	Leu	Gln	Ala	Arg	Leu	Leu	Ala	Val	Glu	Arg	Tyr	Leu	Lys 560
Asp	Gln	Gln	Ile	Leu	Gly	Leu	Trp	Gly	Cys	Ser	Gly	Lys	Ala	Ile	Cys
Tyr	Thr	Asn	Val	Pro	Trp	Asn	Arg	Asn	Trp	Thr	Asn	Ser	Ser	Gln	Asp
Tyr	Asn	Glu	Ile	Trp	Asn	Asn	Leu	Thr	Trp	Asn	Glu	Trp	Asp	Lys	Gln
Val 610	Ser	Asn	Tyr	Thr	Thr	Glu	Ile	Phe	Lys	Ala	Leu	Glu	Glu	Ala	Gln
Gly	Gln	Gln	Glu	Lys	Asn	Glu	Lys	Glu	Leu	Leu	Glu	Leu	Asp	Lys	Trp
Ser	Ser	Leu	Trp	Ser	Trp	Leu	Asp	Ile	Thr	Gln	Trp	Leu	Trp	Tyr	Ile

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Lys Ile Phe Leu Met Val Val Gly Ala Ile Ile Gly Leu Lys Ile Val
 660 665 670
 Met Thr Val Val Ser Leu Ile Lys Arg Val Arg Gln Gly Tyr Ser Pro
 675 680 685
 Leu Ser Leu Gln Thr Leu Ile Pro Ala Ala Gly Glu Arg Gly Pro Pro
 690 695 700
 Glu Glu Lys Glu Glu Gly Val Gly Asp Asn Gly Arg Thr Arg Ser Ile
 705 710 715 720
 Arg Leu Ala Asn Gly Phe Phe Ala Leu Phe Trp Glu Asp Leu Arg Ser
 725 730 735
 Leu Phe Leu Phe Ser Tyr His Arg Leu Arg Asp Leu Thr Ser Ile Leu
 740 745 750
 Ser Arg Ile Leu Gln Leu Leu Tyr Arg Gln Ala Val Thr Gly Leu Gln
 755 760 765
 Thr Val Arg Lys Ala Leu Ser Leu Leu Lys Ala His Ile Ala Tyr Trp
 770 775 780
 Gly Ser Glu Leu Lys Ser Ser Ala Ile Ser Leu Leu Asp Ser
 785 790 795

<210> SEQ ID NO 342

<211> LENGTH: 864

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 342

Met Arg Arg Leu Ile Gly Thr Ile Leu Thr Leu Ser Ile Thr Ile Leu
 1 5 10 15
 Gly Ile Gly Tyr Ser Val His Tyr Ala Thr Val Phe Tyr Gly Val Pro
 20 25 30
 Val Trp Lys Asp Ala Lys Pro Gln Leu Phe Cys Ala Ser Asp Ala Asp
 35 40 45
 Ile Thr Ser Arg Asp Pro His Asn Ile Trp Ala Thr His Asn Cys Val
 50 55 60
 Pro Leu Asp Pro His Pro Tyr Glu Val Pro Leu Ser Asn Val Ser Val
 65 70 75 80
 Asp Phe Asp Met Glu Asn Asn Tyr Met Val Glu Glu Met Lys Lys Asp
 85 90 95
 Leu Ile Ala Leu Phe Gln Gln Ser Phe Lys Pro Cys Val Lys Leu Thr
 100 105 110
 Pro Phe Cys Val Thr Met Asn Cys Val Lys Leu Lys Glu Thr Thr Thr
 115 120 125
 Thr Ile Ala Thr Ile Thr Asn Pro Leu Val Asn Leu Thr Gly Ile Glu
 130 135 140
 Asp Tyr Lys Val Tyr Asn Cys Ser Phe Asn Gln Thr Thr Glu Phe Arg
 145 150 155 160
 Asp Lys Lys Lys Gln Ile Tyr Ser Leu Phe Tyr Arg Glu Asp Ile Met
 165 170 175
 Lys Thr Asp Asn Asn Asp Ser Tyr Tyr Leu His Asn Cys Asn Thr Ser
 180 185 190
 Ala Ile Thr Gln Glu Cys Glu Lys Ser Ser Phe Glu Pro Ile Pro Ile
 195 200 205
 Arg Tyr Cys Ala Pro Ala Gly Tyr Ala Met Leu Lys Cys Asn Ser Ala
 210 215 220
 Asn Phe Thr Gly Val Gly Thr Cys Asn Asn Val Ser Val Val His Cys
 225 230 235 240

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Thr	His	Gly	Ile	Tyr	Pro	Met	Ile	Ala	Thr	Ala	Leu	His	Leu	Asn	Gly	
			245						250					255		
Thr	Leu	Glu	Lys	His	Asn	Thr	Thr	Ala	Tyr	Phe	Ala	Thr	Pro	Thr	His	
			260					265					270			
Asn	Lys	Pro	Leu	Leu	Ile	Lys	Phe	Asn	Lys	Ser	Val	Asp	Met	Asn	Cys	
		275					280					285				
Thr	Arg	Thr	Gly	Asn	Asn	Ser	Arg	Gly	Gln	Val	Gln	Ile	Gly	Pro	Gly	
	290					295					300					
Met	Thr	Phe	Tyr	Asn	Ile	Glu	Asn	Ile	Val	Gly	Asn	Thr	Arg	Lys	Ala	
305				310						315					320	
Tyr	Cys	Thr	Val	Asn	Tyr	Gln	Glu	Trp	Ser	His	Ala	Ile	Gly	Glu	Ala	
			325					330						335		
Lys	Lys	Val	Ala	Glu	Glu	Ala	Leu	Lys	Lys	Asn	Ile	Thr	Phe	Arg	Trp	
		340						345					350			
Asn	Gln	Gly	Gly	Asp	Leu	Glu	Val	Thr	Asn	Phe	Trp	Phe	Asn	Cys	Gln	
		355				360						365				
Gly	Glu	Phe	Phe	Tyr	Cys	Asn	Leu	Thr	Asn	Trp	Thr	Asn	Ile	Thr	Trp	
	370					375					380					
Leu	Glu	Thr	Trp	Leu	Lys	Thr	Asn	Asn	Ser	Ser	Arg	Pro	Leu	Val	Ala	
385				390					395						400	
Pro	Cys	Lys	Leu	Arg	Gln	Ile	Val	Asn	His	Trp	Gly	Ile	Val	Ser	Lys	
			405					410						415		
Gly	Ile	Tyr	Leu	Pro	Pro	Arg	Arg	Gly	Asn	Leu	Thr	Cys	Arg	Ser	Asn	
			420					425					430			
Ile	Thr	Gly	Phe	Ile	Thr	Thr	Trp	Asp	Asn	Asp	Thr	Glu	Asn	Val	Leu	
		435					440					445				
Val	Thr	Phe	Ser	Ala	Lys	Val	Glu	Asp	Tyr	Trp	Lys	Val	Glu	Met	Ser	
	450				455						460					
Arg	Tyr	Lys	Val	Val	Glu	Ile	Gln	Pro	Leu	Ala	Ile	Ala	Pro	Thr	Thr	
465				470					475						480	
Gly	Lys	Arg	Pro	Glu	Ile	Arg	Ala	Asn	His	Thr	Arg	Ser	Lys	Arg	Asp	
			485					490						495		
Val	Gly	Ile	Gly	Leu	Leu	Phe	Leu	Gly	Phe	Leu	Ser	Ala	Ala	Gly	Ser	
		500						505					510			
Thr	Met	Gly	Ala	Ala	Ser	Ile	Ala	Leu	Thr	Ala	Gln	Ala	Arg	Gly	Leu	
	515						520					525				
Leu	Ser	Gly	Ile	Val	Gln	Gln	Gln	His	Asn	Leu	Leu	Gln	Ala	Ile	Glu	
	530				535						540					
Ala	Gln	Gln	His	Leu	Leu	Gln	Leu	Ser	Val	Trp	Gly	Ile	Lys	Gln	Leu	
545				550						555					560	
Gln	Ala	Arg	Met	Leu	Ala	Val	Glu	Lys	Tyr	Ile	Arg	Asp	Gln	Gln	Leu	
			565					570						575		
Leu	Ser	Val	Trp	Gly	Cys	Ala	Asn	Lys	Leu	Val	Cys	His	Ser	Asn	Val	
		580						585					590			
Val	Trp	Asn	Met	Thr	Trp	Ala	Glu	Asn	Gly	Thr	Thr	Cys	Pro	Gln	Gly	
		595					600					605				
Ser	Glu	Asp	Tyr	Tyr	Asn	Cys	Ile	Trp	Asn	Asn	Leu	Thr	Trp	Gln	Gln	
	610				615						620					
Trp	Glu	Lys	Leu	Val	Ala	Asn	Ser	Thr	Glu	Glu	Ile	Tyr	Thr	Leu	Leu	
625				630						635					640	
Glu	Lys	Ala	Gln	Val	Gln	Gln	Glu	Thr	Asn	Lys	Lys	Glu	Leu	Phe	Glu	
			645					650						655		

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Leu Asp Lys Trp Ser Thr Leu Trp Asp Trp Phe Asp Ile Thr Gln Trp
    660                      665                      670

Leu Trp Tyr Ile Lys Ile Ala Ile Leu Ile Val Ala Gly Leu Val Gly
    675                      680                      685

Leu Arg Ile Val Met Phe Ile Val Asn Val Ile Arg Gln Val Arg Gln
    690                      695                      700

Gly Tyr Met Pro Leu Phe Ser Gln Ile Pro Thr Gln Ala His Gln Asp
    705                      710                      715                      720

Pro Glu Gln Pro Glu Glu Thr Ala Gly Gly Gly Gly Gly Thr Gly Asn
    725                      730                      735

Val Arg Trp Thr Pro Leu His Arg Gly Phe Phe Ile Val Val Trp Glu
    740                      745                      750

Asp Leu Arg Thr Leu Leu Leu Trp Ile Tyr Gln Thr Cys Gln Asn Phe
    755                      760                      765

Gly Trp Leu Leu Trp Thr Ile Leu Lys Thr Leu Arg Gln Gly Ile Ile
    770                      775                      780

Ser Leu Thr Gln Arg Leu Ile Ile Val Gln Arg Asp Ile Ile Leu Lys
    785                      790                      795                      800

Ser Arg Gln Leu Phe Asp Trp Leu Ser Asn Thr Tyr Ser Ile Leu Arg
    805                      810                      815

Thr Ser Leu Ile Gln Ala Ile Asp Arg Leu Ala Asn Phe Thr Gly Trp
    820                      825                      830

Trp Thr Asp Leu Val Ile Ala Gly Val Ala Phe Val Ala Gln Gly Ile
    835                      840                      845

Arg Asn Ile Pro Arg Arg Ile Arg Gln Gly Leu Glu Ile Ala Leu Asn
    850                      855                      860

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<210> SEQ ID NO 343

<211> LENGTH: 875

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 343

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Met Arg Asn Leu Ile Gly Ile Thr Leu Thr Leu Ile Ile Thr Thr Leu
 1          5          10          15

Gly Ile Gly Phe Ser Gln Tyr Tyr Ala Thr Val Phe Tyr Gly Val Pro
20          25          30

Ile Trp Lys Asp Ala Lys Pro Ser Leu Phe Cys Ala Ser Asp Ala Asp
35          40          45

Val Thr Ser Arg Asp Glu His Asn Ile Trp Ala Thr His Asn Cys Val
50          55          60

Pro Leu Asp Pro Asn Pro Tyr Glu Val Pro Leu Ala Asn Met Thr Val
65          70          75          80

Asp Phe Asn Met Glu Glu Asn Tyr Met Val Gln Glu Met Lys Glu Asp
85          90          95

Leu Ile Ser Leu Phe Gln Gln Ser Phe Lys Pro Cys Val Lys Leu Thr
100         105         110

Pro Phe Cys Val Ser Met Ser Cys Asn Leu Thr Val Pro Ser Thr Thr
115         120         125

Val Ser Pro Ala Thr Lys Val Ser Ser Thr Thr Lys Glu Pro Glu Ile
130         135         140

Lys Arg Leu Gly Lys Asn Ile Thr Asn Tyr Glu Val Tyr Lys Cys Ser
145         150         155         160

Phe Asn Gln Thr Thr Glu Phe Arg Asp Lys Lys Lys Gln Ile Tyr Ser
165         170         175

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Leu Phe Tyr Lys Glu Asp Ile Met Lys Asn Asn Asn Thr Asn Asn Asn
 180 185 190
 Asn Ser Tyr Tyr Leu His Asn Cys Asn Thr Ser Ala Ile Thr Gln Glu
 195 200 205
 Cys Ala Lys Ser Thr Phe Glu Pro Ile Pro Ile Arg Tyr Cys Ala Pro
 210 215 220
 Ala Gly Phe Ala Met Leu Lys Cys Lys Asp Gln Asn Phe Thr Gly Lys
 225 230 235 240
 Gly Pro Cys Ser Asn Val Ser Val Val His Cys Thr His Gly Ile Tyr
 245 250 255
 Pro Met Ile Ala Thr Ala Leu His Leu Asn Gly Thr Leu Glu Glu Asn
 260 265 270
 Lys Thr Thr Ala Tyr Phe Ala Asp Ser Thr His Asn Gln Pro Leu Leu
 275 280 285
 Val Lys Phe His Lys Ala Val Asn Leu Thr Cys Val Arg Thr Gly Asn
 290 295 300
 Asn Thr Arg Gly Gln Val Gln Ile Gly Pro Gly Met Thr Phe Tyr Asn
 305 310 315 320
 Ile Glu Asn Val Val Gly Asp Thr Arg Lys Ala Phe Cys Ile Val Asn
 325 330 335
 Ala Thr Glu Trp Ser Asn Ala Leu Asn Glu Ser Lys Ile Ala Ile Asn
 340 345 350
 Glu Thr Leu Arg Lys Tyr Asn Lys Thr Ser Tyr Gly Asn Phe Thr Trp
 355 360 365
 Asn Thr Gly Asp Pro Glu Val Ser Asn Phe Trp Phe Asn Cys Gln Gly
 370 375 380
 Glu Phe Phe Tyr Cys Asn Leu Thr Asn Trp Thr Ser Pro Asp Tyr Ile
 385 390 395 400
 Gly Lys Asn Gly Ser Ile Val Ala Gln Cys Arg Leu Arg Gln Ile Val
 405 410 415
 Asn His Trp Gly Ile Val Ser Lys Gly Ile Tyr Leu Pro Pro Arg Arg
 420 425 430
 Gly Thr Ile Lys Cys Thr Ser Asn Ile Thr Gly Leu Ile Leu Thr Ala
 435 440 445
 Glu Gly Gly Asn Glu Thr Ile Thr Pro Thr Phe Ser Ser Lys Val Glu
 450 455 460
 Asp Tyr Trp Lys Val Glu Leu Ala Arg Tyr Lys Val Val Glu Ile Gln
 465 470 475 480
 Pro Leu Ser Val Ala Pro Arg Pro Gly Lys Arg Pro Glu Ile Lys Ala
 485 490 495
 Asn His Thr Arg Ser Lys Arg Asp Val Gly Ile Gly Leu Leu Phe Leu
 500 505 510
 Gly Phe Leu Ser Ala Ala Gly Ser Thr Met Gly Ala Ala Ser Ile Ala
 515 520 525
 Leu Thr Ala Gln Ala Arg Gly Leu Leu Ser Gly Ile Val Gln Gln Gln
 530 535 540
 Gln Asn Leu Leu Gln Ala Ile Ala Ala Gln Gln His Leu Leu Gln Leu
 545 550 555 560
 Ser Val Trp Gly Ile Lys Gln Leu Gln Ala Arg Met Leu Ala Ile Glu
 565 570 575
 Lys Tyr Ile Arg Asp Gln Gln Leu Leu Ser Ile Trp Gly Cys Ala Asn
 580 585 590

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Lys Leu Val Cys His Ser Ser Val Pro Trp Asn Ile Thr Trp Ala Gly
 595 600 605
 Asn Ser Thr Lys Cys Asn His Ser Asp Thr Met Asp Ala Lys Tyr Tyr
 610 615 620
 Glu Cys Ile Trp Asn Asn Leu Thr Trp Gln Glu Trp Asp Lys Leu Val
 625 630 635 640
 Lys Asn Ser Ser Glu Thr Ile Tyr Ser Leu Leu Glu Ile Ala Gln Thr
 645 650 655
 Gln Gln Glu Lys Asn Lys Gln Asp Leu Leu Glu Leu Asp Lys Trp Ser
 660 665 670
 Ser Leu Trp Asp Trp Phe Asp Ile Thr Gln Trp Leu Trp Tyr Ile Lys
 675 680 685
 Ile Ala Ile Ile Ile Val Ala Gly Leu Ile Gly Leu Arg Ile Leu Met
 690 695 700
 Phe Ile Ile Asn Val Val Arg Gln Val Arg Gln Gly Tyr Met Pro Leu
 705 710 715 720
 Phe Ser Gln Thr Pro Thr Gln Ala Glu Gln Asp Pro Glu Gln Pro Gly
 725 730 735
 Gly Ile Ala Glu Gly Gly Gly Gly Gly Lys Leu Arg Trp Thr Pro
 740 745 750
 Ser Pro Ala Gly Phe Phe Thr Ile Val Trp Val Asp Leu Arg Asn Leu
 755 760 765
 Leu Leu Trp Ile Tyr Gln Thr Cys Arg Asn Phe Ile Trp Ile Leu Trp
 770 775 780
 Ser Ile Leu Gln Ala Leu Lys Gln Gly Ile Ile Ser Leu Val Asp Asn
 785 790 795 800
 Leu Val Ile Ile His Arg Ser Ile Val Val Gly Val Arg Gln Ala Leu
 805 810 815
 Glu Trp Ser Gly Asn Thr Tyr Ala Ser Ile Arg Ala Ser Leu Ile Gln
 820 825 830
 Ala Ile Asp Arg Leu Ala Glu Phe Thr Gly Trp Trp Thr Asp Ile Val
 835 840 845
 Ile Glu Ala Val Val Tyr Ile Ala Arg Gly Ile Arg Asn Ile Pro Arg
 850 855 860
 Arg Ile Arg Gln Gly Leu Glu Ile Ala Leu Asn
 865 870 875

<210> SEQ ID NO 344

<211> LENGTH: 835

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 344

Met Lys Val Thr Glu Met Gln Lys Asn Trp Leu Ile Cys Cys Leu Leu
 1 5 10 15
 Ile Gly Leu Ile Lys Ile Ile Gly Ser Glu Leu Trp Val Thr Val Tyr
 20 25 30
 Tyr Gly Val Pro Val Trp Arg Asp Ala Glu Thr Val Leu Phe Cys Ala
 35 40 45
 Ser Asp Ala Lys Ala His Ser Thr Glu Ala His Asn Ile Trp Ala Thr
 50 55 60
 Gln Ala Cys Val Pro Thr Asp Pro Asn Pro Gln Glu Val Leu Ile Pro
 65 70 75 80
 Asn Val Thr Glu Arg Phe Asp Met Trp Lys Asn Asn Met Val Asp Gln
 85 90 95

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Met	Gln	Glu	Asp	Ile	Ile	Ser	Leu	Trp	Glu	Gln	Ser	Leu	Lys	Pro	Cys	
			100					105					110			
Val	Lys	Leu	Thr	Pro	Leu	Cys	Val	Thr	Leu	Ser	Cys	Ser	Ser	Trp	Arg	
		115					120					125				
Ser	Val	Asn	Asn	Ser	Val	Asn	Gln	Thr	Asn	His	Val	Gln	Met	Gln	Asn	
	130					135					140					
Cys	Ser	Phe	Asn	Val	Thr	Thr	Glu	Leu	Arg	Asp	Lys	Lys	Lys	Gln	Val	
145					150					155					160	
Tyr	Ser	Leu	Phe	Tyr	Met	Gly	Asp	Ile	Ile	Pro	Leu	Asp	Thr	Asn	Asn	
			165					170						175		
Ser	Ser	Gly	Asn	Asn	Ser	Gln	Tyr	Arg	Leu	Ile	Asn	Cys	Asn	Thr	Thr	
		180						185					190			
Ala	Val	Thr	Gln	Ala	Cys	Pro	Lys	Ile	Ser	Phe	Glu	Pro	Ile	Pro	Ile	
	195						200					205				
Tyr	Tyr	Cys	Ala	Pro	Pro	Gly	Phe	Ala	Ile	Ile	Lys	Cys	Asn	Asp	Gln	
	210					215					220					
Asp	Phe	Asn	Gly	Thr	Gly	Glu	Cys	Asn	Asn	Val	Ser	Thr	Val	Gln	Cys	
225					230					235					240	
Thr	His	Gly	Ile	Lys	Pro	Val	Ile	Ser	Thr	Gln	Leu	Ile	Leu	Asn	Gly	
			245						250					255		
Ser	Leu	Ala	Thr	Ser	Asn	Ile	Val	Ile	Arg	Asn	Asn	Ser	Lys	Asp	Thr	
		260						265					270			
Leu	Leu	Val	Gln	Leu	Asn	Glu	Ser	Ile	Pro	Ile	Asn	Cys	Thr	Arg	Pro	
		275					280				285					
Gly	Asn	Lys	Thr	Arg	Gly	Gln	Val	Gln	Ile	Gly	Pro	Gly	Met	Thr	Phe	
	290				295					300						
Tyr	Asn	Ile	Glu	Asn	Ile	Ile	Gly	Asp	Thr	Arg	Gln	Ala	Tyr	Cys	Glu	
305					310					315					320	
Val	Asn	Arg	Thr	Trp	Glu	Gln	Ile	Trp	Asn	Thr	Thr	Lys	Gln	Ile	Ile	
			325						330					335		
Ile	Asn	Asn	Arg	Lys	Asn	Ile	Thr	Phe	Ile	Pro	Asn	Pro	Gly	Gly	Asp	
			340					345					350			
Leu	Glu	Val	Thr	Asn	Leu	Met	Ile	Asn	Cys	Gly	Gly	Glu	Phe	Phe	Tyr	
		355				360						365				
Cys	Asn	Thr	Ser	Gln	Leu	Phe	Thr	Asn	Gln	Asn	Gly	Asn	Thr	Thr	Gly	
	370					375					380					
Asn	Ile	Thr	Leu	Gln	Cys	Arg	Ile	Arg	Gln	Ile	Val	Asn	Leu	Trp	Thr	
385					390					395					400	
Arg	Val	Gly	Lys	Gly	Ile	Tyr	Ala	Pro	Pro	Ile	Lys	Gly	Pro	Ile	Asn	
			405					410					415			
Cys	Leu	Ser	Asn	Ile	Thr	Gly	Ile	Ile	Leu	Asp	Tyr	Thr	Lys	Ser	Gly	
			420					425					430			
Thr	Glu	Lys	Tyr	Thr	Ile	Tyr	Pro	Thr	Gly	Gly	Asp	Met	Thr	Asn	Leu	
		435					440					445				
Trp	Arg	Gln	Glu	Leu	Tyr	Lys	Tyr	Lys	Val	Val	Ser	Ile	Glu	Pro	Ile	
	450					455					460					
Gly	Val	Ala	Pro	Gly	Lys	Ala	Lys	Arg	His	Thr	Val	Thr	Arg	Gln	Lys	
465					470					475					480	
Arg	Ala	Ala	Phe	Gly	Leu	Gly	Ala	Leu	Phe	Leu	Gly	Phe	Leu	Gly	Ala	
			485					490					495			
Ala	Gly	Ser	Thr	Met	Gly	Ala	Ala	Ser	Ile	Thr	Leu	Thr	Val	Gln	Ala	
			500					505					510			

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Arg	Lys	Leu	Leu	Ser	Gly	Ile	Val	Gln	Gln	Gln	Asn	Asn	Leu	Leu	Arg
	515						520					525			
Ala	Ile	Glu	Ala	Gln	Gln	His	Leu	Leu	Gln	Leu	Ser	Val	Trp	Gly	Ile
	530					535					540				
Lys	Gln	Leu	Gln	Ala	Arg	Val	Leu	Ala	Ile	Glu	Arg	Tyr	Leu	Arg	Asp
	545				550					555					560
Gln	Gln	Ile	Leu	Gly	Leu	Trp	Gly	Cys	Ser	Gly	Lys	Ser	Val	Cys	Tyr
			565						570					575	
Thr	Asn	Val	Pro	Trp	Asn	Thr	Thr	Trp	Ser	Asn	Asn	Asn	Ser	Tyr	Asp
			580					585					590		
Thr	Ile	Trp	Gly	Asn	Met	Thr	Trp	Gln	Asn	Trp	Asp	Glu	Gln	Val	Arg
		595					600					605			
Asn	Tyr	Ser	Gly	Val	Ile	Phe	Gly	Leu	Leu	Glu	Gln	Ala	Gln	Glu	Gln
	610					615					620				
Gln	Ser	Ile	Asn	Glu	Lys	Ser	Leu	Leu	Glu	Leu	Asp	Gln	Trp	Ser	Ser
	625				630					635					640
Leu	Trp	Asn	Trp	Phe	Asp	Ile	Thr	Lys	Trp	Leu	Trp	Tyr	Ile	Lys	Ile
				645					650					655	
Phe	Ile	Met	Val	Val	Ala	Gly	Ile	Val	Gly	Ile	Arg	Ile	Ile	Ser	Ile
			660					665					670		
Ile	Met	Ser	Met	Val	Ala	Arg	Val	Arg	Gln	Gly	Tyr	Ser	Pro	Leu	Ser
		675					680					685			
Leu	Gln	Thr	Leu	Ile	Pro	Thr	Thr	Arg	Gly	Pro	Asp	Arg	Pro	Glu	Arg
	690					695					700				
Thr	Glu	Glu	Asp	Ala	Gly	Glu	Leu	Asp	Asn	Gly	Arg	Ser	Val	Arg	Leu
	705				710					715					720
Val	Ser	Gly	Phe	Leu	Ala	Leu	Ala	Trp	Glu	Asp	Phe	Arg	Asn	Leu	Leu
				725					730					735	
Leu	Phe	Leu	Tyr	His	Arg	Leu	Thr	Asp	Cys	Leu	Ser	Ile	Leu	Arg	Arg
			740					745					750		
Thr	Leu	Glu	Leu	Leu	Arg	Gln	Asn	Ile	His	Lys	Gly	Leu	Gln	Leu	Leu
		755					760					765			
Asn	Glu	Leu	Arg	Ile	Tyr	Leu	Trp	Gly	Ile	Ile	Ala	Tyr	Trp	Gly	Arg
	770					775					780				
Glu	Leu	Lys	Ile	Ser	Ala	Ile	Asn	Leu	Leu	Asp	Thr	Thr	Ala	Val	Ala
	785				790					795					800
Val	Ala	Glu	Gly	Thr	Asp	Arg	Ile	Ile	Glu	Leu	Val	Gln	Arg	Ile	Gly
				805					810					815	
Arg	Gly	Ile	Leu	His	Ile	Pro	Arg	Arg	Ile	Arg	Gln	Gly	Leu	Glu	Arg
			820					825					830		
Ala	Leu	Leu													
		835													

<210> SEQ ID NO 345

<211> LENGTH: 844

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 345

Met	Lys	Val	Met	Gly	Thr	Gln	Arg	Ser	Trp	Lys	Gln	Phe	Leu	Ser	Leu
1				5					10					15	
Arg	Ile	Val	Met	Cys	Leu	Trp	Leu	Leu	Gly	Val	Ile	Arg	Ser	Glu	Asn
			20					25					30		
Asn	Leu	Trp	Val	Thr	Val	Tyr	Tyr	Gly	Val	Pro	Val	Trp	Lys	Glu	Ala
		35					40					45			

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Lys Thr Thr Leu Phe Cys Ala Ser Asp Ala Lys Ala Gln Ile Ala Glu
 50 55 60
 Ala His Asn Ile Trp Ala Ser Gln Ala Cys Val Pro Thr Asp Pro Asn
 65 70 75 80
 Pro Glu Glu Ile Thr Leu Glu Asn Val Thr Glu Glu Phe Asp Ala Trp
 85 90 95
 Asn Asn Asn Met Val Asp Gln Met Gln Glu Asp Leu Ile Ser Leu Trp
 100 105 110
 Asp Gln Ser Leu Lys Pro Cys Val Lys Leu Thr Pro Leu Cys Val Thr
 115 120 125
 Leu Asn Cys Ser Thr Asn Thr Thr Thr Glu Ala Ser Gln Val Gln Tyr
 130 135 140
 Asn Cys Ser Phe Asn Val Thr Thr Glu Leu Arg Asp Lys Lys Lys Gln
 145 150 155 160
 Val Tyr Ser Leu Phe Tyr Arg Glu Asp Ile Thr Ser Leu Asp Ser Asn
 165 170 175
 Lys Thr Val Lys Asn Gly Thr Tyr Arg Leu Ile Asn Cys Asn Thr Thr
 180 185 190
 Ala Ile Thr Gln Ala Cys Pro Lys Thr Ser Phe Glu Pro Ile Pro Ile
 195 200 205
 Tyr Tyr Cys Ala Pro Ala Gly Phe Ala Leu Leu Lys Cys Asn Asp Gln
 210 215 220
 Asn Phe Lys Gly Lys Gly Thr Cys Arg Asn Val Ser Thr Val His Cys
 225 230 235 240
 Thr His Gly Ile Lys Pro Val Val Ser Thr Gln Phe Leu Leu Asn Gly
 245 250 255
 Ser Leu Ala Glu Gly Asn Lys Thr Val Val Arg Val Arg Ser Lys Ser
 260 265 270
 Asn Thr Glu Thr Ile Ile Val Gln Leu Ala Thr Ala Ile Tyr Ile Asn
 275 280 285
 Cys Thr Arg Leu Gly Asn Lys Thr Ile Glu Gly Ile Pro Ile Gly Pro
 290 295 300
 Gly Gln Ile Phe Tyr Arg Thr Lys Thr Val Val Gly Asp Thr Arg Gly
 305 310 315 320
 Ala Glu Cys Arg Ile Asn Gly Thr Ala Trp Asn Glu Thr Leu Arg Gln
 325 330 335
 Val Lys Glu Ala Leu Asn Asn Thr Tyr Arg Asn Leu Asn Leu Ser Leu
 340 345 350
 Thr Glu Ile Asn Phe Glu Gly Ala Ser Gly Gly Asp Leu Glu Val Thr
 355 360 365
 Thr His Tyr Phe Asn Cys Gly Gly Glu Phe Phe Tyr Cys Asn Thr Ser
 370 375 380
 Asn Leu Phe Asn Gln Ser Thr Ile Asn Asn Ile Thr His Ile Pro Cys
 385 390 395 400
 Arg Ile Arg Gln Ile Val Asn Gln Trp Gln Gly Val Gly Lys Gly Ile
 405 410 415
 Phe Ala Pro Pro Ile Arg Gly Thr Ile Gln Cys Asn Ser Thr Ile Thr
 420 425 430
 Gly Leu Leu Leu Thr Arg Asp Gly Lys Asn Glu Thr Glu Thr Phe Arg
 435 440 445
 Pro Thr Gly Gly Asp Met Arg Asn Asn Trp Arg Ser Glu Leu Tyr Lys
 450 455 460

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Tyr Lys Val Val Lys Ile Glu Pro Leu Gly Leu Ala Pro Thr Lys Ala
465                               470                               475                               480

Lys Arg Arg Thr Val Gln Arg Glu Lys Arg Ala Val Gly Leu Gly Val
                               485                               490                               495

Met Phe Leu Gly Phe Met Gly Ala Ala Gly Ser Thr Met Gly Ala Ala
                               500                               505                               510

Ala Leu Thr Leu Thr Val Gln Ala Arg Asn Leu Leu Ser Gly Ile Val
                               515                               520                               525

Gln Gln Gln Ser Asn Leu Leu Lys Ala Ile Glu Ala Gln Gln His Leu
                               530                               535                               540

Leu Gln Leu Ser Val Trp Gly Ile Lys Gln Leu Gln Ala Arg Val Leu
545                               550                               555                               560

Ala Val Glu Arg Tyr Leu Lys Asp Gln Gln Leu Leu Gly Leu Trp Gly
                               565                               570                               575

Cys Ser Gly Lys Leu Ile Cys Thr Thr Asn Val Pro Trp Asn Asn Thr
                               580                               585                               590

Trp Ser Lys Lys Leu Gly Thr Asn Leu Ser Phe Trp Asp Asn Met Thr
                               595                               600                               605

Trp Met Gln Trp Glu Lys Glu Ile Asp Asn Tyr Thr Glu Thr Ile Tyr
610                               615                               620

Glu Leu Leu Thr Arg Ser Gln Asn Gln Gln Glu Val Asn Glu Gln Glu
625                               630                               635                               640

Leu Leu Ala Leu Asp Lys Trp Ser Ser Leu Trp Asn Trp Phe Asp Ile
                               645                               650                               655

Thr Gln Trp Leu Trp Tyr Ile Lys Leu Phe Val Met Ile Val Gly Gly
                               660                               665                               670

Leu Ile Gly Ile Arg Ile Val Phe Ala Met Leu Ser Ile Val Asn Arg
675                               680                               685

Val Arg Gln Gly Tyr Ser Pro Leu Ser Phe Gln Thr Pro Phe Pro Val
690                               695                               700

Arg Arg Asp Pro Asp Arg Pro Glu Gly Ile Glu Glu Gly Gly Gly Glu
705                               710                               715                               720

Asn Asp Asn Val Arg Ser Val Arg Leu Val Ser Gly Phe Leu Ala Leu
                               725                               730                               735

Ala Trp Asp Asp Leu Arg Asn Leu Cys Leu Phe Ser Tyr His Arg Leu
740                               745                               750

Arg Asp Leu Leu Trp Ile Val Arg Arg Thr Leu Glu Ile Val Gly Gln
755                               760                               765

Lys Phe Leu Ala Gly Leu Arg Leu Val Trp Glu Thr Leu Thr Tyr Leu
770                               775                               780

Lys Ser Ile Leu Gln Tyr Trp Gly Gln Glu Leu Lys Thr Ser Ala Ile
785                               790                               795                               800

Ser Leu Leu Asp Ala Thr Ala Ile Ala Val Ala Glu Gly Thr Asp Arg
                               805                               810                               815

Ile Ile Glu Ile Ala Gln Arg Val Gly Arg Gly Ile Leu His Ile Pro
820                               825                               830

Arg Arg Ile Arg Gln Gly Leu Glu Arg Ala Leu Leu
835                               840

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<210> SEQ ID NO 346

<211> LENGTH: 855

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 346

Met 1	Lys	Val	Met	Gly 5	Met	Lys	Tyr	Tyr	Trp 10	Trp	Arg	Asn	Ser	Cys	Arg 15
Ser	Ile	Ser	Ile 20	Lys	Leu	Ile	Leu	Ile 25	Gly	Trp	Ile	Ala	Ser 30	Cys	Phe
Gly	Glu	Glu	Asn 35	Trp	Trp	Val	Thr	Val 40	Tyr	Tyr	Gly	Val 45	Pro	Val	Trp
Arg	Glu	Ala	Lys 50	Thr	Thr	Leu	Phe	Cys 55	Ala	Ser	Asp 60	Ala	Lys	Ser	Tyr
Ser 65	Thr	Glu	Ala	His 70	Asn	Ile	Trp	Ala	Thr	Gln 75	Ala	Cys	Val	Pro	Thr 80
Asp	Pro	Thr	Pro	Gln 85	Glu	Val	Leu	Leu 90	Pro	Asn	Val	Thr	Glu	Glu 95	Phe
Asn	Met	Trp	Glu 100	Asn	Tyr	Met	Val	Asp 105	Gln	Met	Gln	Glu	Asp 110	Ile	Ile
Ser	Leu	Trp	Glu 115	Gln	Ser	Leu	Lys 120	Pro	Cys	Val	Lys	Leu 125	Thr	Pro	Leu
Cys 130	Val	Thr	Leu	Thr	Cys	Asn 135	Asn	Pro	Thr	Asn	Thr 140	Ser	Cys	Thr	Asn
Ser 145	Thr	Asp	Asp	Arg 150	Leu	Gly	Asp	Met	Arg	Asn 155	Cys	Ser	Phe	Asn	Val 160
Thr	Thr	Glu	Leu	Arg 165	Asp	Lys	Lys	Arg	Gln 170	Val	Tyr	Ser	Leu	Phe 175	Tyr
Val	Glu	Asp	Ile 180	Thr	Ala	Ile	Gly	Asn 185	Asn	Ser	Thr	Tyr 190	Arg	Leu	Ile
Asn 195	Cys	Asn	Thr	Thr	Ala	Ile 200	Thr	Gln	Ala	Cys	Pro	Lys 205	Thr	Ser	Phe
Glu 210	Pro	Ile	Pro	Ile	His 215	Tyr	Cys	Ala	Pro	Ala	Gly 220	Phe	Ala	Leu	Leu
Lys 225	Cys	Asn	Asp	Ile 230	Asp	Tyr	Lys	Gly	Asn 235	Glu	Thr	Cys	Lys	Asn	Val 240
Ser	Thr	Val	His 245	Cys	Thr	His	Gly	Ile	Lys 250	Pro	Val	Ala	Thr	Thr 255	Gln
Leu	Ile	Leu	Asn 260	Gly	Ser	Thr	Ala	Asp 265	Asn	Gln	Thr	Val	Ala	Arg	Ile
Asp	Pro	Ser 275	Glu	Asn	Leu	Ala	Ile 280	Ile	Gln	Leu	Lys	Asp 285	Pro	Val	Lys
Ile 290	Thr	Cys	Arg	Arg	Pro	Gly 295	Asn	Asn	Thr	Arg	Gly 300	Gln	Ile	Gln	Ile
Gly 305	Pro	Ala	Met	Thr	Phe 310	Tyr	Asn	Ile	Glu	Asn 315	Val	Val	Gly	Asp	Thr 320
Arg	Lys	Ala	Tyr 325	Cys	Glu	Ile	Asn	Gly	Thr 330	Gln	Trp	Ala	Lys	Ala	Leu 335
Asn	Glu	Thr 340	Lys	Glu	Val	Leu	Arg	Asn 345	Ile	Leu	Arg	Lys	Asn	Ile	Ser
Phe 355	Met	Val	Pro	Ser	Gly	Gly 360	Asp	Pro	Glu	Val	Thr	Asn 365	His	His	Phe
Asn 370	Cys	Gly	Gly	Glu	Phe	Phe 375	Tyr	Cys	Asn	Thr	Ser 380	Glu	Ile	Ile	Asn
Ile 385	Thr	Lys	Ile	Asn 390	Lys	Thr	Glu	Asn	Met	Thr 395	Ile	Ile	Pro	Cys	Arg 400
Ile	Arg	Gln	Ile 405	Val	Asn	Ser	Trp	Met	Arg 410	Val	Gly	Lys	Gly	Ile	Phe 415

Ala 420	Pro	Pro	Ile 420	Arg	Gly	Asn	Ile 425	Thr 425	Cys	Thr	Ser	Asn	Ile 430	Thr	Gly
Met 435	Leu 435	Leu	Glu	Ile	His	Lys	Asn 440	Arg	Glu	Asp	Gln	Gly 445	Glu	Asp	Gln
Asp 450	Gln 450	Asn	Asn	Thr	Tyr	Val 455	Cys	Leu	Thr	Gly	Gly 460	Asn	Met	Lys	Asp
Ile 465	Trp	Arg	Ser	Glu	Leu 470	Tyr	Lys	Tyr	Lys	Ile 475	Val	Glu	Ile	Gln	Pro 480
Leu	Gly	Val	Ala 485	Pro	Thr	Lys	Ser	Arg	Arg 490	Tyr	Ala	Val	Glu	Lys 495	Gln
His	His	Arg	Glu 500	Lys	Arg	Ala	Leu	Gly 505	Leu	Gly	Ala	Leu	Phe 510	Leu	Gly
Phe 515	Leu 515	Gly	Ala	Ala	Gly	Ser	Thr 520	Met	Gly	Ala	Ala	Ser 525	Val	Val	Leu
Thr 530	Val 530	Gln	Ala	Arg	Gln	Leu 535	Leu	Thr	Gly	Ile	Val 540	Gln	Gln	Gln	Asn
Asn 545	Leu 545	Leu	Arg	Ala 550	Ile 550	Glu	Ala	Gln	Gln	His 555	Leu	Leu	Gln	Leu	Ser 560
Val	Trp	Gly	Ile 565	Lys 565	Gln	Leu	Gln	Ala	Arg 570	Val	Leu	Ala	Val	Glu 575	Arg
Tyr	Leu	Arg	Asp 580	Gln	Gln	Leu	Leu	Gly 585	Leu	Trp	Gly	Cys 590	Thr	Gly	Lys
Thr	Ile 595	Cys 595	Pro	Thr	Ala	Val	Arg 600	Trp	Asn	Lys	Thr	Trp 605	Gly	Asn	Ile
Ser 610	Asp 610	Tyr	Gln	Val	Ile	Trp 615	Asn	Asn	Tyr	Thr	Trp 620	Gln	Gln	Trp	Asp
Arg 625	Glu	Val	Asn	Asn	Tyr 630	Thr	Gly	Leu	Ile	Tyr 635	Thr	Leu	Leu	Glu	Glu 640
Ala	Asn	Thr	Gln 645	Gln 645	Glu	Lys	Asn	Glu	Lys 650	Glu	Leu	Leu	Glu	Leu	Asp 655
Ser	Trp	Ala 660	Asn 660	Leu	Trp	Ser	Trp	Phe 665	Asp	Ile	Thr	Asn	Trp 670	Leu	Trp
Tyr	Ile 675	Lys 675	Met	Phe	Leu	Ile 680	Val	Val	Gly	Gly	Ile	Ile 685	Gly	Leu	Arg
Ile 690	Cys 690	Phe	Ala	Ile	Gly	Ser 695	Leu	Ile	Asn	Arg	Val 700	Arg	Lys	Gly	Tyr
Ser 705	Pro	Leu	Ser	Leu 710	Gln 710	Thr	Leu	Ile	Pro	Ala 715	Asn	Gln	Gly	Pro	Asp 720
Gly	Leu	Gly	Glu 725	Thr 725	Glu	Lys	Gly	Gly	Gly 730	Glu	Asn	Val	Arg	Gly 735	Arg
Ser	Ile	Arg	Leu 740	Val	Ser	Gly	Phe	Leu 745	Pro	Leu	Val	Trp	Glu	Asp	Leu
Arg	Asn	Leu 755	Leu	Ser	Phe	Leu	Tyr 760	His	Gln	Leu	Arg	Asp 765	Cys	Ala	Ser
Leu 770	Ile 770	Trp	Arg	Leu	Leu	Glu 775	Ile	Leu	Gly	Gln	Tyr 780	Ser	Leu	Arg	Gly
Val 785	Gln	Gln	Ile	Gly 790	Thr 790	Leu	Leu	His	Gln 795	Leu	Arg	Gly	Thr	Leu	Gln 800
Tyr	Trp	Thr	Thr 805	Glu 805	Ile	Lys	Asn	Ser	Ala 810	Ile	Ser	Leu	Leu	Asp	Thr 815
Thr	Ala	Ile	Ala 820	Val	Gly	Glu	Gly	Thr 825	Asp	Arg	Ile	Leu	Glu	Ala	Ile
Thr	Arg	Leu	Gly	Arg	Gly	Ile	Leu	His	Ile	Pro	Arg	Arg	Ile	Arg	Gln

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835	840	845
Gly Leu Glu Arg Ala Leu Leu		
850	855	
 <210> SEQ ID NO 347		
<211> LENGTH: 870		
<212> TYPE: PRT		
<213> ORGANISM: Simian immunodeficiency virus		
 <400> SEQUENCE: 347		
Met Lys Asn Ser Ser Gly Leu Ile Leu Thr Leu Ile Thr Thr Ile Leu		
1	5	10 15
Gly Ile Gly Phe Ser Thr Tyr Tyr Ala Thr Val Phe Tyr Gly Val Pro		
	20	25 30
Val Trp Lys Asp Ala Thr Pro Pro Leu Phe Cys Ala Ser Asp Ala Asp		
	35	40 45
Val Ala Ser Thr Gln Ala His Asn Ile Trp Ala Thr His Asn Cys Val		
	50	55 60
Pro Leu Asp Pro His Pro Tyr Glu Val Pro Leu Pro Asn Val Ser Ile		
	65	70 75 80
Glu Phe Asp Met Ser Thr Asn Tyr Met Val Glu Glu Met Lys Thr Asp		
	85	90 95
Leu Ile Ser Leu Phe Gln Gln Ser Phe Lys Pro Cys Val Lys Leu Thr		
	100	105 110
Pro Phe Cys Ile Arg Met Thr Cys Ala Asn Val Thr Glu Ala Thr Thr		
	115	120 125
Thr Thr Pro Pro Thr Thr Ser Thr Ser Thr Thr Glu Pro Pro Thr Ser		
	130	135 140
Lys Thr Thr Val Asn Thr Ser Glu Ile Glu Asp Tyr Lys Val Leu Asn		
	145	150 155 160
Cys Ser Phe Asn Gln Thr Thr Glu Phe Arg Asp Lys Lys Lys Asn Ile		
	165	170 175
Tyr Ser Leu Phe Tyr Arg Glu Asp Val Met Lys Glu Ser Gln Asn Lys		
	180	185 190
Thr Asn Asn Ser Glu Tyr Tyr Tyr Leu His Asn Cys Asn Thr Thr Ala		
	195	200 205
Ile Thr Gln Ser Cys Ser Lys Ser Ser Phe Glu Pro Ile Pro Ile Arg		
	210	215 220
Tyr Cys Ala Pro Ala Gly Tyr Ala Met Leu Lys Cys Asn Asp Ala Asn		
	225	230 235 240
Phe Thr Gly Ala Gly Glu Cys Thr Asn Val Ser Val Val His Cys Thr		
	245	250 255
His Gly Ile Met Pro Met Ile Ala Ser Trp Leu His Leu Asn Gly Thr		
	260	265 270
Tyr Glu Lys Glu Lys Thr Lys Val Tyr His Thr Asn Val Thr Asn Asn		
	275	280 285
Pro Pro Leu Leu Val Lys Phe Asn Glu Thr Ile His Ile Thr Cys Glu		
	290	295 300
Arg Thr Gly Asn Asn Thr Arg Gly Gln Val Gln Ile Gly Pro Gly Met		
	305	310 315 320
Thr Phe Tyr Asn Arg Glu Asn Ile Phe Gly Asp Thr Arg Lys Ala Phe		
	325	330 335
Cys Tyr Val Asn Ala Thr Glu Trp Arg Arg Thr Leu Glu Met Ala Lys		
	340	345 350

Thr	Ala	Leu	Arg	Glu	Ala	Thr	Lys	Lys	Pro	Thr	Leu	Asn	Ile	Thr	Arg
	355						360					365			
Pro	Lys	Gly	Asp	Pro	Glu	Val	Glu	Asn	Phe	Trp	Phe	Asn	Cys	Gln	Gly
	370					375					380				
Glu	Phe	Phe	Tyr	Cys	Asn	Leu	Thr	Lys	Trp	Ile	Asn	Glu	Thr	Trp	Leu
385					390					395					400
Asn	Glu	Thr	Asp	Gln	Thr	Asn	Leu	Val	Ala	Pro	Cys	Arg	Leu	Arg	Gln
				405					410					415	
Ile	Val	Asn	His	Trp	Gly	Ile	Val	Ala	Lys	Ala	Ile	Tyr	Leu	Pro	Pro
			420					425					430		
Arg	Arg	Gly	Thr	Val	Lys	Cys	Val	Ser	Asn	Ile	Thr	Gly	Phe	Leu	Met
		435					440					445			
Thr	Asn	Glu	Glu	Thr	Gly	Ala	Thr	Phe	Ser	Gly	Lys	Val	Glu	Asp	Tyr
	450					455					460				
Trp	Arg	Val	Glu	Leu	Ala	Lys	Tyr	Lys	Ile	Val	Glu	Ile	Gln	Pro	Leu
465					470					475					480
Ser	Val	Ala	Pro	Thr	Thr	Gly	Lys	Arg	Pro	Glu	Ile	Lys	Ala	Asn	His
				485					490					495	
Thr	Arg	Ser	Lys	Arg	Asp	Val	Gly	Ile	Gly	Leu	Leu	Phe	Leu	Gly	Phe
			500					505					510		
Leu	Ser	Ala	Ala	Gly	Ser	Thr	Met	Gly	Ala	Ala	Ser	Ile	Ala	Leu	Val
		515					520					525			
Ala	Gln	Ala	Arg	Gly	Val	Leu	Ser	Gly	Ile	Val	Gln	Gln	Gln	His	Asn
	530					535					540				
Leu	Leu	Gln	Ala	Ile	Glu	Ala	Gln	Gln	His	Leu	Leu	Gln	Leu	Thr	Val
545					550					555					560
Trp	Gly	Val	Lys	Gln	Leu	Gln	Ala	Arg	Leu	Leu	Ala	Met	Glu	Lys	Tyr
				565					570					575	
Ile	Arg	Asp	Gln	Gln	Leu	Leu	Ser	Val	Trp	Gly	Cys	Ala	Asn	Lys	Leu
			580						585				590		
Val	Cys	His	Ser	Thr	Val	Val	Trp	Asn	Asp	Thr	Trp	Ala	Ala	Arg	Lys
		595					600					605			
Pro	Cys	Ser	Asn	Thr	Ser	Gly	Pro	Thr	Asp	Tyr	Asp	Cys	Ile	Trp	Gln
	610					615					620				
Asn	Leu	Thr	Trp	Gln	Gln	Trp	Ser	Glu	Leu	Val	Asp	Asn	Ser	Thr	Asp
625					630					635					640
Thr	Ile	Tyr	Thr	Leu	Leu	Glu	Val	Ala	Gln	Ile	Gln	Gln	Glu	Lys	Asn
				645					650					655	
Lys	Lys	Glu	Leu	Leu	Glu	Leu	Asp	Lys	Trp	Ser	Thr	Leu	Trp	Asp	Trp
			660					665				670			
Phe	Asp	Ile	Thr	Gln	Trp	Leu	Trp	Tyr	Ile	Lys	Leu	Ala	Ile	Ile	Ile
		675					680					685			
Ile	Ala	Gly	Leu	Ile	Gly	Leu	Arg	Ile	Leu	Met	Phe	Ile	Val	Asn	Val
	690					695					700				

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770	775	780
Leu Lys Gln Gly Thr Ile Ser Phe Ile Gln Thr Ile Ile Ile Val His		
785	790	795 800
Arg Asn Leu Val Leu Lys Trp Arg Leu Phe Arg Asp Trp Leu Ala Asn		
	805	810 815
Ala Tyr Val Ile Val Lys Thr Ser Phe Ile Arg Asn Leu Asp Arg Leu		
	820	825 830
Ala Asn Phe Thr Ala Trp Trp Thr Asp Ile Leu Ile Glu Gly Ala Val		
	835	840 845
Asn Ile Phe Arg Gly Ile Arg Asn Ile Pro Thr Arg Ile Arg Gln Gly		
	850	855 860
Leu Glu Ile Ala Leu Asn		
865	870	

<210> SEQ ID NO 348
 <211> LENGTH: 884
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 348

Met Lys Met Ala Leu Leu Ile Gly Trp Ile Leu Thr Ile Ile Phe Cys
1 5 10 15
Cys Lys Leu Gly Ile Gly Thr Tyr Tyr Thr Thr Val Tyr Tyr Gly Val
20 25 30
Pro Val Trp Lys Glu Ala Thr Pro Thr Leu Phe Cys Ala Ser Ser Ala
35 40 45
Lys Val Ala Ser Thr Gln Pro His Asn Ile Trp Ala Thr His Asn Cys
50 55 60
Val Pro Leu Asp Pro Gln Pro Tyr Glu Ile Pro Ile Asn Ile Thr Thr
65 70 75 80
Glu Phe Asn Met Glu Thr Asn Tyr Met Val Glu Glu Met Lys Lys Asp
85 90 95
Leu Ile Ser Leu Phe Gln Gln Ser Leu Lys Pro Cys Val Lys Leu Thr
100 105 110
Pro Phe Cys Val Lys Met Asn Cys Ser Glu Phe Thr Arg Asn Ser Thr
115 120 125
Thr Gly Asn Asn Ala Thr Lys Asn Asn Thr Ser Leu Glu Asp Lys Ile
130 135 140
Lys Ser Asn Leu Thr Ser Leu Glu Met Tyr Glu Cys Phe Phe Asn Gln
145 150 155 160
Thr Thr Glu Phe Arg Asp Lys Lys Lys Gln Ile Tyr Ser Leu Phe Tyr
165 170 175
Lys Glu Asp Ile Met Lys Asp Lys Asn Asp Ser Thr Asn Thr Ser Tyr
180 185 190
Tyr Leu Ile Asn Cys Asn Thr Thr Ala Ile Thr Gln Glu Cys Glu Lys
195 200 205
Ser Ser Phe Glu Pro Val Pro Ile Gln Tyr Cys Ala Pro Pro Gly Tyr
210 215 220
Ala Met Leu Lys Cys Lys Asp Glu Asn Phe Arg Gly Lys Gly Asn Cys
225 230 235 240
Thr Thr Val Arg Ile Val His Cys Thr His Ser Ile Phe Pro Met Ile
245 250 255
Ala Thr Ala Leu His Leu Asn Gly Ser Val Glu Glu Gly Glu Thr Lys
260 265 270

Ala	Tyr	Tyr	Val	Glu	Ala	Lys	His	Asn	Ala	Pro	Leu	Leu	Ile	Lys	Phe
275						280				285					
Asn	Lys	Ala	Thr	Val	Phe	Asn	Ile	Thr	Cys	Val	Arg	Pro	Gly	Asn	Asn
290						295				300					
Thr	Arg	Gly	Gln	Val	Gln	Ile	Gly	Pro	Gly	Met	Thr	Phe	Tyr	Asn	Ile
305				310						315				320	
Arg	Asn	Val	Ile	Gly	Asp	Thr	Arg	Lys	Ala	Phe	Cys	Tyr	Val	Asn	Lys
				325				330						335	
Thr	Leu	Trp	Glu	Asn	Ala	Thr	Glu	Ala	Ala	Lys	Val	Ala	Ile	Asn	Glu
		340						345				350			
Thr	Leu	Lys	Asn	Thr	Thr	Phe	Lys	Asn	Thr	Thr	Tyr	Thr	Phe	Arg	Thr
355						360						365			
Asp	Gly	Asp	Leu	Lys	Trp	Ile	Ser	Gly	Gly	Asp	Gln	Glu	Thr	Gln	Thr
370						375				380					
His	Trp	Phe	Asn	Cys	Gln	Gly	Glu	Phe	Phe	Tyr	Cys	Asp	Leu	Lys	Asn
385				390						395				400	
Trp	Thr	Asn	Asn	Gly	Ile	Glu	Ile	Asp	Gly	Met	Leu	Ile	Ala	Pro	Cys
				405				410						415	
Arg	Leu	Arg	Gln	Ile	Val	Asn	His	Trp	Gly	Ile	Val	Ser	Arg	Gly	Ile
		420						425				430			
Tyr	Leu	Pro	Pro	Arg	Glu	Gly	Gln	Val	Lys	Cys	Val	Ser	Ser	Ile	Thr
435						440						445			
Gly	Phe	Ile	Met	Thr	Ala	Glu	Thr	Asn	Asp	Thr	Asp	Gly	Thr	Ile	Thr
450						455				460					
Pro	Thr	Phe	Ser	Ala	Lys	Val	Glu	Asp	Tyr	Trp	Lys	Val	Glu	Met	Ser
465				470						475				480	
Arg	Tyr	Lys	Val	Val	Glu	Ile	Gln	Pro	Leu	Ser	Val	Ala	Pro	Thr	Lys
		485						490						495	
Gly	Lys	Arg	Pro	Val	Val	Gly	Asp	Pro	Glu	Ala	Lys	Ser	Arg	Ser	Lys
		500						505				510			
Arg	Glu	Ile	Gly	Met	Gly	Met	Ile	Phe	Met	Gly	Phe	Leu	Ser	Ala	Ala
515						520						525			
Gly	Ser	Thr	Met	Gly	Ala	Ala	Ser	Ile	Ala	Leu	Thr	Val	Gln	Ala	Arg
530						535				540					
Gly	Leu	Leu	Ser	Gly	Ile	Val	Gln	Gln	Gln	Ala	Asn	Leu	Leu	Gln	Ala
545				550						555				560	
Ile	Glu	Ala	Gln	Gln	His	Leu	Leu	Gln	Leu	Ser	Val	Trp	Gly	Ile	Lys
				565				570						575	
Gln	Leu	Gln	Ala	Arg	Leu	Leu	Ala	Val	Glu	Lys	Tyr	Leu	Arg	Asp	Gln
		580						585				590			
Gln	Leu	Leu	Gly	Leu	Trp	Gly	Cys	Ala	Asn	Lys	Leu	Ile	Cys	His	Ser
595						600						605			
Ser	Leu	Glu	Trp	Asn	Ser	Ser	Trp	Leu	Asn	Lys	Thr	Leu	Cys	Gly	Asn
610						615				620					
Lys	Thr	Asn	Asn	Tyr	Glu	Asn	Asn	Tyr	Glu	Cys	Ile	Trp	Ser	Gln	Tyr
625				630						635				640	
Thr	Trp</														

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690	695	700
Ser Leu Val Gly Leu Arg Ile Val Met Phe Ile Ile Asn Leu Val Gly		
705	710	715 720
Lys Leu Arg Gln Gly Tyr Ser Leu Leu Ser Pro Gln Ile Pro Ile Gln		
	725	730 735
Thr Ala Arg Gly Pro Gly Gln Pro Glu Gly Thr Glu Glu Gly Gly Gly		
	740	745 750
Glu Ser Gly Arg Asp Arg Trp Arg Gly Tyr Pro Val Gly Phe Leu Gln		
	755	760 765
Leu Ile Trp Glu Asp Cys Arg Gln Leu Leu Leu Trp Ile Tyr Gln Thr		
	770	775 780
Ser Val Asn Leu Leu Trp His Leu Trp Thr Leu Ile Lys Leu Ile His		
	785	790 795 800
Gln Trp Ser Leu Leu Leu Trp Glu Thr Val Arg Gln Leu Leu Arg Arg		
	805	810 815
Leu Ile His Leu Leu Thr Gln Phe Leu His Leu Leu Glu Thr Trp Gly		
	820	825 830
Ile Leu Ile Arg Gln His Cys Ile Val Cys Ile Asp Thr Leu Ala Glu		
	835	840 845
Phe Thr Gly Trp Trp Thr Asp Gly Val Ile Glu Ala Leu Arg Val Ala		
	850	855 860
Val Asp Ile Ile Arg His Ile Pro Thr Arg Ile Arg Gln Gly Leu Glu		
	865	870 875 880
Ile Ala Leu Asn		

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<210> SEQ ID NO 349
<211> LENGTH: 912
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (61)..(61)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (202)..(202)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (267)..(267)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (304)..(304)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (305)..(305)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (307)..(307)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (310)..(310)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (317)..(317)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<400> SEQUENCE: 349

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Met Arg Val Met Gly Thr Lys Lys Ser Cys Val Leu Tyr Tyr Cys Leu
1 5 10 15

Val	Ile	Leu	Ser	Leu	Phe	Ser	Leu	Thr	Ile	Asp	Ser	Lys	Glu	Glu	Leu	
			20				25						30			
Trp	Val	Thr	Val	Tyr	Tyr	Gly	Val	Pro	Val	Trp	Arg	Asp	Ala	Glu	Thr	
			35				40				45					
Thr	Leu	Phe	Cys	Ala	Ser	Asp	Ala	Lys	Ala	Gln	Glu	Xaa	Glu	Ala	His	
			50				55				60					
Asn	Ile	Trp	Ala	Ser	Gln	Ala	Cys	Val	Pro	Thr	Asp	Pro	Asn	Pro	Glu	
65						70				75			80			
Glu	Ile	His	Leu	Pro	Asn	Val	Thr	Glu	Tyr	Phe	Asp	Met	Trp	Lys	Asn	
			85						90			95				
Asn	Met	Val	Glu	Gln	Met	Gln	Glu	Asp	Ile	Ile	Ser	Leu	Trp	Asp	Gln	
			100						105			110				
Ser	Leu	Lys	Pro	Cys	Val	Lys	Leu	Thr	Pro	Leu	Cys	Val	Thr	Leu	Asn	
			115						120			125				
Cys	Thr	Asn	Cys	Ile	Trp	Gly	Asn	Gly	Thr	Phe	Arg	Glu	Trp	Asn	Asp	
130						135						140				
Thr	Arg	Glu	Asp	Leu	Arg	Asn	Cys	Ser	Phe	Asn	Ala	Gly	Asp	Ile	Arg	
145						150			155			160				
Asp	Lys	Lys	Lys	Arg	Val	His	Ser	Leu	Phe	Tyr	Ala	Glu	Asp	Ile	Ile	
			165						170			175				
Leu	Ile	Asp	Arg	Thr	Arg	Thr	Thr	Lys	Gln	Gly	Asn	Ser	Thr	Gln	Ala	
			180						185			190				
Pro	Ala	Thr	Thr	Pro	Ile	Ala	Ser	Thr	Xaa	Val	Pro	Asp	Ser	Asn	Asp	
			195						200			205				
Thr	Cys	Ser	Pro	Asn	Asn	Thr	Leu	Glu	Tyr	Arg	Leu	Ile	Ser	Cys	Asn	
210						215						220				
Thr	Ser	Val	Ile	Thr	Gln	Ala	Cys	Pro	Lys	Thr	Ser	Phe	Glu	Pro	Ile	
225						230			235			240				
Pro	Ile	His	Tyr	Cys	Ala	Pro	Ala	Gly	Phe	Ala	Leu	Leu	Lys	Cys	Asn	
			245						250			255				
Asp	Thr	Asn	Phe	Ser	Gly	Gln	Gly	Ile	Cys	Xaa	Asn	Val	Ser	Thr	Val	
			260						265			270				
His	Cys	Thr	His	Gly	Ile	Arg	Pro	Val	Val	Ser	Thr	His	Leu	Leu	Leu	
			275						280			285				
Asn	Gly	Ser	Leu	Ala	Lys	Gly	Asp	Ile	Met	Ile	Arg	Val	Lys	Asx	Xaa	
290						295						300				
Xaa	Gln	Xaa	Thr	Asp	Xaa	Ile	Ile	Val	Gln	Leu	Asn	Xaa	Ser	Ile	Asn	
305						310			315			320				
Ile	Thr	Cys	Ile	Arg	Pro	Gly	Asn	Asn	Thr	Ile	Lys	Gly	Ile	Pro	Ile	
			325						330			335				
Gly	Pro	Gly	Gln	His	Phe	Tyr	Gly	Thr	Glu	Thr	Val	Val	Gly	Asp	Thr	
			340						345			350				
Arg	Lys	Ala	Tyr	Cys	Lys	Ile	Asn	Gly	Thr	Gln	Trp	Phe	Asp	Thr	Leu	
			355						360			365				
Lys	Arg	Val	Ala	Trp	Glu	Leu	Lys	Glu	Thr	Phe	Asn	Leu	Thr	Arg	Val	
			370			375						380				
Thr	Phe	Asp	Gln	Ala	Ser	Pro	Gly	Asp	Pro	Glu	Ile	Gln	Lys	His	Tyr	
385						390			395			400				
Phe	Asn	Cys	Gly	Gly	Glu	Phe	Phe	Tyr	Cys	Glu	Thr	Ser	Ser	Leu	Phe	
			405						410			415				
Asn	Ala	Thr	Ile	Asp												

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Trp	Asp	Tyr	Pro	Ala	Glu	Asn	Asn	Ile	Asn	Gln	Glu	Met	Ile	Ile	His
	435						440					445			
Cys	Arg	Ile	Lys	Gln	Ile	Val	Ser	Met	Trp	Gln	Arg	Val	Gly	Lys	Gly
	450					455					460				
Ile	Phe	Ser	Pro	Pro	Ile	Arg	Gly	Thr	Ile	Gln	Cys	Asn	Ser	Thr	Ile
465					470					475					480
Thr	Gly	Leu	Leu	Leu	Thr	Arg	Asp	Gly	Asn	Asn	Lys	Ala	Asp	Ser	Glu
			485					490						495	
Asn	Lys	Thr	Arg	Ile	Glu	Thr	Gly	Gln	Asn	Lys	Ala	Glu	Glu	Ile	Asn
			500					505					510		
Arg	Thr	Asp	Ile	Phe	Arg	Pro	Ala	Gly	Gly	Glu	Met	Arg	Asp	Asn	Trp
	515						520					525			
Arg	Ser	Glu	Leu	Tyr	Arg	Tyr	Lys	Val	Val	Lys	Ile	Glu	Pro	Leu	Gly
	530					535					540				
Leu	Ala	Pro	Thr	Lys	Ala	Lys	Arg	Arg	Thr	Val	Asn	Arg	Glu	Lys	Arg
545					550					555					560
Ala	Pro	Gly	Leu	Gly	Leu	Met	Phe	Leu	Gly	Phe	Leu	Gly	Ala	Ala	Gly
			565					570						575	
Ser	Thr	Met	Gly	Ala	Ala	Ser	Leu	Thr	Leu	Thr	Val	Gln	Ala	Arg	Gln
			580					585					590		
Leu	Leu	Thr	Gly	Ile	Val	Gln	Gln	Gln	Ser	Asn	Leu	Leu	Arg	Ala	Ile
	595					600						605			
Glu	Ala	Gln	Gln	His	Leu	Leu	Gln	Leu	Ser	Val	Trp	Gly	Ile	Lys	Gln
	610					615					620				
Leu	Gln	Ala	Arg	Val	Leu	Ala	Ile	Glu	Arg	Tyr	Leu	Lys	Asp	Gln	Gln
625					630					635					640
Leu	Leu	Gly	Ile	Trp	Gly	Cys	Ser	Gly	Lys	Leu	Ile	Cys	Thr	Thr	Asp
			645					650						655	
Val	Lys	Trp	Asn	Glu	Thr	Trp	Ser	Asn	Lys	Ser	Tyr	Asn	Glu	Ile	Trp
			660					665					670		
Asp	Asn	Met	Thr	Trp	Met	Gln	Trp	Asp	Lys	Glu	Val	Arg	Asn	Tyr	Thr
	675						680					685			
Glu	Thr	Ile	Tyr	Asp	Leu	Ile	Glu	Gln	Ala	Gln	Asn	Gln	Gln	Glu	Tyr
	690					695					700				
Asn	Glu	Lys	Ser	Leu	Leu	Glu	Leu	Asn	Gln	Trp	Ala	Ser	Leu	Trp	Asn
705					710					715					720
Trp	Phe	Asp	Ile	Ser	Asn	Trp	Leu	Trp	Tyr	Ile	Lys	Ile	Phe	Ile	Met
			725						730					735	
Ile	Val	Gly	Gly	Leu	Ile	Gly	Cys	Arg	Ile	Val	Phe	Ala	Leu	Leu	Ser
			740					745					750		
Ile	Val	Asn	Arg	Val	Arg	Gln	Gly	Tyr	Ser	Pro	Leu	Ser	Phe	Gln	Thr
	755						760					765			
Leu	Phe	Pro	Ile	Pro	Arg	Gly	Pro	Glu	Gln	Pro	Gly	Glu	Thr	Gly	Glu
	770					775					780				
Glu	Gly	Gly	Glu	Gln	Asp	Asn	Val	Arg	Ser	Ile	Arg	Leu	Leu	Asn	Gly
785					790					795					800
Phe	Leu	Pro	Leu	Val	Trp	Asn	Asp	Leu	Arg	Asp	Leu	Cys	Arg	Phe	Leu
			805					810						815	
Tyr	His	Arg	Leu	Arg	Asp	Phe	Leu	Leu	Ile	Ala	Arg	Arg	Thr	Leu	Glu
			820					825					830		
Leu	Val	Gly	Gln	His	Leu	Leu	Lys	Gly	Leu	Arg	Leu	Val	Trp	Glu	Thr
	835						840					845			
Leu	Leu	Tyr	Leu	Lys	Gly	Val	Leu	Gln	Tyr	Trp	Gly	Arg	Glu	Leu	Lys

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850	855	860
Val Ser Ala Ile Ser Leu Leu Asp Thr Thr Ala Ile Ala Val Ala Glu		
865	870	875 880
Gly Thr Asp Arg Ile Ile Glu Val Ala Gln Arg Phe Gly Arg Gly Ile		
	885	890 895
Leu Asn Ile Pro Thr Arg Ile Arg Gln Gly Leu Glu Arg Ala Leu Leu		
	900	905 910
<210> SEQ ID NO 350		
<211> LENGTH: 889		
<212> TYPE: PRT		
<213> ORGANISM: Simian immunodeficiency virus		
<220> FEATURE:		
<221> NAME/KEY: MISC_FEATURE		
<222> LOCATION: (37)..(37)		
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid		
<400> SEQUENCE: 350		
Met Lys Asp Met Thr Val Met Lys Lys Lys Lys Ser Ser Leu Gly Asn		
1	5	10 15
Trp Gly Ile Cys Leu Ala Leu Ile Ile Tyr Phe Asn Ala Ile Ser Tyr		
	20	25 30
Val Asn Cys Thr Xaa Tyr Val Thr Val Tyr Tyr Gly Val Pro Val Trp		
	35	40 45
Gln Asp Ala Lys Thr Thr Leu Phe Cys Ala Ala Asp Ala Asp Leu Ala		
	50	55 60
Ser Lys Glu Gln His Asn Ile Trp Ala Thr Gln Ala Cys Val Pro Leu		
65	70	75 80
Asp Pro Thr Pro Ile Glu Leu Lys Leu Asn Ile Thr Glu Ser Phe Asn		
	85	90 95
Ile Trp Glu Asn Tyr Met Val Glu Gln Met Gln Glu Asp Ile Val Ser		
	100	105 110
Leu Trp Asp Gln Ser Leu Lys Pro Cys Val Lys Leu Thr Phe Leu Cys		
	115	120 125
Val Thr Met Asn Cys Ser Glu Tyr Lys Gly Tyr Asn Cys Ser Thr Glu		
	130	135 140
Gly Asn Thr Thr Ala Ala Pro Cys Asn Thr Thr Glu Asn Thr Thr Lys		
	145	150 155 160
Glu Asn Ser Thr Gly Met Leu Thr Cys Asn Phe Asn Val Thr Thr Val		
	165	170 175
Leu Lys Asp Lys Lys Glu Gln Lys Gln Ala Leu Phe Tyr Arg Glu Asp		
	180	185 190
Leu Ala Ser Leu Glu Ser Asn Asn Ser Tyr Arg Leu Ile Asn Cys Asn		
	195	200 205
Ser Ser Thr Ile Thr Gln Ala Cys Pro Lys Val Ser Phe Glu Pro Leu		
	210	215 220
Pro Ile Gln Tyr Cys Ala Pro Ala Gly Tyr Ala Leu Met Lys Cys Asn		
	225	230 235 240
Arg Thr Asp Phe Asn Gly Thr Gly Thr Cys Asn Glu Thr Ser Ile Val		
	245	250 255
His Cys Thr His Gly Ile Arg Pro Thr Val Ser Thr Gln Leu Val Leu		
	260	265 270
Asn Gly Thr Leu Ala Lys Gly Lys Pro Leu Val Ile Thr Lys Asn Val		
	275	280 285
Ser Glu Thr Gly Ala Pro Ile Ile Val Lys Leu Ser Ala Ser Thr Ala		
	290	295 300

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Ile Thr Cys Ile Arg Pro Gly Asn Asn Thr Arg Gly Glu Val Gln Leu	
305	310 315 320
Gly Pro Met Thr Trp Tyr Asn Met Arg His Tyr Ile Gly Asp Ile Arg	
	325 330 335
Lys Ala His Cys Thr Val Ser Arg Gly Asn Trp Thr Lys Val Leu Gln	
	340 345 350
Asn Val Ser Ala Ala Leu Trp Glu Ala Tyr Pro Ala Glu Trp Lys Asn	
	355 360 365
Lys Thr Gln Asp Lys Asn His Thr Ile Leu Phe Arg Ala Ser Ser Gly	
	370 375 380
Gly Asp Pro Glu Val Ala Ser Leu His Phe Asn Cys His Gly Glu Phe	
	385 390 395 400
Phe Tyr Cys Asn Thr Ser Ala Leu Phe Asn His Ser Cys Thr Lys Asn	
	405 410 415
Met Thr Lys Asn Glu Trp Ile Cys Thr Pro Asn Asn Ala Thr Gly Thr	
	420 425 430
Leu Arg Leu Pro Cys Arg Leu Lys Gln Val Val Asn Ser Trp Met Arg	
	435 440 445
Val Gly Ser Gly Leu Phe Ala Pro Pro Ile Pro Gly Ser Leu Thr Cys	
	450 455 460
Lys Ser Asn Ile Thr Gly Ile Leu Leu Glu Arg Asp Leu Pro Leu Gly	
	465 470 475 480
Asn Met Thr Asn Thr Thr Leu Arg Pro Ile Gly Gly Asp Met Lys Asn	
	485 490 495
Ile Trp Arg Ser Glu Leu Tyr Pro Tyr Lys Val Val Gln Val Lys Ala	
	500 505 510
Leu Gly Val Ala Pro Thr Lys Ile Ser Arg Pro Thr Ile Met Gly Pro	
	515 520 525
His Arg Glu Lys Arg Gly Ala Gly Leu Gly Met Leu Phe Leu Gly Phe	
	530 535 540
Leu Ser Ala Ala Gly Ser Thr Met Gly Ala Ala Ala Val Thr Leu Thr	
	545 550 555 560
Val Gln Ala Lys Gln Leu Leu His Gly Ile Val Gln Gln Gln Asn Asn	
	565 570 575
Leu Leu Arg Ala Ile Gln Ala Gln Gln Glu Leu Leu Arg Leu Ser Val	
	580 585 590
Trp Gly Ile Arg Gln Leu Arg Ala Arg Leu Leu Ala Ile Glu Thr Tyr	
	595 600 605
Leu Lys Asp Gln Gln Leu Leu Gly Leu Trp Gly Cys Ser Gly Lys Leu	
	610 615 620
Val Cys Tyr Thr Asn Val Pro Trp Gln Lys Asn Trp Thr Thr Tyr Gln	
	625 630 635 640
Ser Asp Ser Gln Leu Asp Ala Ile Trp Asp Asn Leu Thr Trp Gln Glu	
	645 650 655
Trp Asp Lys Gln Val Asn Asn Tyr Thr Asp Leu Ile Phe Leu Glu Ile	
	660 665 670
Gln Ile Ala Gln Glu Gln Gln Glu Gln Asn Gln Lys Lys Leu Leu Glu	
	675 680 685
Leu Asp Gln Trp Ala Gln Leu Trp Ser Trp Leu Asp Ile Thr Gln Trp	
	690 695 700
Leu Trp Tyr Ile Lys Ile Phe Ile Met Ile Val Gly Gly Leu Ile Gly	
	705 710 715 720

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Leu Arg Ile Phe Ile Ala Val Val Asn Val Val Lys Arg Val Arg Gln
      725                      730                      735

Gly Tyr Ser Pro Leu Ser Leu Gln Thr Leu Gly Gln Asn Arg Gly Pro
      740                      745                      750

Ala Gly Ile Ala Thr Gly Thr Asp Glu Glu Gly Gly Gly Ala Gly Ser
      755                      760                      765

Val Arg Ser Ile Arg Leu Leu Asp Gly Phe Leu Pro Leu Ile Trp Asp
      770                      775                      780

Asp Leu Lys Asn Leu Val Val Trp Ile Tyr Gln Ile Leu Val Ser Cys
      785                      790                      795                      800

Ile Leu Gly Ile Lys Asp Leu Leu Ala Ile Leu Trp Ile His Leu Arg
      805                      810                      815

His Leu Leu Asn Lys Gly Leu Asn Cys Val Arg Asp Cys Leu Ala Val
      820                      825                      830

Cys Gly Tyr Trp Val Gln Glu Leu Gln Arg Ser Ala Thr Asn Leu Leu
      835                      840                      845

Asp Thr Thr Ala Val Ala Val Ala Glu Trp Thr Asp Gln Val Ile Leu
      850                      855                      860

Ile Gly Gln Arg Ile Gly Arg Gly Ile Leu Asn Ile Pro Arg Arg Ile
      865                      870                      875                      880

Arg Gln Gly Leu Glu Arg Ser Leu Leu
      885

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<210> SEQ ID NO 351
<211> LENGTH: 892
<212> TYPE: PRT
<213> ORGANISM: Simian immunodeficiency virus
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (248)..(248)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (250)..(250)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (254)..(254)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<400> SEQUENCE: 351

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Met Lys Asp Met Arg Val Met Arg Lys Lys Lys Arg Ser Leu Gly Asn
 1          5          10          15

Trp Gly Leu Ser Leu Ala Leu Leu Ile Ile Cys Phe Asn Thr Gly Ser
 20          25          30

Tyr Val Met Gly Thr Arg Tyr Val Thr Val Tyr Tyr Gly Val Pro Val
 35          40          45

Trp Arg Glu Ala Lys Thr Thr Leu Phe Cys Ala Ala Asp Ala Asn Leu
 50          55          60

Ala Ser Lys Glu Gln His Asn Ile Trp Ala Thr Gln Ala Cys Val Pro
 65          70          75          80

Thr Asp Pro Thr Pro Ile Lys Val Lys Leu Asn Ile Ser Glu Pro Tyr
 85          90          95

Asn Ile Trp Glu Asn Tyr Ile Val Lys Gln Met Gln Lys Asn Ile Ile
100          105          110

Ser Leu Leu Asn Gln Ser Leu Lys Pro Cys Val Lys Leu Thr Val Ile
115          120          125

Cys Val Thr Ile Asn Cys Ser Lys Phe Lys Pro Ala Thr Thr Asn Thr
130          135          140

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Thr	Ser	Ala	Pro	Ile	Asn	Ala	Ser	Leu	Gln	Ser	Ser	Thr	Thr	Ala	Gln
145					150					155					160
Asn	Asn	Leu	Asn	Phe	Tyr	Asn	Cys	Ser	Phe	Asn	Val	Thr	Thr	Val	Leu
			165						170					175	
Lys	Asp	Lys	Lys	Glu	Lys	Lys	Gln	Ala	Leu	Phe	Tyr	Lys	Glu	Asn	Leu
			180					185					190		
Val	Pro	Leu	Asn	Gln	Thr	Ser	Lys	Glu	Lys	Leu	Tyr	Arg	Leu	Ile	Asn
		195					200					205			
Cys	Asn	Thr	Thr	Thr	Ile	Thr	Gln	Ala	Cys	Pro	Lys	Val	Ser	Phe	Lys
	210					215					220				
Pro	Leu	Pro	Ile	Gln	Tyr	Cys	Ala	Pro	Ala	Lys	Tyr	Ala	Leu	Ile	Lys
225					230					235					240
Cys	Lys	Gln	Lys	Gly	Phe	Asn	Xaa	Thr	Xaa	Leu	Cys	Asn	Xaa	Thr	Val
			245					250						255	
Ile	Thr	His	Cys	Thr	His	Gly	Ile	Arg	Pro	Thr	Val	Ser	Thr	Gln	Leu
			260					265					270		
Ile	Phe	Asn	Gly	Ser	Leu	Ala	Glu	Glu	Glu	Ala	Leu	Val	Met	Ser	Lys
		275					280					285			
Asp	Val	Thr	Ala	Thr	Gly	Lys	Asn	Ile	Ile	Ile	Lys	Leu	Ser	Thr	Gly
	290					295					300				
Val	Asn	Ile	Thr	Cys	Ile	Arg	Thr	Gly	Asn	Asn	Thr	Arg	Gly	Gln	Thr
305					310					315					320
Gln	Ile	Gly	Pro	Met	Thr	Trp	Tyr	Asn	Ser	Val	Asn	Tyr	Ile	Gly	Asn
			325						330					335	
Ile	Arg	Arg	Ala	Tyr	Cys	Gln	Val	Glu	Glu	Thr	Asp	Trp	Lys	Gly	Ile
			340					345					350		
Leu	Lys	Asn	Val	Ser	Lys	Ala	Leu	Ile	Lys	Arg	Tyr	Asn	Gly	Lys	Val
		355					360					365			
Asn	Glu	Thr	His	Thr	Phe	Glu	Phe	Lys	Asn	Ala	Ser	Gly	Gly	Asp	Pro
	370					375					380				
Glu	Val	Thr	His	Leu	His	Phe	Asn	Cys	His	Gly	Glu	Phe	Phe	Tyr	Cys
385				390						395					400
Asn	Thr	Ser	Lys	Leu	Phe	Asn	Tyr	Thr	Tyr	Val	Cys	Asn	Gln	Thr	Asp
			405					410						415	
Gln	Gly	Asn	Ala	Asn	Tyr	Ser	Cys	His	Ser	Thr	Ser	Thr	Leu	Glu	Asn
		420					425						430		
Gly	Thr	Leu	Val	Ile	Pro	Cys	Lys	Leu	Arg	Gln	Val	Val	Asn	Ser	Trp
	435					440					445				
Met	Arg	Val	Gly	Ser	Gly	Leu	Phe	Ala	Pro	Pro	Val	Pro	Gly	Ser	Leu
450						455					460				
Thr	Cys	His	Ser	Asn	Ile	Thr	Gly	Leu	Ile	Leu	Gln	Arg	Asp	Trp	Pro
465				470						475					480
Leu	Asn	Asn	Asn	Thr	Thr	Thr	Ile	Leu	Arg	Pro	Glu	Gly	Gly	Asp	Met
			485					490						495	
Lys	Asp	Ile	Trp	Arg	Ser	Glu	Leu	Tyr	Pro	Tyr	Lys	Val	Val	Gln	Val
			500					505					510		
Lys	Ala	Leu	Ala	Val	Ala	Pro	Thr	Lys	Ile	Ser	Arg	Pro	Thr	Ile	Met
		515					520					525			
Ala	Gln	Thr	His	Gln	Arg	Glu	Lys	Arg	Gly	Ala	Gly	Leu	Gly	Met	Leu
	530					535					540				
Phe	Leu	Gly	Phe	Met	Ser	Ala	Ala	Gly	Ser	Thr	Met	Gly	Ala	Ala	Ala
545					550					555					560

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<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (150)..(150)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (202)..(202)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (473)..(473)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (621)..(621)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (661)..(661)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (711)..(711)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (839)..(839)
<223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid

<400> SEQUENCE: 352

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Met Lys Val Met Lys Lys Lys Lys Glu Ser Trp Ile Leu Xaa Ile
1          5          10          15

Val Met Ala Phe Ile Ile Pro Cys Ser Ser Ser Arg Pro Leu Tyr Ala
20        25        30

Thr Val Tyr Tyr Gly Val Pro Val Trp Arg Asp Ala Gln Thr Thr Leu
35        40        45

Phe Cys Ala Ala Asp Ala Lys Met Ala Ser Ser Glu Met His Asn Val
50        55        60

Trp Ala Thr Gln Ala Cys Val Pro Thr Asp Pro Gln Pro Ile Glu Leu
65        70        75        80

Lys Leu Thr Asn Val Thr Glu Thr Phe Asn Ile Trp Lys Ser Gly Met
85        90        95

Val Glu Gln Met Gln Glu Asp Ile Ile Ser Leu Trp Asp Gln Ser Leu
100       105       110

Lys Pro Cys Val Lys Leu Thr Val Met Cys Val Thr Met Asn Cys Xaa
115       120       125

Arg Xaa Thr Pro Asn Thr Thr Thr His Ser Leu Asn Met Thr Gly Pro
130       135       140

Ala Pro Arg Thr Ser Xaa Pro Thr Thr Ser Ser His Asn Met Thr Ser
145       150       155       160

Thr Pro Asn Ile Pro Glu Leu Glu Val Tyr Asn Cys Thr Phe Asn Val
165       170       175

Thr Thr Val Leu Lys Asp Lys Lys Ser Gln Gln Gln Ala Leu Phe Tyr
180       185       190

Arg Glu Asp Leu Ser Gln Ile Gly Asp Xaa Asn Ser Thr Tyr Arg Leu
195       200       205

Ile Asn Cys Asn Thr Ser Thr Ile Ser Gln Ala Cys Pro Lys Val Ser
210       215       220

Phe Glu Pro Leu Pro Ile Gln Tyr Cys Ala Pro Ala Gly Tyr Ala Leu
225       230       235       240

Met Lys Cys Asn Gln Ser Lys Phe Asn Gly Thr Gly Thr Cys Asn Glu
245       250       255

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Thr	Leu	Ile	Thr	His	Cys	Thr	His	Gly	Ile	Arg	Pro	Thr	Val	Ser	Thr	
			260						265						270	
Gln	Phe	Ile	Phe	Asn	Gly	Thr	Leu	Glu	Lys	Glu	Leu	Leu	Met	Leu	Ser	
			275						280						285	
Lys	Asn	Ile	Ser	Asp	Ser	Gly	Lys	Thr	Ile	Ile	Val	Lys	Leu	Lys	Lys	
			290						295						300	
Ala	Val	Thr	Phe	Lys	Cys	Glu	Arg	Thr	Gly	Asn	Asn	Thr	Arg	Gly	Gln	
			305						310						315	
Ile	Gln	Ile	Gly	Pro	Met	Thr	Ile	Tyr	Asn	Ser	Glu	Asn	Ile	Val	Gly	
			320						325						330	
Lys	Thr	Arg	Lys	Ala	Tyr	Cys	Ala	Tyr	Asn	Arg	Thr	Glu	Trp	Glu	Gln	
			335						340						345	
Ala	Leu	Lys	Thr	Ile	Ser	Glu	Ala	Phe	Ala	Lys	Leu	Glu	Asn	Val	Thr	
			350						355						360	
Lys	Val	Gln	Trp	Lys	Asn	Ser	Ser	Gly	Gly	Asp	Leu	Glu	Val	Ala	Met	
			365						370						375	
Leu	His	Phe	Asn	Cys	His	Gly	Glu	Phe	Phe	Tyr	Cys	Asn	Thr	Thr	Thr	
			380						385						390	
Met	Phe	Asn	Tyr	Thr	Tyr	Asn	Cys	Ser	His	His	Ser	Cys	Lys	Ala	Gln	
			395						400						405	
Ser	Lys	Asn	Ile	Ser	Glu	Gly	Ser	Gly	Asp	Ser	Tyr	Ile	Pro	Cys	Lys	
			410						415						420	
Leu	Lys	Gln	Val	Val	Asn	Ser	Trp	Met	Arg	Val	Gly	Ser	Gly	Leu	Phe	
			425						430						435	
Ala	Pro	Pro	Ile	Arg	Gly	Thr	Leu	Lys	Cys	Met	Ser	Asn	Ile	Thr	Gly	
			440						445						450	
Leu	Leu	Leu	Glu	Arg	Asp	Val	Pro	Xaa	Asn	Ile	Thr	Ala	Thr	Asx	Arg	
			455						460						465	
Asn	Ile	Thr	Leu	Arg	Pro	Thr	Gly	Gly	Glu	Met	Lys	Asp	Ile	Trp	Arg	
			470						475						480	
Ser	Glu	Leu	Tyr	Pro	Tyr	Lys	Val	Val	Gln	Val	Lys	Ala	Leu	Ser	Val	
			485						490						495	
Ala	Pro	Thr	Lys	Ile	Lys	Arg	Pro	Ile	Ile	Gly	Leu	Asn	Arg	Glu	Lys	
			500						505						510	
Arg	Gly	Ala	Gly	Leu	Gly	Met	Leu	Phe	Leu	Gly	Phe	Met	Ser	Ala	Ala	
			515						520						525	
Gly	Ser	Thr	Met	Gly	Ala	Ala	Ser	Leu	Thr	Leu	Thr	Val	Gln	Ala	Lys	
			530						535						540	
Gln	Leu	Leu	His	Gly	Ile	Val	Gln	Gln	Gln	Asn	Asn	Met	Leu	Arg	Ala	
			545						550						555	
Ile	Ala	Ala	Gln	Gln	Glu	Leu	Leu	Arg	Leu	Ser	Val	Trp	Gly	Ile	Arg	
			560						565						570	
Gln	Leu	Arg	Ala	Arg	Leu	Leu	Ala	Ile	Glu	Thr	Tyr	Leu	Arg	Asp	Gln	
			575						580						585	
Gln	Leu	Leu	Gly	Leu	Trp	Gly	Cys	Ala	Gly	Lys	Leu	Xaa	Cys	Tyr	Thr	
			590						595						600	
Asn	Val	Pro	Trp	Asn	Asn	Thr	Trp	Thr	Asn	Lys	Ser	Asx	Thr	Glu	Leu	
			605						610						615	
Glu	His	Ile	Trp	Glu	Asn	Leu	Thr	Trp	Gln	Glu	Trp	Asp	Lys	Leu	Val	
			620						625						630	
Asp	Asn	Tyr	Thr	Xaa</												

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Gln Gln Glu Val Asn Glu Lys Lys Leu Leu Glu Leu Asp Lys Trp Ala
 675 680 685
 Asp Leu Trp Ser Trp Phe Asp Ile Thr Lys Trp Leu Trp Tyr Ile Lys
 690 695 700
 Ile Ala Ile Met Ile Val Xaa Ala Leu Ile Gly Leu Arg Val Val Met
 705 710 715 720
 Val Val Leu Ser Leu Val Lys Asn Ile Arg Lys Gly Tyr Gln Pro Leu
 725 730 735
 Ser Leu Gln Thr Pro Ser Pro His Pro Thr Glu Pro Gly Thr Pro Glu
 740 745 750
 Lys Thr Gly Gly Glu Gly Gly Glu Gly Asp Lys His Lys Trp Thr Pro
 755 760 765
 Leu Pro Pro Gly Phe Leu His Leu Leu Tyr Leu Asp Leu Arg Thr Ile
 770 775 780
 Ile Leu Trp Ser Tyr His Leu Leu Ser Ser Leu Val Ser Arg Ile Gln
 785 790 795 800
 Arg Ile Leu Ser Leu Leu Gly Leu Gly Leu Arg Ile Leu Gly Gln Lys
 805 810 815
 Thr Phe Glu Ala Cys Lys Thr Leu Lys Ala Thr Ala Gln Tyr Trp Leu
 820 825 830
 Gln Glu Leu Gln Arg Ser Xaa Thr Asn Leu Leu Asp Thr Val Ala Val
 835 840 845
 Ala Val Ala Asn Trp Thr Asp Ser Ile Ile Leu Gly Val Gln Arg Phe
 850 855 860
 Gly Arg Gly Ile Leu Asn Ile Pro Arg Arg Ile Arg Gln Gly Leu Glu
 865 870 875 880
 Leu Ser Leu Leu

<210> SEQ ID NO 353
 <211> LENGTH: 858
 <212> TYPE: PRT
 <213> ORGANISM: Simian immunodeficiency virus
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (99)..(99)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (482)..(482)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (538)..(538)
 <223> OTHER INFORMATION: Xaa can be any naturally occurring amino acid
 <400> SEQUENCE: 353

Met Arg Lys Pro Ile His Ile Ile Trp Gly Leu Ala Leu Leu Ile Gln
 1 5 10 15
 Phe Ile Glu Lys Gly Thr Asn Glu Asp Tyr Val Thr Val Phe Tyr Gly
 20 25 30
 Val Pro Val Trp Arg Asn Ala Thr Pro Thr Leu Phe Cys Ala Thr Asn
 35 40 45
 Ala Ser Met Thr Ser Thr Glu Val His Asn Val Trp Ala Thr Thr Ser
 50 55 60
 Cys Val Pro Ile Asp Pro Asp Pro Ile Val Val Arg Leu Asn Thr Ser
 65 70 75 80
 Val Trp Phe Asn Ala Tyr Lys Asn Tyr Met Val Glu Ser Met Thr Glu
 85 90 95

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Asp	Met	Xaa	Gln	Leu	Phe	Gln	Gln	Ser	His	Lys	Pro	Cys	Val	Lys	Leu
			100					105					110		
Thr	Pro	Met	Cys	Ile	Lys	Met	Asn	Cys	Thr	Gly	Tyr	Asn	Gly	Thr	Pro
		115					120					125			
Thr	Thr	Pro	Ser	Thr	Thr	Thr	Ser	Thr	Val	Thr	Pro	Lys	Thr	Thr	Thr
	130					135					140				
Pro	Ile	Val	Asp	Gly	Met	Lys	Leu	Gln	Glu	Cys	Asn	Phe	Asn	Gln	Ser
145					150					155					160
Thr	Gly	Phe	Lys	Asp	Lys	Lys	Gln	Lys	Met	Lys	Ala	Ile	Phe	Tyr	Lys
			165						170					175	
Gly	Asp	Leu	Met	Lys	Cys	Gln	Asp	Asn	Asn	Glu	Thr	Asn	Cys	Tyr	Tyr
		180						185					190		
Leu	Trp	His	Cys	Asn	Thr	Thr	Thr	Ile	Thr	Gln	Ser	Cys	Glu	Lys	Ser
		195					200					205			
Thr	Phe	Glu	Pro	Ile	Pro	Ile	His	Tyr	Cys	Ala	Pro	Ala	Gly	Tyr	Ala
	210					215					220				
Ile	Leu	Arg	Cys	Glu	Asp	Glu	Asp	Phe	Thr	Gly	Val	Gly	Met	Cys	Lys
225					230					235					240
Asn	Val	Ser	Val	Val	His	Cys	Thr	His	Gly	Ile	Ser	Pro	Met	Val	Ala
			245						250					255	
Thr	Trp	Leu	Leu	Leu	Asn	Gly	Thr	Tyr	Gln	Thr	Asn	Thr	Ser	Val	Val
		260						265					270		
Met	Asn	Gly	Arg	Lys	Asn	Glu	Ser	Val	Leu	Val	Arg	Phe	Gly	Lys	Glu
		275					280					285			
Phe	Glu	Asn	Leu	Thr	Ile	Thr	Cys	Ile	Arg	Pro	Gly	Asn	Arg	Thr	Val
	290					295					300				
Arg	Asn	Leu	Gln	Ile	Gly	Pro	Gly	Met	Thr	Phe	Tyr	Asn	Val	Glu	Ile
305					310					315					320
Ala	Thr	Gly	Asp	Thr	Arg	Lys	Ala	Phe	Cys	Thr	Val	Asn	Lys	Thr	Leu
			325						330					335	
Trp	Glu	Gln	Ala	Arg	Asn	Lys	Thr	Glu	His	Val	Leu	Ala	Glu	His	Trp
		340						345					350		
Lys	Lys	Val	Asp	Asn	Lys	Thr	Asn	Ala	Lys	Thr	Ile	Trp	Thr	Phe	Gln
		355					360					365			
Asp	Gly	Asp	Pro	Glu	Val	Lys	Val	His	Trp	Phe	Asn	Cys	Gln	Gly	Glu
	370					375					380				
Phe	Phe	Tyr	Cys	Asp	Ile	Thr	Pro	Trp	Phe	Asn	Ala	Thr	Tyr	Thr	Gly
385					390					395					400
Asn	Leu	Ile	Thr	Asn	Gly	Ala	Leu	Ile	Ala	His	Cys	Arg	Ile	Lys	Gln
			405						410					415	
Ile	Val	Asn	His	Trp	Gly	Ile	Val	Ser	Lys	Gly	Ile	Tyr	Leu	Ala	Pro
		420						425					430		
Arg	Arg	Gly	Asn	Val	Ser	Cys	Thr	Ser	Ser	Ile	Thr	Gly	Ile	Met	Leu
		435					440					445			
Glu	Gly	Gln	Ile	Tyr	Asn	Glu	Thr	Val	Lys	Val	Ser	Pro	Ala	Ala	Arg
	450					455					460				
Val	Ala	Asp	Gln	Trp	Arg	Ala	Glu	Leu	Ser	Arg	Tyr	Gln	Val	Val	Glu
465					470					475					480
Ile	Xaa	Pro	Leu	Ser	Val	Ala	Pro	Thr	Thr	Gly	Lys	Arg	Pro	Glu	Ile
			485						490					495	
Lys	Gln	His	Ser	Arg	Gln	Lys	Arg	Gly	Ile	Gly	Ile	Gly	Leu	Phe	Phe
		500						505					510		
Leu	Gly	Leu	Leu	Ser	Ala	Ala	Gly	Ser	Thr	Met	Gly	Ala	Ala	Ser	Ile

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515	520	525
Ala Leu Thr Ala Gln Thr Arg	Asn Leu Xaa His Gly	Ile Val Gln Gln
530	535	540
Gln Ala Asn Leu Leu Gln Ala Ile Glu Thr Gln Gln His Leu Leu Gln		
545	550	555 560
Leu Ser Val Trp Gly Val Lys Gln Leu Gln Ala Arg Met Leu Ala Val		
	565 570	575
Glu Lys Tyr Leu Arg Asp Gln Gln Leu Leu Ser Leu Trp Gly Cys Ala		
	580 585	590
Asp Lys Val Thr Cys His Thr Thr Val Pro Trp Asn Asn Ser Trp Val		
	595 600	605
Asn Phe Thr Gln Thr Cys Ala Lys Asn Ser Ser Asp Ile Gln Cys Ile		
	610 615	620
Trp Glu Asn Met Thr Trp Gln Glu Trp Asp Arg Leu Val Gln Asn Ser		
	625 630	635 640
Thr Gly Gln Ile Tyr Asn Ile Leu Gln Ile Ala His Glu Gln Gln Glu		
	645 650	655
Arg Asn Lys Lys Glu Leu Tyr Glu Leu Asp Lys Trp Ser Ser Leu Trp		
	660 665	670
Asn Trp Phe Asp Ile Thr Gln Trp Leu Trp Tyr Ile Lys Ile Phe Ile		
	675 680	685
Met Ile Val Gly Ala Ile Val Gly Leu Arg Ile Leu Leu Val Leu Val		
	690 695	700
Ser Cys Leu Arg Lys Val Arg Gln Gly Tyr His Pro Leu Ser Phe Gln		
	705 710	715 720
Ile Pro Thr Gln Asn Gln Gln Asp Pro Glu Gln Pro Glu Glu Ile Arg		
	725 730	735
Glu Glu Gly Gly Arg Lys Asp Arg Ile Arg Trp Arg Ala Leu Gln His		
	740 745	750
Gly Phe Phe Ala Leu Leu Trp Val Asp Leu Thr Ser Ile Ile Gln Trp		
	755 760	765
Ile Tyr Gln Ile Cys Arg Thr Cys Leu Leu Asn Leu Trp Ala Val Leu		
	770 775	780
Gln His Leu Cys Arg Ile Thr Phe Arg Leu Cys Asn His Leu Glu Asn		
	785 790	795 800
Asn Leu Ser Thr Leu Trp Thr Ile Ile Arg Thr Glu Ile Ile Lys Asn		
	805 810	815
Ile Asp Arg Leu Ala Ile Trp Val Gly Glu Lys Thr Asp Ser Ile Leu		
	820 825	830
Leu Ala Leu Gln Thr Ile Val Arg Ile Ile Arg Glu Val Pro Arg Arg		
	835 840	845
Ile Arg Gln Gly Leu Glu Ile Ala Leu Asn		
	850 855	

<210> SEQ ID NO 354

<211> LENGTH: 854

<212> TYPE: PRT

<213> ORGANISM: Simian immunodeficiency virus

<400> SEQUENCE: 354

Met Lys Val Met Glu Lys Lys Lys Arg Asp Trp Asn Ser Leu Ser Ile
1 5 10 15

Ile Thr Ile Ile Thr Ile Ile Leu Leu Thr Pro Cys Leu Thr Ser Glu
20 25 30

Leu	Trp	Val	Thr	Val	Tyr	Tyr	Gly	Val	Pro	Val	Trp	His	Asp	Ala	Asp
35							40					45			
Pro	Val	Leu	Phe	Cys	Ala	Ser	Asp	Ala	Lys	Ala	His	Ser	Thr	Glu	Ala
50						55					60				
His	Asn	Ile	Trp	Ala	Thr	Gln	Ala	Cys	Val	Pro	Thr	Asp	Pro	Ser	Pro
65					70					75					80
Gln	Glu	Val	Phe	Leu	Pro	Asn	Val	Ile	Glu	Ser	Phe	Asn	Met	Trp	Lys
				85					90					95	
Asn	Asn	Met	Val	Asp	Gln	Met	His	Glu	Asp	Ile	Ile	Ser	Leu	Trp	Asp
			100					105					110		
Gln	Ser	Leu	Lys	Pro	Cys	Val	Lys	Leu	Thr	Pro	Leu	Cys	Val	Thr	Leu
		115					120					125			
Gln	Cys	Ser	Lys	Ala	Asn	Phe	Ser	Gln	Ala	Lys	Asn	Leu	Thr	Asn	Gln
		130				135					140				
Thr	Ser	Ser	Pro	Pro	Leu	Glu	Met	Lys	Asn	Cys	Ser	Phe	Asn	Val	Thr
145					150					155					160
Thr	Glu	Leu	Arg	Asp	Lys	Lys	Lys	Gln	Val	Tyr	Ser	Leu	Phe	Tyr	Val
				165					170					175	
Glu	Asp	Val	Val	Asn	Leu	Gly	Asn	Glu	Asn	Asn	Thr	Tyr	Arg	Ile	Ile
			180					185					190		
Asn	Cys	Asn	Thr	Thr	Ala	Ile	Thr	Gln	Ala	Cys	Pro	Lys	Thr	Ser	Phe
		195					200					205			
Glu	Pro	Ile	Pro	Ile	His	Tyr	Cys	Ala	Pro	Ala	Gly	Phe	Ala	Ile	Leu
		210				215					220				
Lys	Cys	Asn	Asp	Lys	Asp	Phe	Ser	Gly	Lys	Gly	Lys	Cys	Thr	Asn	Val
		225			230					235				240	
Ser	Thr	Val	His	Cys	Thr	His	Gly	Ile	Lys	Pro	Val	Val	Thr	Thr	Gln
				245					250					255	
Leu	Leu	Ile	Asn	Gly	Ser	Leu	Ala	Glu	Gly	Asn	Ile	Thr	Val	Arg	Val
			260					265					270		
Glu	Asn	Lys	Ser	Lys	Asn	Thr	Asp	Val	Trp	Ile	Val	Gln	Leu	Val	Glu
		275					280					285			
Ala	Val	Ser	Leu	Asn	Cys	His	Arg	Pro	Gly	Asn	Asn	Thr	Arg	Gly	Glu
		290				295					300				
Val	Gln	Ile	Gly	Pro	Gly	Met	Thr	Phe	Tyr	Asn	Ile	Glu	Asn	Val	Val
		305			310					315					320
Gly	Asp	Thr	Arg	Ser	Ala	Tyr	Cys	Lys	Ile	Asn	Gly	Thr	Thr	Trp	Asn
				325					330					335	
Arg	Thr	Val	Glu	Glu	Val	Lys	Lys	Ala	Leu	Ala	Thr	Ser	Ser	Asn	Arg
			340					345					350		
Thr	Ala	Ala	Asn	Ile	Thr	Leu	Asn	Arg	Ala	Ser	Gly	Gly	Asp	Pro	Glu
		355					360					365			
Val	Thr	His	His	Met	Phe	Asn	Cys	Gly	Gly	Glu	Phe	Phe	Tyr	Cys	Asn
		370				375					380				
Thr	Ser	Gln	Ile	Phe	Thr	Asp	Asn	Ile							

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450	455	460
Arg Ser Glu Leu Tyr Lys Tyr Lys Val Val Arg Ile Glu Pro Leu Ser		
465	470	475 480
Val Ala Pro Thr Lys Ala Arg Arg His Thr Val Ala Arg Gln Lys Asp		
	485	490 495
Arg Gln Lys Arg Ala Ala Phe Gly Leu Gly Ala Leu Phe Leu Gly Phe		
	500	505 510
Leu Gly Ala Ala Gly Ser Thr Met Gly Ala Ala Ala Val Thr Leu Thr		
	515	520 525
Val Gln Ala Arg Gln Leu Leu Ser Gly Ile Val Gln Gln Gln Asn Asn		
	530	535 540
Leu Leu Lys Ala Ile Glu Ala Gln Gln His Leu Leu Gln Leu Ser Ile		
545	550	555 560
Trp Gly Val Lys Gln Leu Gln Ala Arg Leu Leu Ala Val Glu Arg Tyr		
	565	570 575
Leu Gln Asp Gln Gln Ile Leu Gly Leu Trp Gly Cys Ser Gly Lys Ala		
	580	585 590
Val Cys Tyr Thr Thr Val Pro Trp Asn Asn Ser Trp Pro Gly Ser Asn		
	595	600 605
Ser Thr Asp Asp Ile Trp Gly Asn Leu Thr Trp Gln Gln Trp Asp Lys		
	610	615 620
Leu Val Ser Asn Tyr Thr Gly Lys Ile Phe Gly Leu Leu Glu Glu Ala		
625	630	635 640
Gln Ser Gln Gln Glu Lys Asn Glu Arg Asp Leu Leu Glu Leu Asp Gln		
	645	650 655
Trp Ala Ser Leu Trp Asn Trp Phe Asp Ile Thr Lys Trp Leu Trp Tyr		
	660	665 670
Ile Lys Ile Phe Leu Met Ala Val Gly Gly Ile Ile Gly Leu Arg Ile		
	675	680 685
Ile Met Thr Val Phe Ser Val Val Arg Arg Val Arg Gln Gly Tyr Ser		
	690	695 700
Pro Leu Ser Leu Gln Thr Leu Ile Pro Val Gln Arg Glu Gln Gly Arg		
705	710	715 720
Leu Gly Glu Ile Asp Glu Gly Gly Gly Glu Gln Asp Arg Ser Arg Ser		
	725	730 735
Val Arg Leu Val Glu Gly Cys Leu Pro Leu Ile Trp Asp Asp Leu Arg		
	740	745 750
Asn Leu Gly Ile Trp Ser Tyr Gln Ser Leu Thr Ser Leu Ala Cys Asn		
	755	760 765
Val Trp Arg Gln Leu Lys Thr Leu Gly His Leu Ile Leu His Ser Leu		
	770	775 780
Arg Leu Leu Arg Glu Arg Leu Cys Leu Leu Gly Gly Ile Ile Gln Tyr		
785	790	795 800
Trp Gly Lys Glu Leu Lys Ile Ser Ala Ile Ser Leu Leu Asp Ala Thr		
	805	810 815
Ala Ile Ala Val Ala Glu Gly Thr Asp Arg Ile Ile Glu Ala Phe Gln		
	820	825 830
Val Thr Leu Arg Ile Ile Arg Asn Ile Pro Arg Arg Ile Arg Gln Gly		
	835	840 845
Leu Glu Arg Ala Leu Leu		
850		

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<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: C K-rich region

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<400> SEQUENCE: 355

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Arg Asp Lys Lys Gln Lys
1          5

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<210> SEQ ID NO 356
<211> LENGTH: 4
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: V3 loop protein backbone residues

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<400> SEQUENCE: 356

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Gly Asp Ile Arg
1

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<210> SEQ ID NO 357
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Linker

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<400> SEQUENCE: 357

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Gly Gly Gly Gly Ser
1          5

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<210> SEQ ID NO 358
<211> LENGTH: 4
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Cleavage site

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<400> SEQUENCE: 358

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Arg Glu Lys Arg
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<210> SEQ ID NO 359
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Furin inducible R6 site

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<400> SEQUENCE: 359

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```

Arg Arg Arg Arg Arg Arg
1          5

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What is claimed is:

1. An engineered or non-naturally occurring molecule that binds to a Human Immunodeficiency Virus (HIV-1) broadly neutralizing antibody (bnAb),

wherein the molecule comprises the amino acid sequence of SEQ ID NO: 28.

2. The molecule of claim 1,

wherein the HIV-1 bnAb comprises three complementarity determining regions (CDRs) of a heavy chain variable domain, a first CDR of which comprising the amino acid sequence of SEQ ID NO: 33, a second CDR of which comprising the amino acid sequence of SEQ

ID NO: 34, and a third CDR of which comprising the amino acid sequence of SEQ ID NO: 35, and three CDRs of a light chain variable domain, a first CDR of which comprising the amino acid sequence of SEQ ID NO: 42, a second CDR of which comprising the amino acid sequence of 43, and a third CDR of which comprising the amino acid sequence of SEQ ID NO: 44.

3. The molecule of claim 1, wherein the HIV-1 bnAb comprises three complementarity determining regions (CDRs) of a heavy chain variable domain, a first CDR of which comprising the amino acid sequence of SEQ ID NO: 50, a second CDR of which comprising the amino acid

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sequence of SEQ ID NO: 51, and a third CDR of which comprising the amino acid sequence of SEQ ID NO: 52, and three CDRs of a light chain variable domain, a first CDR of which comprising the amino acid sequence of SEQ ID NO: 59, a second CDR of which comprising the amino acid sequence of SEQ ID NO: 60, and a third CDR of which comprising the amino acid sequence of SEQ ID NO: 61.

4. An engineered or non-naturally occurring molecule that binds to a Human Immunodeficiency Virus (HIV-1) broadly neutralizing antibody (bnAb), wherein the bnAb binds to the V2 apex region of HIV-1 envelope, wherein the molecule comprises an amino acid sequence comprising SEQ ID NO: 28 or an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 28.

5. A nucleic acid encoding the molecule of claim 1.

6. A vector comprising a regulatory element operable in a eukaryotic cell operably linked to the nucleic acid of claim 5, wherein the vector comprises a viral vector.

7. A method of eliciting an immune response in a mammal or stimulating an HIV-1 broadly neutralizing antibody (bnAb) comprising administering the molecule of claim 1.

8. The method of claim 7, wherein the mammal is a mouse, non-human primate, or a human.

9. The engineered or non-naturally occurring molecule of claim 4, wherein the HIV bnAb comprises (i) three complementarity determining regions (CDRs) of a heavy chain variable domain, a first CDR of which comprises an amino acid sequence comprising SEQ ID NO: 33 or an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 33, a second CDR of which comprises an amino acid sequence comprising SEQ ID NO: 34 or an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 34, and a third CDR of which comprises an amino acid sequence comprising SEQ ID NO: 35 or an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 35; and, (ii) three CDRs of a light chain variable domain, a first CDR of which comprises an amino acid sequence comprising SEQ ID NO: 42 or an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 42, a second CDR of which comprises an amino acid sequence comprising SEQ ID NO: 43 or an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 43, and a third of which comprises an amino acid sequence comprising SEQ ID NO: 44 or an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 44.

10. The molecule of claim 1, wherein the HIV-1 bnAb comprises three complementarity determining regions (CDRs) of a heavy chain variable domain, a first CDR of which comprising the amino acid sequence of SEQ ID NO: 50 or an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 50, a second CDR of which

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comprising the amino acid sequence of SEQ ID NO: 51 or an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 51, and a third CDR of which comprising the amino acid sequence of SEQ ID NO: 52 or an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 52, and three CDRs of a light chain variable domain, a first CDR of which comprising the amino acid sequence of SEQ ID NO: 59 or an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 59, a second CDR of which comprising the amino acid sequence of SEQ ID NO: 60 or an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 60, and a third CDR of which comprising the amino acid sequence of SEQ ID NO: 61 or an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 61.

11. The method of claim 7, wherein the molecule is administered with an adjuvant.

12. The method of claim 11,

wherein the adjuvant comprises a lecithin or

wherein the adjuvant is an immune stimulating complex-forming adjuvant or a lipid-based vaccine adjuvant.

13. The method of claim 12, wherein the lecithin is (a) combined with an acrylic polymer, (b) in a coated oil droplet in an oil-in-water emulsion or (c) in an acrylic polymer in an oil-in-water emulsion.

14. A method of eliciting an immune response in a mammal or stimulating an HIV-1 broadly neutralizing antibody (bnAb) comprising administering the molecule of claim 1,

wherein the molecule is administered in a liposome or in a nanoparticle or

wherein the molecule is fixed, wherein the molecule is fixed in glutaraldehyde, or

wherein the molecule is quenched with glycine.

15. A method of identifying an HIV-1 binder which comprises contacting a candidate binder with the molecule of claim 1 and identifying a candidate binder that binds to the molecule.

16. A method of identifying a broadly neutralizing antibody (bnAb) against HIV-1 which comprises contacting a candidate antibody with the molecule of claim 1 and identifying an antibody that binds to the molecule as a bnAb, wherein the bnAb is a germline or germline reverted bnAb.

17. The vector of claim 6, wherein the viral vector is selected from the group comprising adenoviruses, adeno-associated viruses (AAV), alphaviruses, herpesviruses, retroviruses, and poxviruses.

18. The vector of claim 17, wherein the viral vector is AAV.

* * * * *